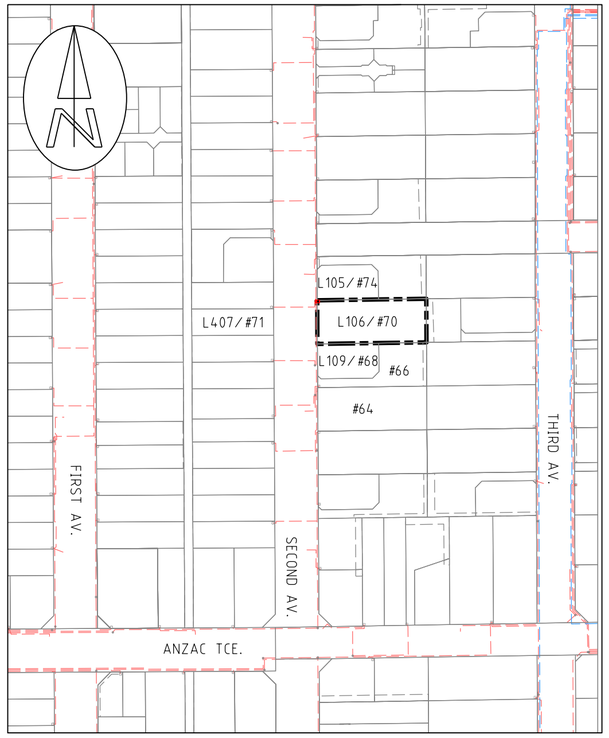
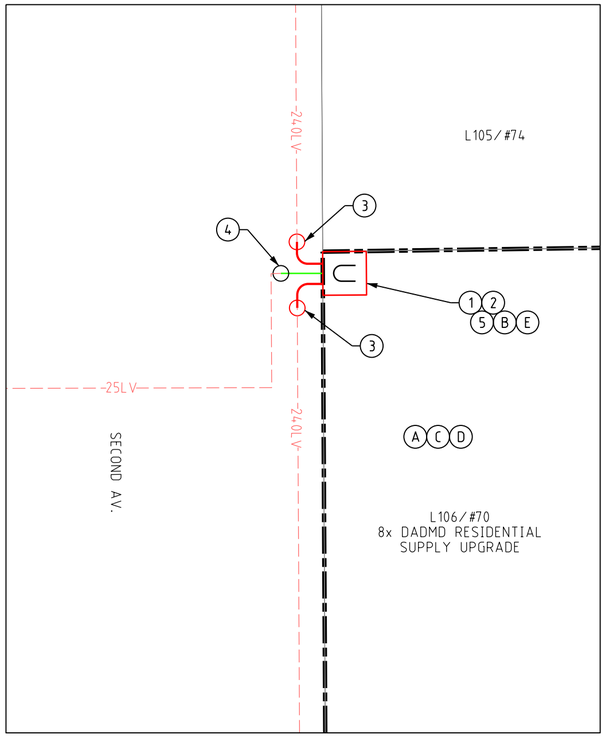


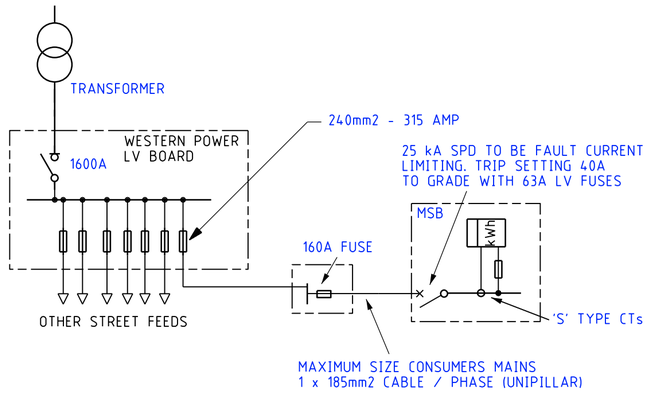
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

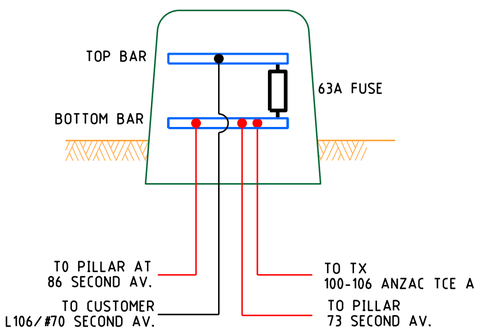
1:100 2 1 0 2 4 6 8



NOTE: TRANSFORMER IS OVERLOADED. WP TO ASSESS THE NETWORK.

APPROVED FOR CONSTRUCTION

UNI-PILLAR L106/#70
CONNECTION DETAILS

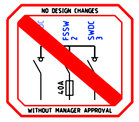


CRITICAL INFRASTRUCTURES
IN VICINITY OF WORKS

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED





CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR TOP BAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING 25LV CABLE AND CUSTOMER MAINS FROM EXISTING UNI-PILLAR. REMOVE EXISTING MINI-PILLAR.

2. INSTALL NEW UNI PILLAR ON NORTH CORNER OF L106/#70 SECOND AV, AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU11 x 1
DCSH: U9

3. INSTALL NEW 2x240LV CABLE FROM NEW UNI-PILLAR AND STRAIGHT JOINT TO EXSITING 240 LV CABLE.
CU: CN45 x 10m, CN73_LVMAINS x 2, LU2 x 2

4. INSTALL NEW 25LV CABLE FROM NEW UNI-PILLAR AND STRAIGHT JOINT TO EXISTING 240 LV CABLE
CU: CN42 x 5m, CN73_LV25 x 1, LU8 x 1, CN54 x 3

5. AT NEW UNI-PILLAR (L106/#70)
-TERMINATE 240LV STREET MAINS CABLE.
-TERMINATE EXISTING 25LV CABLE TO BACK BLOCKS.
-TERMINATE CUSTOMER'S CABLE AT TOP BAR VIA 63A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR.
CU: LU7 x 2, LU36_63A x 1
DCSH: U9

6. ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT ADN LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
42W CFL; 80W MV; 125W MV; 70W HPS; 150W HPS; 250W HPS; 250W HPS AERODROME; 70W MH; 150W MH; 250W MH
1L LED; 2L LED; 3L LED; 4L LED; 5L LED; 6L LED
1E LED; 2E LED; 3E LED; 4E LED; 5E LED; 6E LED
LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

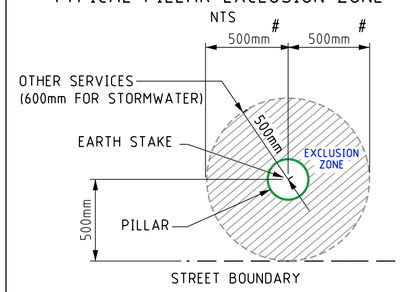
PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
SUDI Serviced Uncleared De-energised Lot

TRANSFORMER (T) EXISTING SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CONSULTANT CONTACT DETAILS

COMPANY NAME: SWAN CONSTRUCTION
CONTACT: evan.rocke@swangroup.com.au
TEL: 0450601362

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 868494698
HAZARD MANAGEMENT REGISTER EDM#: 68494576

8x DADMD RESIDENTIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

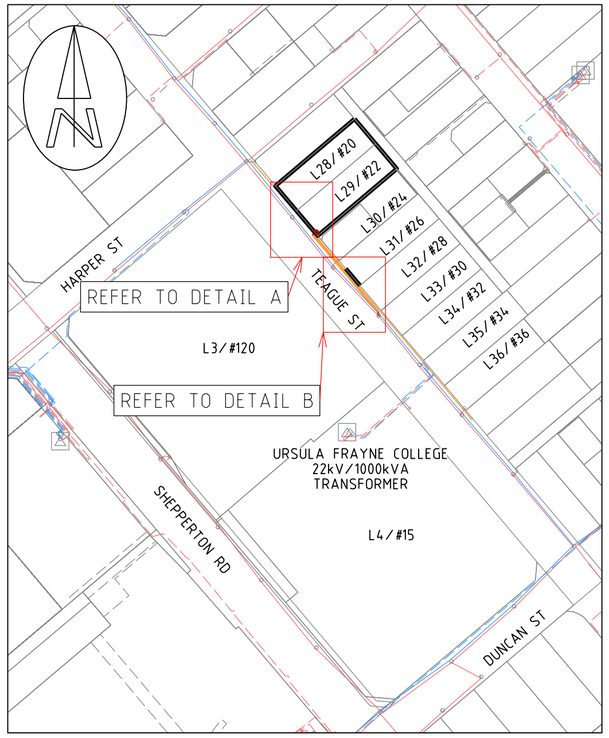
70 SECOND AVENUE- BASSENDEAN WA 6054

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 | SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 53' 51" S, Long. - 115° 56' 56" E

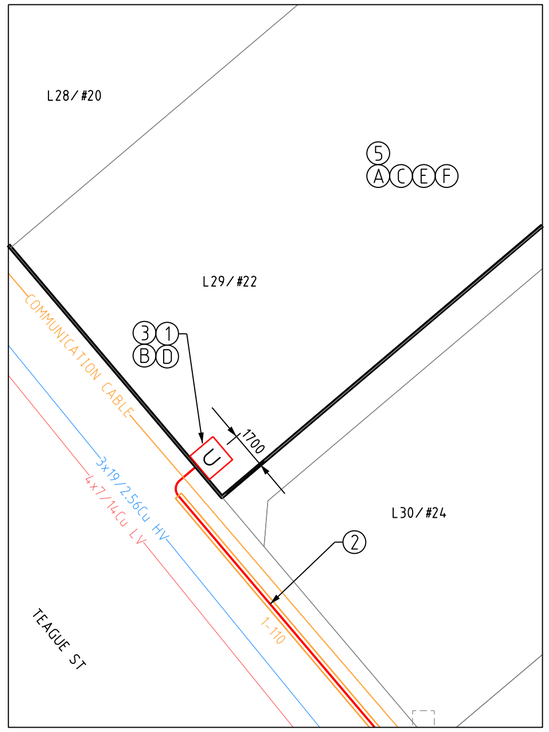
westernpower

DRAWING NUMBER
MP204744
SHEET 01 OF 01
REV: A | ORIGINAL SIZE - A2



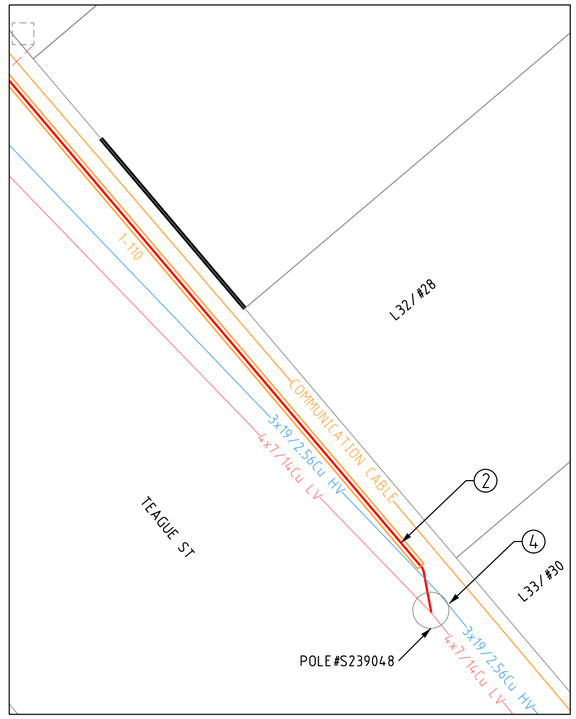
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



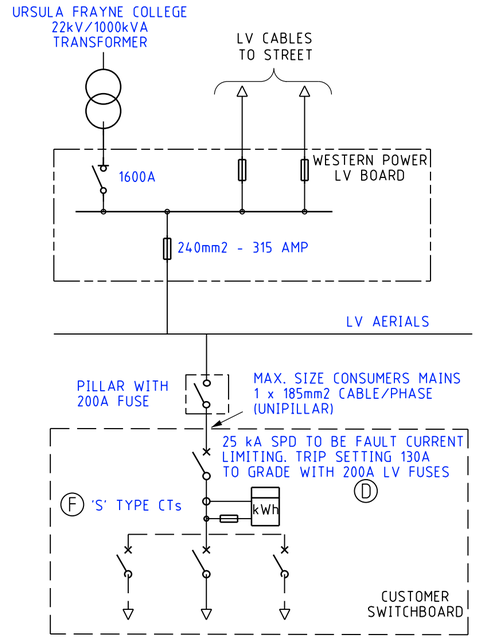
DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16



SIMPLIFIED SINGLE LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L29/#22 TEAGUE ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11x1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S239048 AT #30 TEAGUE ST VIA POLYPIPE.
   CU: CN45x65m, CN54x4, CN73_LVMAINSx3
   DCSH: U16-1
3. AT UNI-PILLAR (L29/#22 TEAGUE ST)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_200Ax1
   DCSH: U9
4. AT POLE#S239048
   - TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
   CU: LU6_240O x 1, LU6_240U x 1
   DCSH: U16-1
5. INSTALL CT METER.
   CU: LM1_M x 1
6. ENVIRONMENTAL FILTER CHECKED.
   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
   IN AREAS WHERE THIS IS A RISK YOU NEED TO:
   BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
   IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED
   REFER TO environmf@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

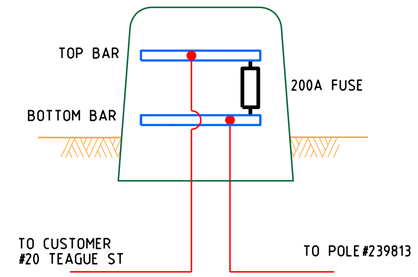
C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

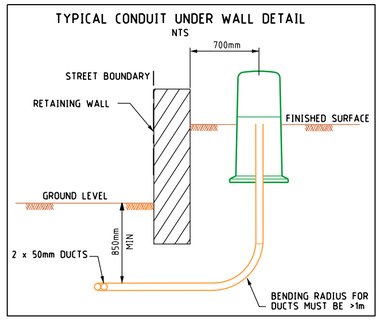
E. CUSTOMER TO OBTAIN WORK ZONE PERMIT FROM THE TOWN OF VICTORIA PARK AT NO COST TO WESTERN POWER.

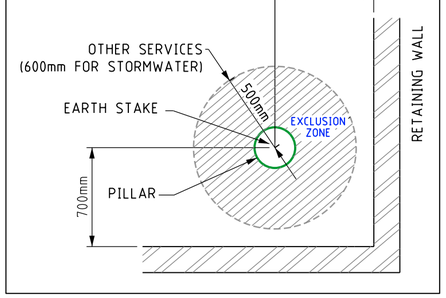
FOLLOW PERMIT CONDITIONS FROM THE TOWN OF VICTORIA PARK.

F. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.



NEW UNI-PILLAR L29/#22 TEAGUE ST.





GAS PIPE
IN VICINITY OF WORKS

COMMUNICATION CABLES
IN VICINITY OF WORKS

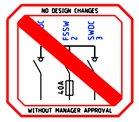
W.A.P.C. REFERENCE No. 162326

CUSTOMER CONTACT DETAILS
COMPANY NAME: DB SURVEYS
CONTACT: DAVID BUSSELL TEL: 0419 930 950

ISSUED FOR CONSTRUCTION

LEGEND
- RETAINING WALL
- LV CABLES
  - CN45 - 240mm2 (1x3c)
  - EXISTING LV CABLES (SIZE AND TYPE INDICATED)
- PILLARS / PITS / ETC
  - MINI (WORKING END)
  - CONDUIT / POLY PIPE
  - U UNIVERSAL
  - UTILITIES PIT
- AERIAL CONDUCTORS AND POLES
  - NEW POLE (ANY TYPE AS INDICATED)
  - EXISTING POLE (ANY TYPE AS INDICATED)
  - HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

EXCAVATION WARNING:
EXCAVATION TO BE CONDUCTED IN CONJUNCTION WITH CIVIL ENGINEERING ADVICE CONTAINED IN EDM #68762030

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

SHE ASSESSMENT REPORT EDM#: 68745457
HAZARD MANAGEMENT REGISTER EDM#: 68762030
FILE REFERENCE DETAILS

415V/130A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV

20 TEAGUE ST, VICTORIA PARK WA 6100

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 06/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31° 58' 17" S
Long. - 115° 53' 55" E

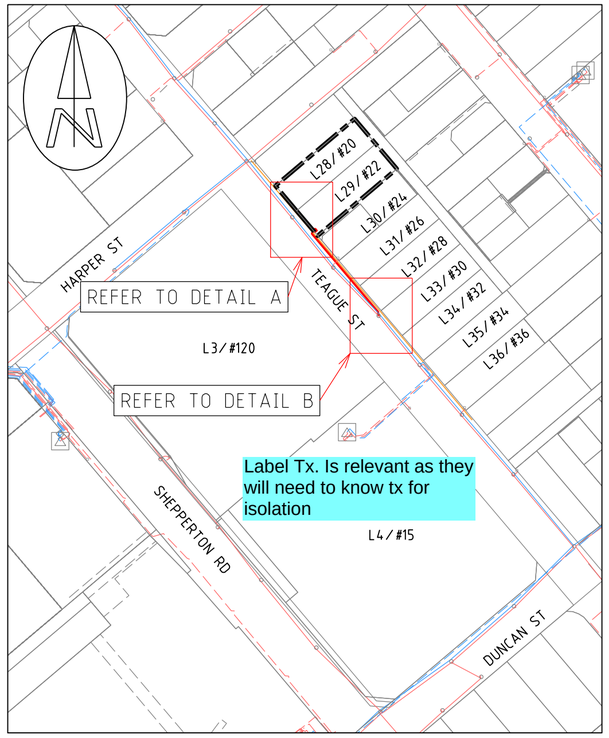
westernpower

DRAWING NUMBER

MP206362

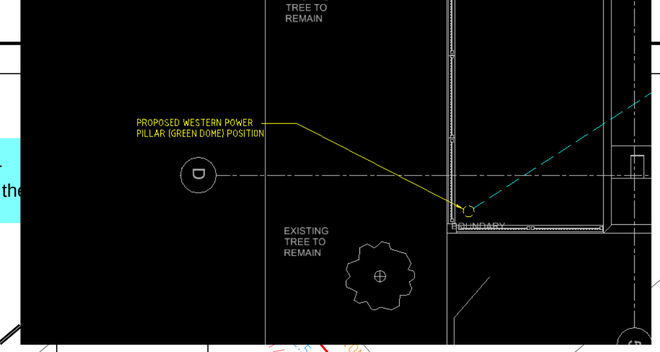
SHEET 01 OF 01

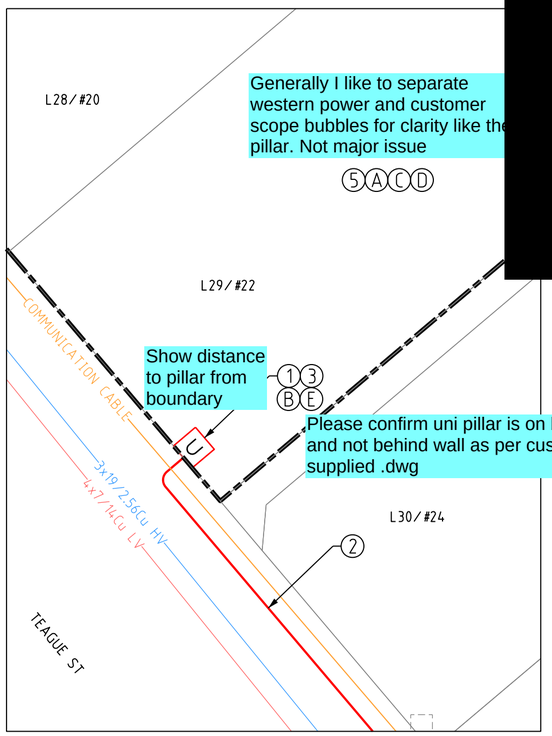
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

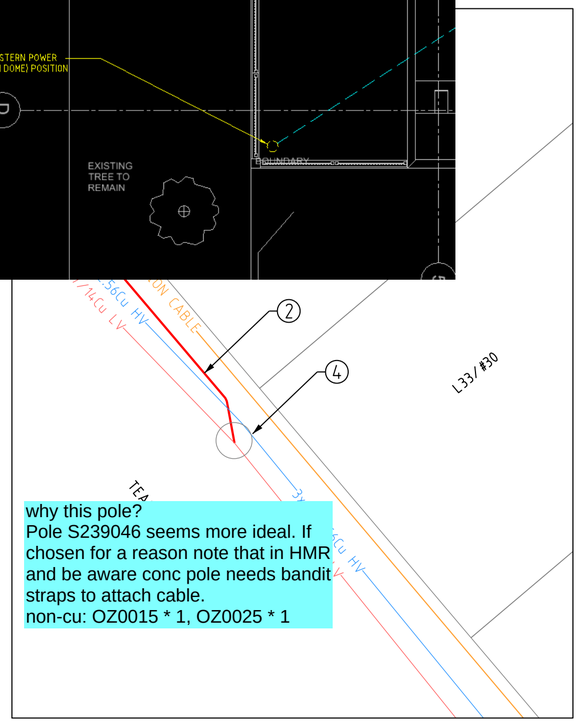
1:2000 40 20 0 40 80 120 160





DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL A

1:200 4 2 0 4 8 12 16

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

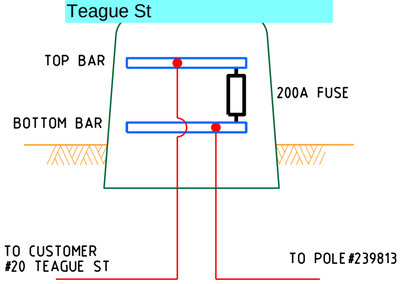
HV CABLES
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

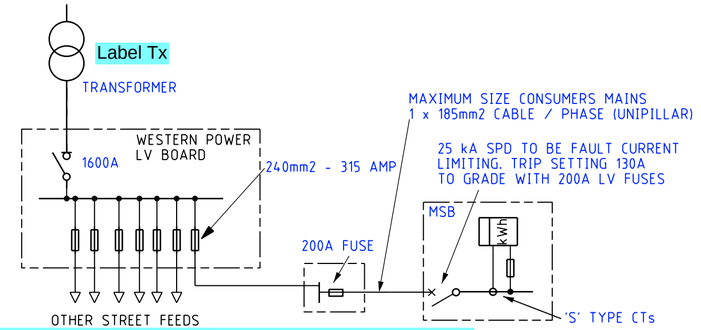
LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UMS PIT
U UNIVERSAL
UTILITIES PIT
MINI
SP BGSP PIT
L LINK
100/200AMP CUTOUT
CABLE MARKER U/G
CONDUIT / POLY PIPE
SUD Serviced Unclearad De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

Label e.g. new uni pillar L29/#22 Teague St





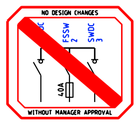
Separate into individual services e.g., communication assets and another for gas pipelines

CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

APPROVED FOR CONSTRUCTION

LVDesign Scaling is not per your screenshot 31% max. Also proposed load was added before scaling took place. First scale the network to resemble current config. Then add proposed load.





WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L29/#22 TEAGUE ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11x1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO Pole# S239048 3 AT #30 TEAGUE ST.
   CU: CN45x65m, CN54x26, CN73_LVMAINSx3
   DCSH: U16-1
   why so many slabs?
   1 for each end of cable and 1 for cable drum
3. AT UNI-PILLAR (L29/#22 TEAGUE ST)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   Good stuff
   CU: LU7_240x1, LU36_200Ax1
   DCSH: U9
4. AT POLE#239048
   - TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
   CU: LU6_240 x 1, LU6_240U x 1
   DCSH: U16-1
5. INSTALL CT METER.
   CU: LM1_M x 1
   Remove Scope item F.
   Only WP can terminate to Uni Pillar top bar and it's cover under scope item 3.
6. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.
   Follow permit conditions from the town of victoria park
   Acid sulfate standard note

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

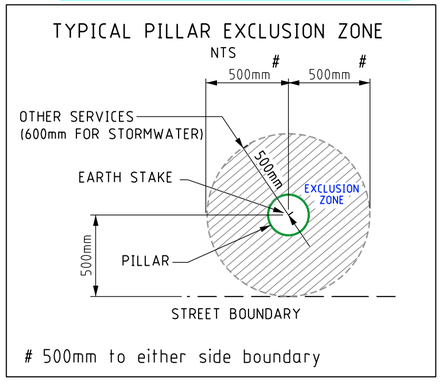
C. THIS YOU THIS — Scope item C not required. Covered under sticker and has historically created confusion with customers

D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

F. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR TOP BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 185mm2/PHASE.
new scope item. Customer to obtain Work Zone Permit from the town of victoria park at no cost to western power

Update if on retaining pole as on cust .dwg



CUSTOMER CONTACT DETAILS
COMPANY NAME: DB SURVEYS
CONTACT: DAVID BUSSELL TEL: 0419 930 950

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68745457
HAZARD MANAGEMENT REGISTER EDM#: edm reference for HMR needed

130A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 415V

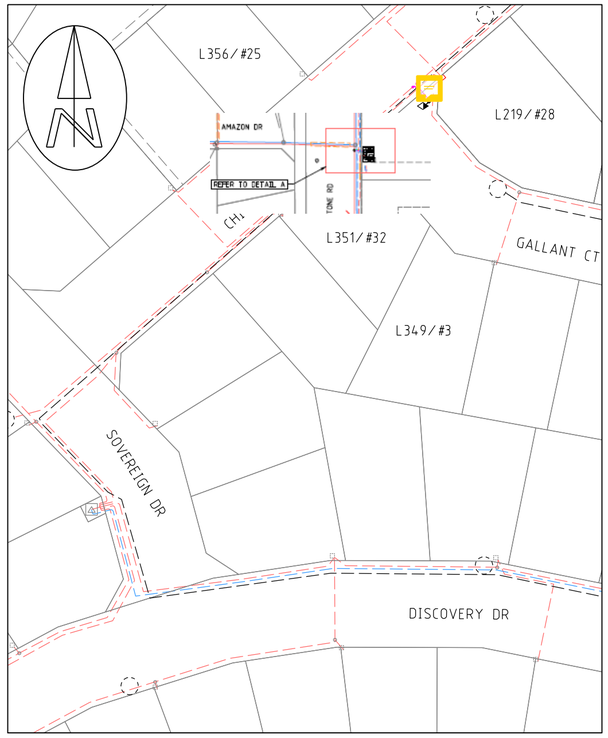
20 TEAGUE ST, VICTORIA PARK — HV on pole, need HV voltage for safe approach distances

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 58' 17" S Long. - 115° 53' 55" E

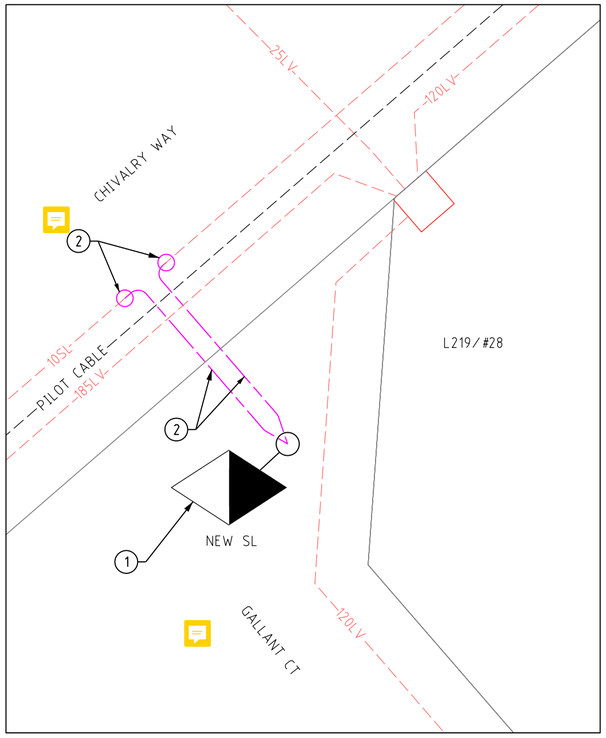
westernpower

DRAWING NUMBER
MP206362
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:100 2 1 0 2 4 6 8

LEGEND

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING

42W CFL	70W HPS	70W MH	1L LED, 1E LED
80W MV	150W HPS	150W MH	2L LED, 2E LED
125W MV	250W HPS	250W MH	3L LED, 3E LED
	250W HPS AERODROME		4L LED, 4E LED
			5L LED, 5E LED
			6L LED, 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 6.5m STANDARD POLE WITH 36W LED STEETLIGHT AS PER POLE SCHEDULE TABLE.
   CU: SL8_3L x 1
   DCSH: S08
2. INSTALL 2x NEW 16SL CABLE FROM NEW SL TO EXISTING 10SL CABLE ON CHIVALRY WAY.
   CU: CN41 x 15m, CN73_10/16 x 4, CN54 x 3
3. CUT EXISTING 10SL CABLE AND 2 x STRAIGHT JOINT NEW 16SL CABLES.
   LU13 x 2
4. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

7. AT NEW STREETLIGHT POLE, TERMINATE NEW 16SL CABLE.
   NON CU: FE0032 x 1, FE0071 x 1

CUSTOMER SCOPE OF WORK

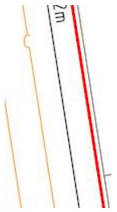
A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE POSITION OF NEW STREETLIGHTS IN ACCORDANCE WITH AS1158 AND SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER.

C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

POLE SCHEDULE				AS CON CHECK
POLE ID	POLE DETAILS	POLE LOCATION		
		EASTING	NORTHING	
NEW SL	6.5m STANDARD POLE	TBC BY CUSTOMER	TBC BY CUSTOMER	

POLE SCHEDULE



CRITICAL WATER PIPELINE IN VICINITY OF WORKS

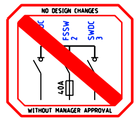
FIBRE OPTIC CABLES IN VICINITY OF WORKS

CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

APPROVED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF GOSNELLS
CONTACT: LINCOLN McDERMOTT TEL: 0482518998

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

HAZARD MANAGEMENT REGISTER EDM#:

FILE REFERENCE DETAILS

STREET LIGHT INSTALLATION

NETWORK VOLTAGE: 415V

28 CHIVALRY WAY, THORNLIE WA 6108

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 04' 00" S Long. - 115° 57' 53" E

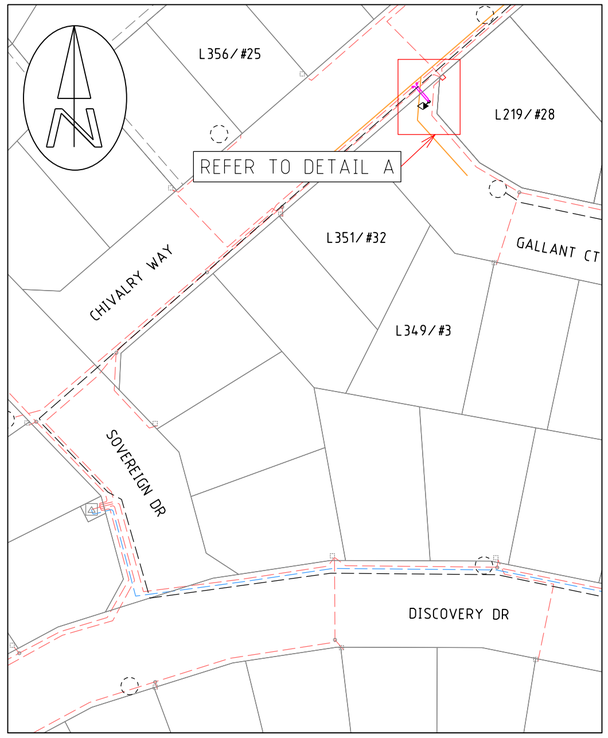
westernpower

DRAWING NUMBER

MP206943

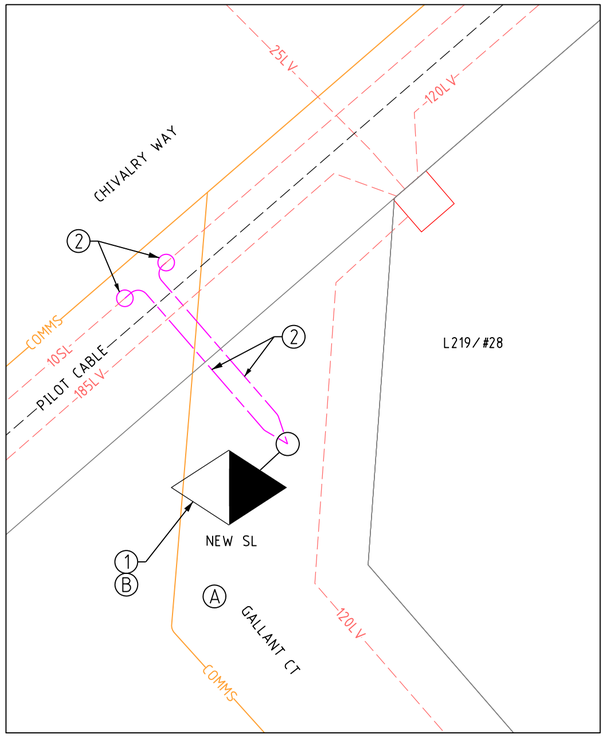
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:100 2 1 0 2 4 6 8

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING

42W CFL	70W HPS	70W MH	1L LED	1E LED
80W MV	150W HPS	150W MH	2L LED	2E LED
125W MV	250W HPS	250W MH	3L LED	3E LED
	250W HPS AERODROME		4L LED	4E LED
			5L LED	5E LED
			6L LED	6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 6.5m STANDARD POLE WITH 36W 4000k LED STEETLIGHT AS PER POLE SCHEDULE TABLE.
   CU: SL8_3L x 1
   DCSH: S08
2. INSTALL 2x NEW 16SL CABLE FROM NEW SL TO EXISTING 10SL CABLE ON CHIVALRY WAY.
   CU: CN41 x 30m, CN73_10/16 x 4, CN54 x 3
3. CUT EXISTING 10SL CABLE AND 2 x STRAIGHT JOINT NEW 16SL CABLES.
   CU: LU13 x 2
4. AT NEW STREETLIGHT POLE, TERMINATE NEW 16SL CABLE.
   NON CU: FE0032 x 1, FE0071 x 1
5. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE POSITION OF NEW STREETLIGHTS IN ACCORDANCE WITH AS1158 AND SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER.

POLE SCHEDULE				AS CON CHECK
POLE ID	POLE DETAILS	POLE LOCATION		
		EASTING	NORTHING	
NEW SL	6.5m STANDARD POLE	TBC BY CUSTOMER	TBC BY CUSTOMER	

POLE SCHEDULE

APPROVED FOR CONSTRUCTION

☐ ☒ ☒ ☒ ☐ ☐ ☒ ☒ ☒

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF GOSNELLS
CONTACT: LINCOLN McDERMOTT TEL: 0482518998

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68704079
HAZARD MANAGEMENT REGISTER EDM#: 68703972

STREET LIGHT INSTALLATION

NETWORK VOLTAGE: 415V

28 CHIVALRY WAY, THORNLIE WA 6108

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 04' 00" S Long. - 115° 57' 53" E

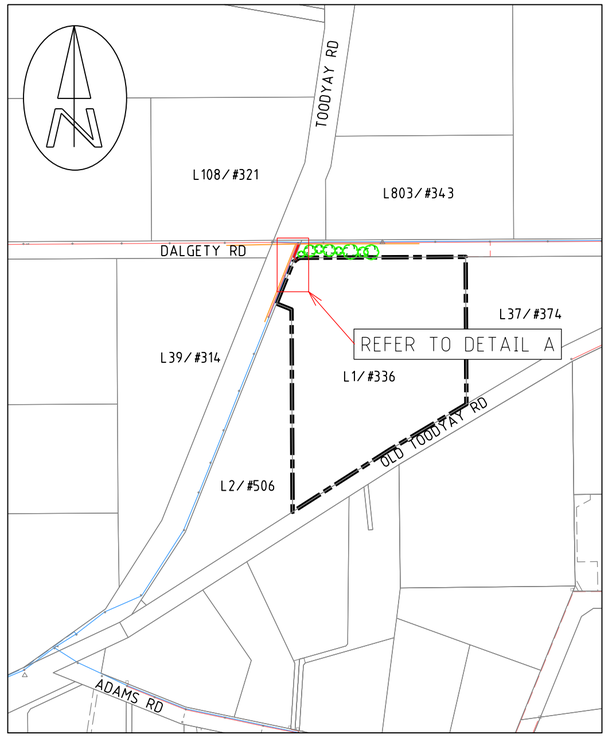
westernpower

DRAWING NUMBER

MP206943

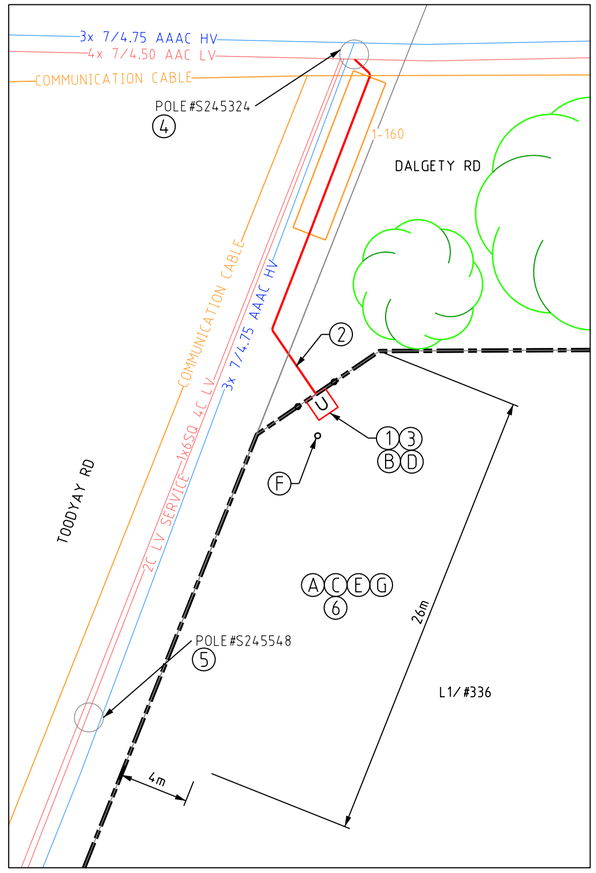
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:5000 100 50 0 100 200 300 400



DETAIL A

1:250 5 2.5 0 5 10 15 20

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER TO OBTAIN WORK ZONE PERMIT FROM THE TOWN OF VICTORIA PARK AT NO COST TO WESTERN POWER.

FOLLOW PERMIT CONDITIONS FROM THE TOWN OF VICTORIA PARK.

(F) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

(G) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW UNI-PILLAR ON L1/#336 DALGETY RD.
AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11x1
DCSH: U9

(2) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S245324
CU: CN45x35m, CN54x4, CN73_LVMAINSx3
DCSH: U16-1

(3) AT UNI-PILLAR (L1/#336 DALGETY RD.)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
- TERMINATE CUSTOMER CABLE AT BACK BLOCK IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU7_240x1
DCSH: U9

(4) AT POLE#S245324
- TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
- REMOVE 2C AERIAL SERVICE TO #245548
CU: LU6_240O × 1, LU6_240U × 1
DCSH: U16-1

(5) AT POLE#S245548
- REMOVE 2C AERIAL SERVICE TO #336 DALGETY RD.

(6) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

THREATENED ECOLOGICAL COMMUNITIES
NATIVE VEGETATION REGS ESAS CLEARING
PRIORITY FLORA
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

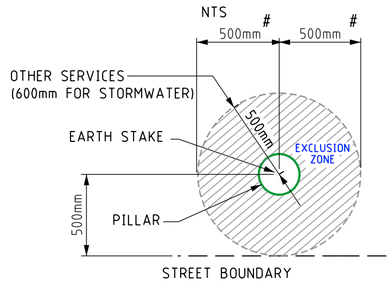
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

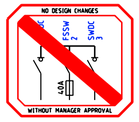
NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

APPROVED FOR CONSTRUCTION



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SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: VENTIA SOLUTIONS PTY LTD
CONTACT: FRANK WANG TEL: 0404723814

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68769269
HAZARD MANAGEMENT REGISTER EDM#: 68803432

415V/80A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV

336 DALGETY RD, RED HILL WA 6056

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 50' 60" S Long. - 116° 03' 13" E

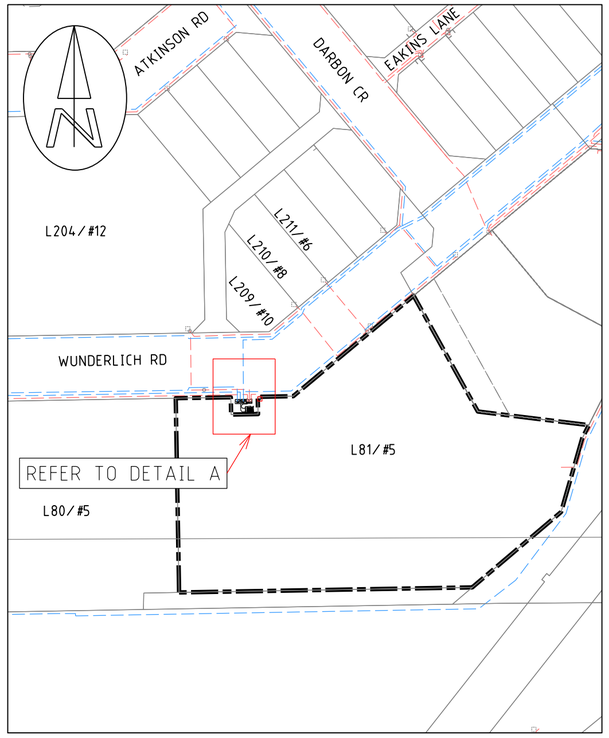
westernpower

DRAWING NUMBER

MP207924

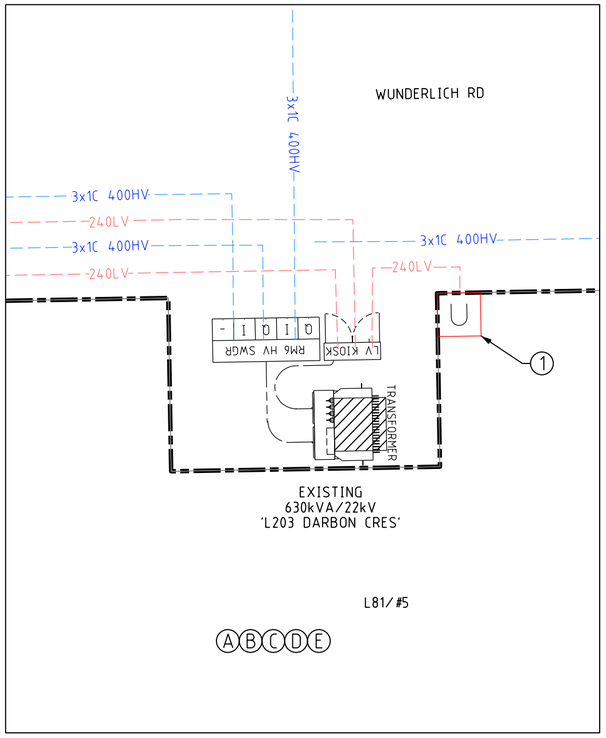
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:100 — 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

(1) AT UNI-PILLAR (L81/#5 WUNDERLICH RD)
- TERMINATE CUSTOMER CABLE AT TOP BAR WITH 315A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR

CU: LU36_315A x 1
DCSH: U9

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER TO OBTAIN WORK ZONE PERMIT FROM THE TOWN OF VICTORIA PARK AT NO COST TO WESTERN POWER.

FOLLOW PERMIT CONDITIONS FROM THE TOWN OF VICTORIA PARK.

(E) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

LEGEND

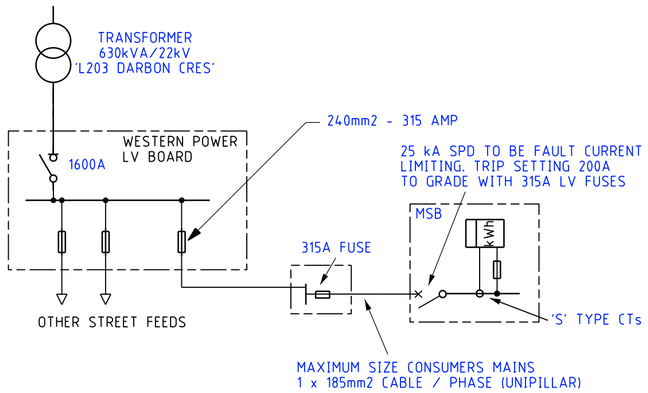
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



APPROVED FOR CONSTRUCTION

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: IDEAL COMMERCIAL AUSTRALIA PTY LTD
CONTACT: MICHAEL LOUREY TEL: 0448 354 506

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS: HAZARD MANAGEMENT REGISTER EDM#: 68840790

200A TEMPORARY SUPPLY

NETWORK VOLTAGE: 415V

5 WUNDERLICH RD, SUBIACO WA 6008

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 56' 47" S Long. - 115° 49' 08" E

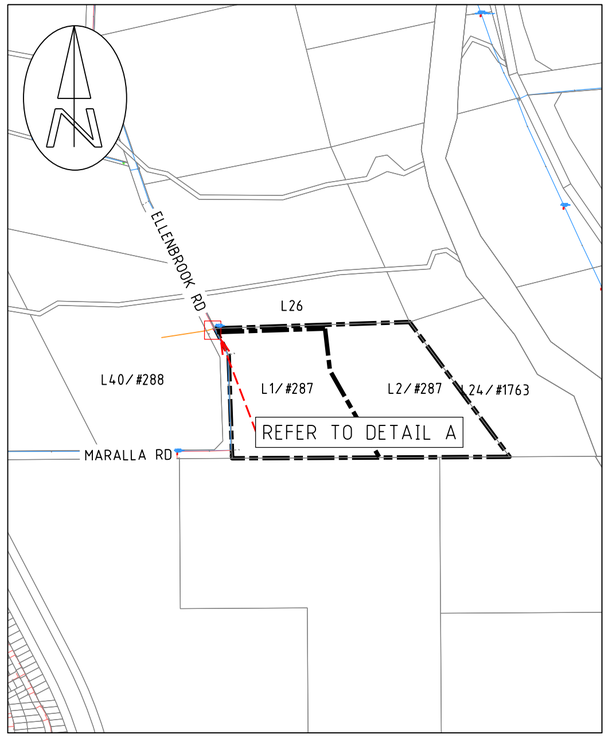
westernpower

DRAWING NUMBER

MP208032

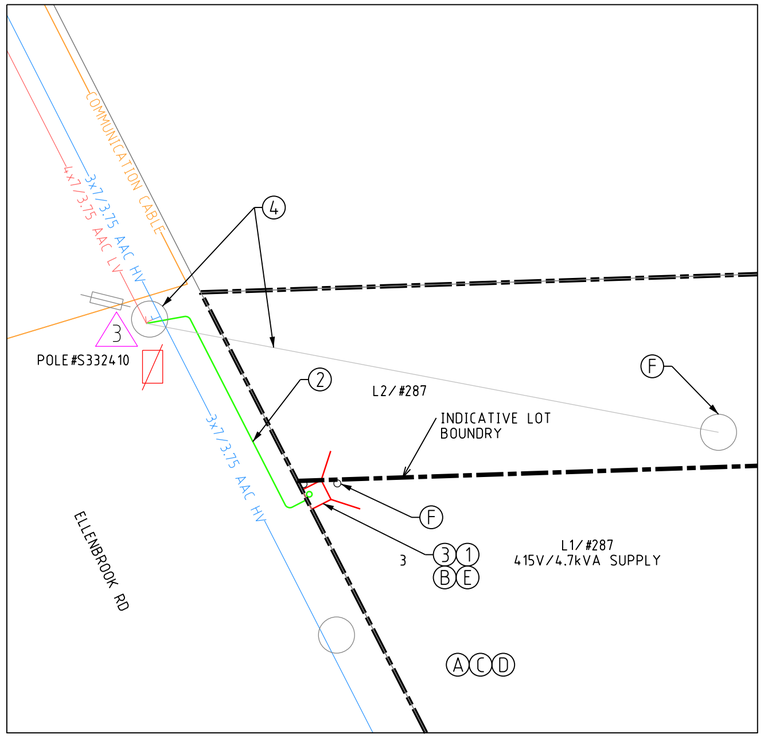
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L1/#287 ELLENBROOK RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 1
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S332410
   CU: CN42 x 30m, CN73_LV25 x 3, CN54 x 2
3. AT MINI-PILLAR (L1/#287 ELLENBROOK RD)
   - TERMINATE NEW 25LV CABLE.
   - TERMINATE CUSTOMER'S CABLE IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
   DCSH: U08-04
4. AT POLE#S332410
   - TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
   - REMOVE CUSTOMER AERIAL SERVICE TO #287 ELLENBROOK RD.
   CU: LU51_25ALO x 1, LU51_25ALU x 1
   DCSH: U19-1
5. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   HERITAGE COUNCIL OF WA
   MUNICIPAL INVENTORY
   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT ADN LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 41507941)

   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
   IN AREAS WHERE THIS IS A RISK YOU NEED TO:
   BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
   IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT, AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

   NATIVE VEGETATION REGS ESAS CLEARING
   ENVIRONMENTALLY SENSITIVE AREA
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

COMMUNICATION CABLES IN VICINITY OF WORKS

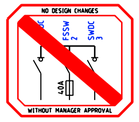
W.A.P.C. REFERENCE No.	162412

CUSTOMER CONTACT DETAILS
CUSTOMER NAME: MICHAEL KINGHAM
E-MAIL: osirisponystud@hotmail.com TEL: 0407083522

ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE





CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN POWER AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER MUST INSTALL SMSB WITHIN 30m OF THE POINT OF SUPPLY

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

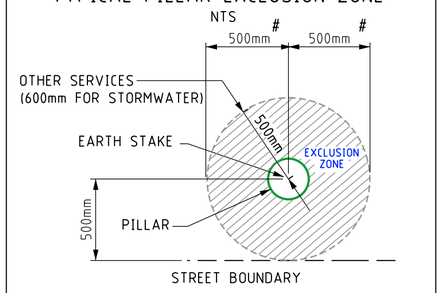
(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(F) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

(G) CUSTOMER RESPONSIBLE TO REMOVE CUSTOMER'S POLES AS REQUIRED AFTER COMPLETION OF WORK.

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE FILTER ASSESSMENT EDM#: 69149154
A	ISSUED FOR CONSTRUCTION	I.P.	07/24	HAZARD MANAGEMENT REGISTER EDM#: 69148855

OVERHEAD TO UNDERGROUND CONVERSION

NETWORK VOLTAGE: 22kV/415V

287 ELLENBROOK ROAD, BULLSBROOK, WA, 6084

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 07/24	SCALE: AS NOTED @ A2
GEO REF:	Lat. - 31° 44' 22" S Long. - 116° 01' 38" E

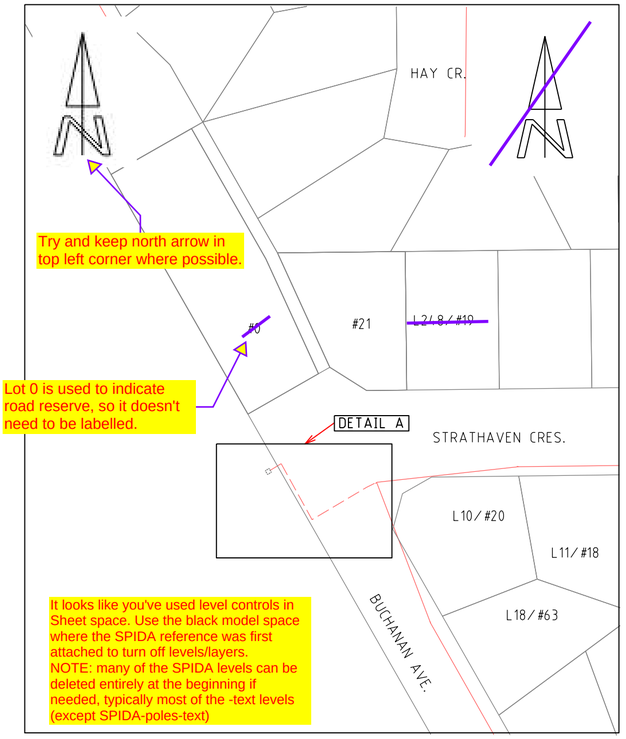
westernpower

DRAWING NUMBER

MP208126

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



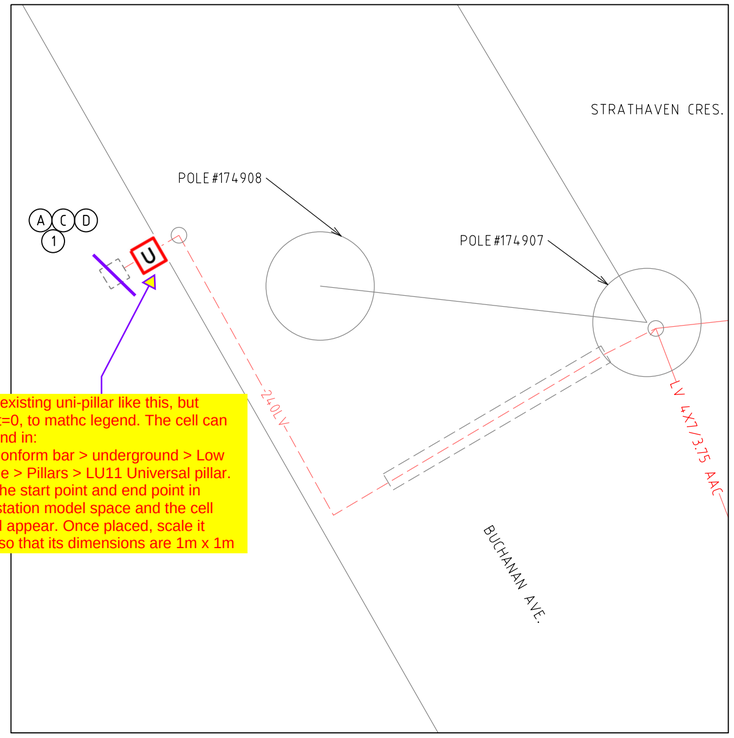
Try and keep north arrow in top left corner where possible.

Lot 0 is used to indicate road reserve, so it doesn't need to be labelled.

It looks like you've used level controls in Sheet space. Use the black model space where the SPIDA reference was first attached to turn off levels/layers.
NOTE: many of the SPIDA levels can be deleted entirely at the beginning if needed, typically most of the -text levels (except SPIDA-poles-text)

LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



Show existing uni-pillar like this, but weight=0, to mathc legend. The cell can be found in:
CADConform bar > underground > Low Voltage > Pillars > LU11 Universal pillar.
Click the start point and end point in Microstation model space and the cell should appear. Once placed, scale it down so that its dimensions are 1m x 1m

DETAIL A

1:200 — 4 2 0 4 8 12 16

Printer and Paper Size
pdf-wpc.pltcfg
Bentley PDF printer driver
ISO A2
Usable area is 594 x 420 mm

Use the 'pdf-wpc'config file when printing to PDF, more detail in Sharepoint doc "Guideline - Distribution Design Specification" section 19.

MP208315 GREENWOOD Supply Upgrade
SPIDAGreenwood- Original
SPIDAGreenwood-Updated

Start file names with project ID, eg.
MP208315 - Design Drawing.dgn
MP208315 - SPIDA.dgn
Original CAD export doesnt need renaming. Refer to "Guideline - Distribution Design Specification" section 1.6

LEGEND

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

TRANSFORMER (T) EXISTING SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

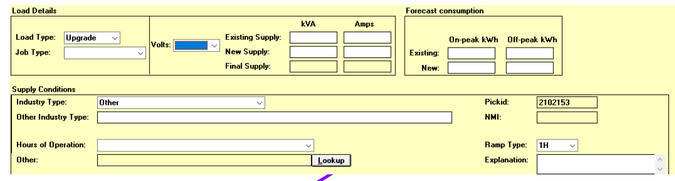
Display Overrides:	Allow
New Level Display:	Use MS_REF_NEWLEVELDISPLAY Configuration
Global LineStyle Scale:	Master

<-- In MODEL SPACE (to attach SPIDA dgn)

Display Overrides:	Never
New Level Display:	Use MS_REF_NEWLEVELDISPLAY Configuration
Global LineStyle Scale:	Reference

<-- In SHEET SPACE (to ref MODEL VIEWS)

Using the above attachment method, I've found the level controls can be done in the model space (to turn off redundant SPIDA levels), and the sheet space will automatically update.



Update the DQM > Load Details/Capcon screen (use bottom left button on Quote Screen) - fill using details in Validation Summary

CUSTOMER SCOPE OF WORK

- (A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.
- (B) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.
- (B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.
- (D) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.
- (C) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.
- (D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.
- (E) THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME
- (F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI / UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

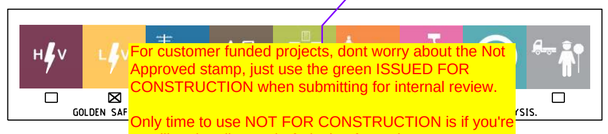
WESTERN POWER SCOPE OF WORK

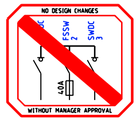
- (1) CAPACITY CHARGE ONLY, NO CONSTRUCTION IS REQUIRED

NOT APPROVED for CONSTRUCTION

For customer funded projects, dont worry about the Not Approved stamp, just use the green ISSUED FOR CONSTRUCTION when submitting for internal review.

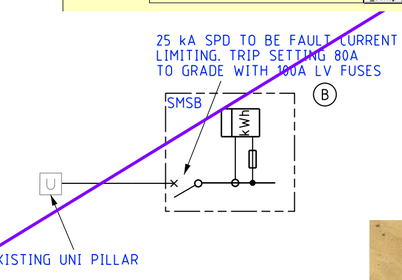
Only time to use NOT FOR CONSTRUCTION is if you're emailing the client a draft design for review.



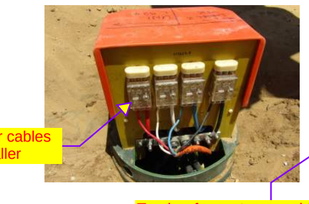


SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



Just general knowledge:
New Max Demand load is 80A, and is under 100A so it doesn't need a SLD.
1 thing to confirm with customer whether the attached MSB shares a supply with Main Roads WA - ie. determine what the total load is (Main Roads load + 80A) that will be supplied via the existing uni-pillar.
Also, the customer SLD PDF shows a 25mm2 mains connecting to the uni-pillar. this size of cable does not need to be fused (ie. connects to the back blocks instead of the top bar on other side of panel)



Back blocks for cables 35mm2 or smaller



Top bar for customer cables larger than 35mm2 up to 185mm2; requires fuse or link to connect to bottom bar.
Alternatively, a WP 120/185/240 cable can also terminate to top bar to create a network open point.

Bottom bar for WP 120/185/240LV cable terminations

CONTACT DETAILS
COMPANY NAME: ALL TELECOS ENGINEERING
CONTACT: vinay.rao@alltelecos.com.au TEL: 0426201397

Rev.	REVISIONS	DRN	DATE
A	UNDER ENGINEERING REVIEW	I.P.	05/24

Always use ISSUED FOR CONSTRUCTION unless issuing a draft, as per above

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT - EDM# XXXXX
HAZARD MANAGEMENT REGISTER EDM#:

Note: when editing these lines, use the 'Enter Data Field' button (pictured right) in Microstation to edit the blank field. Alternatively, double click the line, then double click again in the text window to open Data field

80A/415V COMMERCIAL SUPPLY

NETWORK VOLTAGE: 415V

9322 DARNLEY AV - GREENWOOD WA 6024

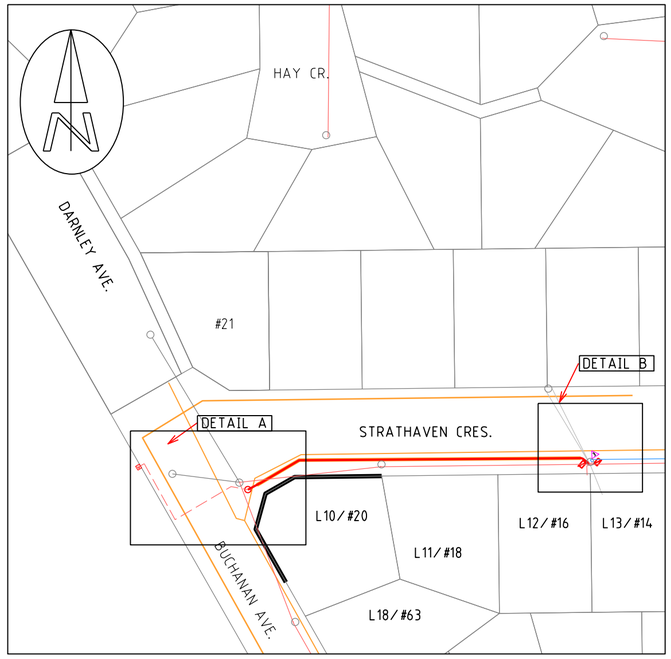
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 05/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 49' 39" S Long. - 115° 47' 14" E

westernpower

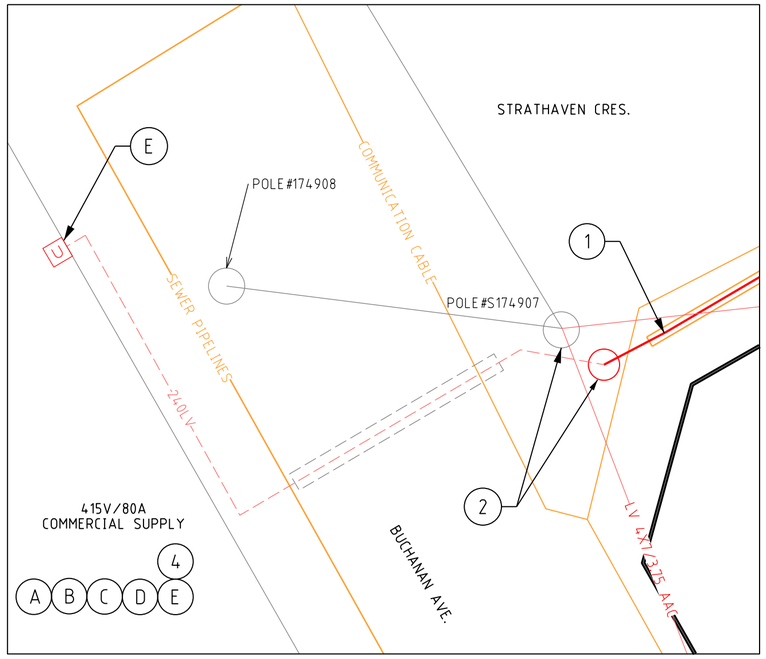
DRAWING NUMBER
MP208315
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



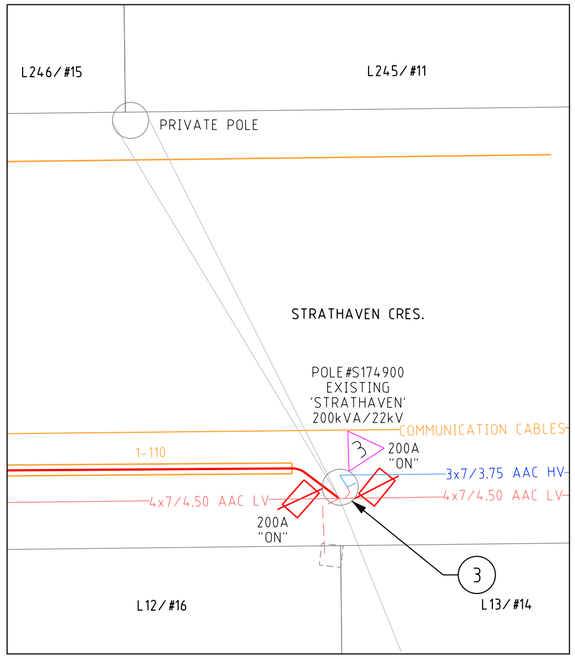
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- U UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- SUDI Serviced Uncleared De-energised Lot
- UMS PIT
- UTILITIES PIT
- SP BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

TRANSFORMER (T) EXISTING SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(C) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.
CUSTOMER MUST INSTALL SMSB WITHIN 30m OF POINT OF SUPPLY.

(D) THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

WESTERN POWER SCOPE OF WORK

(1) INSTALL 240LV CABLE FROM POLE#S174900 TO EXISTING 240LV CABLE VIA 110mm POLYPIPE IN 2.7m ALIGNMENT.
CU: CN45 x 100m, CN54 x 4, CN73_LVMAINS x 3

(2) DISCONNECT EXISTING 240LV CABLE FROM POLE#S174907. CUT AND SWING EXISTING 240LV CABLE AND STRAIGHT JOINT TO NEW 240LV CABLE.
CU: LU2 x 1

(3) AT POLE#S174900
- REPLACE EXISTING POLE WITH NEW 11m/6kN.
- INSTALL NEW HV TERMINATION CROSSARM.
- INSTALL NEW LV STRAIN CROSSARM.
- REATTACH EXISTING Tx TO NEW POLE.
- REATTACH EXISTING AERIAL STAY AND 3x AERIAL SERVICES TO NEW POLE.
- INSTALL MASTER KRONE FUSE BOX WITH 630A LINKS AS PER DCSH R12-3
- INSTALL 2 x NEW KRONE FUSE BOX WITH 200A FUSES.
- TAP PTTX LV BUSHING ONTO MASTER KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE WEST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE EAST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TERMINATE EXISTING 25LV CABLE TO LV BARE WEST.
- TERMINATE NEW 240LV ONTO POLE AND TAP TO LV BARE EAST.

CU: P06_M x 1, HV05_3MN x 1, LV3_W3 x 1, CN26_7/475 x 4, CN39_150/4M x 3, CN11_150 x 8, CN70_150 x 7, CN9_7/475AA x 7, LV8_1P x 1, LV8_3P x 2, LU6_240O x 1, LU6_240U x 1
NON CU: GF1508 x 3, GF1096 x 6, GS0111 x 2, GS0109 x 2, CB0204 x 2, FC0110 x 10, AB6733 x 6, AB6742 x 2, CC0110 x 8
DCSH: U17, R12-3, L03, H11-1, U19-1, R13-3

(4) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

PUBLIC DRINKING WATER SOURCE AREA
THERE SHOULD BE NO REFUELING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

THREATENED ECOLOGICAL COMMUNITIES
NATIVE VEGETATION REGS ESAS CLEARING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

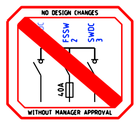
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

SIMPLIFIED SINGLE LINE DIAGRAM

ISSUED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

SEWER PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: All TELECOS ENGINEERING
CONTACT: vinay.rao@alltelecos.com.au TEL: 0426201397

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/24

SHE ASSESSMENT EDM#: 69603732
HAZARD MANAGEMENT REGISTER EDM#: 68478469
FILE REFERENCE DETAILS

80A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 415V/22kV

9322 DARNLEY AV - GREENWOOD WA 6024

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 49' 39" S Long. - 115° 47' 14" E

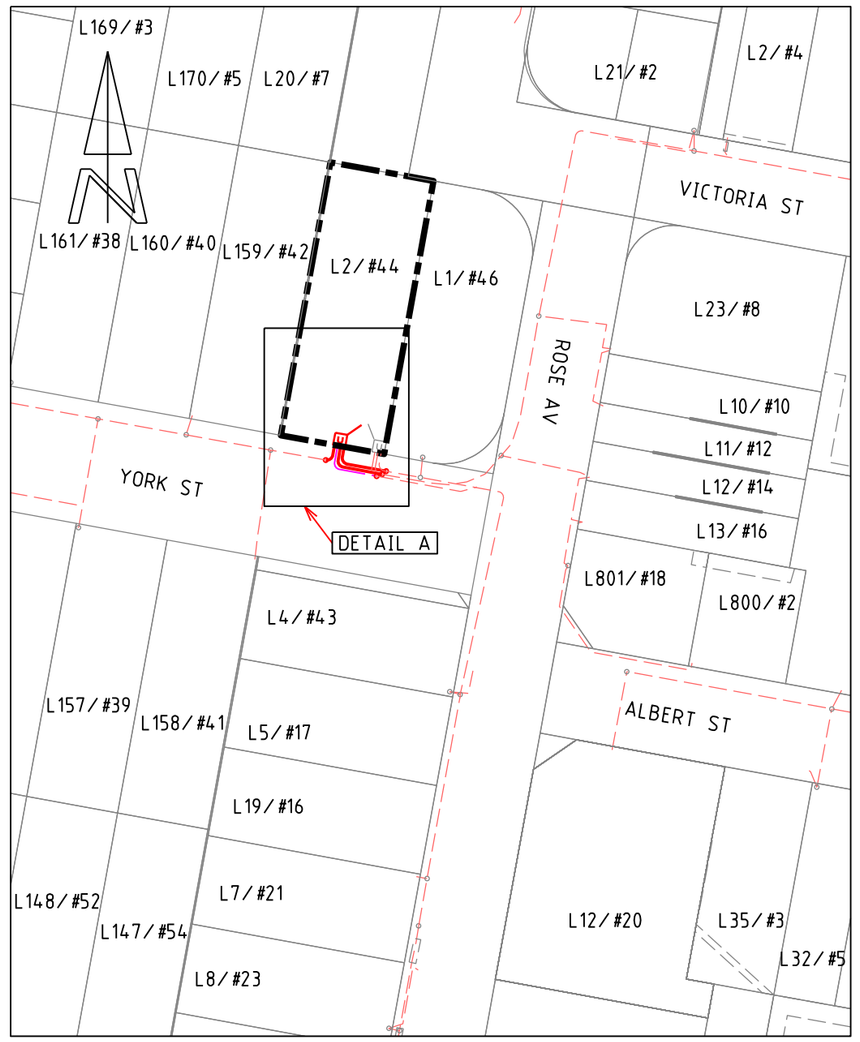
westernpower

DRAWING NUMBER

MP208315

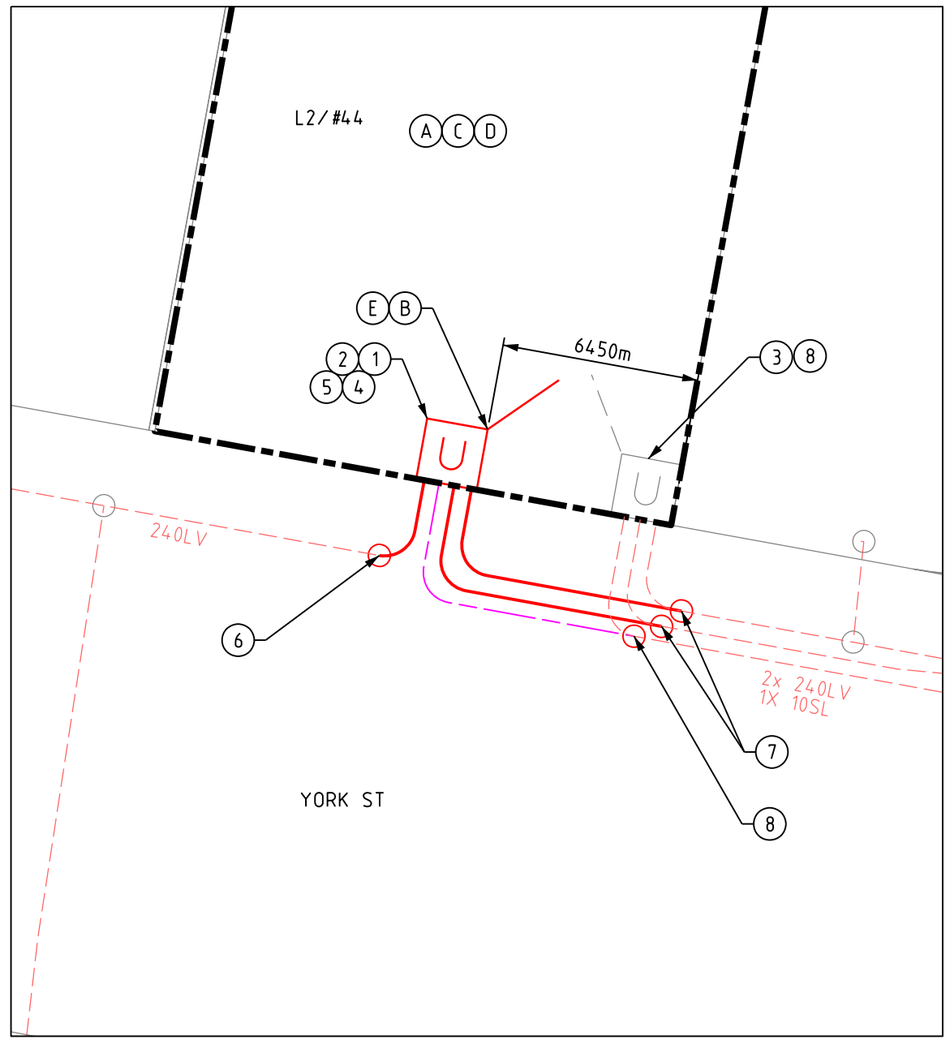
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



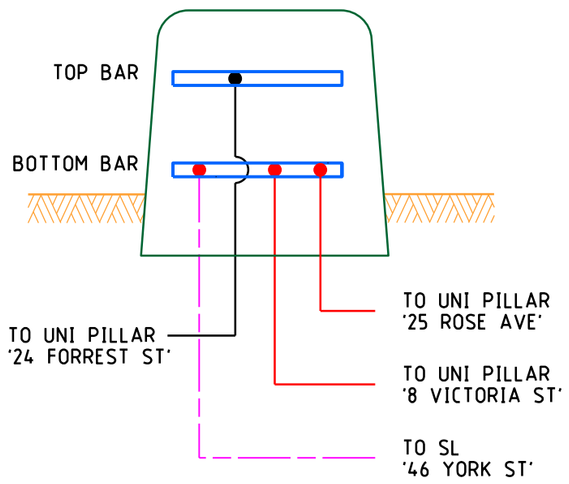
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16



WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI PILLAR ON L2/#44 YORK ST AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU11
   DCSH: U9
2. INSTALL 2x240LV CABLE FROM UNI PILLAR ON L2/#44 YORK ST TO THE EXISTING 240LV CABLES.
   CU: CN45x10m
3. DISCONNECT EXISTING CONSUMER MAINS SUPPLYING L2/#44 YORK ST FROM THE EXISTING UNI PILLAR LOCATED ON L2/#44 YORK ST IN CONJUNCTION WITH THE CUSTOMERS ELECTRICAL CONTRACTOR.
4. TERMINATE EACH 240LV CABLE TO THE NEWLY INSTALLED UNI PILLAR L2/#44 YORK ST.
   CU: 3x LU7
   DCSH: U9
5. TERMINATE CUSTOMER'S CABLE TO THE NEWLY INSTALLED UNI PILLAR LOCATED ON L2/#44 YORK ST. CONSUMER MAINS TO BE TERMINATED TO THE BACK BLOCKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
   DCSH: U9
6. CUT AND SWING EXISTING 240LV CABLE INTO THE NEWLY INSTALLED UNI PILLAR.
   CU: 1x LU2
7. INSTALL NEW 240LV CABLE FROM EXISTING UNI-PILLAR TO NEW UNI-PILLAR AND STRAIGHT JOINT TO EXSITING 240 LV CABLE
   CU: CN45 × 25m, CN73_LVMAINS × 2, LU2 × 1
8. STRAIGHT NEW 16SL CABLE AND EXTEND TO NEW UNI-PILLAR AT L2/#44 YORK ST WITH RED SPOT FUSE
   CU: CN41 × 15m
8. DISCONNECT EXISTING 240LV CABLES FROM EXISTING UNI-PILLAR

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED, 2L LED, 3L LED, 4L LED, 5L LED, 6L LED
- 1E LED, 2E LED, 3E LED, 4E LED, 5E LED, 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

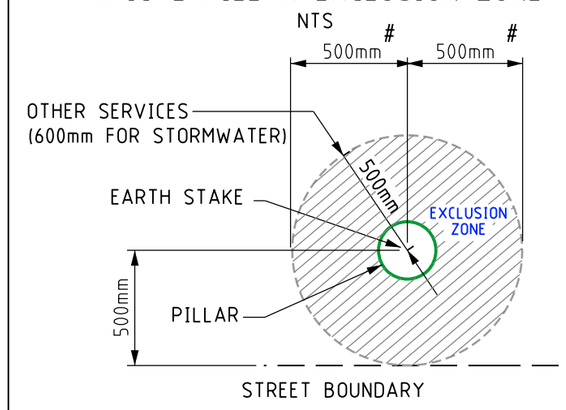
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

TRANSFORMER — EXISTING — SWITCHGEAR

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



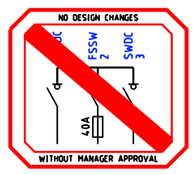
500mm to either side boundary

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



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APPROVED FOR CONSTRUCTION

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: WESLEY VAN DER SPUY TEL: 0458020012

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	IP	05/24

FILE REFERENCE DETAILS
- SHE ASSESSMENT REPORT EDM#68367012
- HAZARD MANAGEMENT REGISTER EDM#68355489

RELOCATE UNI PILLAR

NETWORK VOLTAGE: 415V

L2/#44 YORK ST - SOUTH PERTH WA 6151

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 05/24	SCALE: AS NOTED @ A2
GEO REF:	Lat. - 31° 58' 46" S Long. - 115° 51' 41" E

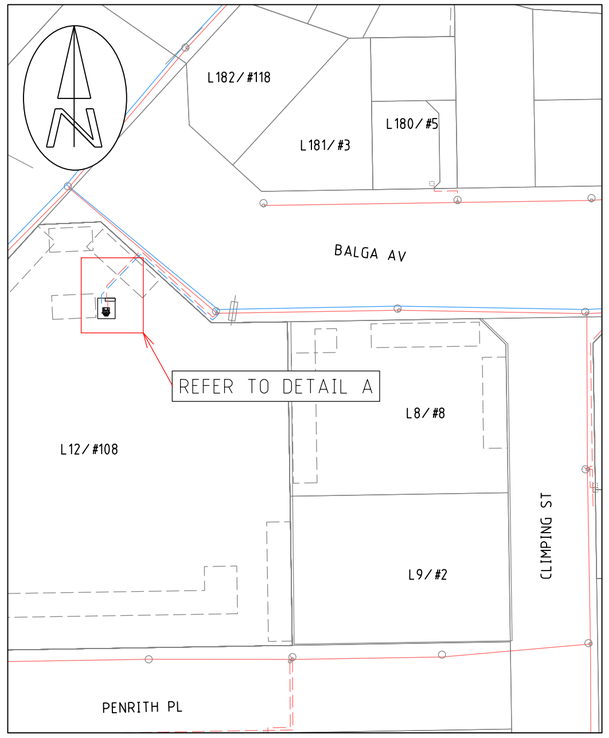
westernpower

DRAWING NUMBER

MP208328

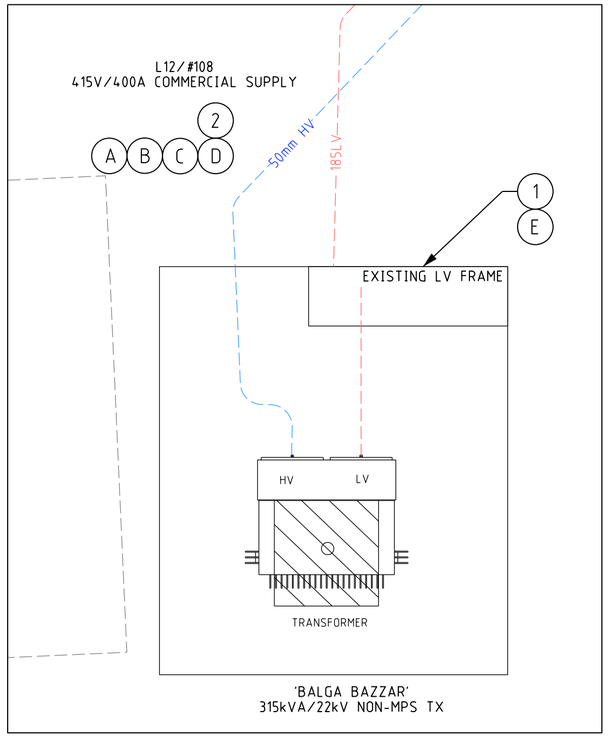
SHEET 01 OF 01

REV: A — ORIGINAL SIZE - A2



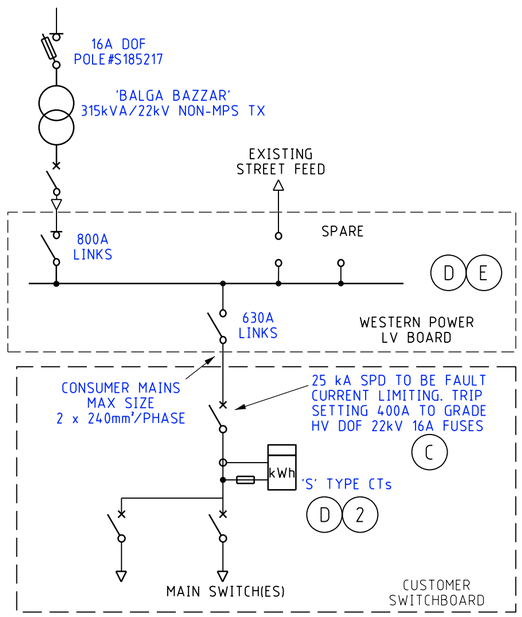
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:50 1 0.5 0 1 2 3 4



SIMPLIFIED SINGLE LINE DIAGRAM

LEGEND

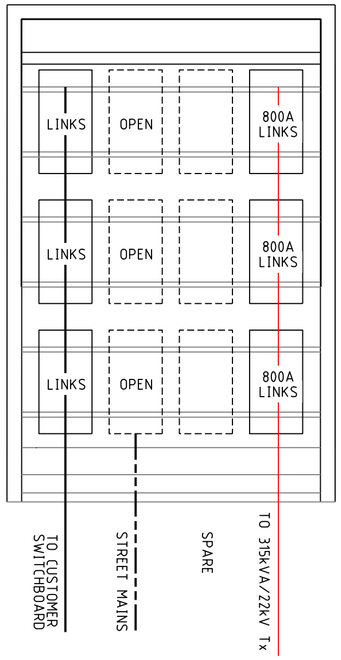
- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

HV CABLES

- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)



LV KIOSK
108 PRINCESS RD
SUBSTATION

WESTERN POWER SCOPE OF WORK

(1) AT EXISTING LV FRAME (BALGA BAZZAR Tx)
- DISCONNECT EXISTING CONSUMER MAINS
- TERMINATE NEW CUSTOMER CABLE VIA LINKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.

NON-CU: GF1200 x 4

(2) INSTALL CT METER.
CU: LM1_M x 1

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

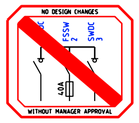
(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(E) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: RAIDEN CENTAURI
CONTACT: RANGANA GUNASEKARA TEL: 0492 903 530

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	08/24	HAZARD MANAGEMENT REGISTER EDM#: 69215400

400A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

108 PRINCESS RD, BALGA WA 6061

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 08/24 | SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 51' 20" S Long. - 115° 50' 17" E

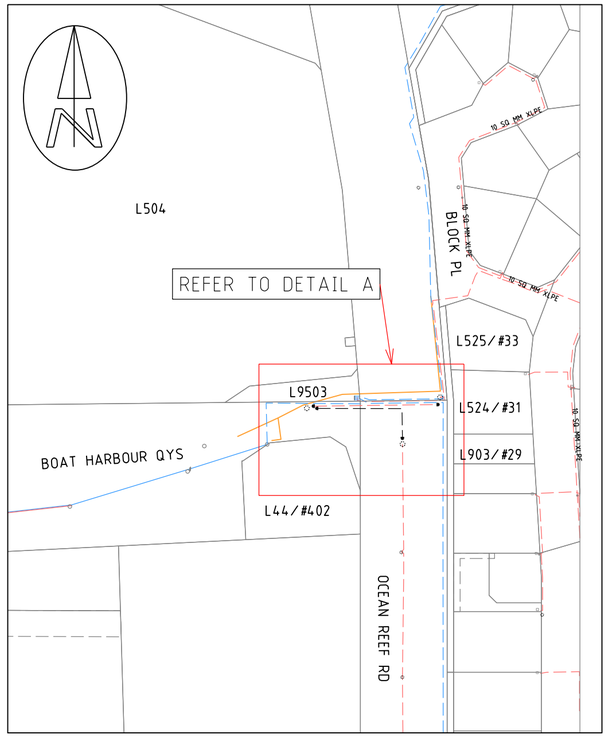
westernpower

DRAWING NUMBER

MP208516

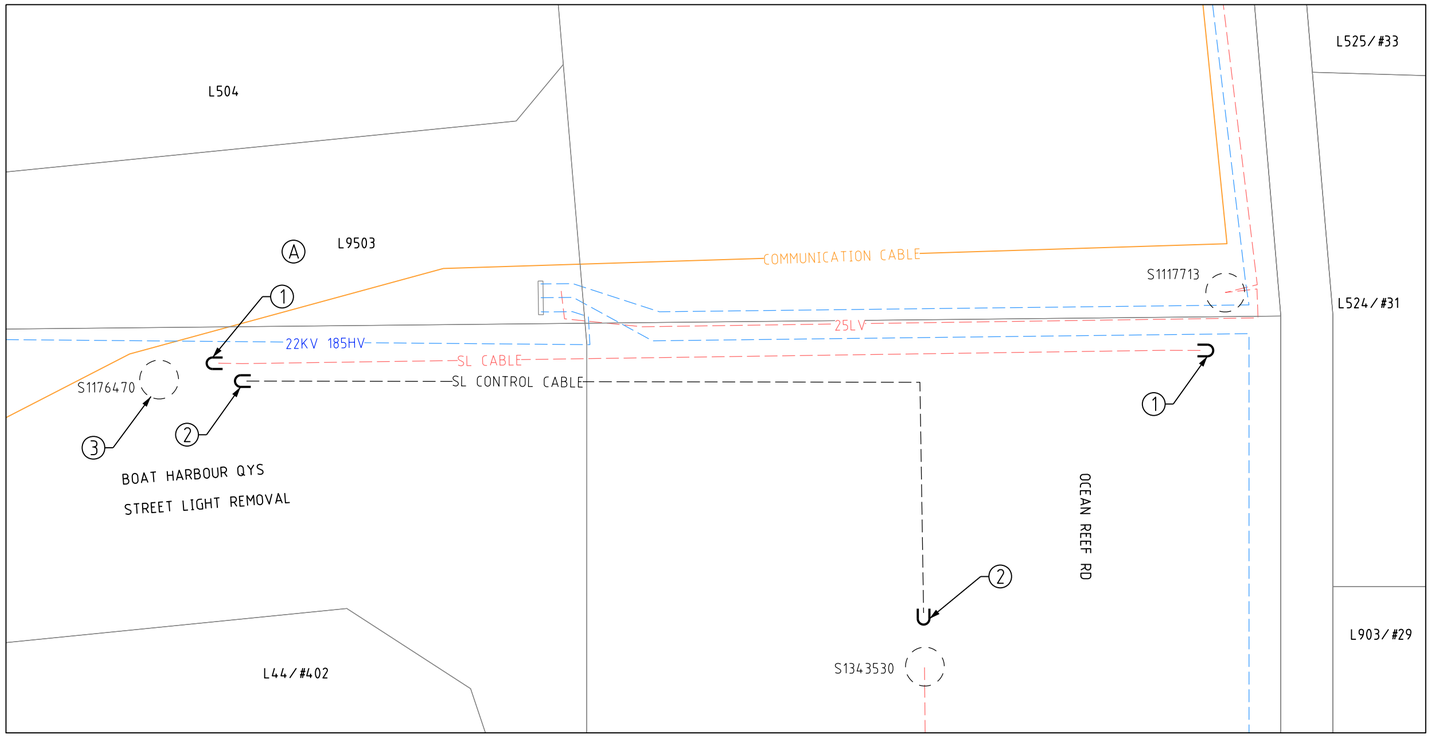
SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:250 5 2.5 0 5 10 15 20

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING SL CABLE FROM S1176470 AND S1117713. CUT, CAP AND ABANDON EXISTING SL CABLES.
   CU: CN73_LV10/16 x 3
2. DISCONNECT EXISTING SL CONTROL CABLE FROM S1176470 AND S1343530. CUT, CAP AND ABANDON EXISTING SL CONTROL CABLES.
   CU: CN73_LV10/16 x 2
3. REMOVE EXISTING STREETLIGHT.
4. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   SYSTEM RESERVE

   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT ADN LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

   NATIVE VEGETATION REMAINING
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

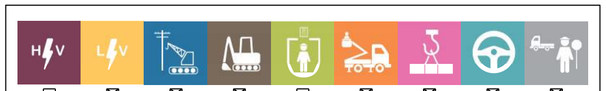
A. CUSTOMER MAY BE REQUIRED TO REMOVE SECTIONS OF FENCE TEMPORARILY TO ALLOW FOR STREET LIGHT REMOVAL.

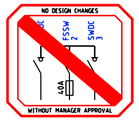
NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

APPROVED FOR CONSTRUCTION

COMMUNICATION CABLES IN VICINITY OF WORKS





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: STANTEC AUSTRALIA PTY LTD
CONTACT: SHARNI MILLMAN TEL: 0418 925 609

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	08/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT REPORT EDM#: 69457003
HAZARD MANAGEMENT REGISTER EDM#: 69457002

STREETLIGHT REMOVAL

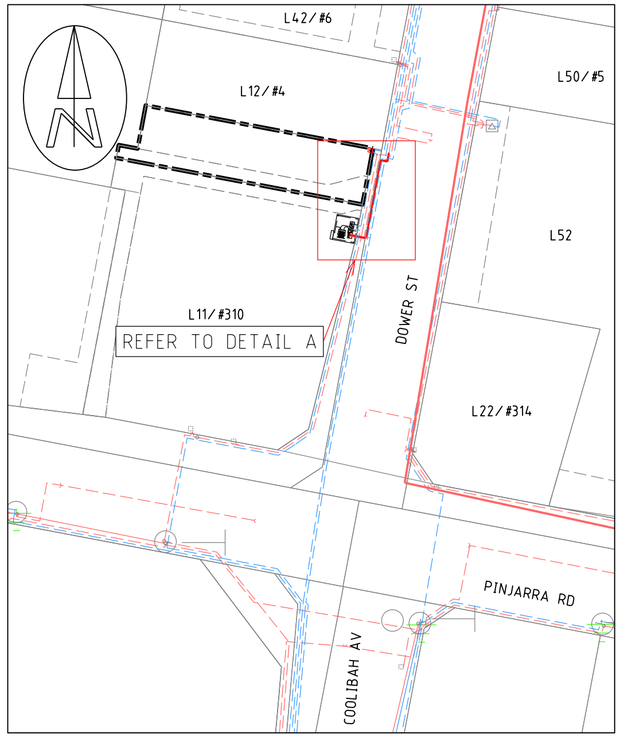
NETWORK VOLTAGE: 22kV/415V

LOT 9503 BOAT HBR QUAYS, OCEAN REEF WA 6027
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 08/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 45' 42" S Long. - 115° 43' 58" E

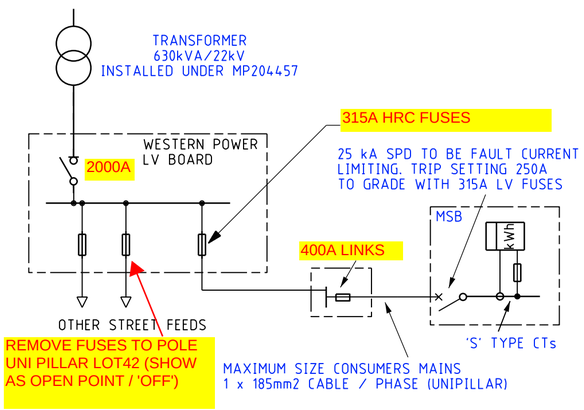
westernpower

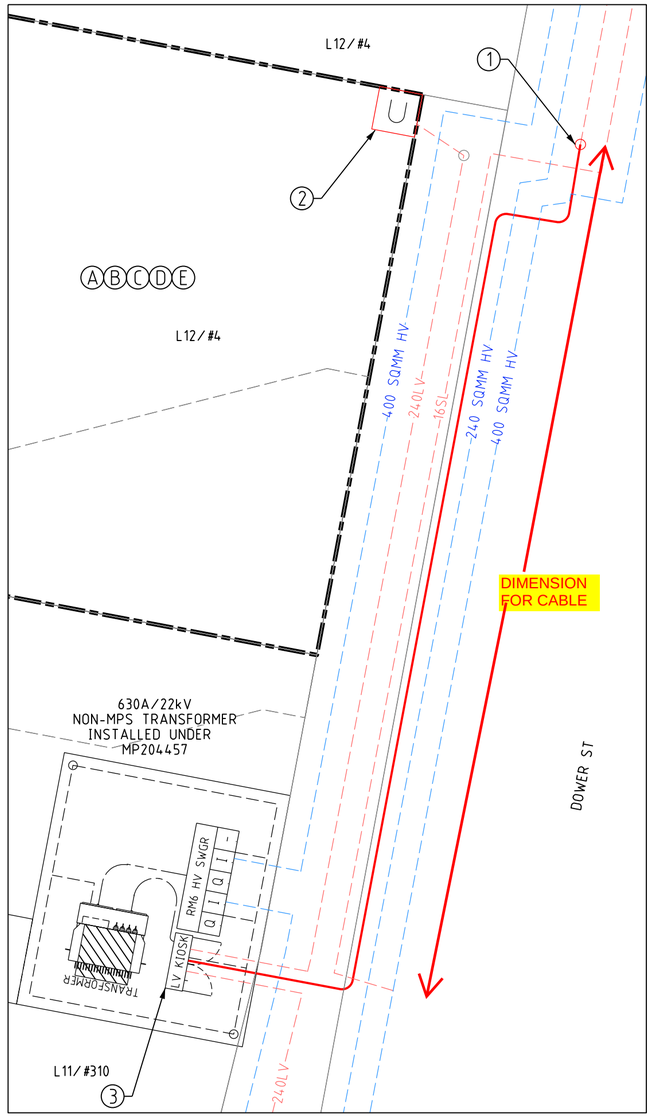
DRAWING NUMBER
MP208649
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80





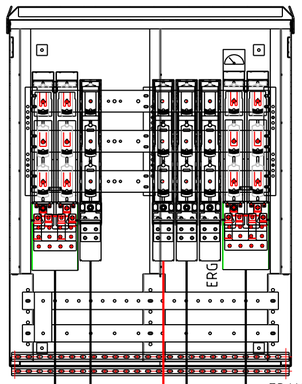
DETAIL A

1:100 — 2 1 0 2 4 6 8

TYPE 2.1 PENDA CIRCUIT TABLE

	LEFT	NAME/LOCATION
CIRCUIT 1 (C1)	LU67(ON)	CONTIGUOUS CUSTOMER
CIRCUIT 2 (C2)	LU69(ON)	TO POLE S80519
CIRCUIT 3 (C3)		
CIRCUIT 4 (C4)		
CIRCUIT 5 (C5)		
CIRCUIT 6 (C6)	LU69(OFF)	TO UNI PILLAR LOT 42 / #6 DOWER ST
CIRCUIT 7 (C7)	LU69(ON)	TO UNIPILLAR LOT 12/#4 DOWER ST
CIRCUIT 8 (C8)	LU68(OFF)	TO EMERGENCY GENERATOR
CIRCUIT 9 (C9)	LU66	TRANSFORMER
CIRCUIT 10 (C10)		

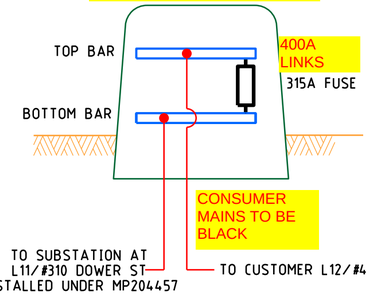
TYPE 2.1 PENDA ARRANGEMENT



LEGEND

- SCHEME BDY.
- HV CABLES
  - EXISTING HV CABLES (SIZE AND TYPE INDICATED)
- LV-SL CABLES
  - CN45 - 240mm2 (1x3c)
  - EXISTING LV CABLES (SIZE AND TYPE INDICATED)
- PILLARS / PITS / ETC
  - MINI (WORKING END)
  - UNIVERSAL
  - MINI
  - LINK
  - CABLE MARKER U/G
  - UMS PIT
  - UTILITIES PIT
  - BGSP PIT
  - 100/200AMP CUTOUT
  - CONDUIT / POLY PIPE
  - SWD Serviced Uncleared De-energised Lot

UNI PILLAR L12/#14 CONNECTION DETAILS



INCLUDE RED TEXT ABOVE FIRST SCOPE:
NOTE TO PNO: CONSTRUCT IN CONJUNCTION WITH MP204457.

OF WORK

1. INSTALL 240LV CABLE FROM LV PENDA INSTALLED UNDER MP204457 AND STRAIGHT JOINT TO EXISTING 240LV CABLE AT #4 DOWER ST
   CU: CN45 x 35m, CN54 x 1, CN73_LVMAINS x 4 LU2
   DCSH: U16-1
2. AT UNI-PILLAR (L12/#4 DOWER ST)
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 400A LINKS FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   - DISCONNECT AND REMOVE EXISTING 240LV CABLE TOWARDS POLE#S8380689 AS INDICATED IN DETAIL A
   CU: LU36_315A x 1, LU7_240 x 1
   DCSH: U9
3. AT LV PENDA INSTALLED UNDER MP204457
   - INSTALL 315A FUSES AT EXISTING SWITCH TO TO UNI PILLAR LOT 12/#4 DOWER ST (ON)
   - INSTALL NEW FUSE SWITCH AND TERMINATE 240LV CABLE (OFF)
   CU: LU69 x 1, LU14 x 1
4. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS.
   THREATENED ECOLOGICAL COMMUNITIES ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   DO NOT IMPACT ON NATOVE VEGETATION...

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

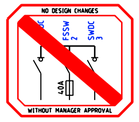
C. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

D. CUSTOMER TO OBTAIN WORK ZONE PERMIT FROM THE TOWN OF VICTORIA PARK AT NO COST TO WESTERN POWER.
FOLLOW PERMIT CONDITIONS FROM THE TOWN OF VICTORIA PARK.

E. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED





APPROVED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE

COMMUNICATION CABLES IN VICINITY OF WORKS

CRITICAL WATER PIPELINE IN VICINITY OF WORKS

DO NOT IMPACT NATIVE VEGETATION WARNING NOTE

CONSULTANT CONTACT DETAILS
COMPANY NAME: ELECTRICAL CONSULTANCY WA
CONTACT: MATT SCANTLEBURY TEL: 0427 760 146

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
- SHE ASSESSMENT REPORT EDM#: 68852496
- MP204457- Design Drawing- IFC EDM#: 68851883
- HAZARD MANAGEMENT REGISTER EDM#: 68851885

250A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

4 DOWER ST, MANDURAH WA 6210

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 32' 07" S Long. - 115° 44' 18" E

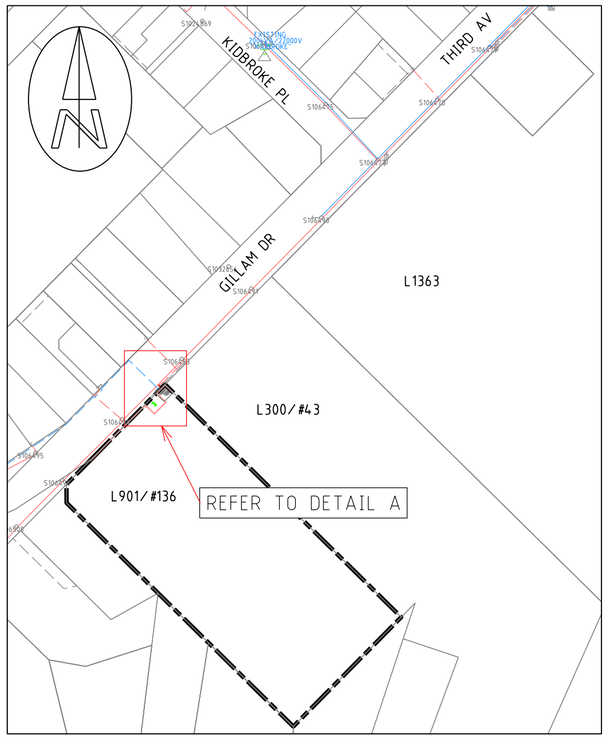
westernpower

DRAWING NUMBER

MP208751

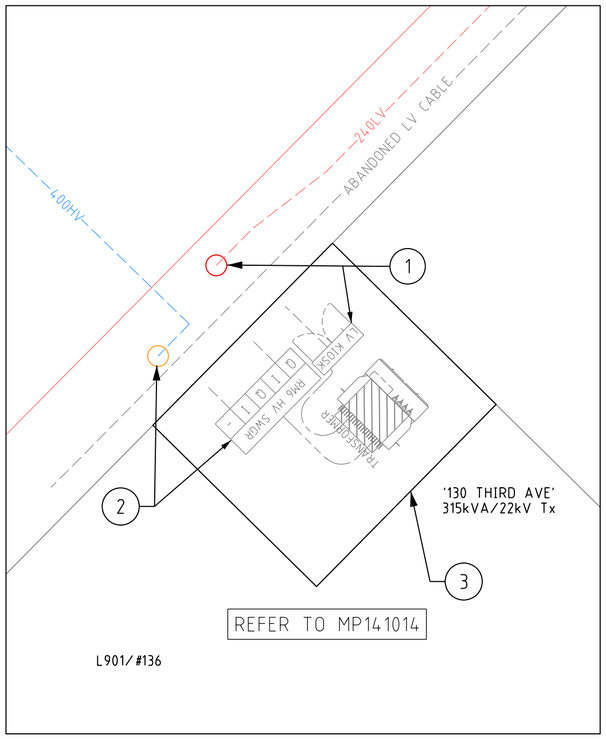
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING LV CABLES FROM EXISTING LV KIOSK AT '130 THIRD AVE' AND PULL BACK TO WP ALLIGNMENT. LIVE END SEAL EXISTING HV CABLE. REMOVE EXCESS CABLE. CU: LU62_185_240x1
2. DISCONNECT EXISTING HV CABLES FROM EXISTING RMU AT '130 THIRD AVE' AND PULL BACK TO WP ALLIGNMENT. LIVE END SEAL EXISTING HV CABLE. REMOVE EXCESS CABLE. CU: HU3_22_400x1
3. DECOMISSION AND REMOVE EXISTING SUBSTATION SITE.
4. ENVIRONMENTAL FILTER CHECKED. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

UNEXPLODED ORDNANCE
NO FURTHER ACTIONS REQIUIRED FOR THIS CONSTRAINT. HOWEVER PLEASE BE MINDFUL OF ANY UNEXPECTED FINDS AND MANAGE IN ACCORDANCE WITH EXCAVATION ANO TRENCHING PROCEDURE (EDMS# 43740956)

THREATENED ECOLOGICAL COMMUNITIES
NATIVE VEGETATION REGS ESAS CLEARING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

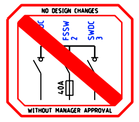
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME:
CONTACT: THOMAS KONG TEL: 08 93533380

CONSULTANT CONTACT DETAILS
COMPANY NAME: INFINITE ENGINEERING
CONTACT: MITCHELL WHITBREAD TEL: +61 477 999 674

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	12/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70791865
HAZARD MANAGEMENT REGISTER EDM#: 70791242

SUBSTATION DECOMISSION- PHASE 1

NETWORK VOLTAGE: 22kV/415V

136 THIRD AVE, KELMSCOTT WA 6111

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 12/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 07' 31" S Long. - 116° 00' 12" E

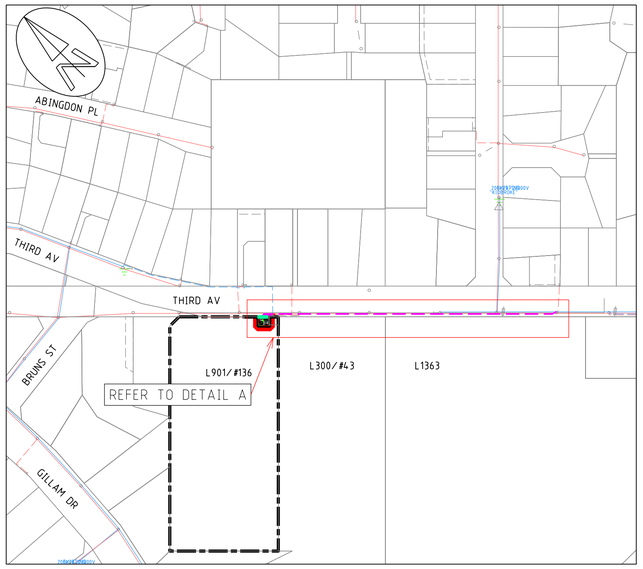
westernpower

DRAWING NUMBER

MP208787

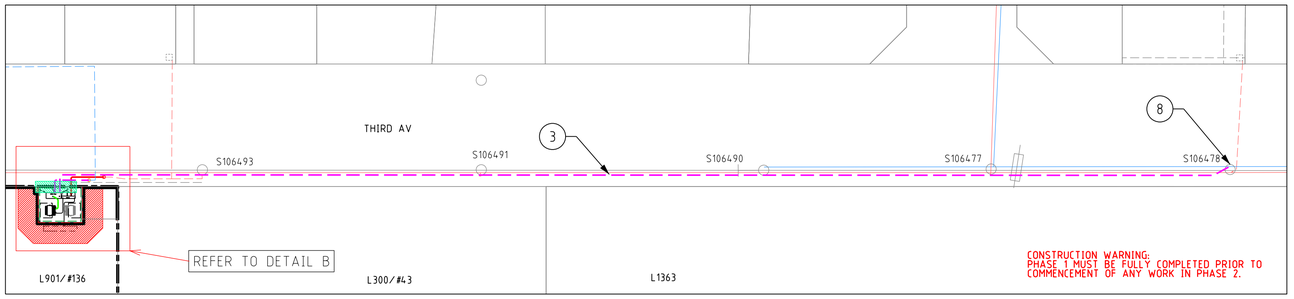
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



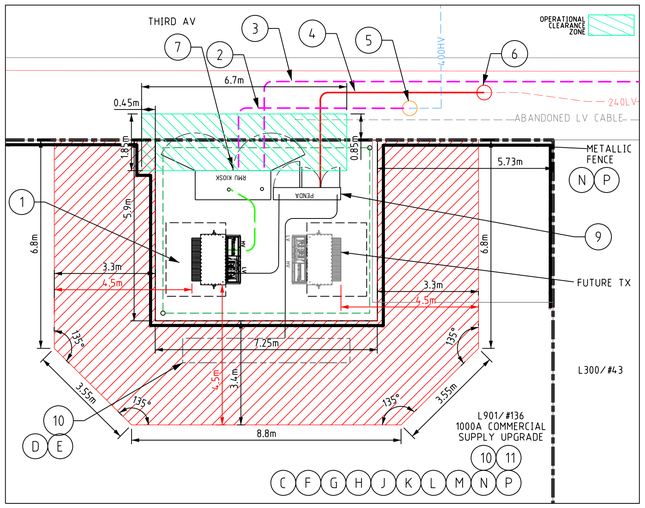
LOCATION PLAN

1:2000 (40 20 0 40 80 120 160)



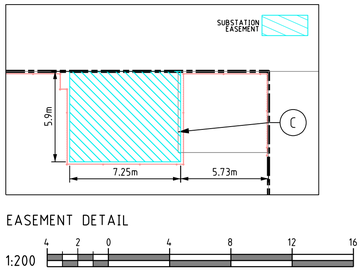
DETAIL A

1:500 (10 5 0 10 20 30 40)



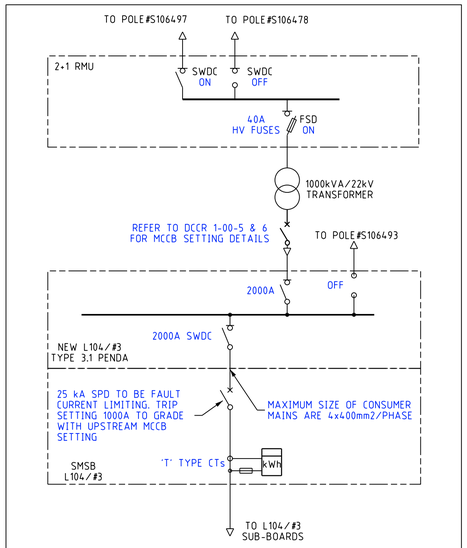
DETAIL B

1:100 (2 1 0 2 4 6 8)



EASEMENT DETAIL

1:200 (4 2 0 4 8 12 16)



SIMPLIFIED SINGLE LINE DIAGRAM

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

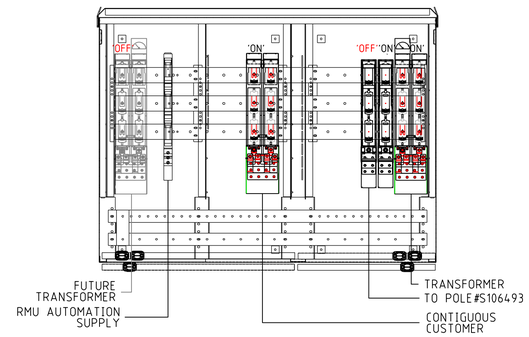
GOLDEN SAFETY RULES ARE INDICATIVE ONLY

ALWAYS PERFORM A DETAILED HAZARD ANALYSIS

TYPE 3.1 PENDA CIRCUIT TABLE

	LEFT	NAME/LOCATION
Circuit 1 (C1)		FUTURE TRANSFORMER 'OFF'
Circuit 2 (C2)		
Circuit 3 (C3)		
Circuit 4 (C4)	LU70	RMU AUTOMATION SUPPLY
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)		
Circuit 8 (C8)		
Circuit 9 (C9)	LU66_CUST	CONTIGUOUS CONNECTION
Circuit 10 (C10)		
Circuit 11 (C11)		
Circuit 12 (C12)		
Circuit 13 (C13)	LU69	POLE#S106493 'OFF'
Circuit 14 (C14)	LU68	ERG
Circuit 15 (C15)	LU66_TX	TRANSFORMER CONNECTION
Circuit 16 (C16)	RIGHT	

TYPE 3.1 PENDA ARRANGEMENT



WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
- THESE WORKS ARE DEPENDENT ON THE COMPLETION OF MP208787 PHASE 1 WORKS.
- PNO TO ORGANISE PURCHASE OF BUND FROM SUPPLIER (MACRETE PRECAST) AND ARRANGE CONSTRUCTION CONTRACTOR TO PICK UP BUND FROM SUPPLIER FOR THE DELIVERY AND INSTALLATION.

(1) INSTALL 1000kVA 22kV SUBSTATION WITH 1x TRANSFORMER 2+1 AUTOMATED SWITCHGEAR AND A TYPE 3.1 PENDA TO DSPM-3-06 AS NOTED ON DRAWING.
INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER.
INSTALL OIL CONTAINMENT BUND WITH A SOAK WELL IN ACCORDANCE WITH WASIR 14.6.5 AND DRAWING DSPM-4-06 (NON-MPS).
CU: HU59_1000/22x1, HU66_22x1, HU6_A400x1, HU55_1000/22x1, CN46x15m, CN60_400CUx80m, LU16_400CU12x1, LU16_400CU16x1, LU43x1, LU65_0x1, LU66_CUSTx1, LU68x1, LU69x1, LU70x1, CN73_35/185x3
DSPM: DSPM-3-04, DSPM-4-06

(2) INSTALL NEW 400 HV CABLES FROM NEW RMU TO EXISTING 400HV CABLE IN ROAD RESERVE.
CU: CN53x20m, CN54x2, CN73_240x3

(3) INSTALL NEW 400 HV CABLES FROM NEW RMU TO EXISTING POLE#S106478 in 0.5m ALIGNMENT.
CU: CN53x210m, CN54x2, CN73_240x3

(4) INSTALL 240LV CABLE FROM NEW LV PENDA TO EXISTING 240LV CABLE IN ROAD RESERVE.
CU: CN45x20m, CN54x2, CN73_LVMAINSx3

(5) CUT AND REMOVE EXISTING 400HV CABLE LIVE END SEAL. STRAIGHT JOINT EXISTING 400HV CABLE TO NEW 400HV CABLES.
CU: HU2 × 1

(6) CUT AND REMOVE EXISTING 240LV CABLE LIVE END SEAL. STRAIGHT JOINT EXISTING 240LV CABLE TO NEW 240LV CABLES.
CU: LU2 × 1

(7) TERMINATE NEW 400HV CABLES AND 35HV TO NEW RMU AS PER SINGLE LINE DIAGRAM.

(8) AT POLE#S106478.
- INSTALL 400HV CABLE TERMINATION & TERMINATE CABLE ONTO EXISTING BARE HV CONDUCTORS.
- LOWER EXISTING LV CROSSARM TO ACCOMODATE HV CABLE TERMINATION.

CU: HU16_0x1, HU16_U_22x1, CN9_7/475AAx3, CN70_150x3
DCSH: H08-3, L01

(9) AT NEW PENDA.
- TERMINATE NEW 240LV CABLE AND TO BE 'OFF'.
- TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT & CIRCUIT TABLE.

REFER TO PENDA TYPE 3.1 CIRCUIT LAYOUT.
CU: LU14_240x1

(10) INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
CU: LM1_M x 1

(11) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

UNEXPLODED ORDNANCE
NO FURTHER ACTIONS REQUIRED FOR THIS CONSTRAINT. HOWEVER PLEASE BE MINDFUL OF ANY UNEXPECTED FINDS AND MANAGE IN ACCORDANCE WITH EXCAVATION AND TRENCHING PROCEDURE (EDMS# 43740956)

THREATENED ECOLOGICAL COMMUNITIES
NATIVE VEGETATION REGS ESAS CLEARING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING, COMPACTION AND LEVELLING.
2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

(C) CUSTOMER TO ARRANGE FOR AN 'EASEMENT IN GROSS' UNDER THE ENERGY OPERATORS POWERS ACT 1979 COVERING THE ENTIRE SUBSTATION SITE AND CABLE ROUTE WITHIN CUSTOMER PROPERTY. CUSTOMER TO SUBMIT THE DEPOSITED PLAN TO WESTERN POWER PRIOR TO THE COMMENCEMENT OF WORKS. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS

(D) CUSTOMER MUST INSTALL MSB CONTIGUOUS TO THE SUBSTATION.

(E) CUSTOMER TO SUPPLY AND INSTALL 'T' TYPE METERING CT's.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(G) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(H) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(J) NO BUILDINGS ALLOWED WITHIN 4.5m OF THE SUBSTATION UNLESS 2 HOUR FIRE RATED. SEE WESTERN POWER DRAWING DSM-5-01 FOR DETAILS.

(K) CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

(L) CUSTOMER TO CONSTRUCT METALLIC FENCES ON THE LOT BOUNDARY AROUND THE BACK OF THE SUBSTATION SO THAT THE FENCE IS NOT IN THE OPERATIONAL CLEAR ZONE AND THE RMU DOORS ARE ABLE TO BE OPENED.

(M) CUSTOMER TO CONTACT WESTERN POWER PROPERTY SPECIALIST AT conveyancin@westernpower.com.au TO ARRANGE THE PREPARATION AND REGISPRATION OF THE REQUIRED 129BA RESTRICTIVE COVENANT. RESTRICTIVE COVENANT TO BE REGISTERED COMMENCEMENT OF WORKS OR ENERGISATION

(N) THE CUSTOMER PROPOSES TO INSTALL SCREENING. REFER TO DETAIL B. APPROVAL SHALL BE OBTAINED FROM WESTERN POWER PRIOR TO ITS INSTALLATION. THE SCREENING SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE. REFER TO THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSES 14.4.10.3.2, 14.4.11 AND 14.5.4 FOR DETAILS.

(P) SUBSTATION TO BE DESIGNED MINIMUM DIMENSIONS SHOWN ON THE DRAWING AND IN-CONJUNCTION WITH WESTERN POWER'S DRAWING MODIFIED DSPM-03-06.
NOTE: STANDARD DSPM-3-06 IS REDUCED LAND SIZE AND DOESN'T SHOW BUNDING INFO.

CONSULTANT CONTACT DETAILS
COMPANY NAME: INFINITE ENGINEERING
CONTACT: MITCHELL WHITBREAD TEL: +61 477 999 674

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

APPROVED FOR CONSTRUCTION

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

SAFETY - MAKE IT YOUR PRIMARY VALUE

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM#: 70791865
A	UNDER ENGINEERING REVIEW	I.P.	12/24	HAZARD MANAGEMENT REGISTER - EDM# 70791242

1000A COMMERCIAL SUPPLY UPGRADE- PHASE 2

NETWORK VOLTAGES: 22kV/415V

136 THIRD AVE, KELMSCOTT WA 6111

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 12/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 07' 31" S Long. - 116° 00' 12" E

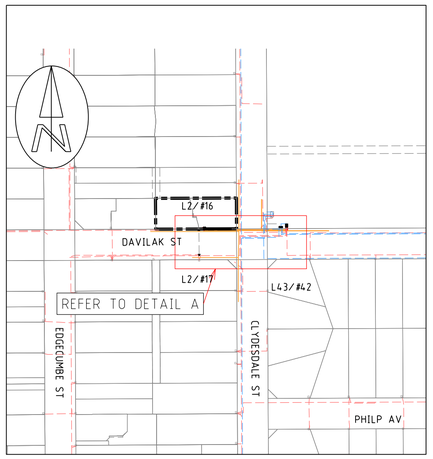
westernpower

DRAWING NUMBER

MP208787

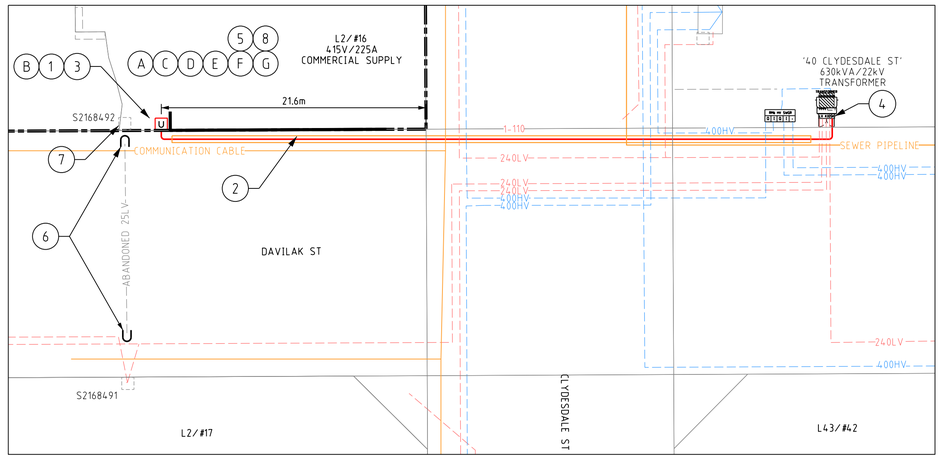
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L2/#16 DAVILAK ST, AS INDICATED ON DETAIL A AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE LV KIOSK AT '40 CLYDESDALE ST' SUBSTATION VIA 110mm POLYPIPE.
   CU: CN45 x 65m, CN54 x 4, CN73_LVMAINS x 3
3. AT UNI-PILLAR (L5/#56 FORREST AV)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240 x 1, LU36_LINK x 1
   DCSH: U9
4. AT LV KIOSK '40 CLYDESDALE ST' SUBSTATION
   - TERMINATE NEW 240LV CABLE VIA 315A FUSES.
   CU: LU14_240 x 1
   DCSH: U27-1
5. INSTALL CT METER.
   CU: LM1_M x 1
6. DISCONNECT EXISTING 25LV CABLE FROM EXISTING PILLAR#S2168492 AND S2168491. PULL BACK TO WP ALIGNMENT, CUT, CAP AND ABANDON THE EXISTING 25V CABLE. REMOVE EXCESS CABLE.
   CU: CN73_LV25 x 3
7. REMOVE EXISTING MINI-PILLAR #S2168492 AT L2/#16 DAVILAK ST.
8. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

E. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

G. CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

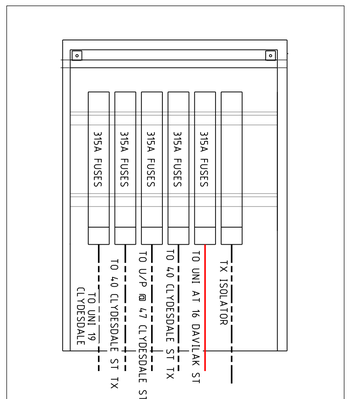
LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

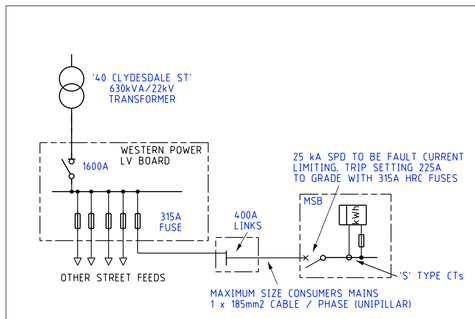
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

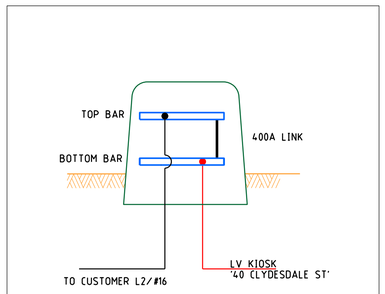
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot



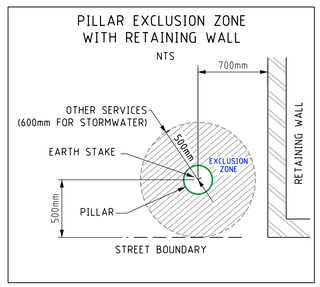
LV KIOSK
'40 CLYDESDALE ST'



SIMPLIFIED SINGLE LINE DIAGRAM



UNI-PILLAR L2/#16
CONNECTION DETAILS



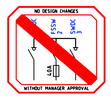
COMMUNICATION CABLES IN VICINITY OF WORKS

WATER Co. PIPELINES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: ADCO ELECTRICS
CONTACT: K.BABUREJENDRAN TEL: 0449863053

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM# 70492531
A	ISSUED FOR CONSTRUCTION	I.P.	11/24	HAZARD MANAGEMENT REGISTER - EDM# 70491245

225A MULTI-RESIDENTIAL SUPPLY UPGRADE

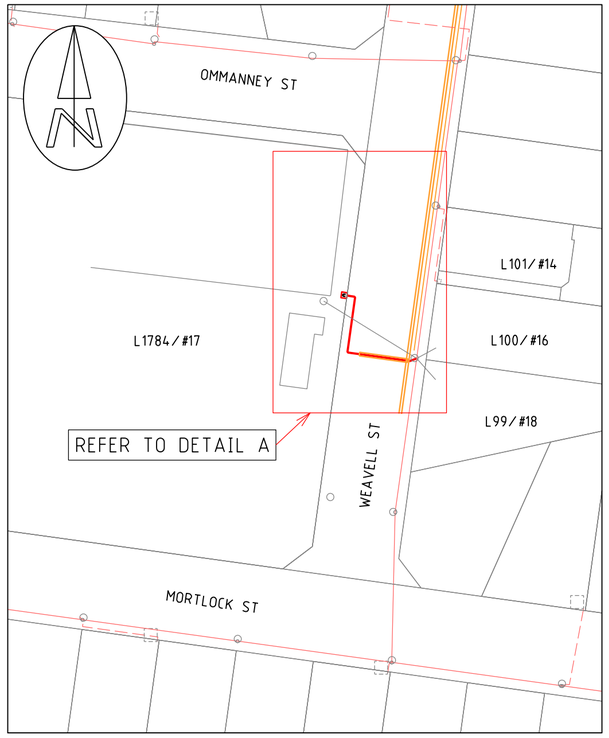
NETWORK VOLTAGES: 22kV/415V

16 DAVILIK STREET, COMO 6152 WA

DESIGN DRAWING

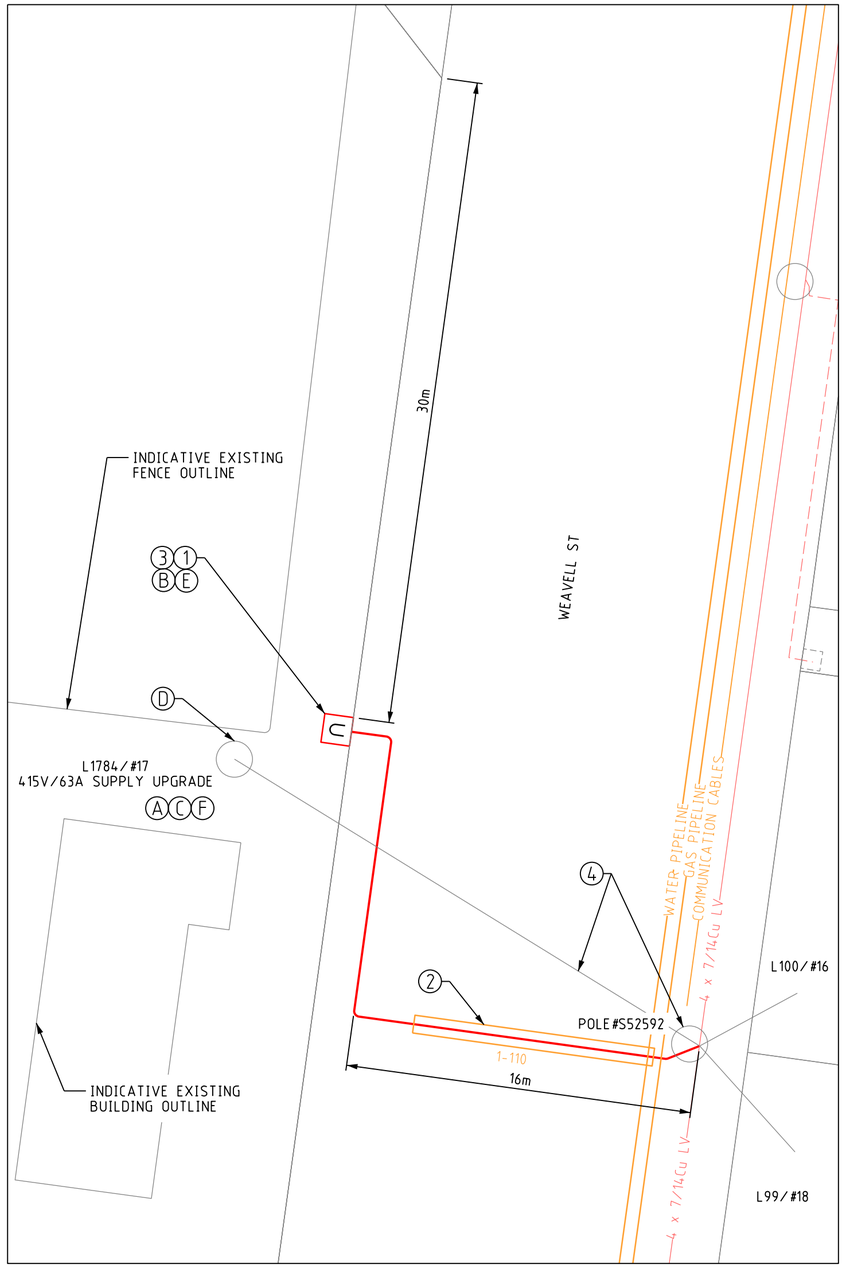
DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 00' 35" S Long. - 115° 51' 43" E

DRAWING NUMBER
MP208802
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16

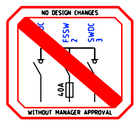
GAS PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO REMOVE CUSTOMER POLE AFTER COMPLETION OF WORK

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW UNI-PILLAR ON L1784/#17 WEAVELL ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11 x 1
DCSH: U09

(2) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S52592 VIA 110mm POLYPIPE.
CU: CN45 x 45m, CN54 x 8, CN73_LVMAINS x 3

(3) AT UNI-PILLAR (L1784/#17 WEAVELL ST.)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
CU: LU7_240 x 1
DCSH: U09

(4) AT POLE#S52592
- REPLACE THE POLE WITH NEW 9.5m/5kN POLE.
- REATTACH EXISTING CROSSARM WITH STREETLIGHT AND OH SERVICES TO NEW POLE
- REMOVE AERIAL SERVICE TO L1784/#17 WEAVELL ST
- TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
CU: P02 x 1, CN27_CULV x 4, LU6_240O x 1, LU6_240U x 1
NON-CU's: CB3025 x 1, AB6732 x 2, CE7541 x 2
DCSH: L01, R13-3, U16-1

(5) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS.

LEGEND

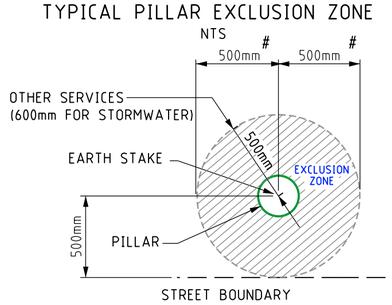
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CUSTOMER CONTACT DETAILS

COMPANY NAME: NORTH LAKE ELECTRICAL
CONTACT: TONY MANGANO TEL: 0413 940 557

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

SHE ASSESSMENT EDM#: 68991780
HAZARD MANAGEMENT REGISTER EDM#: 68971794
FILE REFERENCE DETAILS

415V/63A SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

L1784/#17 WEAVELL ST, HAMILTON HILL WA 6163

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 04' 46" S Long. - 115° 46' 35" E

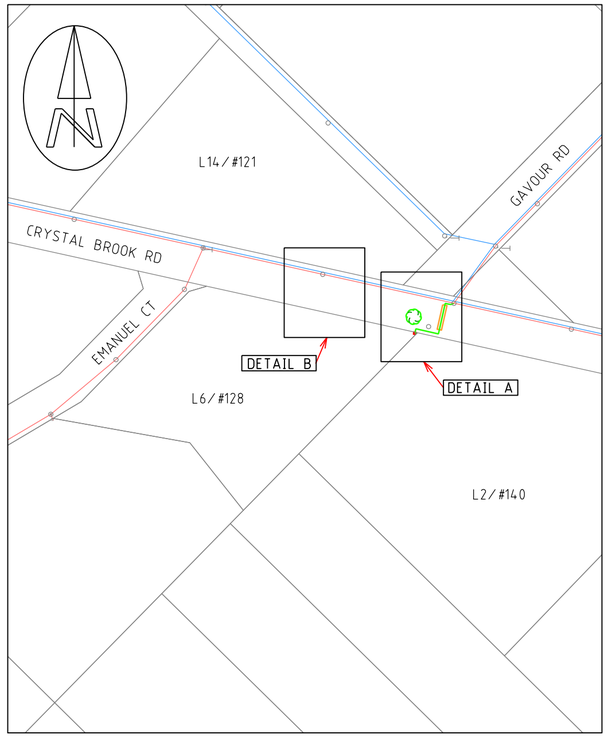
westernpower

DRAWING NUMBER

MP208859

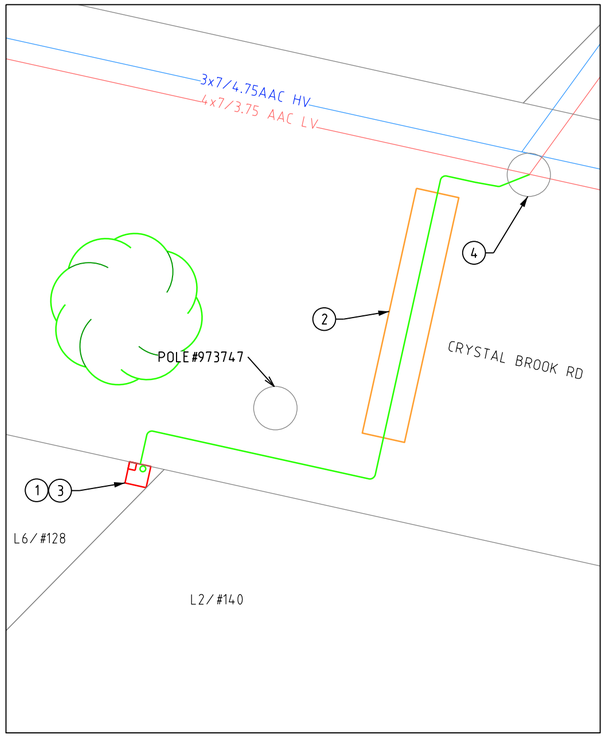
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



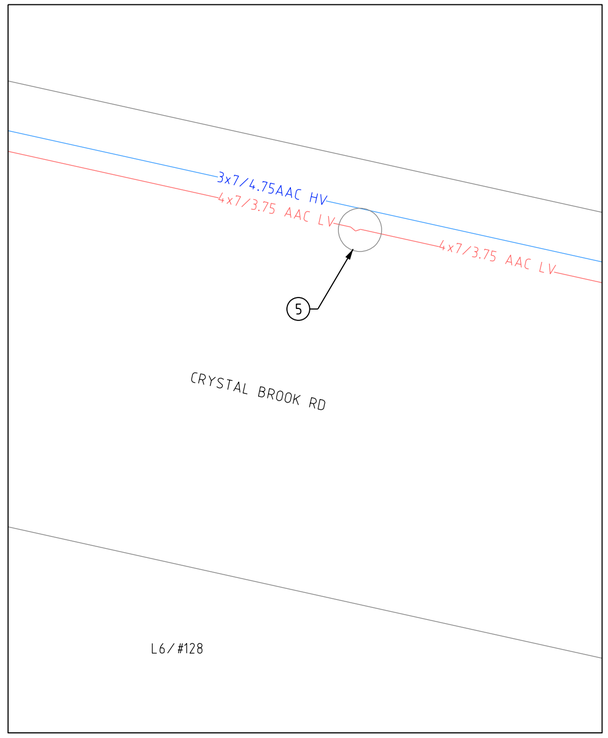
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16



DETAIL B

1:200 — 4 2 0 4 8 12 16

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV - AERIAL CONDUCTORS
- 19/3.25 AAC/AAAC
- 7/4.75 AAC/AAAC
- 7/2.5 AAC/AAAC
- 3/2.75 SCAC
- 6/1/3.00 AACSR/AC
- 150 HENDRIX
- EXISTING HV CONDUCTORS (SIZE AND TYPE INDICATED)

LV - AERIAL CONDUCTORS
- 19/3.25 AAC/AAAC
- 7/4.75 AAC/AAAC
- 7/2.5 AAC/AAAC
- LV ABC 95
- LV ABC 150
- SERVICE 16mm
- EXISTING LV CONDUCTORS (SIZE AND TYPE INDICATED)

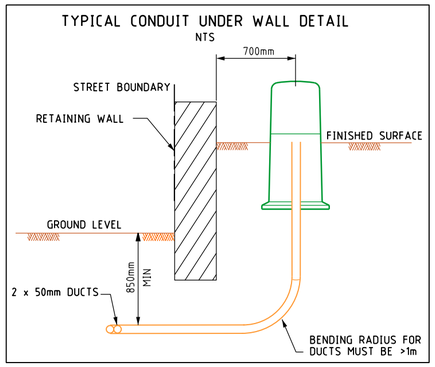
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SWM Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL CONDUIT UNDER WALL DETAIL

NTS



WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L6/#128 CRYSTAL BROOK RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#149638 VIA 63mm POLYPIPE.
   CU: CN42 x 45m, CN73_LV25 x 2, CN54 x 5
3. AT MINI-PILLAR (L6/#128 CRYSTAL BROOK RD)
   - TERMINATE NEW 25LV CABLE.
   - TERMINATE CUSTOMER'S CABLE TO THE BACK BLOCKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
   DCSH: U8-4
4. AT POLE#149638
   - TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
   CU: LU51
   DCSH: U19-1
5. AT POLE#149577
   - REMOVE CUSTOMER AERIAL SERVICE TO L6/#128.
6. ENIVRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

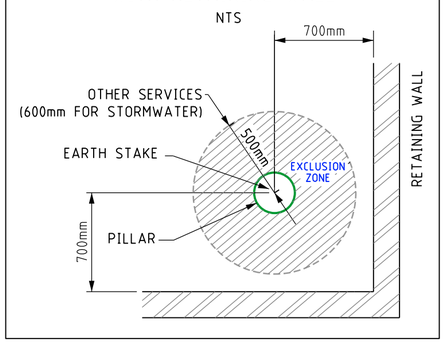
C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

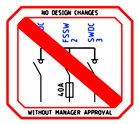
TYPICAL PILLAR EXCLUSION ZONE WITH RETAINING WALL

NTS



APPROVED FOR CONSTRUCTION





CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

VENDOR CONTACT DETAILS

COMPANY NAME: WESCO RESIDENTIAL

CONTACT: EMMA BANDURY TEL: 08 6363 8744

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT EDM#: 68642484
HAZARD MANAGEMENT REGISTER EDM#: 68643001

MINI-PILLAR INSTALLATION

NETWORK VOLTAGE: 415V

128 CRYSTAL BROOK RD - WATTLE GROVE WA, 6107

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 00' 43" S Long. - 116° 00' 58" E

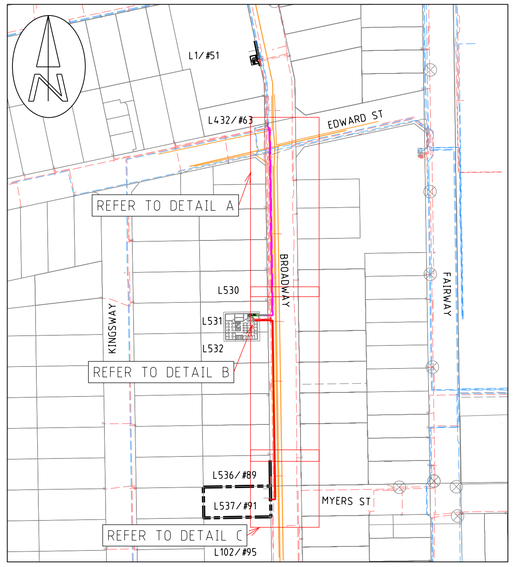
westernpower

DRAWING NUMBER

MP208863

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



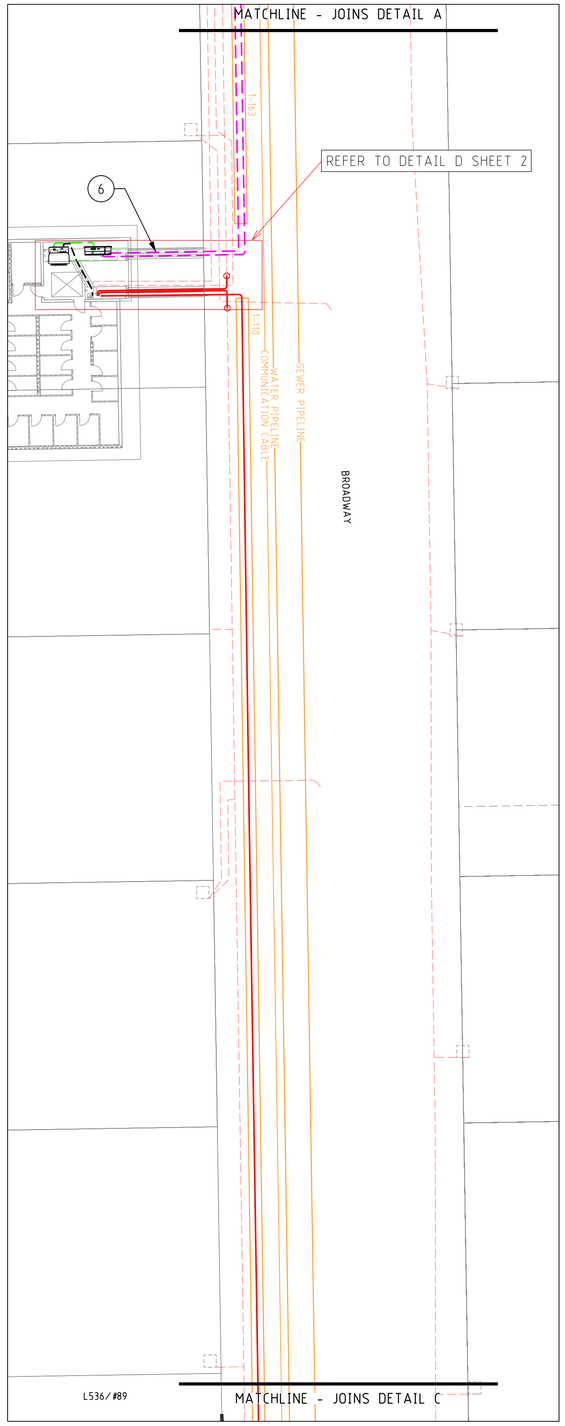
LOCATION PLAN

1:2000 — 40 20 0 40 80 120 160



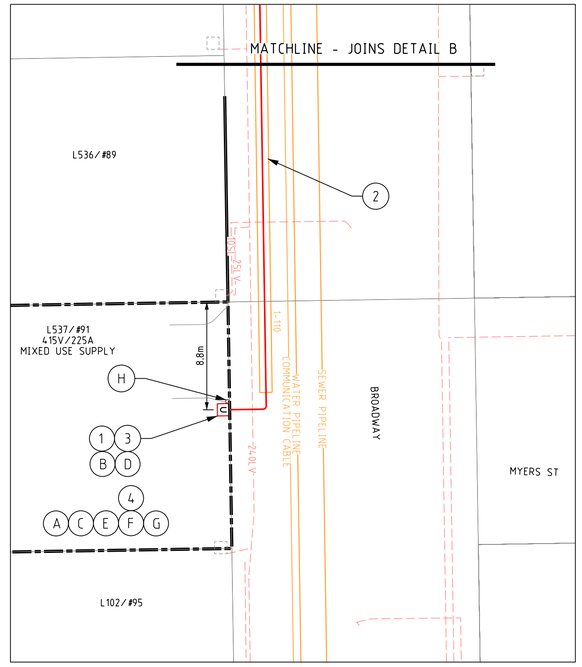
DETAIL A

1:250 — 5 2.5 0 5 10 15 20



DETAIL B

1:250 — 5 2.5 0 5 10 15 20



DETAIL C

1:250 — 5 2.5 0 5 10 15 20

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- Serviced Uncleared De-energised Lot
- HMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

CUSTOMER CONTACT DETAILS

COMPANY NAME: ADCO ELECTRICS
CONTACT: KALAIVANI BABURAJENDRAN TEL: (08) 9459 9600

ISSUED FOR CONSTRUCTION

WATER PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

SEWER PIPELINES IN VICINITY OF WORKS

WESTERN POWER SCOPE OF WORK

REFER TO SHEET 2.

CUSTOMER SCOPE OF WORK

REFER TO SHEET 2.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

D.A.P. REFERENCE No. DAP/21/02110

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT- EDM#: 69479893
A	ISSUED FOR CONSTRUCTION	I.P.	09/24	HAZARD MANAGEMENT REGISTER - EDM# 69479394

225A MIXED USE SUPPLY

NETWORK VOLTAGES: 22kV/415V

91 BROADWAY, NEDLANDS, WA 6009

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 09/24	SCALE: AS NOTED @ A1
GEO REF:	Lat. - 31° 58' 56" S Long. - 115° 48' 52" E

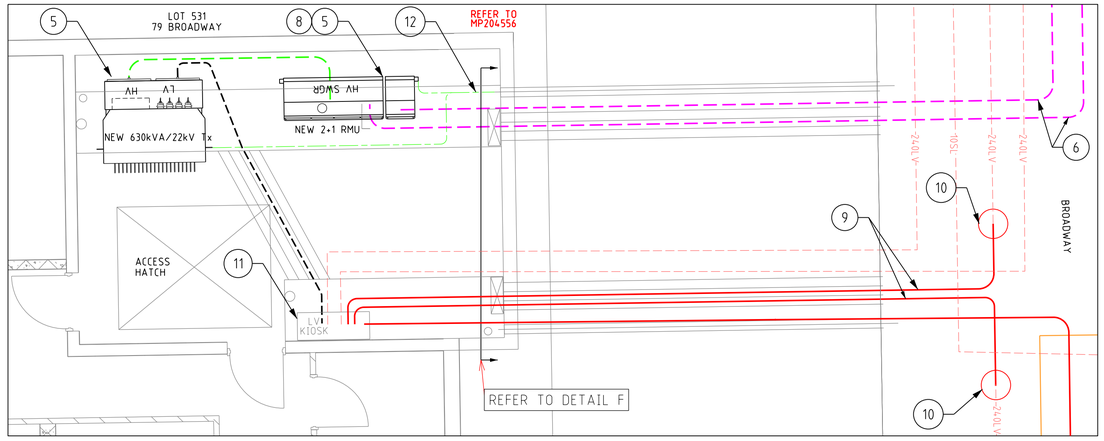
westernpower

DRAWING NUMBER

MP208917

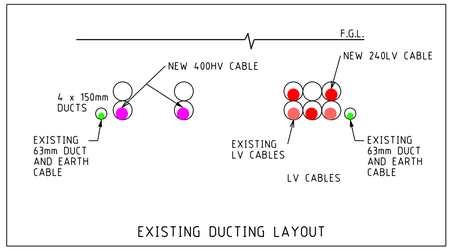
SHEET 01 OF 02

REV: A | ORIGINAL SIZE - A1



DETAIL E

1:50



EXISTING DUCTING LAYOUT

DETAIL F NOT TO SCALE

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L537/#91 BROADWAY AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU11x1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING LV KIOSK AT 55 BROADWAY VIA 110mm POLYPIPE.
   CU: CN45x140m, CN73_LVMAINSx3, CN54x4
3. AT NEW UNI-PILLAR L537/#91 BROADWAY.
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR VIA LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_LINKx1
   DCSH: U9
4. INSTALL CT METER IN CUSTOMER SWITCHBOARD.
   CU: LM1_M x 1
5. INSTALL 630kVA 22kV SUBSTATION WITH 2+1 AUTOMATED SWITCHGEAR TO DSPM-3-08 AS NOTED ON DRAWING. INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER. REFER TO DETAIL E.
   CU: HU59_63011 x1, HU66_22x1, HU55_630/11x1, HU23_M400x1
   CN44x15m, CN60_400CU x80m, LU16_400CU12 x 1, LU16_400CU16 x 1
   DSPM: DSPM-3-08, DSPM-4-06
6. INSTALL 2x 400HV CABLE FROM NEW SWITCHGEAR TO EXISTING 240HV CABLE (FEEDER ID:MCE 333.0) VIA 163mm POLYPIPE.
   CU: CN53x290m, CN54x4, CN73_240x6
7. CUT EXISTING 240HV AND REMOVE EXISTING TRANSITION JOINT (FEEDER ID:MCE 333.0) AND STRAIGHT JOINT TO NEW 400HV CABLES.
   CU: HU2x1, HU30x1
8. TERMINATE 2x 400HV CABLE AND 35HV TO NEW 2+1 SWITCHGEAR AS PER SIMPLIFIED LINE DIAGRAM.
9. INSTALL 2 x 240LV CABLE FROM EXISTING LV PENDA TO EXISTING 240LV CABLE ON BROADWAY VIA EXISTING CONDUITS. REFER TO DETAIL E
   CU: CN45 x 40m, CN73_LVMAINS x 6 , CN54 x 2
10. CUT EXISTING 240LV CABLE AND 2 x STRAIGHT JOINT TO 2 x NEW 240LV CABLES.
    CU: LU2 x 2
11. AT EXISTING LV KIOSK LOCATED AT '79 BROADWAY':
    - TERMINATE NEW 240LV CABLE TO "CCT2" AND INSTALL 315A FUSES.
    - TERMINATE NEW 240LV CABLE TO "CCT3" IN THE "OFF" POSITION.
    - TERMINATE NEW 240LV CABLE TO "CCT4" AND INSTALL 315A FUSES.
    - RE-TERMINATE EXISTING 240LV CABLE FROM EXISTING TRANSFORMER DISCONNECT TO "CCT6" IN THE "OFF" POSITION.
    - TERMINATE NEW TRANSFORMER CU CABLES TO THE EXISTING TRANSFORMER DISCONNECT.
    - RE-LABEL LV FRAME.
    REFER TO EXISTING PENDA TYPE 2.1 CIRCUIT LAYOUT.
    CU: LU14_240 x 3, LU69 x 4
    NON CU: GF1101 x 6
    DCSH: U27/1
12. ATTACH SUBSTATION EARTH TO EXISTING REMOTE EARTH.
13. ENVIRONMENTAL FILTER CHECKED.
    THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

    HERITAGE COUNCIL OF WA
    MUNICIPAL INVENTORY
    IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 41507941)

    REFERAL TO THE LOCAL COUNCIL PLANNING OFFICER (OR HERITAGE OFFICER IN LARGER LGAS) WILL ASSIST IN DETERMINING WHETHER THERE ARE ANY LOCAL HERITAGE MATTERS WHICH REQUIRE CONSIDERATION PRIOR TO COMMENCEMENT OF WORKS.

    REFER TO environment@westernpower.com.au

LEGEND

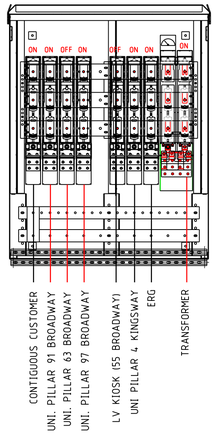
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

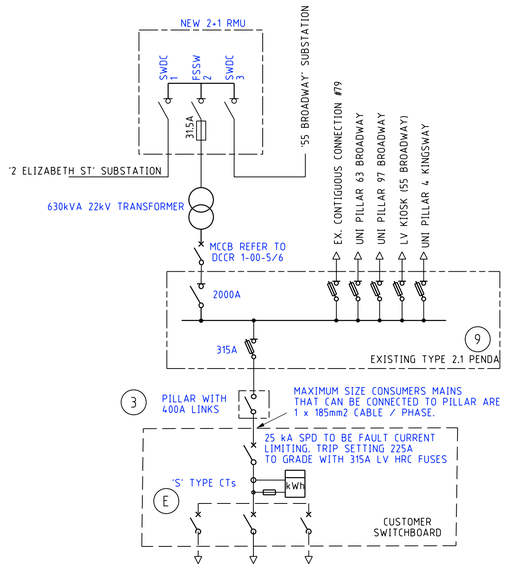
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUM Serviced Uncleared De-energised Lot

TYPE 2.1 PENDA ARRANGEMENT





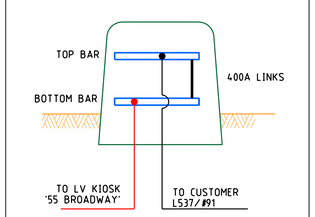
SIMPLIFIED LINE DIAGRAM

	LEFT	NAME/LOCATION
Circuit 1 (C1)	ON	CONTIGUOUS CONNECTION
Circuit 2 (C2)	LU69 'ON'	UNI. PILLAR 91 BROADWAY
Circuit 3 (C3)	LU69 'OFF'	UNI. PILLAR 63 BROADWAY
Circuit 4 (C4)	LU69 'ON'	UNI. PILLAR 97 BROADWAY
Circuit 5 (C5)	-	SPARE
Circuit 6 (C6)	LU69 'OFF'	LV KIOSK (55 BROADWAY)
Circuit 7 (C7)	'ON'	UNI. PILLAR 4 KINGSWAY
Circuit 8 (C8)	'ON'	ERG
Circuit 9 (C9)	'ON'	TRANSFORMER CONNECTION
Circuit 10 (C10)		

RIGHT

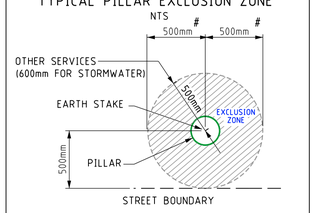
EXISTING TYPE 2.1 PENDA CIRCUIT LAYOUT

UNI-PILLAR L537/#91
CONNECTION DETAILS



TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

E. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

G. CUSTOMER TO SUPPLY AND INSTALL THE SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

H. CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

SEWER PIPELINES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS



ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: ADCO ELECTRICS
CONTACT: KALAIVANI BABURAJENDRAN TEL: (08) 9459 9600



Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT- EDM#: 69479893
A	ISSUED FOR CONSTRUCTION	I.P.	09/24	HAZARD MANAGEMENT REGISTER - EDM# 69479394

225A MIXED USE SUPPLY

NETWORK VOLTAGES: 22kV/415V

91 BROADWAY, NEDLANDS, WA 6009

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 09/24	SCALE: AS NOTED @ A1
GEO REF:	Lat. - 31° 58' 56" S Long. - 115° 48' 52" E

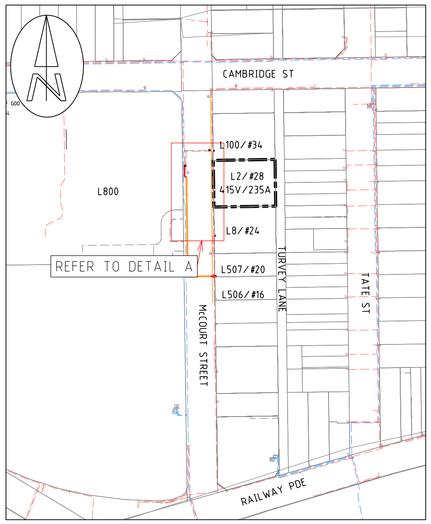
westernpower

DRAWING NUMBER

MP208917

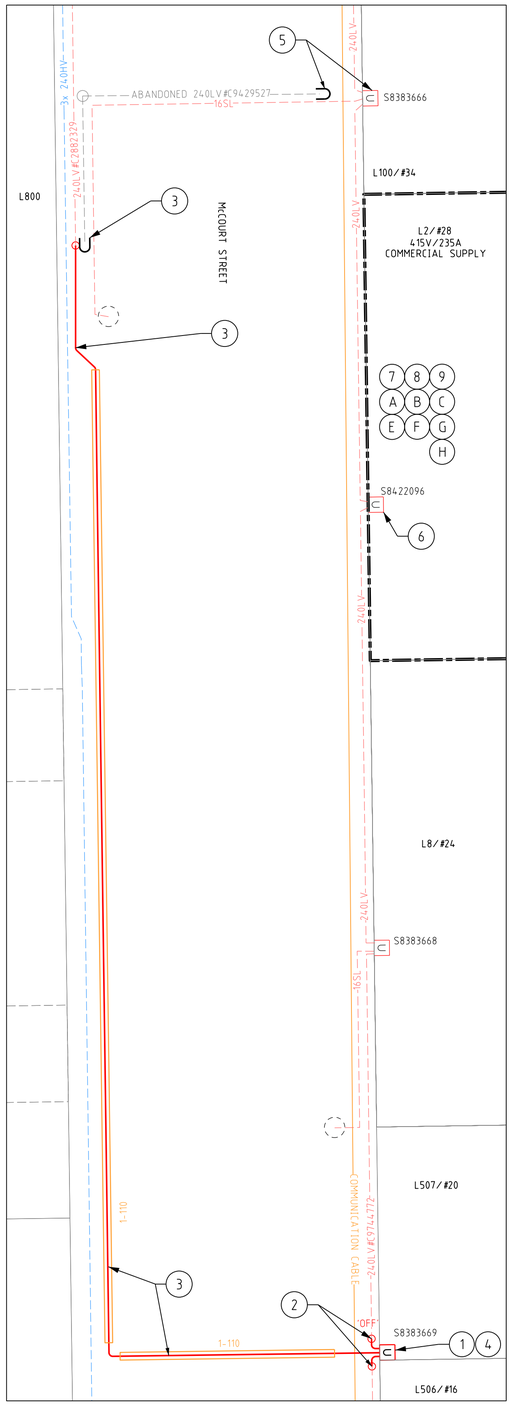
SHEET 01 OF 02

REV: A | ORIGINAL SIZE - A1



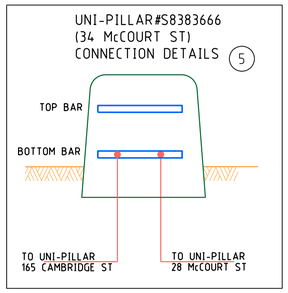
LOCATION PLAN

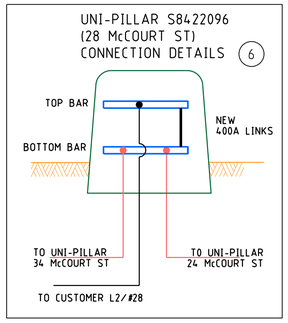
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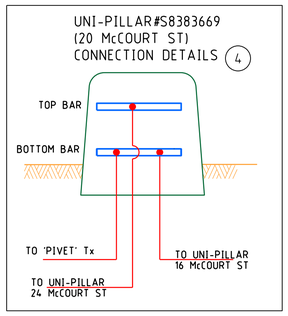


DETAIL A

1:200 — 4 2 0 4 8 12 16





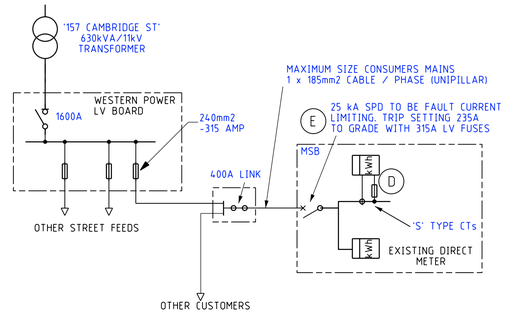


WESTERN POWER SCOPE OF WORK

1. REMOVE EXISTING MINI-PILLAR#S8383669 AND INSTALL NEW UNI-PILLAR ON #20 McCOURT ST AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 2 x 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING 240LV CABLE#C9744772.
   CUT EXISTING 240LV CABLE AND 2 x STRAIGHT JOINT TO 2 x NEW 240LV CABLES.
   CU: CN45 x 20m, CN73_LVMAINS x 4, CN54 x 2, LU2 x 2
3. INSTALL NEW 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING 240LV CABLE VIA 110mm POLYPIPE.
   CUT EXISTING 240LV CABLE#C2882329 AND STRAIGHT JOINT TO NEW 240LV CABLE
   CUT, CAP AND ABANDON EXISTING LV CABLE#C9429527.
   REMOVE EXCESS CABLE
   CU: CN45 x 100m, CN73_LVMAINS x 4, CN54 x 5, LU2 x 1
4. AT NEW UNI-PILLAR (#20 McCOURT ST)
   - TERMINATE NEW 240LV FROM #24 McCOURT ST TO TOP BAR.
   - TERMINATE NEW 240LV FROM PIVET Tx TO BOTTOM BAR.
   - TERMINATE NEW 240LV FROM #16 McCOURT ST TO BOTTOM BAR.
   CU: LU7_240 x 3
   DCSH: U9
5. AT UNI-PILLAR#S8383666 (#34 McCOURT ST)
   - DISCONNECT EXISTING 240LV CABLE FROM PIVET Tx AND PULL BACK TO WESTERN POWER ALIGNMENT.
   - CUT THAN CAP AND ABANDON EXISTING LV CABLE#C9429527. REMOVE EXCESS CABLE.
   - RE-TERMINATE EXISTING 240LV CABLE FROM #165 CAMBRIDGE ST. TO BOTTOM BAR.
   CU: CN73_LVMAINS x 1
6. AT UNI-PILLAR#S8422096 (#28 McCOURT ST)
   - DISCONNET EXISTING CUSTOMER MAINS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR.
   - TERMINATE NEW CUSTOMER MAINS CABLE AT TOP BAR WITH LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR.
   - UPDATE UNI-PILLAR PICK 1D.
   CU: LU36_LINK x 1
7. INSTALL CT METER.
   CU: LM1_M x 1
8. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.
9. CM TO CONFIRM RETAILER CONSENT HAS BEEN OBTAINED AS PER CUSTOMER SCOPE OF WORK H. METERING POINT CHANGES CANNOT PROCEED UNTIL CONFIRMED.

CUSTOMER SCOPE OF WORK

- (A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.
- (B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.
- (C) CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.
- (D) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.
- (E) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.
- (F) CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY.
- (G) CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.
- (H) CUSTOMER TO PROVIDE RETAILER CONSENT TO METERING CHANGES. PROJECT WILL NOT BE SCHEDULED UNTIL THE ETAF IS RECEIVED BY WESTERN POWER.



SIMPLIFIED SINGLE LINE DIAGRAM

LEGEND

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

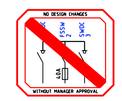
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

ISSUED FOR CONSTRUCTION



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: McCOURT ST DAY SURGERY
CONTACT: Lynne Durkin TEL: 0418 372 117

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM#: 70185002
A	ISSUED FOR CONSTRUCTION	I.P.	11/24	HAZARD MANAGEMENT REGISTER - EDM# 70181627

235A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGES: 22kV/415V

28 McCOURT ST, WEST LEEDERVILLE WA 6007

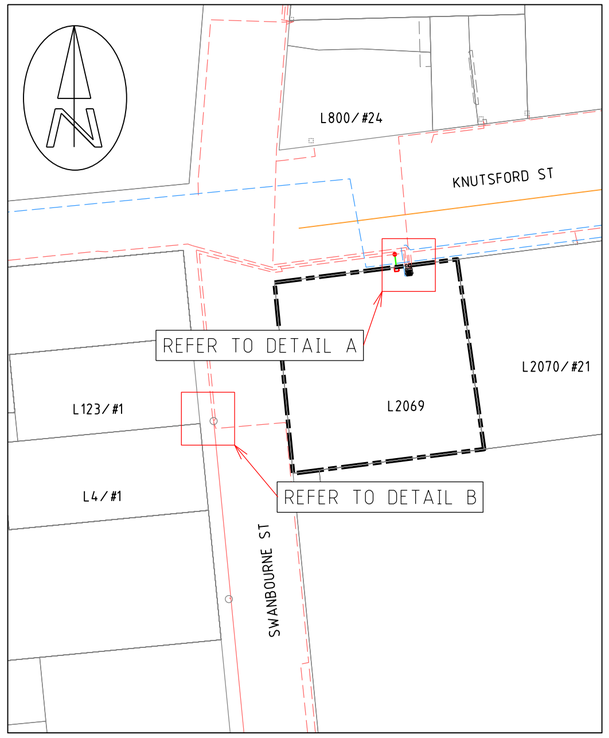
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 | SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 56' 28" S, Long. - 115° 49' 39" E

westernpower

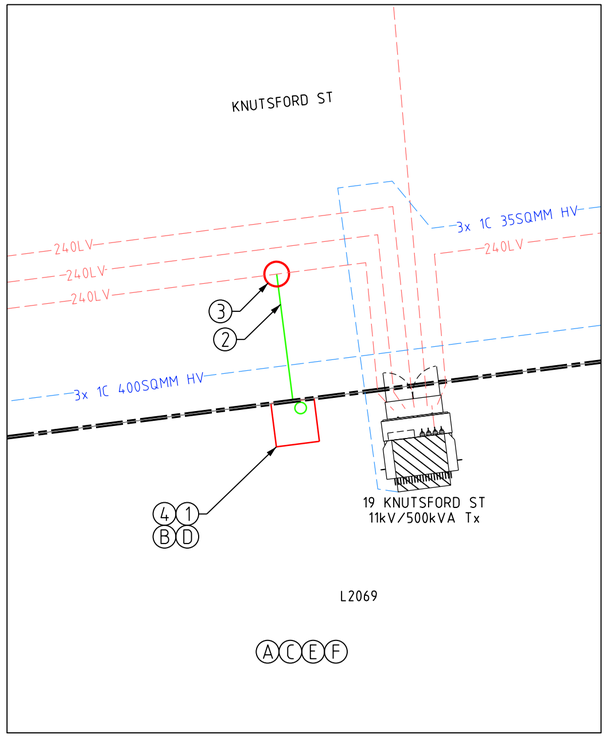
DRAWING NUMBER
MP208958
SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A1



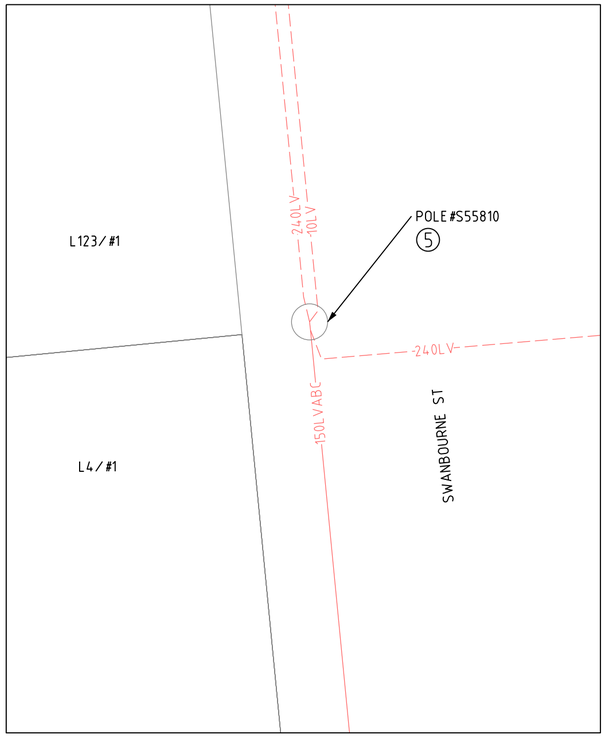
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:100 2 1 0 2 4 6 8



DETAIL B

1:200 4 2 0 4 8 12 16

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L2069 KNUTSFORD ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU10 x 1
2. INSTALL 25LV CABLE FROM NEW MINI-PILLAR TO 250LV CABLES
   CU: CN42 x 15m, CN73_LV25 x 3
3. TEE-JOINT 25LV CABLE WITH EXISITNG 240LV CABLE
   CU: L15 x 1
4. AT MINI-PILLAR (L2069 KNUTSFORD ST.)
   - TERMINATE NEW 25LV CABLE.
   - TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   DCSH: U8-04
5. AT POLE#55810
   - REMOVE OH SERVICE TO L2069 KNUTSFORD ST.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

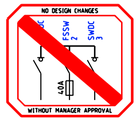
E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER TO OBTAIN WORK ZONE PERMIT FROM THE TOWN OF VICTORIA PARK AT NO COST TO WESTERN POWER.

FOLLOW PERMIT CONDITIONS FROM THE TOWN OF VICTORIA PARK.

APPROVED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

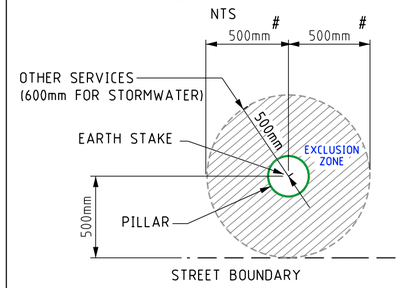
DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CONSULTANT CONTACT DETAILS
COMPANY NAME: NORTHLAKE ELECTRICAL PTY. LTD.
CONTACT: TONY MANGANO TEL: 0413 940 557

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68781548
HAZARD MANAGEMENT REGISTER EDM#: 68796537

415V/32A SUPPLY UPGRADE

NETWORK VOLTAGE: 11kV

SWANBOURNE ST, FREMANTLE WA 6160

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 03' 12" S Long. - 115° 45' 30" E

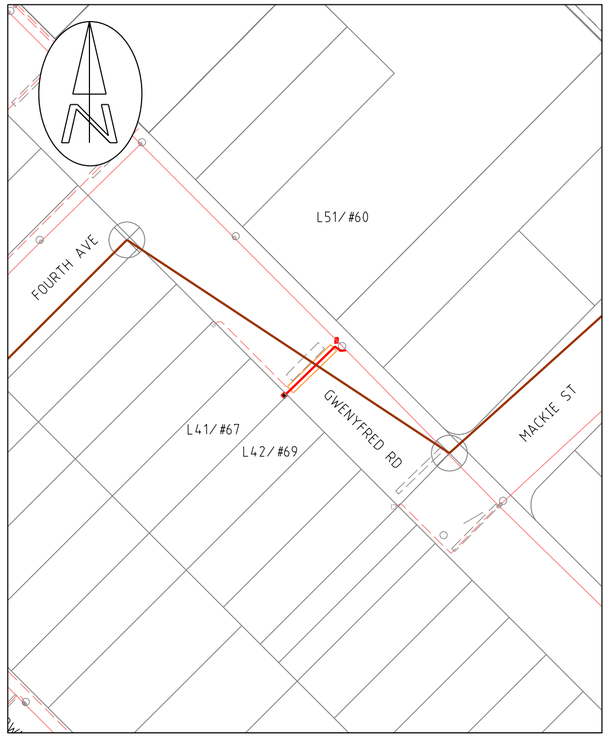
westernpower

DRAWING NUMBER

MP209024

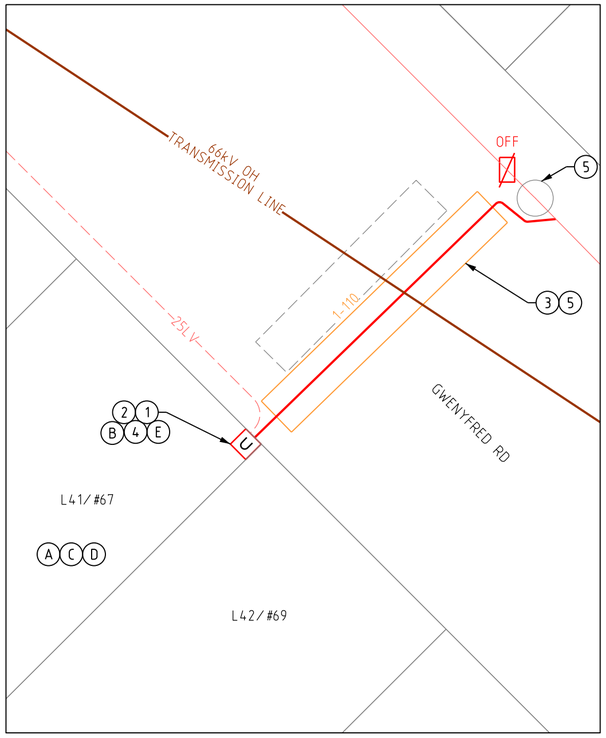
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING CUSTOMER MAINS, 25LV AND REMOVE EXISTING MINI-PILLAR
2. INSTALL NEW UNI PILLAR ON L41/#67 GWENYFRED RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU11
   DCSH: U9
3. INSTALL NEW 240LV CABLE FROM NEW UNI-PILLAR TO POLE#239216 VIA 110mm POLYPILE
   CU: CN45 x 30m, CN54 x 8, CN73_LVMAINS x 2
   DCSH: U16-1
4. AT UNI-PILLAR(L41/#67 GWENYFRED RD)
   - TERMINATE NEW 240LV STREET MAINS CABLE
   - TERMINATE CUSTOMER'S CABLE FROM L42/#69 AND L41/#67 GWENYFRED RD & 25LV STREET CABLE TO THE BACK BLOCKS.
   CU: 1x LU7
   DCSH: U9
5. AT POLE#239216
   - TERMINATE NEW 240LV CABLE.
   - REMOVE EXISTING 25LV CABLE.
   - ABANDON EXISTING 50mm CONDUIT.
   CU: LU6_2400 x 1, LU6_240U x 1
   DCSH: U16-1
6. ENVIRONMENTAL FILTERS CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

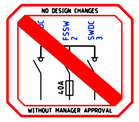
D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.



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GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

OVERHEAD TRANSMISSION ASSETS IN VICINITY OF WORKS

APPROVED FOR CONSTRUCTION

CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV - AERIAL CONDUCTORS
19/3.25 AAC/AAAC
7/4.75 AAC/AAAC
7/2.5 AAC/AAAC
LV ABC 95
LV ABC 150
SERVICE 16mm
EXISTING LV CONDUCTORS (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
42W CFL
80W MV
125W MV
70W HPS
150W HPS
250W HPS
250W HPS AERODROME
70W MH
150W MH
250W MH
1L LED
2L LED
3L LED
4L LED
5L LED
6L LED
1E LED
2E LED
3E LED
4E LED
5E LED
6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

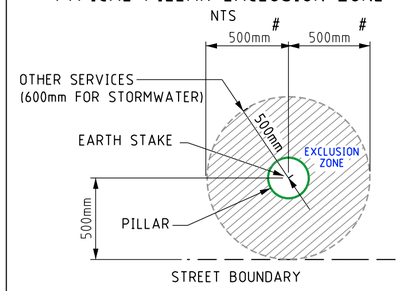
PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

TRANSFORMER EXISTING SWITCHGEAR

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CUSTOMER CONTACT DETAILS
COMPANY NAME: THE TRUSTEE FOR RICCIARDELLO
CONTACT: PETER ROMITI TEL: 0438100296

Rev.	REVISIONS	DRN	DATE
A	UNDER ENGINEERING REVIEW	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68526612
HAZARD MANAGEMENT REGISTER EDM#: 68580258

59.3A/415V RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 415V

67 GWENYFRED RD - KENSINGTON WA 6151

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 58' 42" S Long. - 115° 53' 08" E

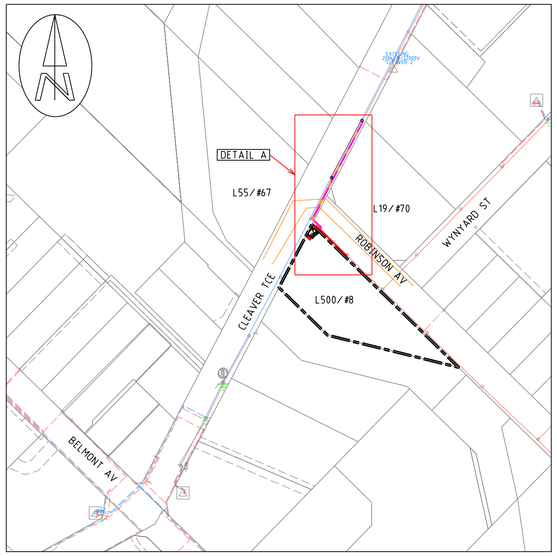
westernpower

DRAWING NUMBER

MP209186

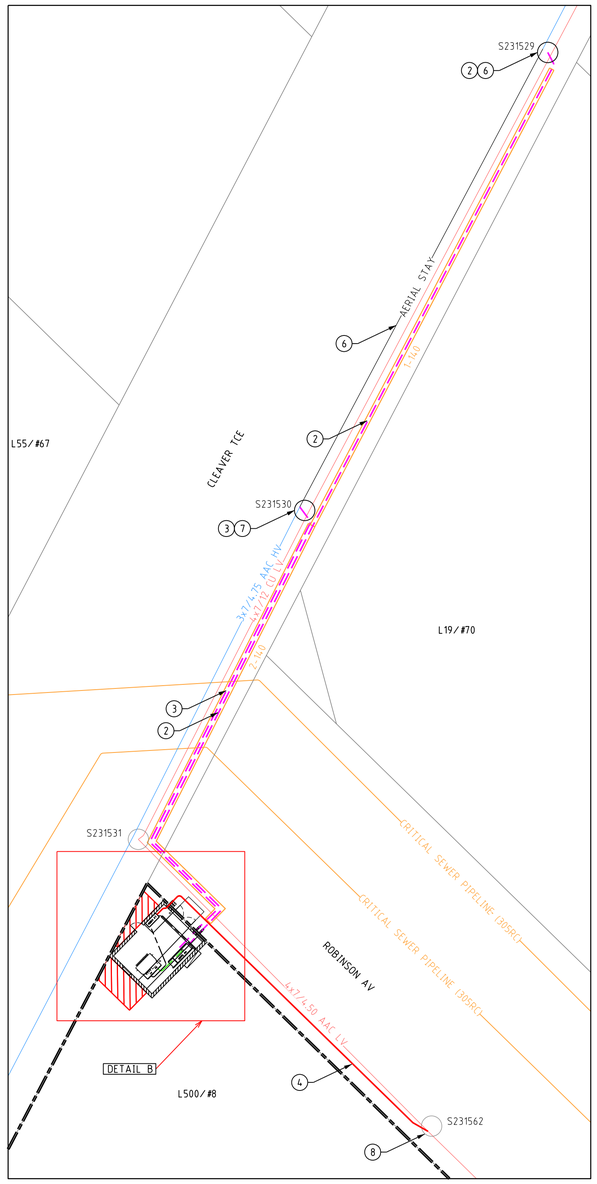
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:250 5 2.5 0 5 10 15 20

CUSTOMER SCOPE OF WORK

PLEASE REFER TO SHEET 2 FOR CUSTOMER SCOPE OF WORK.

WESTERN POWER SCOPE OF WORK

1. INSTALL 1000kVA 22kV INDOOR DISTRICT SUBSTATION TO DSPM-3-08 AS NOTED ON DRAWING WITH 2+1 SWITCHGEAR. REMOTE EARTHING TO BE INSTALLED IN ROAD RESERVE IF DEEP EARTHING IF REQUIRED.
   CU: HU23_A400x1, HU55_1000/22x1, HU59_10002x1, HU66x1, LU16_400CU12x1, LU16_400CU16x1, CN60_400CUx90m, LU64_1x1, LU67x1, LU68x1, LU69x1, CN46x15m, CN81x12, LU70x1, DA6_LVSx1, LU14_16/160Ax1, CN41x10m
   DSPM: DSPM-3-08, DSPM-4-06
2. INSTALL 400 HV CABLE FROM NEW 2+1 HV SWITCHGEAR TO THE EXISTING POLE S231529.
   UTILISE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 400HV CABLE IN THE 0-500mm ALIGNMENT.
   CU: CN53x95m, CN54x5, CN73_240x3
3. INSTALL 400 HV CABLE FROM NEW 2+1 HV SWITCHGEAR TO THE EXISTING POLE S231530.
   UTILISE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 400HV CABLE IN THE 0-500mm ALIGNMENT.
   CU: CN53x60m, CN54x5, CN73_240x3
4. INSTALL 240LV CABLE FROM NEW TYPE 2.1 PENDA KIOSK TO EXISTING 240LV CABLE.
   CU: CN45x45m, CN54x2, CN73_LVMAINSx3
5. TERMINATE 2x400HV CABLE TO NEW 2+1 HV SWITCHGEAR AS PER SIMPLIFIED LINE DIAGRAM.
6. AT POLE#S231529.
   - REPLACE EXISTING POLE WITH NEW 12.5m/6kN POLE.
   - REMOVE EXISTING 3x7/4.75 AAC HV TOWARDS S231530.
   - INSTALL HV TERMINATION CROSSARM.
   - INSTALL LV INTERMEDIATE CROSSARM.
   - TERMINATE NEW 400HV CABLE TO HV BARE.
   - INSTALL AERIAL STAY TOWARDS POLE S231530.
   - REATTACH EXISTING AERIAL SERVICES TO NEW POLE.
   CU: P08_R×1, HV5_3MN×1, LV1_W3×1, ST2_19/200W×1, HU16_U_22×1, HU16_0×1, CN1_7/475×3, CN9_7/475AA×3
   DCSH: H09-1, L01, R13-3, R14-2, R14-3
7. AT POLE#S231530.
   - REPLACE EXISTING POLE WITH NEW 12.5m/6kN POLE.
   - INSTALL HV TERMINATION CROSSARM.
   - INSTALL LV INTERMEDIATE CROSSARM.
   - TERMINATE NEW 400HV CABLE TO HV BARE. TOWARDS SOUTH WEST.
   - ATTACH AERIAL STAY FROM POLE S231529.
   - RE-ATTACH EXISTING AERIAL SERVICES TO NEW POLE.
   CU: P08_R×1, HV5_3MN×1, LV1_W3×1, HU16_U_22×1, HU16_0×1, CN1_7/475×3, CN9_7/475AA×3
   DCSH: H09-1, L01, R13-3, R14-3
8. AT POLE S231562, TERMINATE 240LV CABLE
   CU: LU6_240Ox1, LU6_240Ux1, CN9_7/475AAx4
   DCSH: U16
9. AT NEW TYPE 2.1 PENDA.
   - TERMINATE 1x240LV CABLE IN 'OFF' POSITION.
   - TERMINATE CUSTOMER'S CABLE TO THE TYPE 2.1 PENDA IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT AND CIRCUIT TABLE.
   CU: LU14_240x1
   NON CU: GF1102x6
10. INSTALL 2x 'S' TYPE CT METER IN CUSTOMERS SWITCHBOARD.
    CU: LM1x2
11. ENVIRONMENTAL FILTER CHECKED.
    THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

    ACID SULPHATE SOILS (ASS)
    EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO: BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING. IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

    ABORIGINAL HERITAGE SITES
    IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

    REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

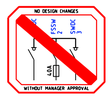
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION





WATER CORP PIPELINE IN VICINITY OF WORKS

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: ECWA
CONTACT: MICHAEL ANILE TEL: 0439 988 739

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	05/25	ENVIRONMENTAL ASSESSMENT
				HAZARD MANAGEMENT REGISTER

NEW 550A COMMERCIAL SUPPLY

NETWORK VOLTAGES: 22kV/415V

8 ROBINSON AVE, BELMONT WA 6104

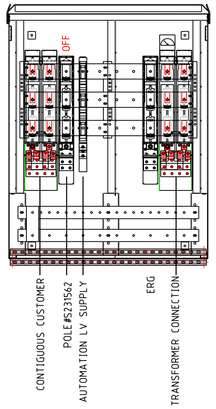
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 05/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 57' 06" S Long. - 115° 55' 13" E

westernpower

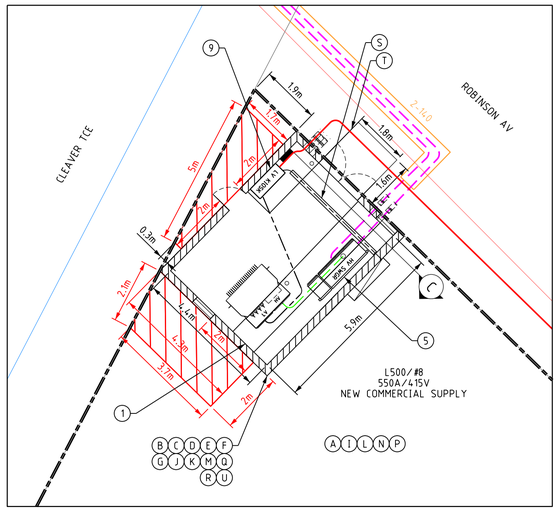
DRAWING NUMBER
MP209285
SHEET 01 OF 02
REV: A ORIGINAL SIZE - A1

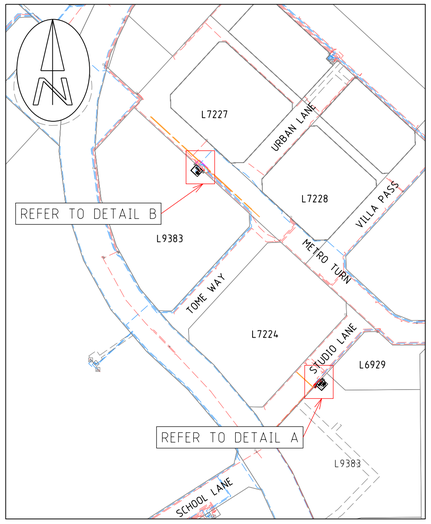
TYPE 2.1 PENDA ARRANGEMENT



	LEFT	NAME/LOCATION
Circuit 1 (C1)	LU67	CONTIGUOUS CONNECTION #8 ROBINSON AVE
Circuit 2 (C2)	LU69 "OFF"	POLE#S231562
Circuit 3 (C3)	LU70	AUTOMATION SUPPLY
Circuit 4 (C4)		
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)		
Circuit 8 (C8)	LU68	ERG
Circuit 9 (C9)	LU66	TRANSFORMER CONNECTION
Circuit 10 (C10)	RIGHT	

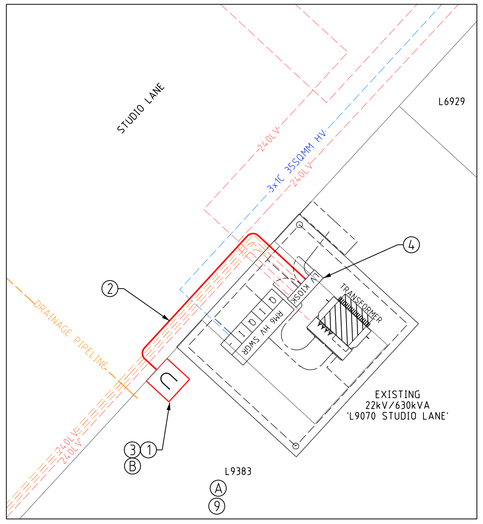
DETAIL E





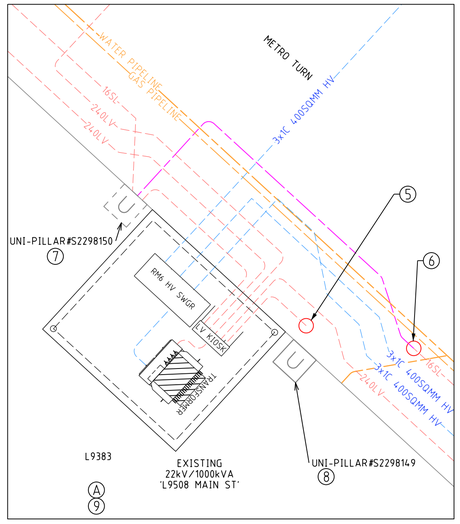
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

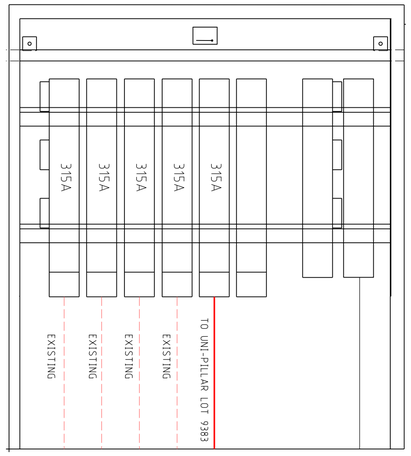
1:100 2 1 0 2 4 6 8



DETAIL B

1:100 2 1 0 2 4 6 8

EXISTING KIOSK ARRANGEMENT L9070 STUDIO LANE



WESTERN POWER SCOPE OF WORK

DETAIL A

1. INSTALL NEW UNI-PILLAR ON L9383 STUDIO LANE AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING LV KIOSK.
   CU: CN45 x 20m, CN73_LVMAINS x 3, CN54 x 2
3. TERMINATE 240LV CABLE TO THE NEWLY INSTALLED UNI-PILLAR AT L9383 STUDIO LANE.
   CU: LU7_240 x 1
   DCSH: U9
4. AT EXISTING LV KIOSK, 'L9070 STUDIO LANE'
   - TERMINATE NEW 240LV CABLE
   - INSTALL 315A FUSE
   CU: LU14_240 x 1
   NON CU: GF1101 x 3
   DCSH: U27-1

DETAIL B

5. LIVE END SEAL EXISTING 240LV
   CU: LU62 x 1
6. INSTALL 16SL CABLE FROM EXISTING UNI-PILLAR S2298150 AND ST JOINT TO EXISTING 16SL CABLE.
   CU: CN41 x 20m, CN73_10/16 x 3, CN54 x 2, LU13 x 1
7. AT UNI-PILLAR (S2298150)
   - TERMINATE 16SL AT THE BACK BLOCK VIA RED SPOT FUSE.
8. AT UNI-PILLAR (S2298149)
   - DISCONNECT EXISTING 240LV CABLES
   - REMOVE EXISTING UNI-PILLAR
9. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   PUBLIC DRINKING WATER SOURCE AREA
   THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
   IN AREAS WHERE THIS IS A RISK YOU NEED TO:
   - BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
   - IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

   NATIVE VEGETATION REGS ESAS CLEARING
   PRIORITY FLORA
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

LEGEND

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

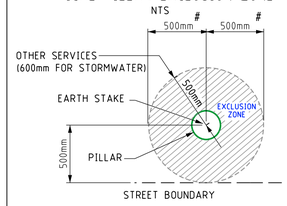
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SDE Serviced Uncleared De-energised Lot

TRANSFORMER EXISTING SWITCHGEAR

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary

APPROVED FOR CONSTRUCTION

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DRAINAGE PIPELINES IN VICINITY OF WORKS

GAS PIPELINES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: UPD
CONTACT: BILL CARMODY TEL: 0402 110 209

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION		06/24 MM/YY

REFERENCE FILE DETAILS
- MP203173 Design Drawing REV B - ISSUED FOR CONSTRUCTION EDM#: 68800633
- SHE ASSESSMENT REPORT EDM#: 68796905
- HAZARD MANAGEMENT REGISTER - EDM# 68818050

UNI-PILLAR INSTALLATION AND REMOVAL (FOR CLEARANCE PURPOSE ONLY)

NETWORK VOLTAGES: 22kV/415V

LOT 9383 STUDIO LANE, ELLENBROOK, WA 6069

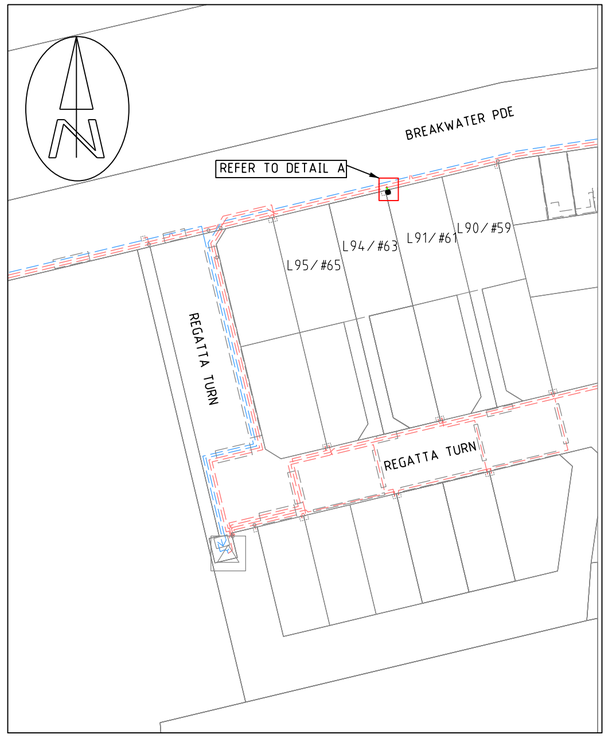
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 47' 07" S Long. - 115° 58' 01" E

westernpower

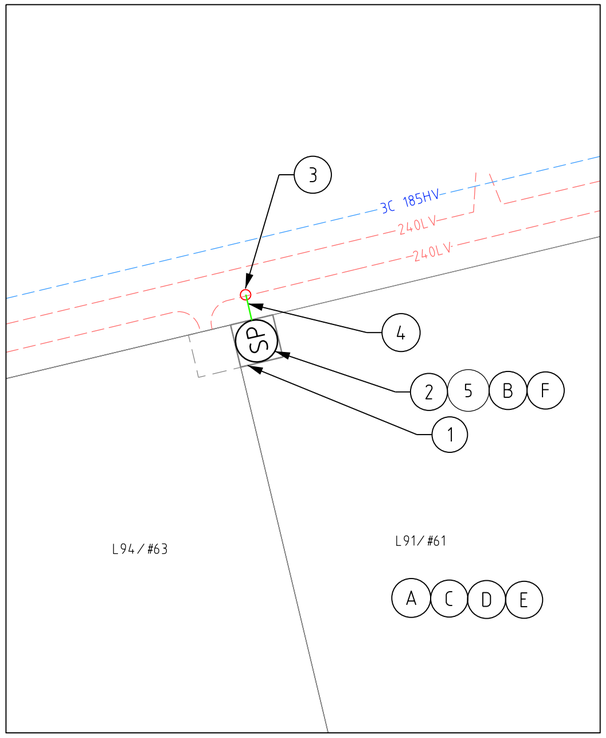
DRAWING NUMBER
MP209386
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A1



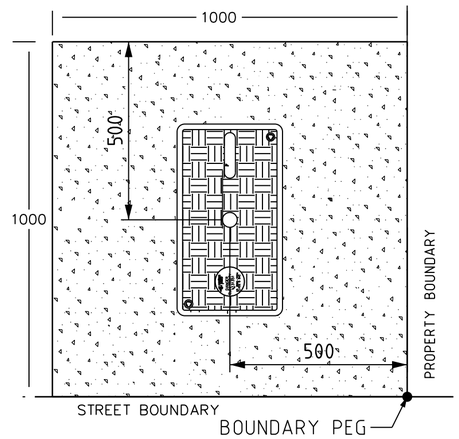
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:100 — 2 1 0 2 4 6 8



TYPICAL BGSP EXCLUSION ZONE

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

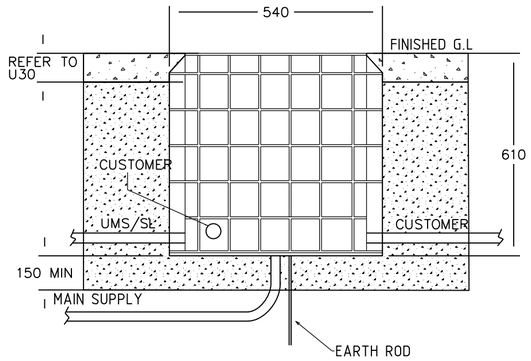
PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CONSUMER MAINS TO BE CONNECTED BY WESTERN POWER ONLY.



BGSP DETAIL

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING 240LV CABLES FROM EXISTING UNI-PILLAR.
   REMOVE EXISITNG UNIPILLAR
2. INSTALL NEW BELOW GROUND SERVICE PIT AND N25 (MINIMUM) GREY CONCRETER OVER THE EXCLUSION ZONE IN SOUTH-EAST CORNER OF L4.
   CU: LU52 × 1
   DCSH: U30
3. STRAIGHT JOINT EXISTING 240LV CABLE WITH 25LV CABLE TEE OFF
   CU: LU15 × 1
4. INSTALL NEW 25LV CABLE FROM EXISTING 240LV CABLE TO NEW BELOW GROUND SERVICE PIT.
   CU: CN42 × 10m, CN73_LV25 × 2
5. AT NEW BELOW GROUND SERVICE PIT.
   -TERMINATE NEW 25LV CABLE.
   -TERMINATE CUSTOMER'S CABLE IN CONJUNCTION WITH CUSTOMER ELECTRICAL CONTRACTOR.
   DCSH: U30
6. ENVIRONMENTAL FILTER CHECKED
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. BELOW GROUND SERVICE PIT EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE BGSP LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. CLEAR OF ALL SERVICES.

3. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

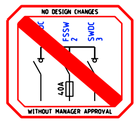
D. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

E. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

F. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



CRITICAL INFRASTRUCTURES IN VICINITY OF WORKS

APPROVED FOR CONSTRUCTION

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: MY REALTY PLUS
CONTACT: megan@myrealtyplus.com.au

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	05/24

SHE ASSESSMENT REPORT EDM#. 68498983
HAZARD MANAGEMENT REGISTER EDM#: 68500513
FILE REFERENCE DETAILS

REPLACE PILLAR WITH BGSP

NETWORK VOLTAGE: 415V

61 BREAKEWATER PDE - MANDURAH WA 6210

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 05/24 | SCALE: AS NOTED @ A2
GEO REF: Lat. - 32°31'15" S Long. - 115°42'57" E

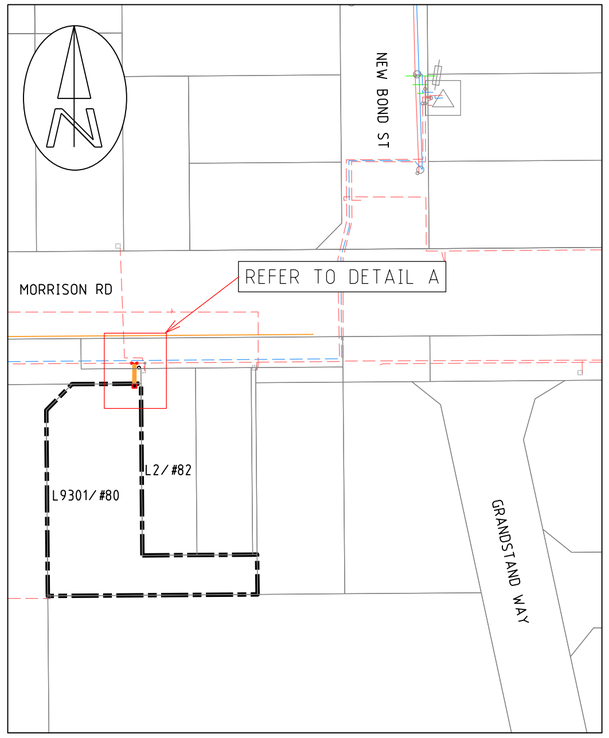
westernpower

DRAWING NUMBER

MP209449

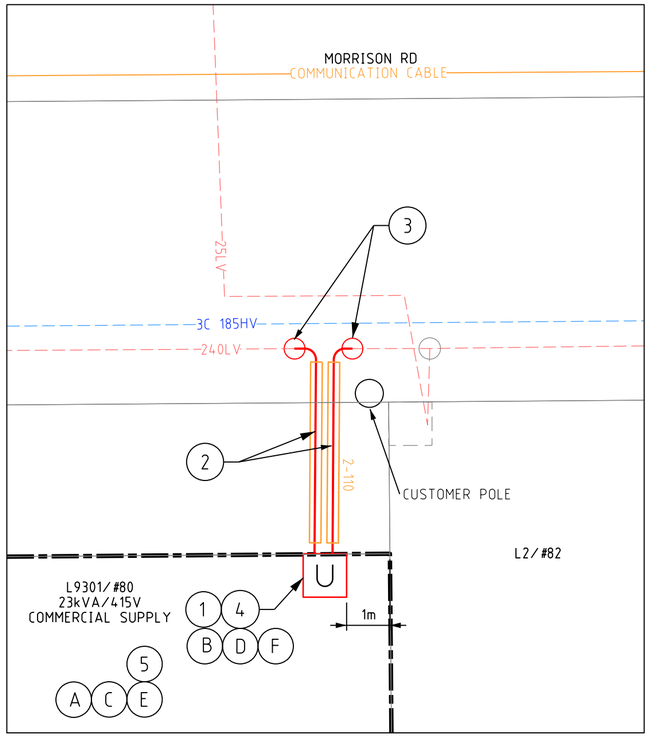
SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L9301/#80 MORRISON RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U09
2. INSTALL 2 x 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING 240LV CABLE VIA 110mm POLYPIPE.
   CU: CN45 x 25m, CN73_LVMAINS x 5 , CN54 x 2
3. CUT EXISTING 240LV CABLE AND 2 x STRAIGHT JOINT TO 2 x NEW 240LV CABLES.
   CU: LU2 x 2
4. AT UNI-PILLAR (L9301/#80 MORRISON RD)
   - TERMINATE 2 x NEW 240LV CABLE TO BOTTOM BAR.
   CU: LU7_240 x 2
   DCSH: U09
5. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT ADN LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO TERMINATE ALL CONSUMER MAINS TO THE UNI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

G. CUSTOMER TO SUPPLY AND INSTALL SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY.
CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

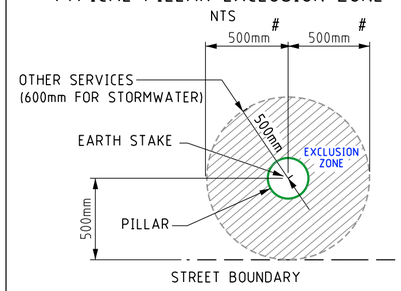
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

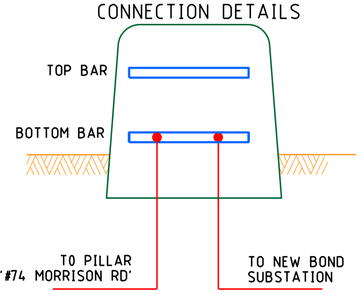
TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

UNI-PILLAR L9301/#80
CONNECTION DETAILS

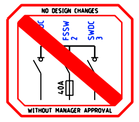


COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: LECE GROUP
CONTACT: PHIL STEVENSON TEL: 0408 930 151

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	08/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69436256
HAZARD MANAGEMENT REGISTER EDM#: 69432748

23kVA COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

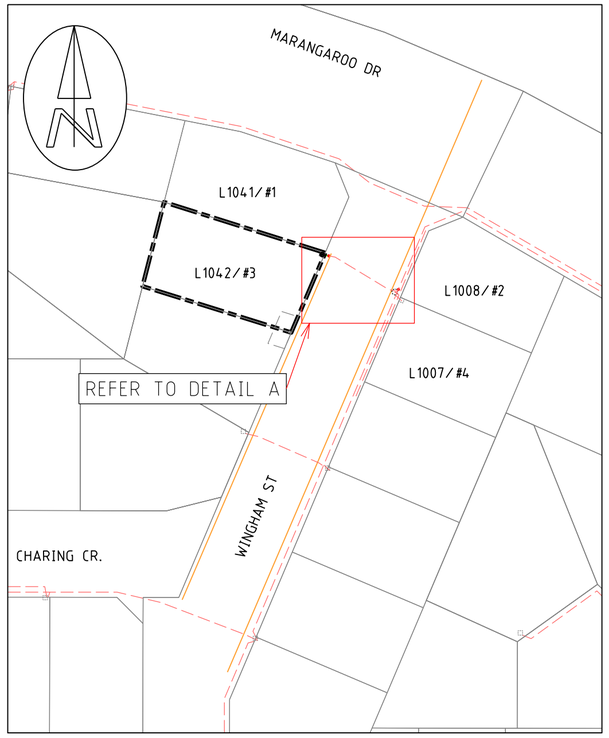
80 MORRISON RD, MIDLAND WA 6056

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 08/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 53' 10" S Long. - 116° 00' 26" E

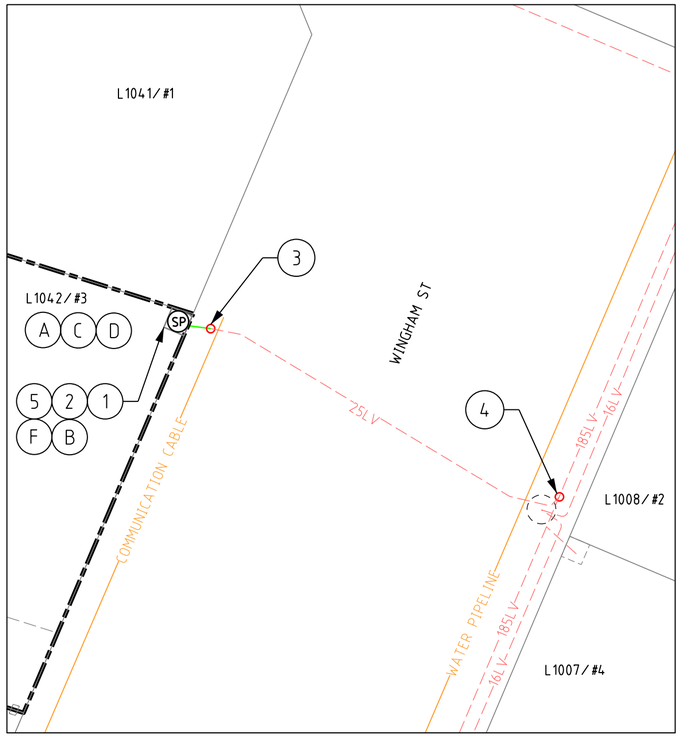
westernpower

DRAWING NUMBER
MP209459
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

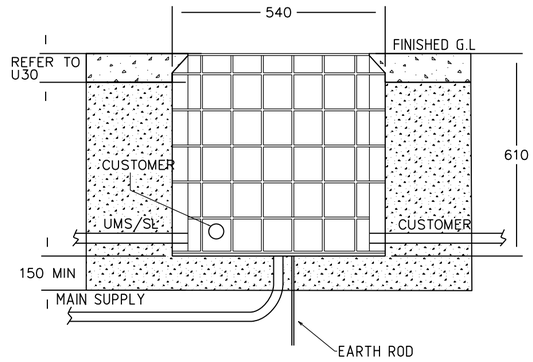
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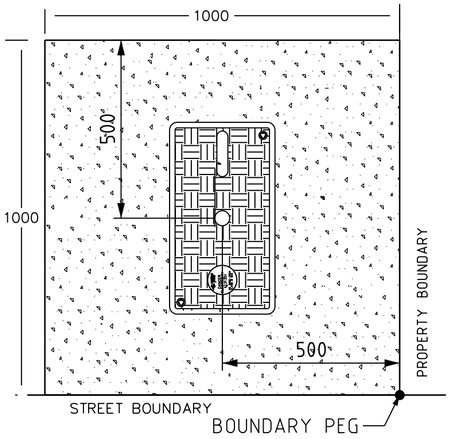
DETAIL A

1:200 (4 2 0 4 8 12 16)

CONSUMER MAINS TO BE CONNECTED BY WESTERN POWER ONLY.



BGSP DETAIL



TYPICAL BGSP EXCLUSION ZONE

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING

- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING 25LV CABLE AND CUSTOMER MAINS FROM EXISTING PILLAR AT L1042/#3 WINGHAM ST. REMOVE EXISITNG PILLAR.
2. INSTALL NEW BELOW GROUND SERVICE PIT AND N25 (MINIMUM) GREY CONCRETER OVER THE EXCLUSION ZONE IN NORTH-EAST CORNER OF L1042/#3 WINGHAM ST.
   CU: LU52 × 1
   DCSH: U30
3. INSTALL NEW 25LV CABLE FROM BGSP AND STRAIGHT JOINT TO EXISTING 25LV CABLE
   CU: CN42 x 10, CN73_LV25 x 3, LU08 × 1
4. CUT AND SWING EXISTING 25LV CABLE FROM PILLAR AT #4 WINGHAM ST (CABLE BETWEEN PILLAR AT L1042/#3 & L1007/#4) AND TEE JOINT TO EXISTING 185LV CABLE.
   CU: LU04 × 1
5. AT NEW BELOW GROUND SERVICE PIT.
   -TERMINATE NEW 25LV CABLE.
   -TERMINATE CUSTOMER'S CABLE FROM #1 & #3 IN CONJUNCTION WITH CUSTOMER ELECTRICAL CONTRACTOR.
   DCSH: U30
6. ENVIRONMENTAL FILTER CHECKED
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

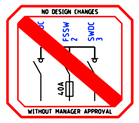
- (A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN POWER AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.
- (B) BELOW GROUND SERVICE PIT EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:
  1. PEGGING THE BGSP LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
  2. CLEAR OF ALL SERVICES.
  3. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
- (C) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.
- (D) CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.
- (E) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.
- (F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, TO TERMINATE ALL CONSUMER MAINS TO THE BGSP. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

ISSUED FOR CONSTRUCTION



☐ ☒ ☐ ☒ ☐ ☐ ☐ ☒ ☒

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

W.A.P.C. REFERENCE No. 342-23

CUSTOMER CONTACT DETAILS
COMPANY NAME: PLANFARM
CONTACT: ROSE COLIN TEL: 0413343888

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM# 69141697
HAZARD MANAGEMENT REGISTER EDM#: 69125552

REPLACE PILLAR WITH BGSP

NETWORK VOLTAGE: 22kV/415V

3 WINGHAM ST, MARANGAROO WA 6064

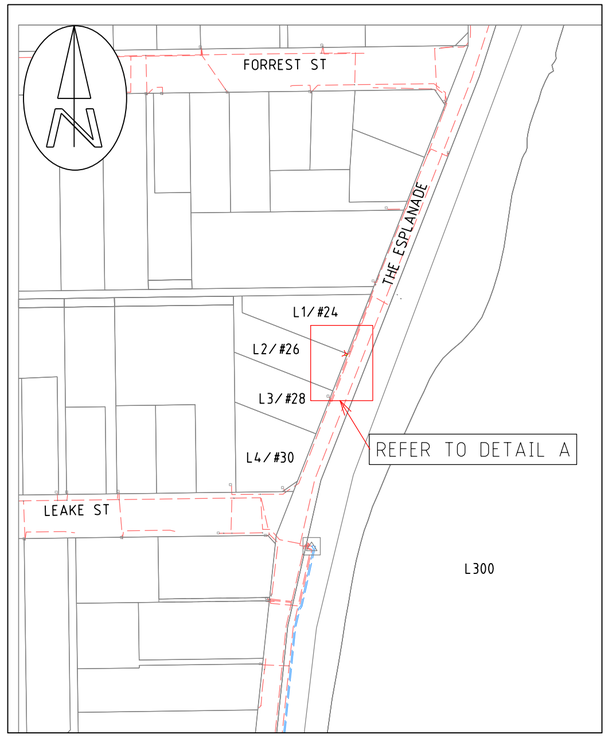
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 49' 54" S Long. - 115° 49' 28" E

westernpower

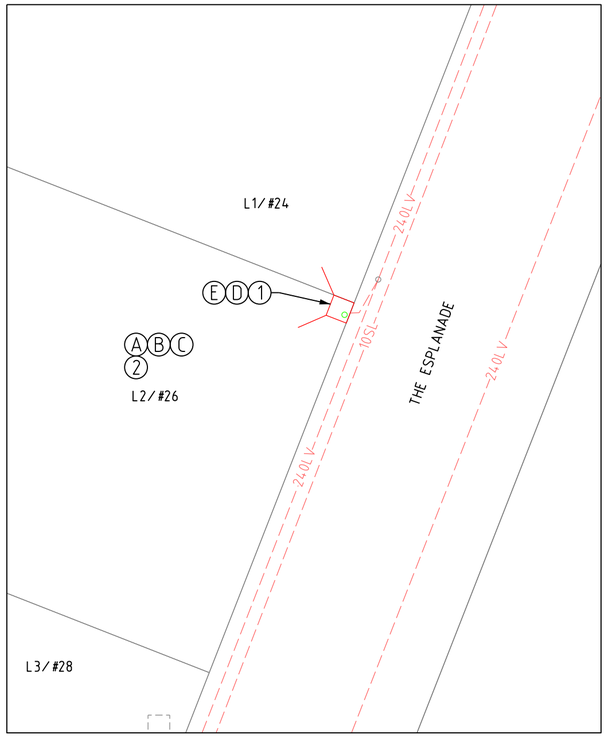
DRAWING NUMBER
MP209503
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16

LEGEND

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

WESTERN POWER SCOPE OF WORK

1. AT EXISTING MINI-PILLAR (L2/#26 THE ESPLANADE)
   - TERMINATE NEW CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   DCSH: U8-4
2. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

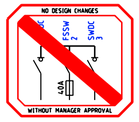
C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

APPROVED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: CB ELECTRICS
CONTACT: JUSTIN DONNELLY TEL: 0417 942 488

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

FILE REFERENCE DETAILS: HAZARD MANAGEMENT REGISTER EDM#: 68868214

63.3A RESIDENTIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

26 THE ESPLANADE, PEPPERMINT GROVE WA 6011

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 59' 50" S Long. - 115° 46' 16" E

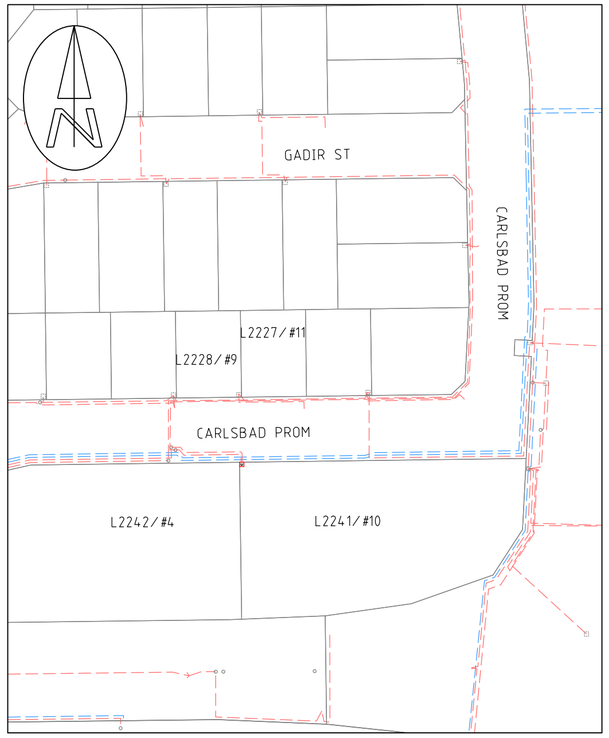
westernpower

DRAWING NUMBER

MP209506

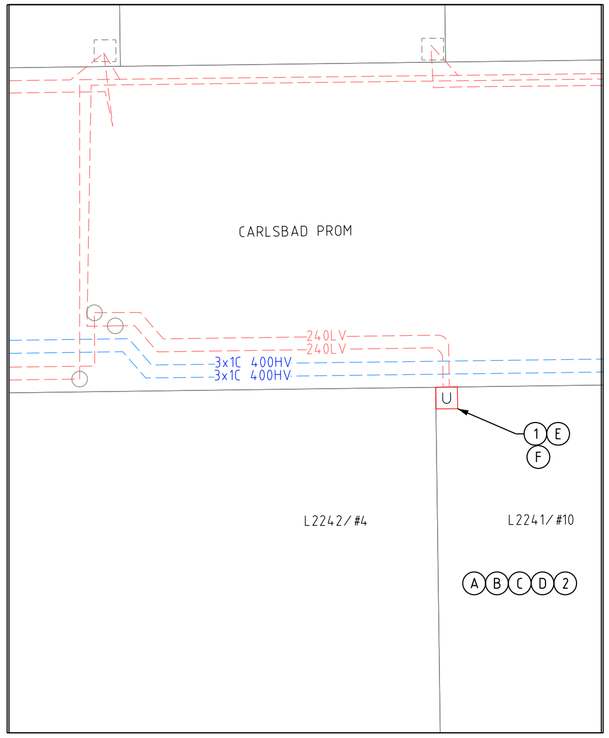
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



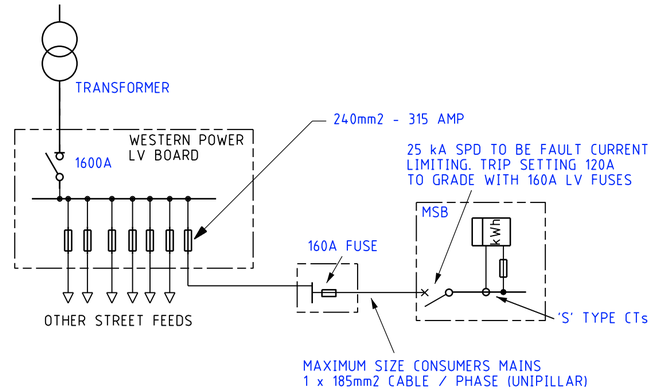
LOCATION PLAN

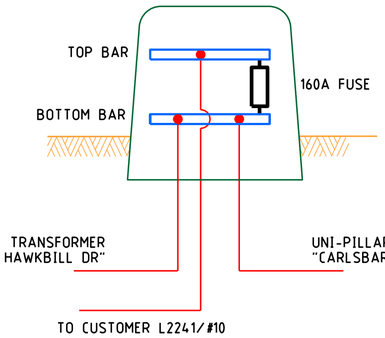
1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16





LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- Serviced Uncleared De-energised Lot
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

TRANSFORMER — EXISTING — SWITCHGEAR

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WESTERN POWER SCOPE OF WORK

1. AT EXISTING UNI-PILLAR (L2241/#10 CARLSBAD PROM.) TERMINATE CUSTOMER'S CABLE TO THE TOP BAR WITH 160A FUSE IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
   CU: LU36_160A x 1
   DCSH: U9
2. INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1_M x 1
3. ENVIRONMENTAL FILTERS CHECKED.

   NO ENVIRONMENTAL ISSUES IDENTIFIED IN IMMEDIATE WORK AREA.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST INSTALL SMSB WITHIN 30m OF THE POINT OF SUPPLY.

C. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

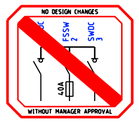
E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR TOP BAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 185mm2/PHASE.

F. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

APPROVED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

COMMUNICATION CABLES IN VICINITY OF WORKS

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: SWAN GROUP
CONTACT: EVAN ROCKE TEL: 0450 601 362

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	06/24	HAZARD MANAGEMENT REGISTER EDM#: 68598863

120A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

10 CARLSBAD PROMENADE, ALKIMOS WA 6038

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 37' 41" S Long. - 115° 41' 22" E

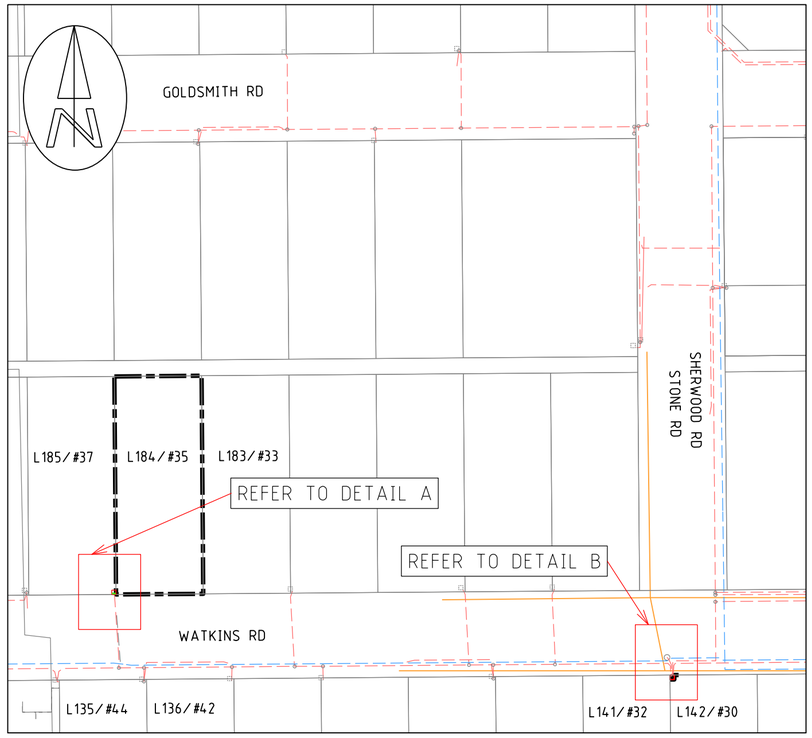
westernpower

DRAWING NUMBER

MP209515

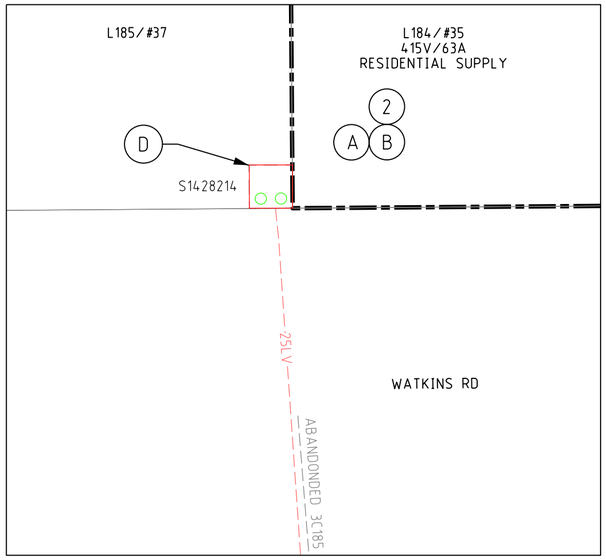
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



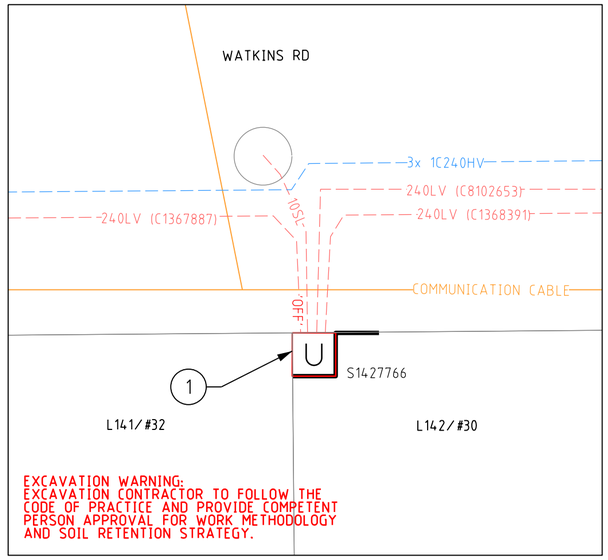
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:100 2 1 0 2 4 6 8



DETAIL B

1:100 2 1 0 2 4 6 8

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

STREET LIGHTS

	EXISTING
POLE BRACKET	POLE BRACKET
SINGLE OUTREACH	SINGLE OUTREACH
DOUBLE OUTREACH	DOUBLE OUTREACH
4-WAY	4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE.
REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

WESTERN POWER SCOPE OF WORK

(1) AT EXISTING UNI-PILLAR#S1427766 ON 30 WATKINS RD
- REPLACE WITH NEW UNI-PILLAR.
- RETERMINATE EXISTING 240LV CABLE#C1367887 FROM WEST TO TOP BAR.
- RETERMINATE EXISTING 240LV CABLES#C8102653 FROM NORTH TO BOTTOM BAR.
- RETERMINATE EXISTING 240LV CABLES#C1368391 FROM EAST TO BOTTOM BAR.
- RETERMINATE EXISITNG SL CABLES TO NEW PILLAR.

CU: LU11 x 1, LU7_240 x 1
DCSH: U9

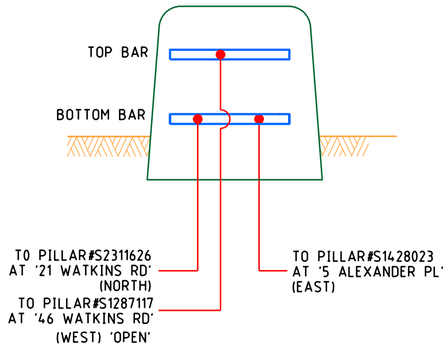
(2) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

UNI-PILLAR#S1427766 (1) CONNECTION DETAILS

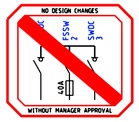


COMMUNICATION CABLE
IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: MONDOS ELECTRICAL
CONTACT: PETER MONDELLO TEL: 0439 932 983

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	11/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70298348
HAZARD MANAGEMENT REGISTER EDM#: 70298903

63A RESIDENTIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 11kV/415V

35 WATKINS RD, DALKEITH WA 6009

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 10/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31° 59' 31" S
Long. - 115° 47' 35" E

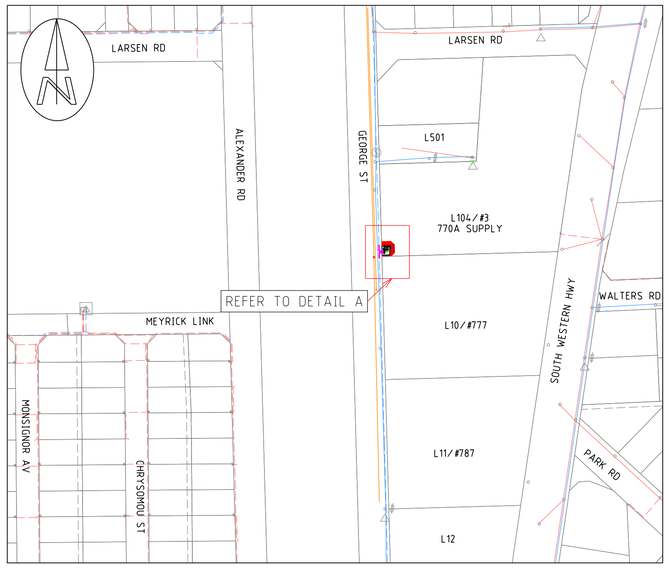
westernpower

DRAWING NUMBER

MP209614

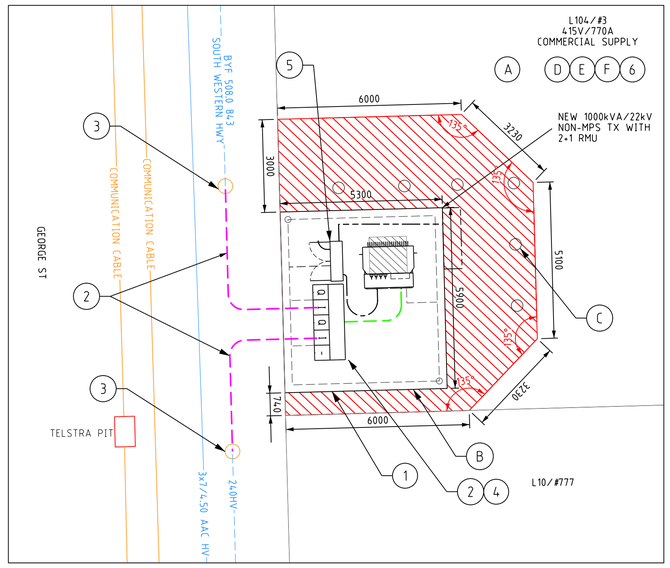
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

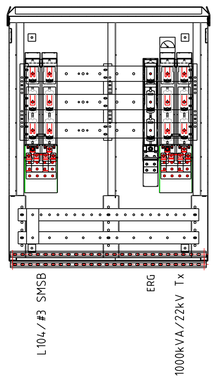
1:2000 — 40 20 0 40 80 120 160



DETAIL 1: NEW DSPM-3-04 SUBSTATION

1:100 — 2 1 0 2 4 6 8

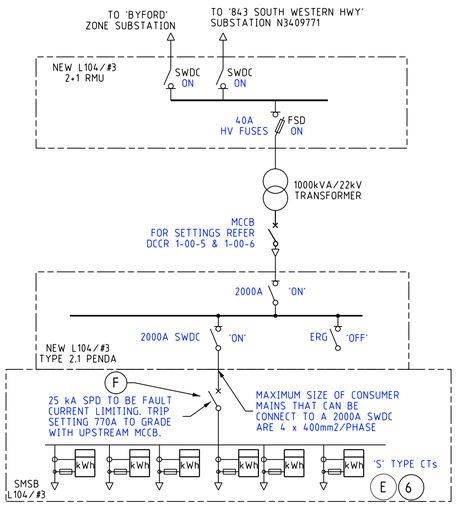
TYPE 2.1 PENDA ARRANGEMENT



TYPE 2.1 PENDA CIRCUIT TABLE

	LEFT	NAME/LOCATION
Circuit 1 (C1)	LU66_CUST	L104/#3 SMSB
Circuit 2 (C2)		
Circuit 3 (C3)		
Circuit 4 (C4)		
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)		
Circuit 8 (C8)	LU68	ERG
Circuit 9 (C9)	LU66	TO TRANSFORMER
Circuit 10 (C10)		

RIGHT



SIMPLIFIED SINGLE LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. INSTALL 1000kVA 22kV SUBSTATION WITH 2+1 AUTOMATED SWITCHGEAR AND A TYPE 2.1 PENDA TO DSPM-3-04 AS NOTED ON DRAWING. INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER. INSTALL OIL CONTAINMENT PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5 AND DRAWING DSPM-4-06 (NON-MPS).
   CU: HU59_1000/22 x1, HU66_22x1, HU76x1, HU6_A400x1, HU55_1000/22x1, CN46x15m, CN60_400CU x80m, LU16_400CU12 x 1, LU16_400CU16 x 1, LU42_Mx1, LU64_0x1, LU66_CUSTx1, LU66_Tx X1, LU68x1
   DSPM: DSPM-3-04, DSPM-4-06
2. INSTALL 2x400 HV CABLES FROM NEW HV SWITCHGEAR TO EXISTING 400HV CABLE IN ROAD RESERVE.
   CU: CN53x30m, CN54x8, CN73_240x5
3. CUT EXISTING 400HV CABLE AND STRAIGHT JOINT TO NEW 400HV CABLES.
   CU: HU2 x 2
4. TERMINATE 2x 400HV CABLE AND 35HV TO NEW 2+1 SWITCHGEAR AS PER SINGLE LINE DIAGRAM.
5. AT NEW PENDA, TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT & CIRCUIT TABLE.
6. INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1_M x 6
7. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING, COMPACTION AND LEVELLING.
2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

C. CUSTOMER TO INSTALL NON-CONDUCTIVE BOLLARDS ADJACENT TO THE SUBSTATION SITE AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6.

D. CUSTOMER MUST INSTALL MSB CONTIGUOUS TO THE SUBSTATION.

E. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

G. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

H. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

J. NO BUILDINGS ALLOWED WITHIN 4m OF THE SUBSTATION UNLESS 2 HOUR FIRE RATED. SEE WESTERN POWER DRAWING DSM-5-01 FOR DETAILS.

K. CUSTOMER TO PROVIDE A 129BA RESTRICTIVE COVENANT BURDENING THE LOT/S AROUND THE SUBSTATION SITE FOR FIRE PROTECTION CLEARANCE REQUIREMENTS TO FUTURE BUILDINGS AS PER UNDERGROUND DISTRIBUTION SCHEMES MANUAL CLAUSE 5.3.18.1. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS.

L. IF THE CUSTOMER PROPOSES TO INSTALL SCREENING APPROVAL SHALL BE OBTAINED FROM WESTERN POWER PRIOR TO ITS INSTALLATION. THE SCREENING SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE. REFER TO THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSES 14.4.10.3.2. 148.11 AND 14.58 FOR DETAILS.

M. IF THE CUSTOMER PROPOSES TO INSTALL METALLIC FENCE, IT SHOULD BE INSTALLED MINIMUM 1m FROM THE SUBSTATION SITE BOUNDARY.

N. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 4.0 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

TRANSFORMER (T) EXISTING (dashed) SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)





ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY: FOCUS WA CONSULTING
CONTACT: MARK GORMAN
MOBILE: 0421 889 180
EMAIL: mark@focuswa.com.au

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM# 69923790
A	ISSUED FOR CONSTRUCTION	I.P.	10/24	HAZARD MANAGEMENT REGISTER - EDM# 69928906

770A COMMERCIAL SUPPLY

NETWORK VOLTAGES: 22kV/415V

3 LARSEN RD, BYFORD WA 6122

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 10/24 SCALE: AS NOTED @ A1

GEO REF: Lat. - 32° 12' 46" S
Long. - 116° 00' 28" E

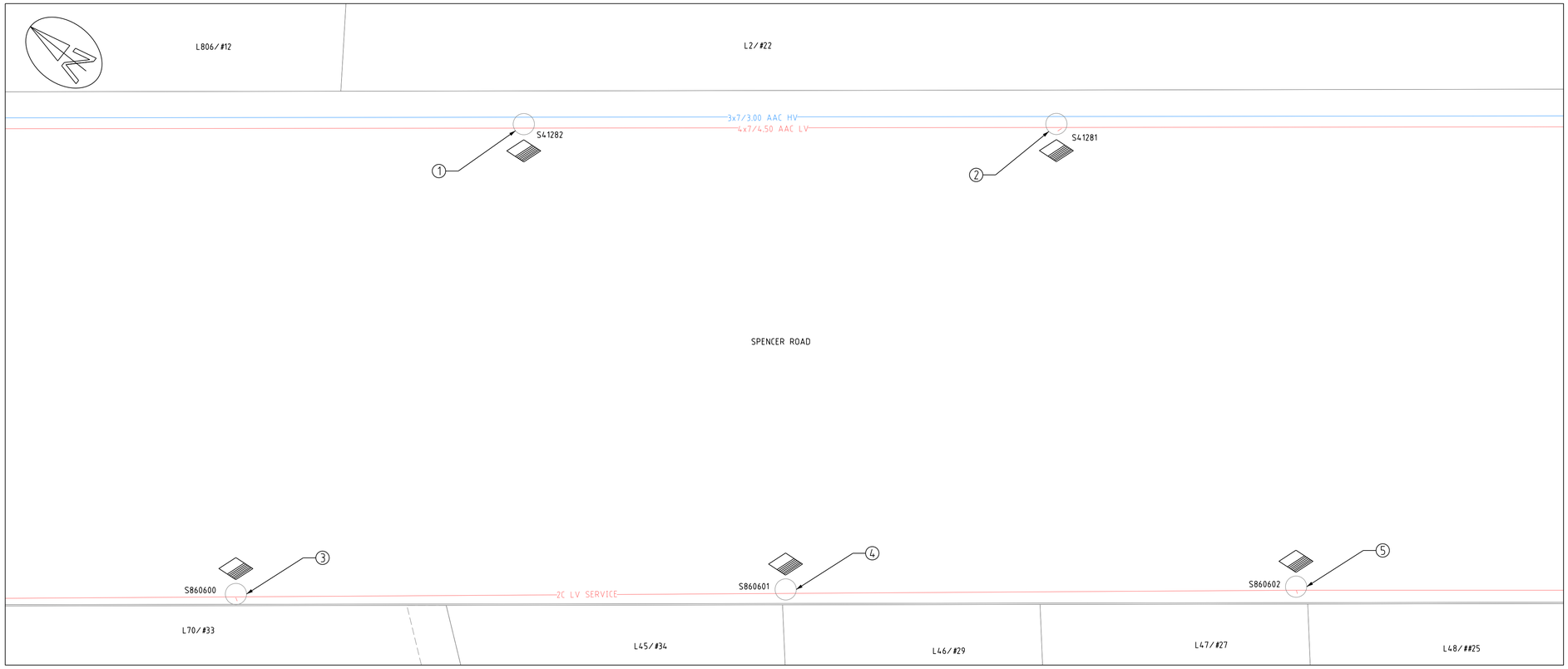
westernpower

DRAWING NUMBER

MP209652

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A1



LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — **EXISTING**

- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SLAD Serviced Unclearad De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

LOCATION PLAN

1:200 — 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. AT POLE S41282
   - INSTALL NEW 135W 4000k LED STREETLIGHT.
   CU: SL3_5L x 1
   DCSH: S03
2. AT POLE S41281
   - REPLACE EXISTING LUMINARE WITH NEW 135W 4000k LED ON EXISITNG OUTREACH.
   NON CU: HL6612 x 1
   DCSH: S03
3. AT POLE S860600
   - REPLACE EXISTING STREETLIGHT WITH NEW 135W 4000k LED STREETLIGHT AND GLARESHEILD.
   CU: SL3_5L x 1, SL21_FLR_5L x 1
   DCSH: S03
4. AT POLE S860601
   - INSTALL NEW 135W 4000k LED STREETLIGHT AND GLARESHEILD.
   CU: SL3_5L x 1, SL21_FLR_5L x 1
   DCSH: S03
5. AT POLE S860602
   - REPLACE EXISTING STREETLIGHT WITH NEW 135W 4000k LED STREETLIGHT AND GLARESHEILD.
   CU: SL3_5L x 1, SL21_FLR_5L x 1
   DCSH: S03

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

ISSUED FOR CONSTRUCTION

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: SAGE CONSULTING ENGINEERS PTY LTD
CONTACT: ANDREEA PUIU TEL: 0431 318 900

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	07/24	HAZARD MANAGEMENT REGISTER - EDM# 69009086

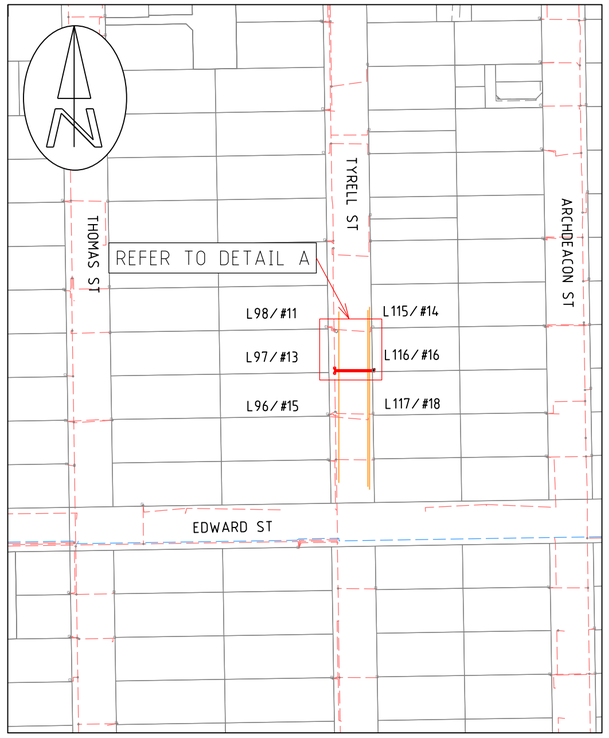
STREET LIGHT INSTALLATION AND UPGRADE

NETWORK VOLTAGES: 22kV/415V

SPENCER ROAD, LANGFORD WA 6147
DESIGN DRAWING

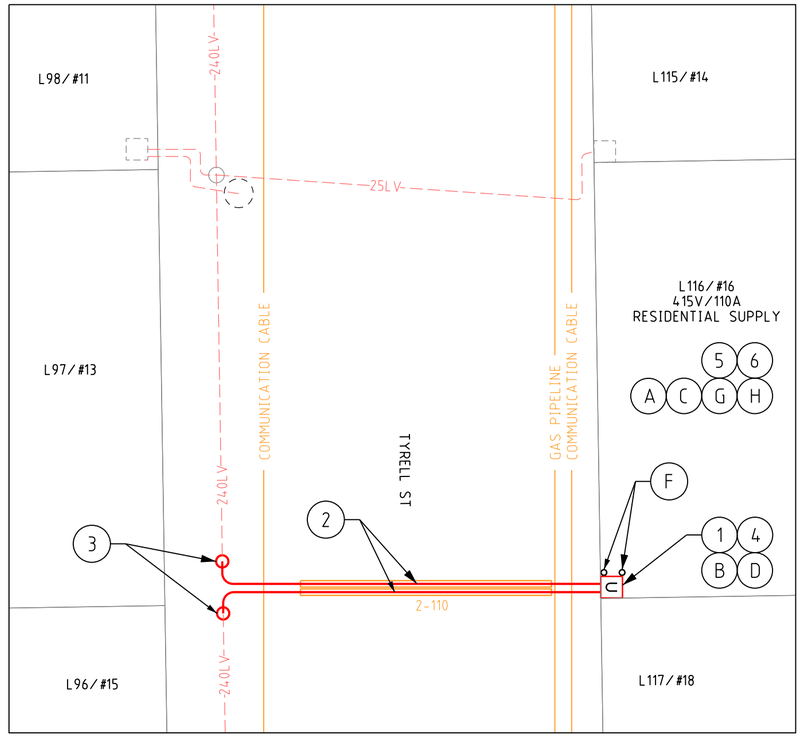
DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 | SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 01' 58" S, Long. - 115° 56' 24" E

DRAWING NUMBER
MP209731
SHEET 01 OF 01
REV: A | ORIGINAL SIZE - A1



LOCATION PLAN

1:2000 (40 20 0 40 80 120 160)



DETAIL A

1:200 (4 2 0 4 8 12 16)

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L116/#16 TYRELL ST AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 2 x 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING 240LV CABLE ON TYRELL ST VIA POLYPIPE.
   CU: CN45 x 55m, CN73_LVMAINS x 6 , CN54 x 4
3. CUT EXISTING 240LV CABLE AND 2 x STRAIGHT JOINT TO 2 x NEW 240LV CABLES.
   CU: LU2 x 2
4. AT UNI-PILLAR (L116/#16 TYRELL ST)
   - TERMINATE 2 x NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR VIA 160A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240 x 2, LU36_160A x 1
   DCSH: U9
5. INSTALL CT METER.
   CU: LM1_M x 1
6. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
   5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

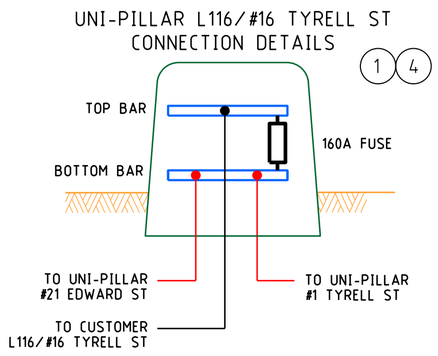
D. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

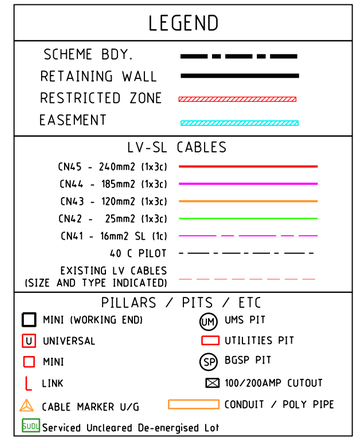
E. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

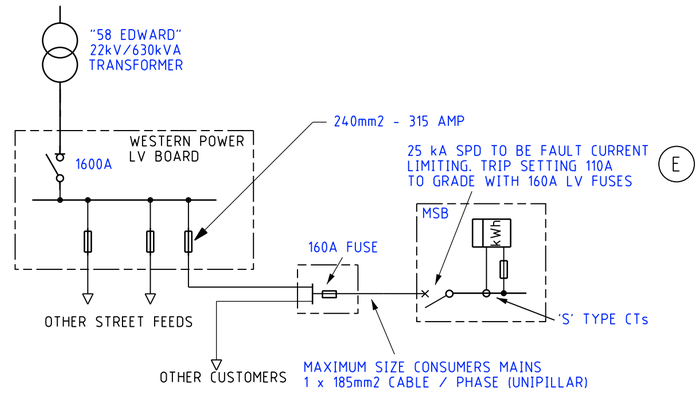
F. CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

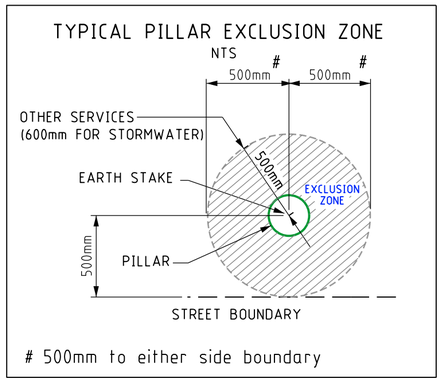
G. CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

H. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.









W.A.P.C. REFERENCE No. 965-22

COMMUNICATION CABLES IN VICINITY OF WORKS

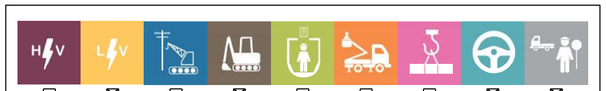
GAS PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

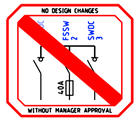
CUSTOMER CONTACT DETAILS
COMPANY NAME: DECOBU PTY LTD
CONTACT: JAY SIDHU TEL: 0430 192 257

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

SAFETY - MAKE IT YOUR PRIMARY VALUE



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 69257370
HAZARD MANAGEMENT REGISTER EDM#: 69227061

415V/110A RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 22kV/ 415V

16 TYRELL ST, NEDLANDS, 6009, WA, PERTH

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 10/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 58' 49" S Long. - 115° 48' 32" E

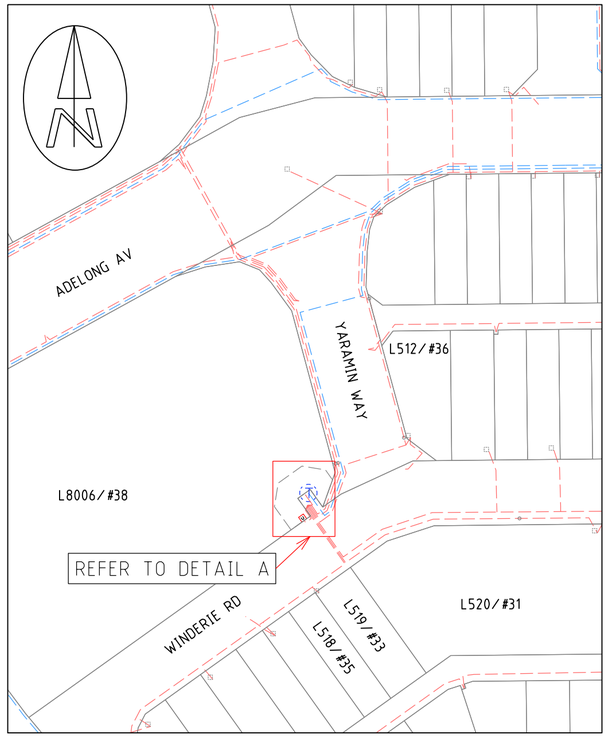
westernpower

DRAWING NUMBER

MP209842

SHEET 01 OF 01

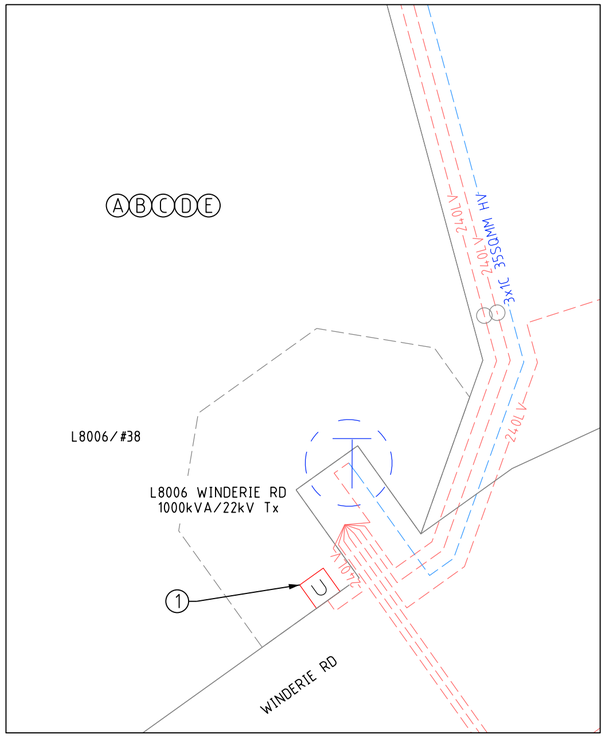
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

20 10 0 20 40 60 80

1:1000



DETAIL A

4 2 0 4 8 12 16

1:200

WESTERN POWER SCOPE OF WORK

1. AT UNI-PILLAR (L29/#22 TEAGUE ST)
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 315A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU36_LINK x 1
   DCSH: U9
2. INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1 x 1

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

C. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

D. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

E. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

LEGEND

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

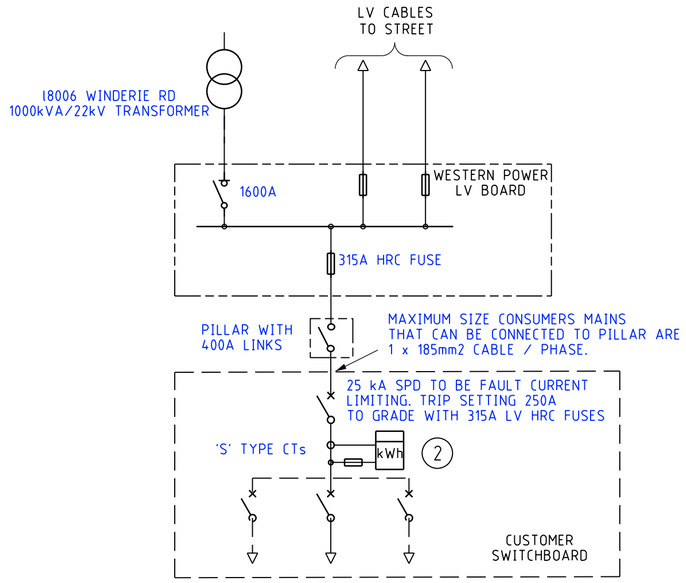
LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

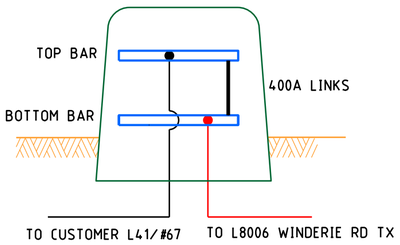
PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

TRANSFORMER (T) EXISTING SWITCHGEAR (S)

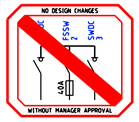


UNI-PILLAR L8806/#38
CONNECTION DETAILS



APPROVED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: PARRY AND WHYTE ARCHITECTS
CONTACT: ALEX WHYTE TEL: 0413 655 300

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

HAZARD MANAGEMENT REGISTER EDM#: 68871623

FILE REFERENCE DETAILS

250A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

38 WINDERIE RD, GOLDEN BAY WA 6174

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 25' 13" S
Long. - 115° 45' 45" E

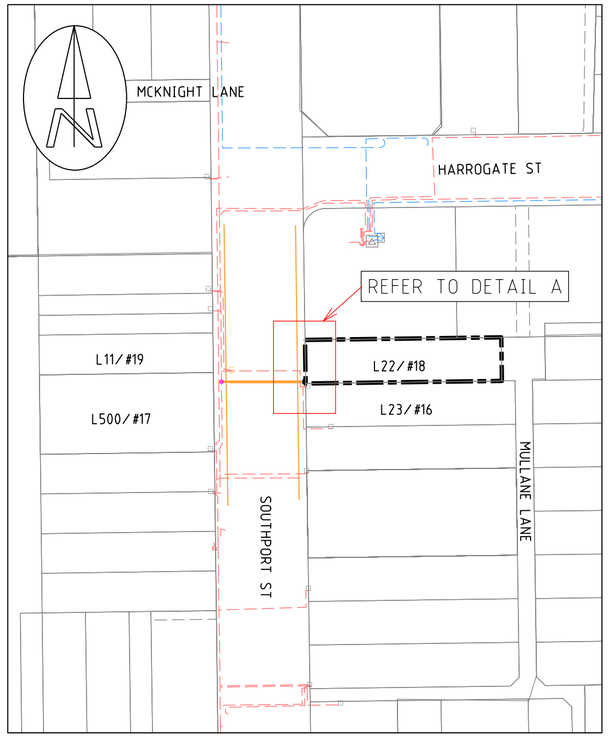
westernpower

DRAWING NUMBER

MP209945

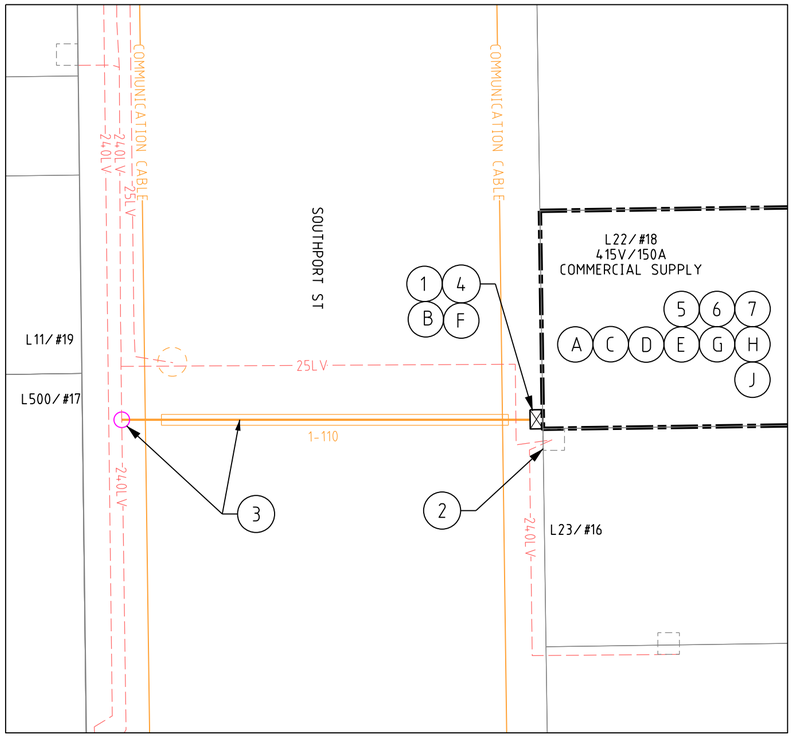
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



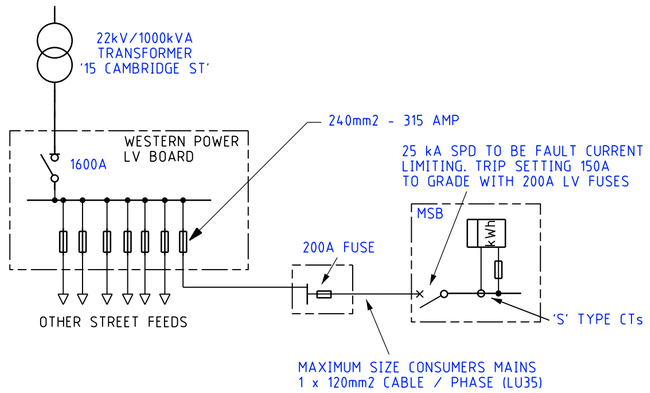
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16



SIMPIFIED SINGLE LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. WESTERN POWER TO SUPPLY 200A WALL MOUNTED BOX TO CUSTOMER'S ELECTRICAL CONTRACTOR AND CUSTOMER IS RESPONSIBLE FOR THE INSTALLATION AS PER DCSH U32.
   CU: LU35 x 1
   DCSH: U32, U21
2. AT EXISTING PILLAR #16SOUTHPORT ST
   - DISCONNECT, PULL BACK TO VERGE, CUT AND ABANDON EXISTING 25LV TOWARDS #18 SOUTHPORT ST.
3. INSTALL NEW 120LV CABLE FROM NEW WALL MOUNTED BOX TO EXISTING 240LV CABLE AT #17 SOUTHPORT ST VIA 110mm POLYPIPE AND BREACH JOINT TO EXISTING 240LV CABLE.
   CU: CN43 x 30m, CN54 x 4, CN73_LVMAINS x 3, LU1 x 1
4. AT WALL MOUNTED BOX (L22/#18 SOUTHPORT ST)
   - TERMINATE NEW 120LV CABLE VIA 200A FUSE.
   - TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7 x 1
   DCSH: U32
5. WP SCHEDULER TO CONFIRM RECEIPT OF ETAF PRIOR TO SCHEDULING WORKS.
   WP PMO TO OBTAIN FINAL METER READ DATE FROM WP METERING.
6. INSTALL CT METER.
   CU: LM1_M x 1
7. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. INSTALL THE WESTERN POWER SUPPLIED 200A WALL MOUNTED BOX. WALL MOUNTED BOX TO BE PREPARED AS PER THE DISTRIBUTION CONSTRUCTION STANDARDS HANDBOOK U32.

   1. SHADED AREA MUST BE KEPT CLEAR FOR OPERATIONAL PURPOSES AND WP PERSONNEL (TO A HEIGHT OF 2500).

C. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLU. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

D. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

E. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

F. CUSTOMER ELECTRICAL CONTRACTOR TO PROVIDE SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

G. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

H. CUSTOMER'S ELECTRICAL CONTRACTOR TO REMOVE EXISTING 25LV CABLE FROM MSB IN CONJUNCTION WITH WP.

J. WESTERN POWER WILL REQUIRE OUTGOING RETAILER AGREEMENT (VIA SUBMISSION OF AN ETAF) PRIOR TO SCHEDULING OF THE PROJECT WORKS.
   CUSTOMER REQUESTED TO CONTACT THE OUTGOING RETAILER AND PROVIDE RELEVANT PROJECT DETAILS TO FACILITATE ETAF.
   CUSTOMER TO CONTACT WP PNO PRIOR TO COMMENCEMENT OF METERING CHANGES.

LEGEND

SCHEME BOY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

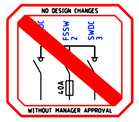
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

COMMUNICATION CABLES IN VICINITY OF WORKS

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: MATEVAL PTY LTD
CONTACT: HAROLD BONG TEL: 0439 988 739

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69646742
HAZARD MANAGEMENT REGISTER EDM#: 69647183

150A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

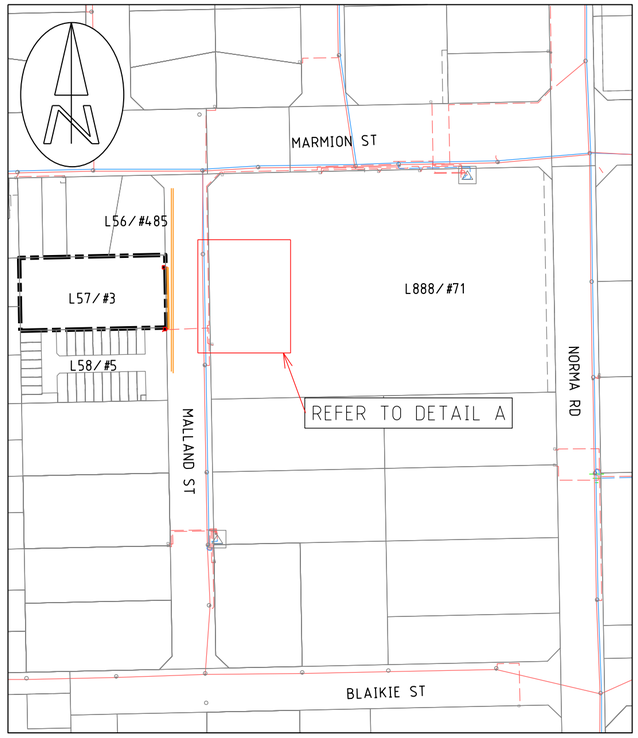
18 SOUTHPORT ST, WEST LEEDERVILLE, 6007, WA

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 10/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 56' 30" S Long. - 115° 50' 24" E

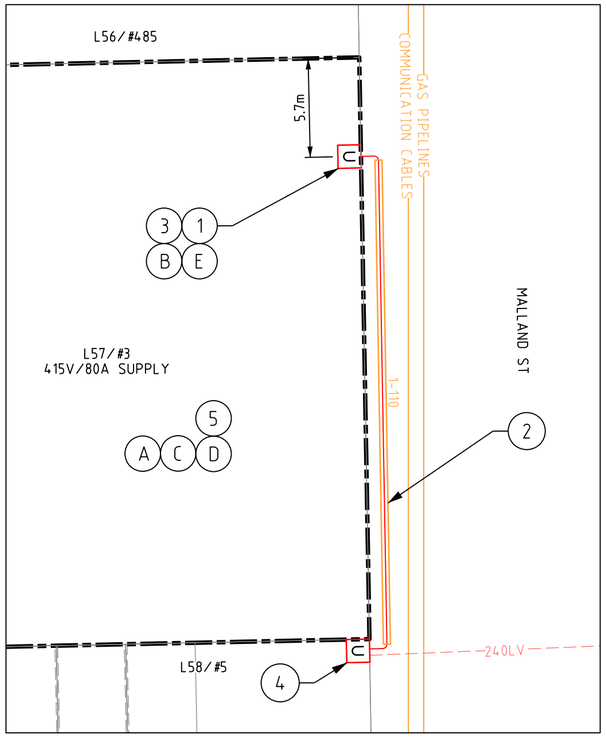
westernpower

DRAWING NUMBER
MP210001
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



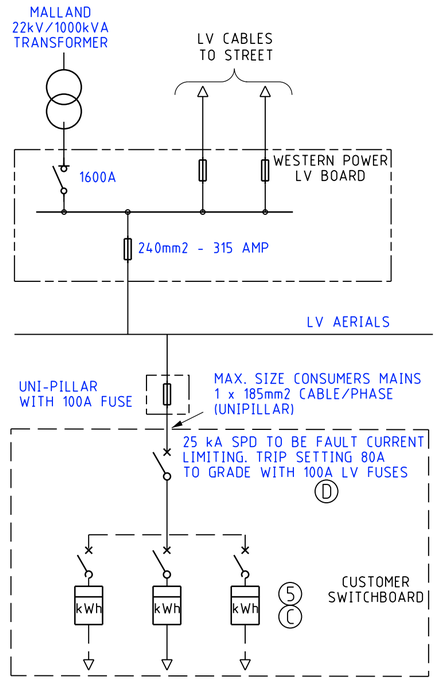
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:250 5 2.5 0 5 10 15 20



SIMPLIFIED SINGLE LINE DIAGRAM

LEGEND

SCHEME BDY.

LV CABLES

CN45 - 240mm2 (1x3c)

EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

MINI (WORKING END)

CONDUIT / POLY PIPE

U UNIVERSAL

UTILITIES PIT

AERIAL CONDUCTORS AND POLES

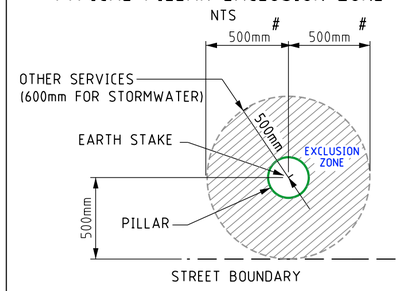
NEW POLE (ANY TYPE AS INDICATED)

EXISTING POLE (ANY TYPE AS INDICATED)

HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L57/#3 MALLAND ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING UNI-PILLAR AT L58/#5 MALLAND ST VIA 110mm POLYPIPE.
   CU: CN45 x 45m, CN54 x 2, CN73_LVMAINS x 3
3. AT NEW UNI-PILLAR (L29/#22 TEAGUE ST)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 100A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240 x 1, LU36_100A x 1
   DCSH: U9
4. AT EXISTING UNI-PILLAR (L58/#5 MALLAND ST)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   CU: LU7_240 x 1
5. INSTALL DIRECT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1_M x 19
6. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   THREATENED ECOLOGICAL COMMUNITIES
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

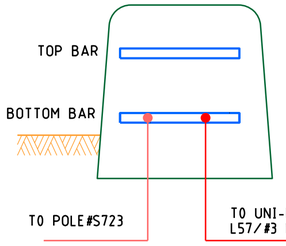
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

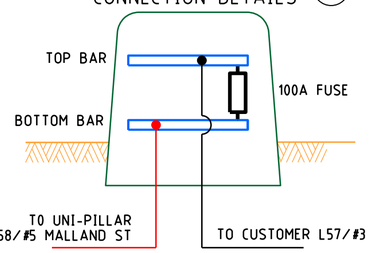
D. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR TOP BAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

EXISTING UNI-PILLAR L58/#5 CONNECTION DETAILS (4)



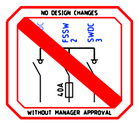
NEW UNI-PILLAR L57/#3 CONNECTION DETAILS (1)



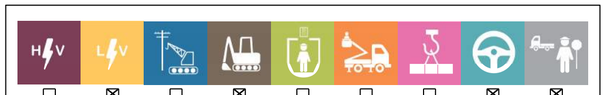
COMMUNICATION CABLES IN VICINITY OF WORKS

GAS PIPELINES IN VICINITY OF WORKS

APPROVED FOR CONSTRUCTION



SAFETY - MAKE IT YOUR PRIMARY VALUE



☐ ☒ ☐ ☒ ☐ ☐ ☐ ☒ ☒

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

CUSTOMER CONTACT DETAILS
COMPANY NAME: PRO COM CONSTRUCTION
CONTACT: SMITH GAELLE TEL: 0439 988 739

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE FILTER ASSESSMENT EDM#: 69225778
A	ISSUED FOR CONSTRUCTION	I.P.	07/24	HAZARD MANAGEMENT REGISTER EDM#: 69219707

80A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

3 MALLAND ST, MYAREE WA 6154

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 08/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 02' 25" S Long. - 115° 49' 12" E

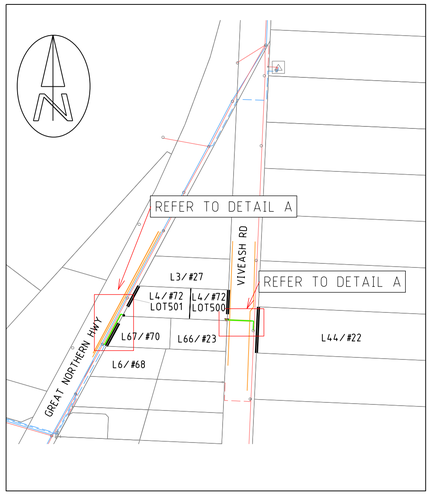
westernpower

DRAWING NUMBER

MP210066

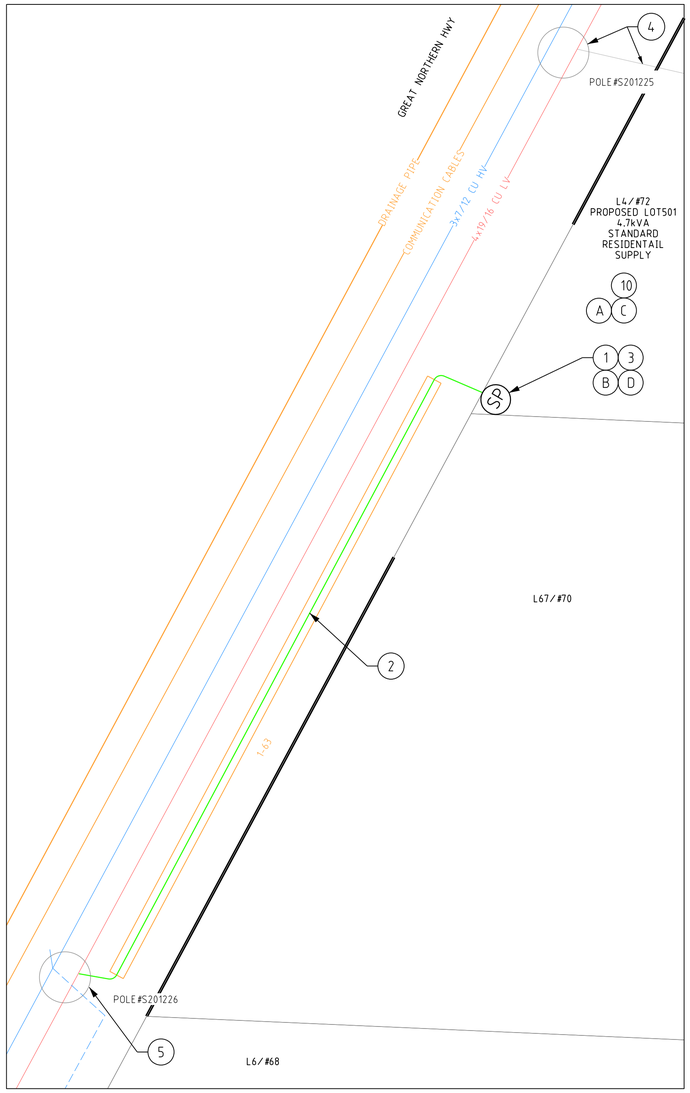
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



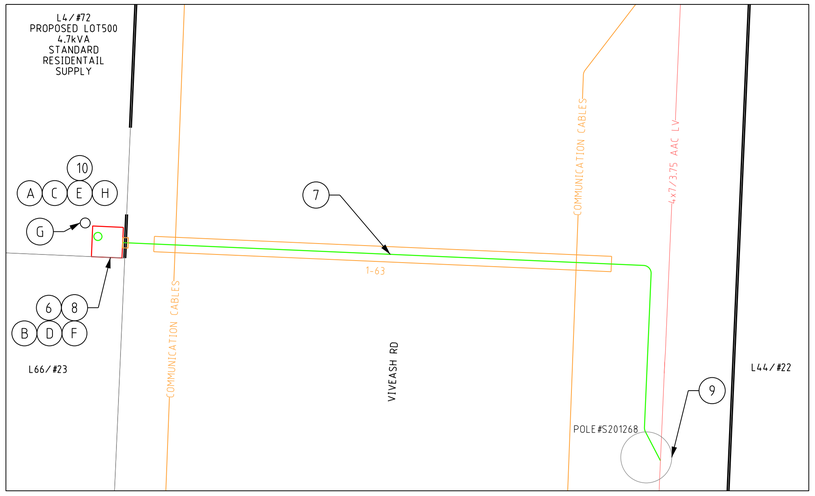
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



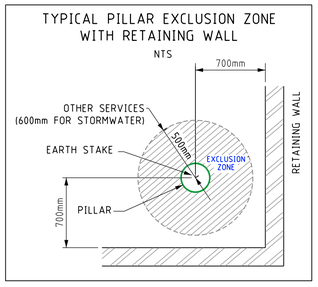
DETAIL A

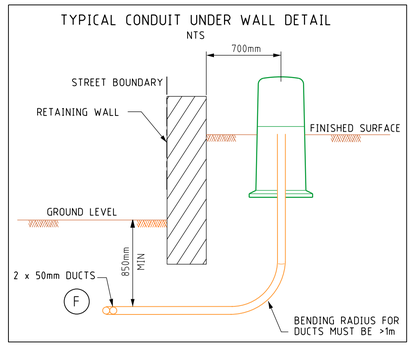
1:100 2 1 0 2 4 6 8

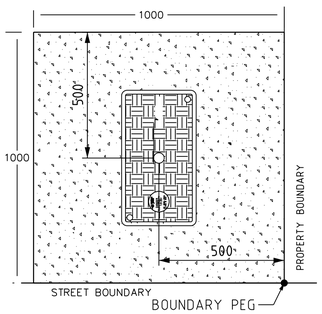


DETAIL B

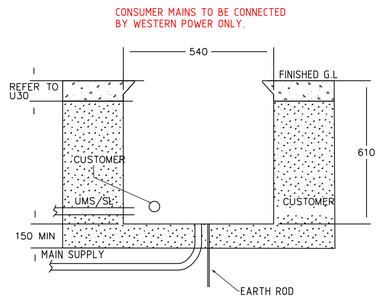
1:100 2 1 0 2 4 6 8







TYPICAL BGSP EXCLUSION ZONE



BGSP DETAIL

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- Serviced Uncleared De-energised Lot
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

COMMUNICATION CABLES IN VICINITY OF WORKS

DRAINAGE PIPE IN VICINITY OF WORKS

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR AND BELOW GROUND SERVICE PIT(BGSP) EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR/BGSP LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR/BGSP BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.
CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM THE POINT OF SUPPLY.

(F) CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

(G) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

(H) CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY.

WESTERN POWER SCOPE OF WORK

(1) AT L4/#72 GREAT NORTHERN HWY LOT501, INSTALL NEW BELOW GROUND SERVICE PIT(BGSP) AND N25 (MINIMUM) GREY CONCRETER OVER THE EXCLUSION ZONE IN SOUTH-WEST CORNER OF LOT501.
CU: LU52 x 1

(2) INSTALL NEW 25LV CABLE FROM NEW BGSP TO POLE#S201226 VIA 63mm POLYPIPE.
CU: CN42 x 40m, CN73_LV25 x 3, CN54 x 3

(3) AT BGSP (L4/#72 GREAT NORTHERN HWY, LOT 501)
-TERMINATE NEW 25LV CABLE.
-TERMINATE CUSTOMER'S CABLE IN CONJUNCTION WITH CUSTOMER ELECTRICAL CONTRACTOR.
DCSH: U30

(4) AT POLE#S201225
- REMOVE CUSTOMER AERIAL SERVICE TO L4/#72.

(5) AT POLE#S201226
- TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
CU: LU51_25CUU x 1, LU51_25CUO x 1, CN9_CU x 4
DCSH: U19-1

(6) AT L4/#72 GREAT VIVEASH RD, LOT 500, INSTALL NEW MINI-PILLAR AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10 x 1

(7) INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S201268. VIA 63mm POLYPIPE.
CU: CN42 x 45m, CN73_LV25 x 3, CN54 x 3

(8) AT MINI-PILLAR (L4/#72 VIVEASH RD, LOT 500)
- TERMINATE NEW 25LV CABLE..
DCSH: U08-04

(9) AT POLE#S201268
- TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS TOWARDS NORTH.
CU: LU51_25ALU x 1, LU51_25ALO x 1, CN9_7/475AA x 4
DCSH: U19-1

(10) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS;

HERITAGE COUNCIL OF WA
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 4150794)

REFER TO environment@westernpower.com.au

W.A.P.C. REFERENCE No. 164115

CUSTOMER CONTACT DETAILS
COMPANY NAME: TOTAL SURVEY SOLUTIONS
CONTACT: LEO MANNELLA TEL: 0411 588 277

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM#: 69632854
A	ISSUED FOR CONSTRUCTION	I.P.	09/24	HAZARD MANAGEMENT REGISTER EDM#: 69631974

2 x 4.7 kVA STANDARD RESIDENTIAL SUPPLY

NETWORK VOLTAGES: 22kV/415V

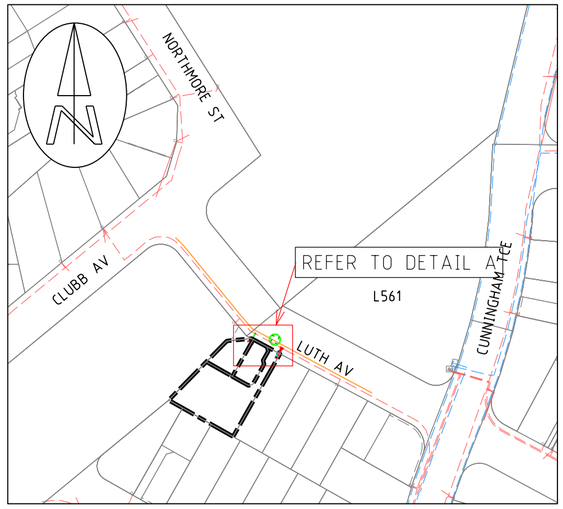
72 GREAT NORTHERN HIGHWAY, MIDDLE SWAN 6065

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 52' 15" S Long. - 116° 00' 50" E

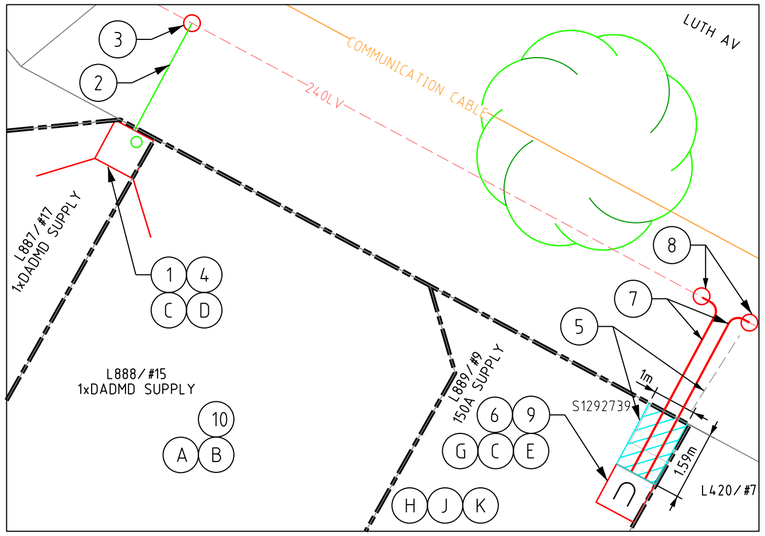
western power

DRAWING NUMBER
MP210099
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

HV CABLES

- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

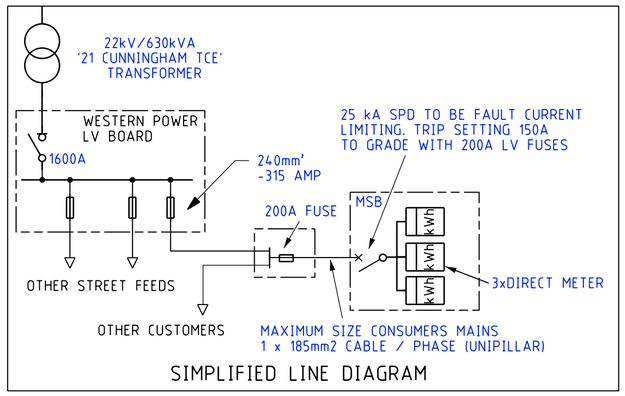
PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDU Serviced Uncleared De-energised Lot

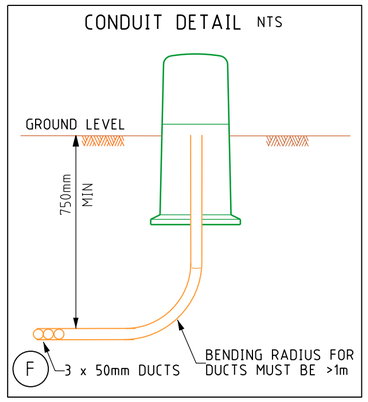
COMMUNICATION CABLES IN VICINITY OF WORKS

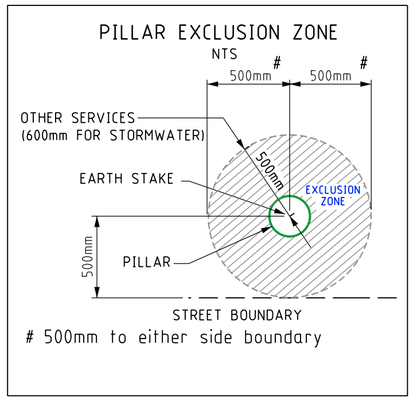
ISSUED FOR CONSTRUCTION

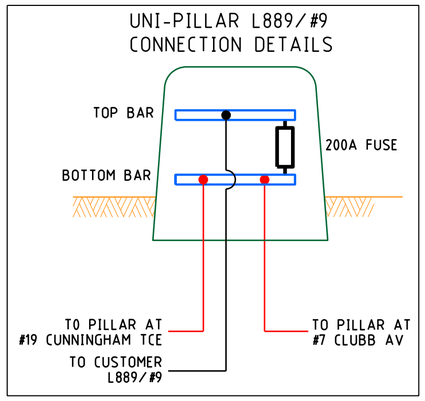
W.A.P.C. REFERENCE No. 161364 & 121721



SIMPLIFIED LINE DIAGRAM







DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

SAFETY - MAKE IT YOUR PRIMARY VALUE

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

C. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE THE CONSUMER MAINS FROM L420/#7 LUTH AV TO THE UNIPILLAR BACK BLOCKS.

F. CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

G. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

H. CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

J. CUSTOMER TO ARRANGE FOR A 136C EASEMENT UNDER THE TRANSFER OF LAND ACT. CUSTOMER TO SUBMIT THE DEPOSITED PLAN TO WESTERN POWER PRIOR TO THE COMMENCEMENT OF WORKS. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L887 LUTH AV. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU10x1

2. INSTALL 25LV CABLE FROM NEW MINI-PILLAR TO 250LV CABLES
CU: CN42x15m, CN73_LV25x3, CN54x2

3. TEE-JOINT 25LV CABLE WITH EXISTING 240LV CABLE
CU: LU5x1

4. AT MINI-PILLAR, TERMINATE NEW 25LV CABLE.
DCSH: U8-04

5. AT MINIPILLAR#S1292739'
- DISCONNECT EXISTING 25LV CABLE AND CUSTOMER MAINS.
- REMOVE EXISTING PILLAR#S1292739 AND 25LV CABLE

6. INSTALL NEW UNI-PILLAR ON L889 LUTH AV 1.59m FROM THE PROPERTY BOUNDARY (REFER DETAIL A) AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11x1
DCSH: U9

7. INSTALL 2x240LV CABLE FROM NEW UNI-PILLAR TO EXISTING 240LV CABLE.
CU: CN45x30m, CN73_LVMAINSx5 , CN54x3

8. CUT EXISTING 240LV CABLE AND 2xSTRAIGHT JOINT TO 2xNEW 240LV CABLES.
CU: LU2x2

9. AT NEW UNI-PILLAR
- TERMINATE 2 x NEW 240LV CABLE AT BOTTOM BAR.
- TERMINATE CUSTOMER CABLE AT TOP BAR VIA 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU7_240x2, LU36_200Ax1
DCSH: U9

10. INSTALL DIRECT METER IN CUSTOMERS SWITCHBOARD.
CU: LM1x5

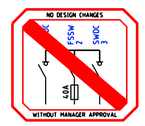
11. ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

MUNICIPAL INVENTORY
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 4150794)

THREATENED ECOLOGICAL COMMUNITIES
DO NOT IMPACT ON THREATENED ECOLOGICAL COMMUNITIES WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au



CONSULTANT CONTACT DETAILS
COMPANY NAME: FULL POWER ELECTRICS
CONTACT: PETER TERPSTRA TEL: 08 6330 880

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE FILTER ASSESSMENT EDM#: 71597633
A	ISSUED FOR CONSTRUCTION	I.P.	04/25	HAZARD MANAGEMENT REGISTER EDM#: 71596627

MIXED RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

15 LUTH AV, DAGLISH, PERTH, WESTERN AUSTRALIA, 6008

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 04/25 SCALE: AS NOTED @ A2

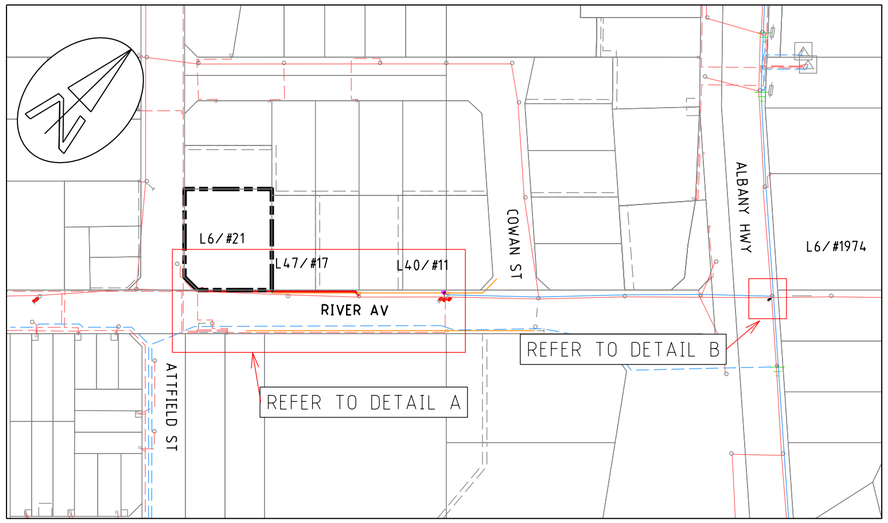
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Long. - 115° 48' 24" E

DRAWING NUMBER

MP210116

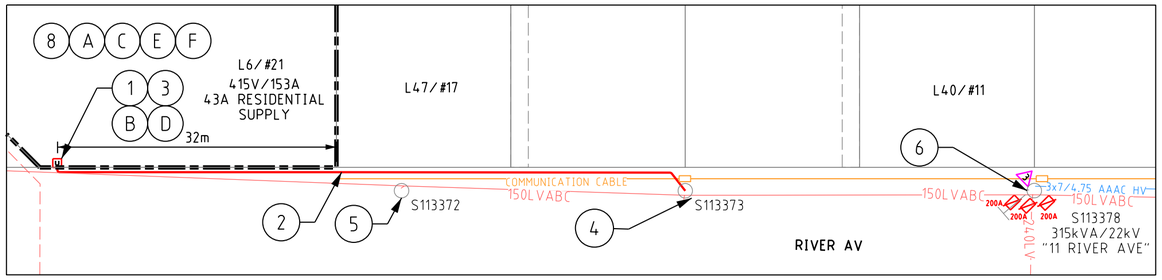
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



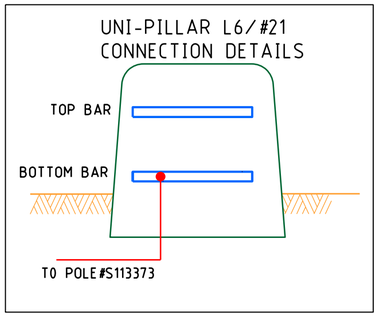
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16



CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(F) CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(G) CUSTOMER TO PROVIDE RETAILER CONSENT TO METERING CHANGES. PROJECT WILL NOT BE SCHEDULED UNTIL THE ETAF IS RECEIVED BY WESTERN POWER.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW UNI-PILLAR ON L6/#21 RIVER AV.
AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11 x 1
DCSH: U9

(2) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S113373.
CU: CN45 x 85m, CN54 x 3, CN73_LVMAINS x 3

(3) AT UNI-PILLAR (L6/#21 RIVER AV.)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
CU: LU7_240x1

(4) AT POLE#S113373
- TAP ONTO LVABC AND TERMINATE NEW 240LV CABLE
CU: LU6_LVABCx1
DCSH: U16-2

(5) AT POLE#S113372
- REMOVE EXISTING AERIAL SERVICE TO L6/#21.

(6) AT POLE S113378:
- DETTACH EXISTING AERIAL SERVICES.
- REPLACE EXISTING TRANSFORMER WITH NEW 315kVA TRANSFORMER AND TAP ONTO HV BARE.
- REATTACH EXISTING HV EARTH TO NEW TRANSFORMER.
- REPLACE EXISTING LVABC INTERMEDIATE WITH LV ABC STRAIN WITH FUSING AND ASSOCIATED EQUIPMENT AS PER THE SIMPLIFIED LINE DIAGRAM.
- REATTACH EXISTING AERIAL SERVICES.
CU: CN9_7/475 x3, TX08_4_315/22 x1, CN39 x2, CN75 x1, AB4_150FS_H
NON-CU: CJ0490 x8, GF1101x3
DCSH: R12-3-1, R13-3-1, R14-2, H04-1, H11-1, A04-1

(7) AT POLE#S113404
REPLACE EXISTING EDOs WITH NEW 16A EDOs.
NON CU: GF1566x3
DCSH: R10

(8) CM TO CONFIRM RETAILER CONSENT HAS BEEN OBTAINED AS PER CUSTOMER SCOPE OF WORK G. METERING POINT CHANGES CANNOT PROCEED UNTIL CONFIRMED.

(9) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO: BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING. IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 4150794)

UNEXPLODED ORDNANCE
NO FURTHER ACTIONS REQIUIRED FOR THIS CONSTRAINT. HOWEVER PLEASE BE MINDFUL OF ANY UNEXPECTED FINDS AND MANAGE IN ACCORDANCE WITH EXCAVATION ANO TRENCHING PROCEDURE (EDMS# 43740956)

REFER TO environment@westernpower.com.au

LEGEND

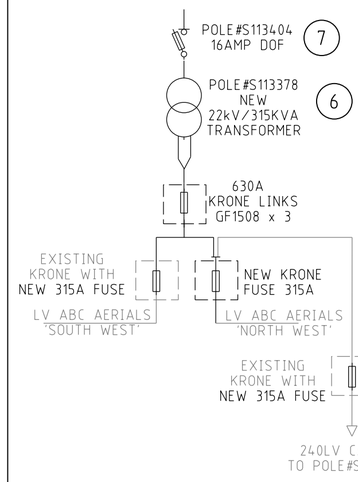
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

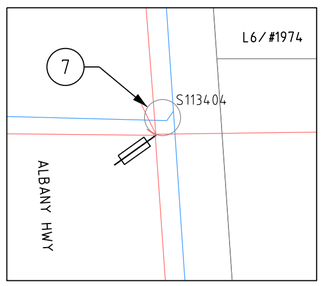
LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
Svcd Serviced Uncleared De-energised Lot
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

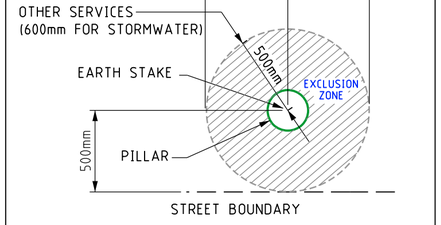


SIMPLIFIED LINE DIAGRAM



DETAIL B

1:200 4 2 0 4 8 12 16

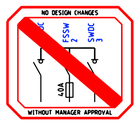


CONSULTANT CONTACT DETAILS
COMPANY NAME: Ghan Electrical Services Pty Ltd
CONTACT: BANESH BOODHOO TEL: 0409165331

COMMUNICATION CABLES IN VICINITY OF WORKS



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



DAP REFERENCE No. BA22/01013

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

ISSUED FOR CONSTRUCTION

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Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
FILE REFERENCE DETAILS

43A RESIDENTIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

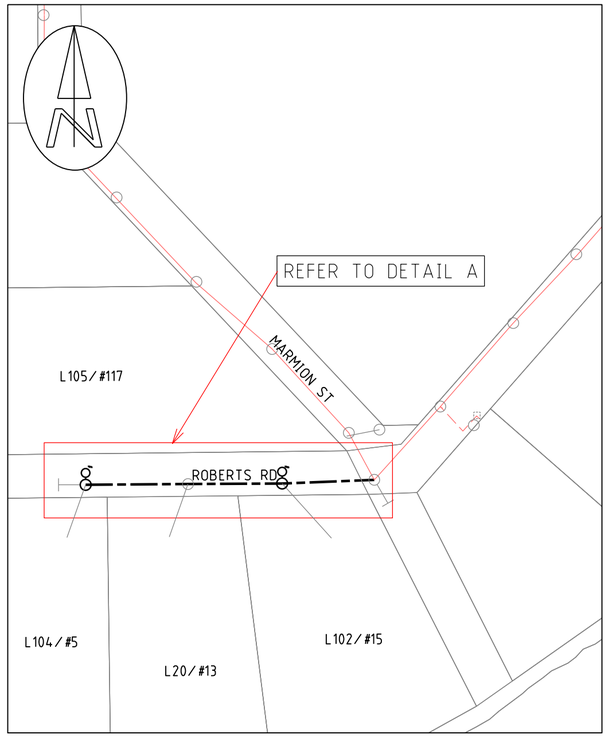
#21 RIVER AVE, MADDINGTON WA 6109

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 06/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 03' 24" S Long. - 115° 59' 16" E

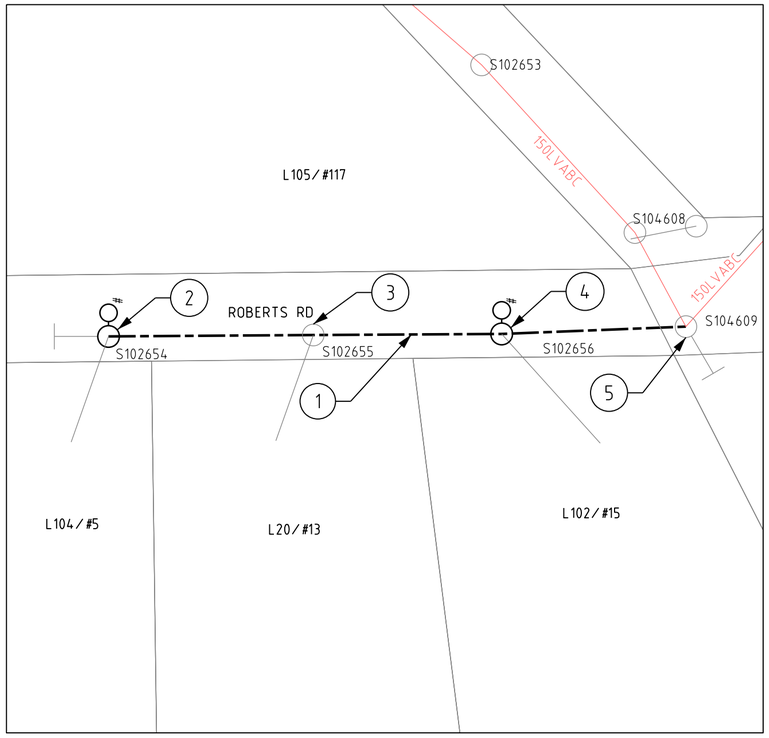
westernpower

DRAWING NUMBER
MP210310
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. REPLACE EXISTING BARE LV CONDUCTOR BETWEEN POLE S102654 AND S102659 WITH 4C150 LVABC. REFER TO CONDUCTOR SCHEDULE.
   CU: CN38_150x 160
2. AT POLE#S102654
   - REPLACE EXISTING POLE WITH 9.5m/5kN WOOD POLE.
   - INSTALL NEW STAY IN SAME LOCATION.
   - INSTALL LVABC TERMINATION.
   - INSTALL NEW 43W 3000K LED STREETLIGHT
   - REATTACH/REPLACE EXISTING SERVICE TO NEW LVABC CONDUCTOR.
   CU: AB3_150 x 1, CN65_150/35 x 4, SL7_2E x 1, P02 x 1, LV8_3P x 1
   DCSH: A03-01, S07, R13-3
3. AT POLE#S102655
   - REPLACE EXISTING INTERMEDIATE CROSSARM WITH LVABC INTERMEDIATE
   - REATTACH/REPLACE 3 x EXISTING SERVICE TO NEW LVABC CONDUCTOR.
   CU: AB1_150 x 1, CN65_150/35 x 4, LV8_3P x 1, LV8_1P x 2
   DCSH: A01
4. AT POLE#S102656
   - REPLACE EXISTING POLE WITH 11m/4kN WOOD POLE.
   - INSTALL NEW LVABC INTERMEDIATE.
   - REATTACH/REPLACE EXISTING SERVICE TO NEW LVABC CONDUCTOR.
   - INSTALL NEW 43W 3000K LED STREETLIGHT
   CU: AB1_150 x 1, CN65_150/35 x 4, LV8_3P x 1, SL7_2E, P04 x 1
   DCSH: A01, S07
5. AT POLE#S102659
   - REPLACE EXISTING LV BARE TERMINATION CROSSARM WITH LVABC T-OFF AND TAP ONTO EXISTING LVABC TAILS.
   CU: AB5_150 x 1
   DCSH: A05
6. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   THREATENED ECOLOGICAL COMMUNITIES
   NATIVE VEGETATION REGS ESAS CLEARING
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

STREET LIGHTS / EXISTING

	STREET LIGHTS	EXISTING
	POLE BRACKET	POLE BRACKET
#A	SINGLE OUTREACH	SINGLE OUTREACH
#A	DOUBLE OUTREACH	DOUBLE OUTREACH
#A	4-WAY	4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV — EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

LV - AERIAL CONDUCTORS

- 19/3.25 AAC/AAAC
- 7/4.75 AAC/AAAC
- 7/2.5 AAC/AAAC
- LV ABC 95
- LV ABC 150
- SERVICE 16mm
- EXISTING LV CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT

- INTERMEDIATE
- PTS
- DOF
- HV ISOLATOR (ISOL)
- INLINE STRAIN
- LV DISCONNECTOR
- PTT
- ISOL TX
- AERIAL STAY
- OUTRIGGER STAY
- RECLOSER
- GROUND STAY
- LOAD BREAK SWITCH (L)
- SECTIONALISER (S)
- CAPACITOR (C)
- LV ABC KRONE
- LBPTS
- STREETLIGHT BOX (SLB)

POLE SCHEDULE

POLE ID	POLE DETAILS	POLE LOCATION EASTING	POLE LOCATION NORTHING	UNSERVICEABLE (Y/N)	AS CON CHECK
S102654	REPLACE WITH 9.5m/5kN POLE WITH NEW 43W 3000K LED	EXISTING	EXISTING	N	
S102656	REPLACE WITH 11m/4kN WITH NEW 43W 3000K LED	EXISTING	EXISTING	N	

CONDUCTOR SCHEDULE

SHEET #	TYPE	CONDUCTOR	STRAIN SECT. FROM	STRAIN SECT. TO	DISTANCE (m)	RULING SPAN (m)	STRINGING TENSION %CBL @EDT	DCSH STRINGING TABLE	COMPATIBLE UNIT	AS-CON CHECK	Wind Region Settings
1	HV	4C150 LVABC	S1002654	S104609	133	44.53	7	CT-0002	CN38_150		AM

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

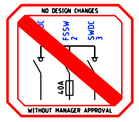
B. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD ON THE WESTERN POWER WEBSITE.

ISSUED FOR CONSTRUCTION

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF ARMADALE
CONTACT: Andy Dela Cruz TEL: 08 9394 5649

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	11/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70421377
HAZARD MANAGEMENT REGISTER EDM#: 70424114

STREETLIGHT INSTALLATION

NETWORK VOLTAGE: 22kV/415V

15 ROBERTS RD, KELMSCOTT WA 6111

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 07' 39" S Long. - 116° 01' 30" E

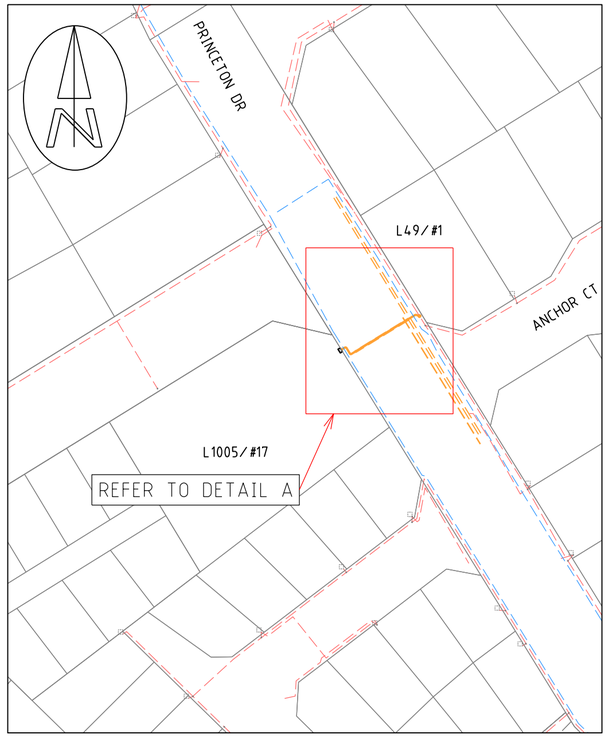
westernpower

DRAWING NUMBER

MP210355

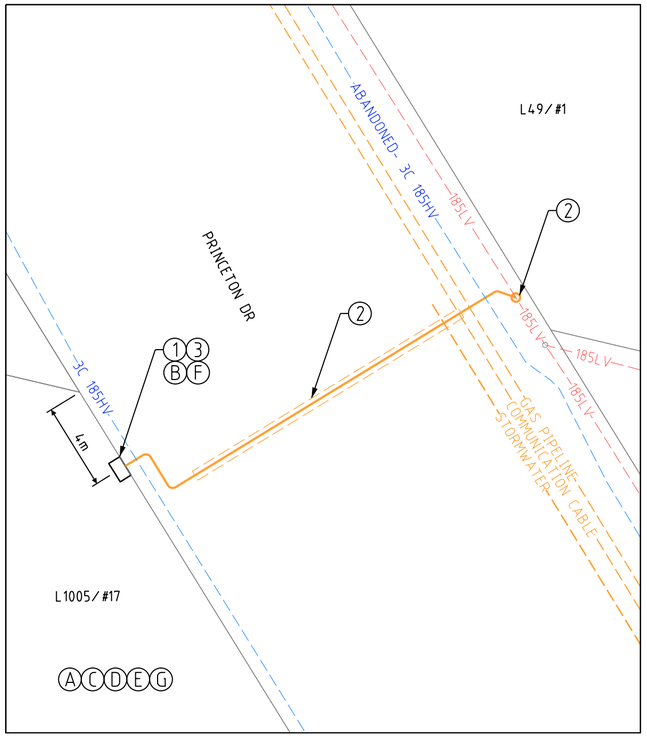
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



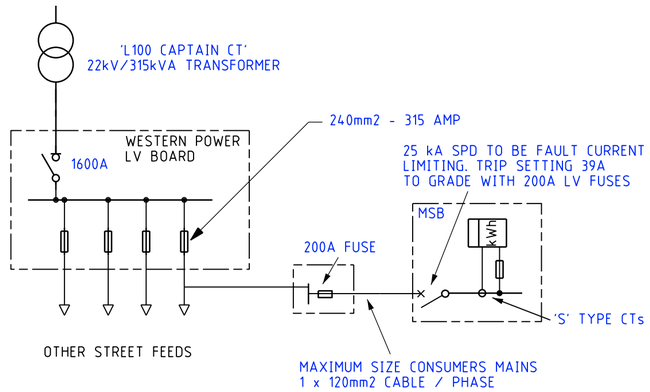
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16



WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO: THIS PROJECT IS REQUIRED TO FINALISE WORKS LEFT INCOMPLETE UNDER MP197852 DUE TO COUNCIL REQUIREMENTS.

(1) WESTERN POWER TO SUPPLY 200A WALL MOUNTED BOX TO CUSTOMER'S ELECTRICAL CONTRACTOR AND CUSTOMER IS RESPONSIBLE FOR THE INSTALLATION AS PER DCSH U33.
CU: LU35 x 1
DCSH: U33, U21

(2) REMOVE AND DISPOSE EXISTING 120LV CABLE.
INSTALL NEW 120LV CABLE FROM NEW WALL MOUNTED BOX TO EXISTING 185LV CABLE AT #1 PRINCETON DR VIA EXSITING CONDUIT AND BREACH JOINT TO EXISITING 185LV CABLE.

CU: CN43 x 30m, CN54 x 4, CN73_LVMAINS x 3, LU1 x 1

(3) AT WALL MOUNTED BOX (L1005/#17 PRINCETON DR)
- TERMINATE NEW 120LV CABLE VIA 200A FUSE.
- TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU7 x 1
DCSH: U9

(4) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) INSTALL THE WESTERN POWER SUPPLIED 200A WALL MOUNTED BOX. WALL MOUNTED BOX TO BE PREPARED AS PER THE DISTRIBUTION CONSTRUCTION STANDARDS HANDBOOK U33.

1. SHADED AREA MUST BE KEPT CLEAR FOR OPERATIONAL PURPOSES AND WP PERSONNEL (TO A HEIGHT OF 2500).

(C) CUSTOMER TO SUPPLY AND INSTALL A STARTA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLU. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(E) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO PROVIDE SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

(G) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

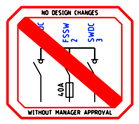
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

APPROVED FOR CONSTRUCTION





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HIGH PRESSURE GAS IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

STORM WATER PIPELINES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: APPOLO ELECTROTECH
CONTACT: SHAYNE STUURMAN TEL: 0408428722

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

FILE REFERENCE DETAILS
- MP197852- AS CONSTRUCTED DRAWING EDM#: 67908352
- SHE ASSESSMENT EDM#: 68969147
- HAZARD MANAGEMENT REGISTER EDM#: 68968010

8 x 3.5kVA DADMD RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 415V

17 PRINCETON DR, WANNANUP WA 6210

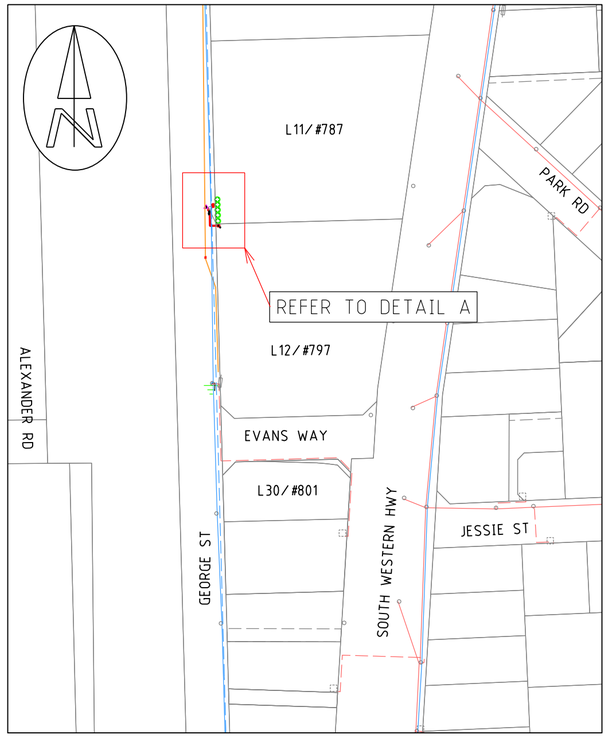
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 36' 07" S Long. - 115° 38' 25" E

westernpower

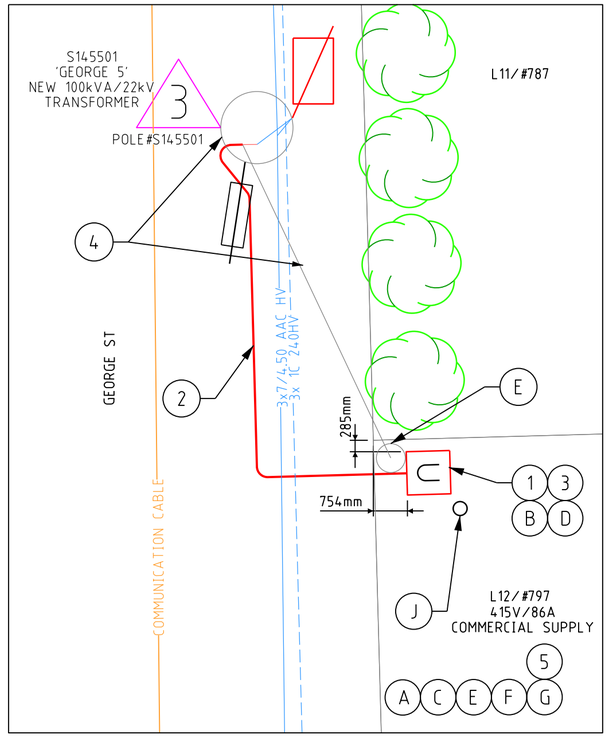
DRAWING NUMBER
MP210377
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 — 40 20 0 40 80 120 160



DETAIL A

1:100 — 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
CONTACT MICHAEL WALLIS FROM BRE ALLIANCE AT 04210530356 (Michael.Wallis@metconnx.com.au) PRIOR TO ENTERING THE AREA AND COMMENCING THE WORK.

(1) INSTALL NEW UNI-PILLAR ON L12/#799 GEORGE ST. BESIDE EXISTING SERVICE POLE AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11 x 1
DCSH: U9

(2) INSTALL 240LV CABLE IN 2.7 ALIGNMENT FROM NEW UNI-PILLAR TO THE POLE#S145501
CU: CN45 x 30m, CN54 x 3, CN73_LVMAINS x 3

(3) AT UNI-PILLAR (L12/#799 GEORGE ST.)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
CU: LU7_240x1

(4) AT POLE#S145501
- REPLACE EXISTING 3A DOF WITH 5A FAULT TAMER
- REPLACE EXISTING 63kVA Tx WITH 100kVA/22kV Tx AND TAP TO HV BARE VIA DOF.
- TERMINATE NEW 240LV CABLE TO NEW Tx VIA 160A KRONE FUSE BOX TO BE 'ON'.
- REMOVE EXISTING AERIAL SERVICE TO L12/#797
CU: LU61_240_3PO x 1, LU61_240_3PU x 1, TX4_100/22 x 1, CN9_7/475AA x 3, HV52x 3
DCSH: U16-1, H10-2

(5) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

(i) CUSTOMER TO ENSURE EXISTING MULTIPLE POINT OF SUPPLY COMPLIES WITH WAER CLAUSE 3.5.11
THE EXISTING POINT OR POINTS OF SUPPLY TO THE UNDISTURBED SECTION MAY REMAIN IN SERVICE PROVIDED THAT:
- THE UNDISTURBED ELECTRICAL INSTALLATION IS SAFE;
- NO ELECTRICAL WORK IS PROPOSED IN THE UNDISTURBED SECTION TO EITHER THE SUPPLY ARRANGEMENT OR SWITCHBOARDS;
- UPDATED ZONE DIAGRAMS AND LABELLING ARE PLACED IN EVERY MAIN SWITCHBOARD IN SERVICE AT THE PROPERTY;
- COPIES OF THE ZONE DIAGRAMS ARE PROVIDED TO DFES AND THE RELEVANT LOCAL AUTHORITY IN AN ACCEPTABLE FORMAT; AND
- EACH POINT OF SUPPLY BEARS SUFFICIENT INDELIBLE LABELLING TO IDENTIFY CLEARLY THE LOCATION AND MEANS OF ISOLATION OF SUPPLY TO THE WHOLE OR PART OF THE UNDISTURBED SECTION TO ENABLE RAPID AND PRECISE DISCONNECTION OF SUPPLY IN THE EVENT OF FIRE.

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5B.

(F) CUSTOMER TO SUPPLY AND INSTALL SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY.

(G) CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(H) CUSTOMER TO REMOVE SERVICE POLE AFTER COMPLETION OF WORK.

LEGEND

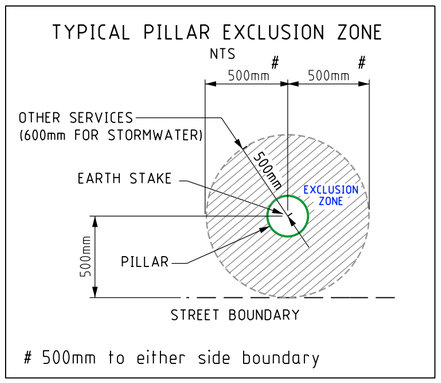
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

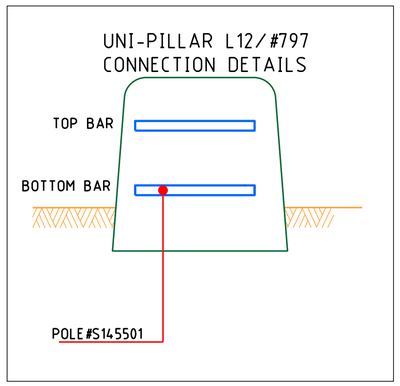
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

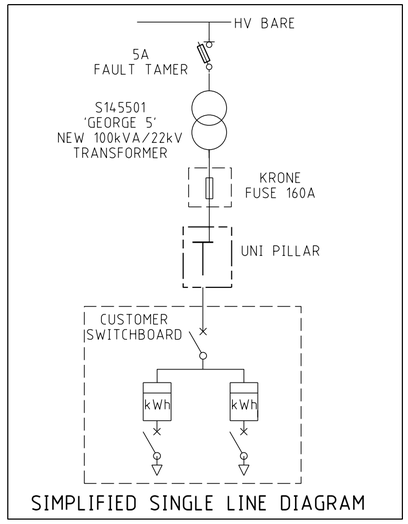
OH EQUIPMENT
- INTERMEDIATE
- PTS
- DOF
- HV ISOLATOR
- INLINE STRAIN
- LV DISCONNECTOR
- PTT
- ISOL TX
- AERIAL STAY
- OUTRIGGER STAY
- RECLOSER
- GROUND STAY
- LOAD BREAK SWITCH
- SECTIONALISER
- CAPACITOR
- LV ABC KRONE
- LBPTS
- STREETLIGHT BOX

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



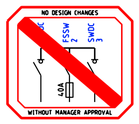




SIMPLIFIED SINGLE LINE DIAGRAM

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: SERVICE STREAM
CONTACT: ABU MOHIUDDIN TEL: +61 405 661 018

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE FILTER ASSESSMENT EDM#: 70153273
A	ISSUED FOR CONSTRUCTION	I.P.	11/24	HAZARD MANAGEMENT REGISTER EDM#: 70150239

86A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

797 SOUTH WESTERN HWY, BYFORD, 6122, WA

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 11/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 32° 12' 51" S
Long. - 116° 00' 28" E

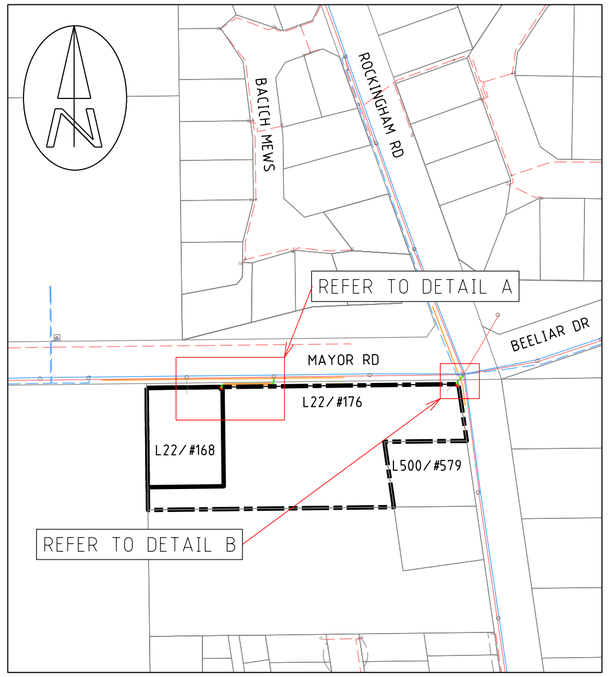
westernpower

DRAWING NUMBER

MP210408

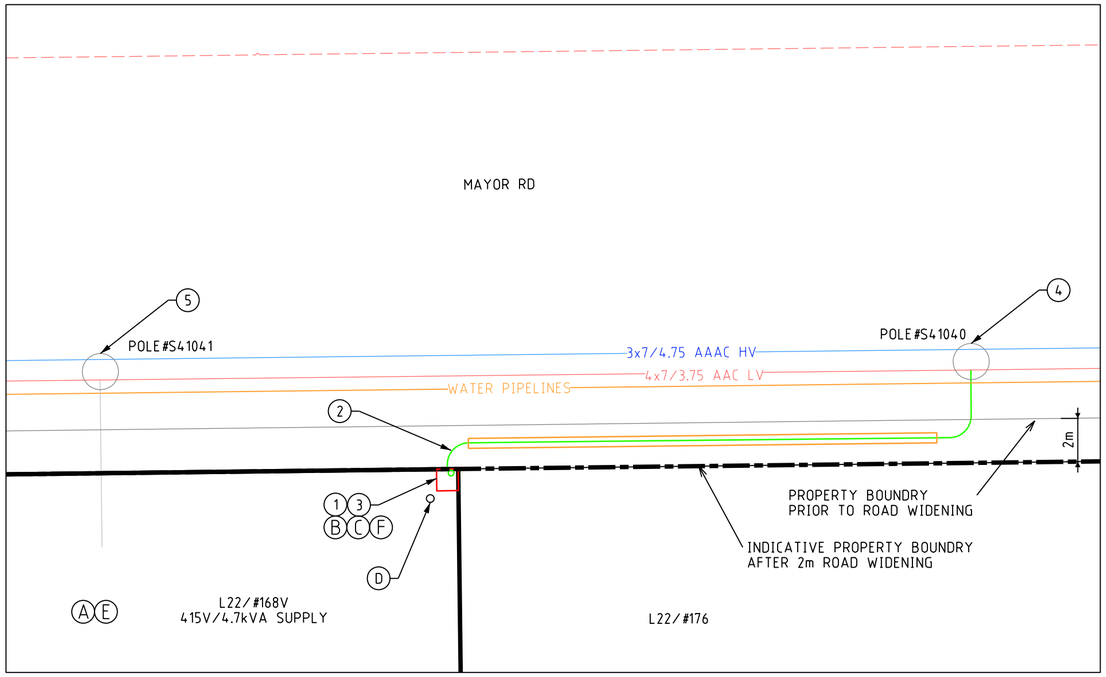
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



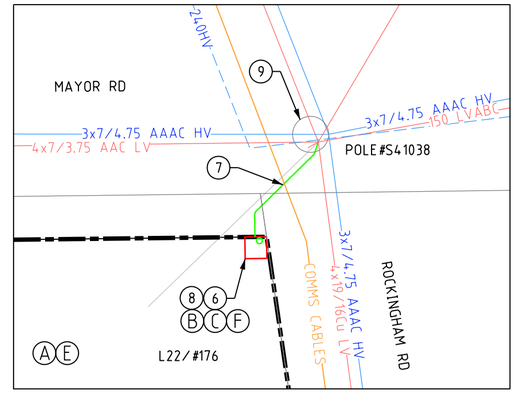
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

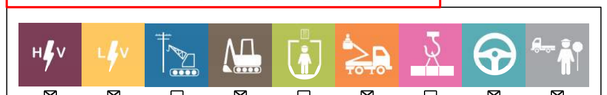
1:200 4 2 0 4 8 12 16

LEGEND

- SCHEME BDY.
- RETAINING WALL
- LV-SL CABLES
  - CN42 - 25mm2 (1x3c)
  - EXISTING LV CABLES (SIZE AND TYPE INDICATED)
- PILLARS / PITS / ETC
  - U UNIVERSAL
  - MINI
  - CABLE MARKER U/G
  - SUDI Serviced Uncleared De-energised Lot
  - UTILITIES PIT
  - SP BGSP PIT
  - CONDUIT / POLY PIPE
- AERIAL CONDUCTORS AND POLES
  - NEW POLE (ANY TYPE AS INDICATED)
  - EXISTING POLE (ANY TYPE AS INDICATED)
  - HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

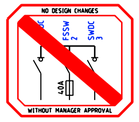
COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS



☒ ☒ ☐ ☒ ☐ ☒ ☐ ☒ ☒

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER TO SUPPLY AND INSTALL CONDUITS AS INDICATED ON THE DETAIL.

(D) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

(E) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

ISSUED FOR CONSTRUCTION

WESTERN POWER SCOPE OF WORK

DETAIL A

(1) INSTALL NEW MINI-PILLAR ON L22/#168 MAYOR RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10 x 1

(2) INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S41040 VIA 63mm POLYPIPE.
CU: CN42 x 45m, CN73_LV25 x 3, CN54 x 2

(3) AT MINI-PILLAR (L22/#168 MAYOR RD)
- TERMINATE NEW 25LV CABLE.
- TERMINATE CUSTOMER'S CABLE TO THE BACK BLOCKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
DCSH: U08-04

(4) AT POLE#S41040
- TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
CU: LU51
DCSH: U19-1

(5) AT POLE#S41041
- REMOVE CUSTOMER AERIAL SERVICE TO L22/#168.

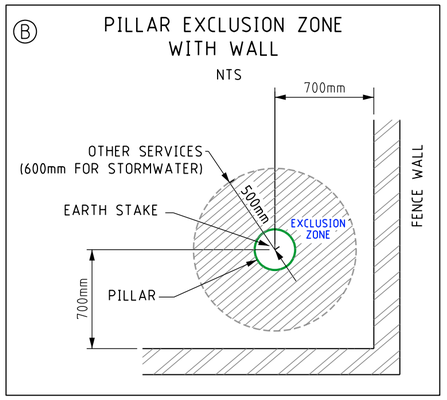
DETAIL B

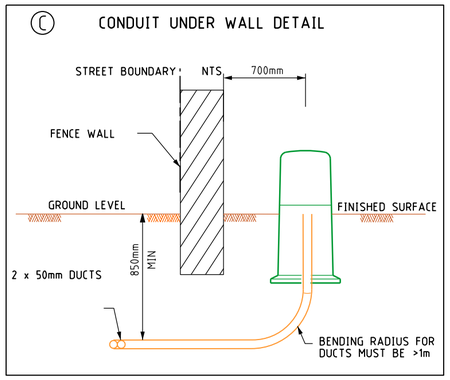
(6) INSTALL NEW MINI-PILLAR ON L22/#176 MAYOR RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10 x 1

(7) INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S41038
CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 2

(8) AT MINI-PILLAR (L22/#176 MAYOR RD)
- TERMINATE NEW 25LV CABLE.
- TERMINATE CUSTOMER'S CABLE IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
DCSH: U08-04

(9) AT POLE#S41038
- TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
- REMOVE CUSTOMER AERIAL SERVICE TO L22/#176
CU: LU51
DCSH: U19-1





W.A.P.C. REFERENCE No. 163511

CUSTOMER CONTACT DETAILS
COMPANY NAME: McMULLEN NOLAN GROUP PTY LTD.
CONTACT: KEN SELLERS TEL: 0477 277 200

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69088892
HAZARD MANAGEMENT REGISTER EDM#: 69089872

OVERHEAD TO UNDERGROUND CONVERSION

NETWORK VOLTAGE: 22kV/415V

168 MAYOR RD, LAKE COOGEE WA 6166

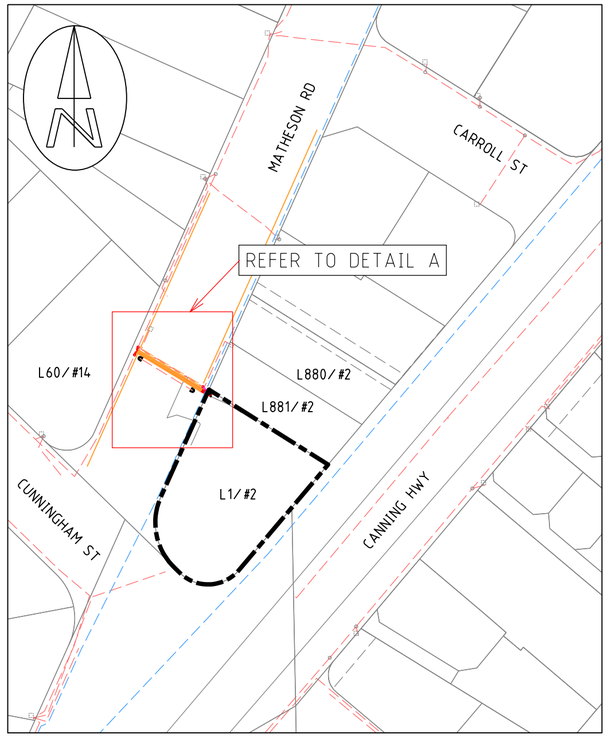
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 07' 41" S Long. - 115° 47' 05" E

westernpower

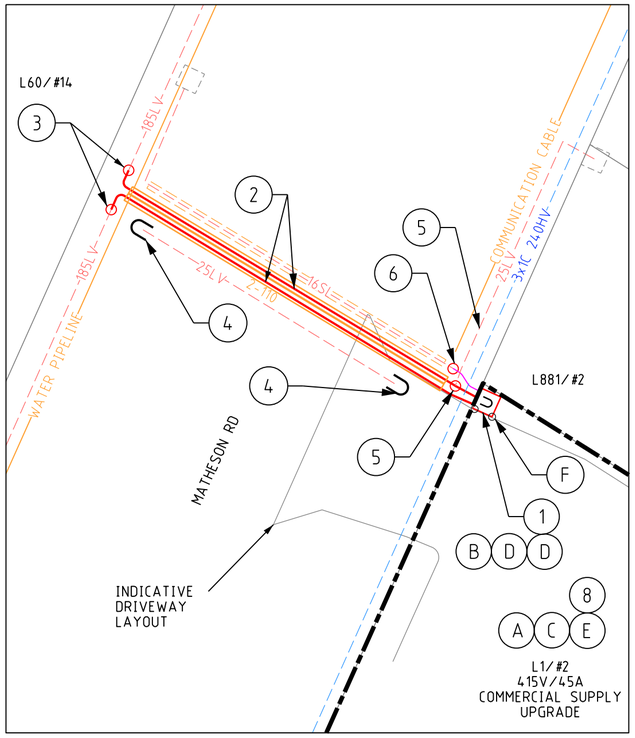
DRAWING NUMBER
MP210427
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING CABLES AND REMOVE EXISTING MINI-PILLAR AT L1/#2 MATHESON RD. INSTALL NEW UNI-PILLAR ON L1/#2 MATHESON RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U09
2. INSTALL 2 x 240LV CABLES FROM NEW UNI-PILLAR TO EXISTING 185LV CABLE ON ACROSS THE MATHESON RD VIA 110mm POLYPIPE.
   CU: CN45 x 55m, CN73_LVMAINS x 7 , CN54 x 4
3. CUT EXISTING 185LV CABLE AND 2 x STRAIGHT JOINT TO 2 x NEW 240LV CABLES.
   CU: LU2 x 2
4. CUT, CAP AND ABANDON EXISTING 25LV CABLE.
   CU: CN73_LV25 x 3
5. CUT EXISTING 25LV CABLE AND T-JOINT TO NEW 240LV CABLE.
   CU: LU05 x 1
6. INSTALL NEW 16SL CABLE FROM NEW UNI-PILLAR TO EXISTING 16SL CABLE. CUT EXISTING 16SL CABLE AND STRAIGHT JOINT TO NEW 16SL CABLE. REMOVE EXCESS CABLE.
   CU: LU13 x 1, CN41 x 5, CN73_10/16 x 3
7. AT UNI-PILLAR (L1/#2 MATHESON RD)
   - TERMINATE 2 x NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE NEW 16LV CABLE AT THE BACK BLOCK VIA 20A RED SPOT FUSE.
   CU: LU7_240 x 2, UM3_R x 1
   DCSH: U09
8. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   MUNICIPAL INVENTORY
   THREATENED ECOLOGICAL COMMUNITIES
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

LEGEND

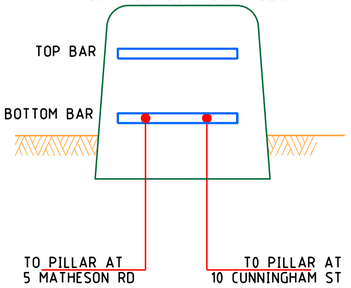
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

UNI-PILLAR L1/#2 CONNECTION DETAILS



NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

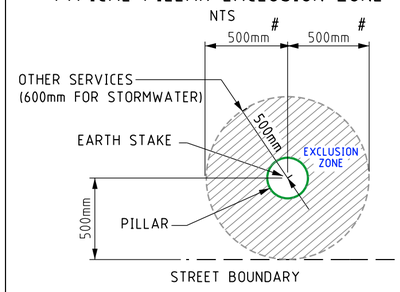
COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

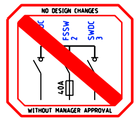
D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.
CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

F. CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: J. L. ELECTRICAL
CONTACT: JAKE RYAN LOFFMAN TEL: 0400 052 852

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/24

FILE REFERENCE DETAILS
- SHE FILTER ASSSESSMENT REPORT EDM#: 69414553
- HAZARD MANAGEMENT REGISTER EDM#: 69424127

45A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

2C MATHESON RD, APPLECROSS WA 6153

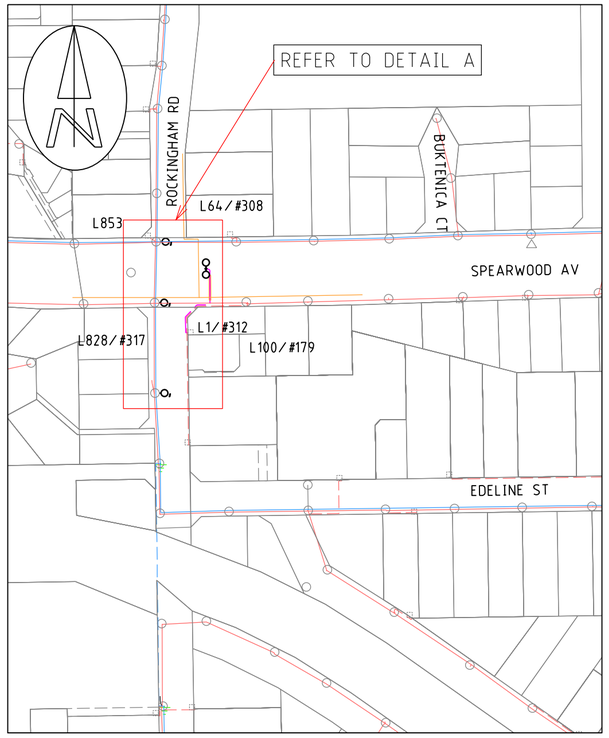
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 | SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 01' 28" S Long. - 115° 49' 42" E

westernpower

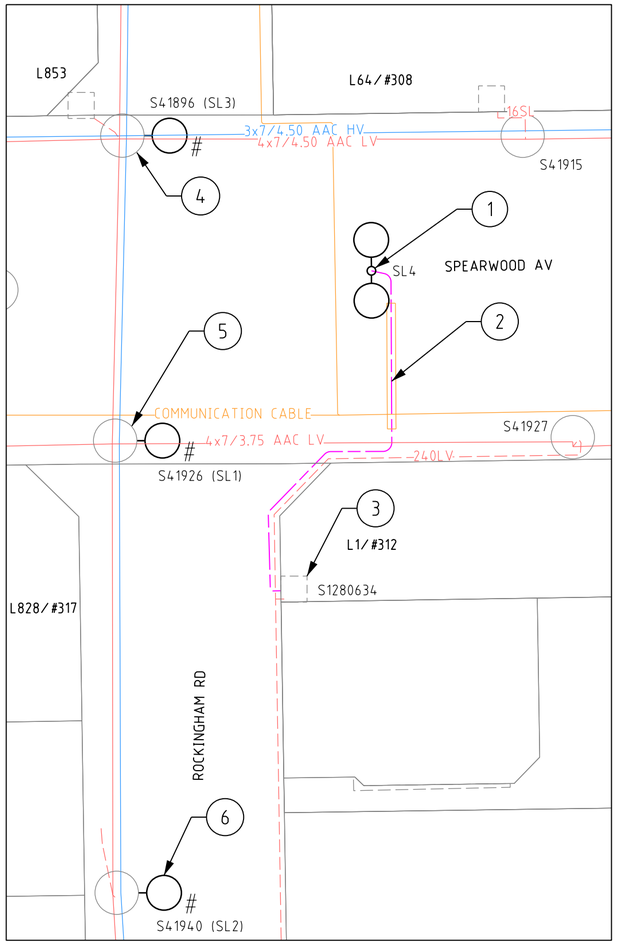
DRAWING NUMBER
MP210438
SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A2



LOCATION PLAN

1:2500 — 50 25 0 50 100 150 200



DETAIL A

1:500 — 10 5 0 10 20 30 40

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 12.5m STREETLIGHT WITH 2x170W 4000k LED STREETLIGHT AS PER POLE SCHEDULE TABLE.
   CU: SL12_6L x 1
   DCSH: S12
2. INSTALL NEW 16SL CABLE FROM NEW SL TO EXISTING MINI-PILLAR #S1280634 AT L1/#312 SPEARWOOD AV.
   CU: CN41 x 65m, CN73_10/16 x 3, CN54 x 3
3. AT MINI-PILLAR#S1280634, TERMINATE NEW 16SL CABLE VIA RED SPOT FUSE.
   CU: UM3_R x 1
   DCSH: U23-2
4. AT EXISTING POLE#S41896, REPLACE EXISTING LUMINARE WITH 68W 4000k LED ON EXISTING SUPPORT BRACKET AS PER POLE SCHEDULE TABLE.
   CU: SL18_FLAIR x 1
   NON CU: HL6309 x 1
   DCSH: S03
5. AT EXISTING POLE#S41926, REPLACE EXISTING LUMINARE WITH 140W 4000k LED ON EXISTING SUPPORT BRACKET AS PER POLE SCHEDULE TABLE.
   CU: SL18_FLAIR x 1
   NON CU: HL6315 x 1
   DCSH: S03
6. AT EXISTING POLE#S41940, REPLACE EXISTING LUMINARE WITH 140W 4000k LED ON EXISTING SUPPORT BRACKET AS PER POLE SCHEDULE TABLE.
   CU: SL18_FLAIR x 1
   NON CU: HL6315 x 1
   DCSH: S03
7. ENVIRONMENTAL FILTER CHECKED

   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING

# POLE BRACKET	POLE BRACKET
# SINGLE OUTREACH	SINGLE OUTREACH
# DOUBLE OUTREACH	DOUBLE OUTREACH
# 4-WAY	4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

PILLARS / PITS / ETC

MINI (WORKING END)	UMS PIT
U UNIVERSAL	UTILITIES PIT
MINI	SP BGSP PIT
LINK	100/200AMP CUTOUT
CABLE MARKER U/G	CONDUIT / POLY PIPE
SUD Serviced Uncleared De-energised Lot	

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE LOCATION OF ALL NEW STREET LIGHTS IN ACCORDANCE WITH THE CUSTOMERS STREET LIGHTING DESIGN, AS1158, WESTERN POWERS 'UNDERGROUND DISTRIBUTION SCHEMES' MANUAL AND ANY SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER. IT IS THE CUSTOMERS RESPONSIBILITY TO ADHERE TO THE FOLLOWING:
- THE LOCATION OF ALL STREET LIGHTS IS CLEAR OF ANY 3RD PARTY ASSETS
- ALL STREET LIGHTS MAINTAIN A MINIMUM SET BACK FROM THE ROAD IN ACCORDANCE WITH AS1158.1.2 AND WESTERN POWERS UDS MANUAL.
- ANY PROPOSED STREET LIGHT LOCATIONS OUTSIDE OF THE WESTERN POWER STANDARD 2.4 - 3.0m POLE ALIGNMENT HAVE THE APPROVAL OF 3RD PARTY UTILITIES IN ACCORDANCE WITH THE UTILITY PROVIDERS CODE OF PRACTICE.
- STREET LIGHTS ARE NOT PERMITTED TO BE INSTALLED WITHIN THE WESTERN POWER STANDARD 0-500mm CABLE ALIGNMENT.
- ALL STREET LIGHTS ARE TO ADHERE TO THE POWER LINE CLEARANCE REQUIREMENTS AS PER AS7000, INCLUDING IN THE EVENT THEY ARE STRUCK BY A VEHICLE AND FALL TOWARDS THE TRANSMISSION LINE.
- CONTRACTOR TO PEG LOCATION OF NEW STREETLIGHT AS NOTED AND IS REQUIRED TO LIAISE WITH PRINCIPAL PRIOR TO PEGGING THE NEW STREETLIGHT TO AVOID CLASHES WITH THE PROPOSED MRWA TRAFFIC SIGNAL POST, CABLING AND JUNCTION PIT.

C. ANY TEMPORARY STREET LIGHTING REQUIRED DURING THE WORKS AFTER REMOVAL OF EXISTING STREET LIGHTS UNTIL NEW STREET LIGHTS HAVE BEEN ENERGISED IS TO BE SUPPLIED AN INSTALLED BY THE CUSTOMER AT NO COST TO WESTERN POWER.

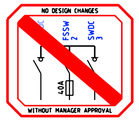
D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT ADDITIONAL STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

POLE SCHEDULE

POLE ID	POLE DETAILS	POLE LOCATION		UNSERVICEABLE	AS CON CHECK
		EASTING	NORTHING	(Y/N)	
S41926 (SL1)	REPLACE EXISTING LUMINAIRE WITH NEW 170W 6L LED	EXISTING	EXISTING	N	
S41940 (SL2)	REPLACE EXISTING LUMINAIRE WITH NEW 170W 6L LED	EXISTING	EXISTING	N	
S41896 (SL3)	REPLACE EXISTING LUMINAIRE WITH NEW 68W 4L LED	EXISTING	EXISTING	N	
SL4	NEW 12.5m STANDARD STEEL POLE WITH 2x 170W 6L LED	385161.647	6447550.754	N	

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: STANTEC AUSTRALIA
CONTACT: SHARNI MILLMAN TEL: +61 418 925 609

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	12/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70664400
HAZARD MANAGEMENT REGISTER EDM#: 70664858

STREET LIGHT UPGRADE

NETWORK VOLTAGE: 22kV/415V

ROCKINGHAM RD, SPEARWOOD WA 6163

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 12/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 06' 09" S Long. - 115° 46' 57" E

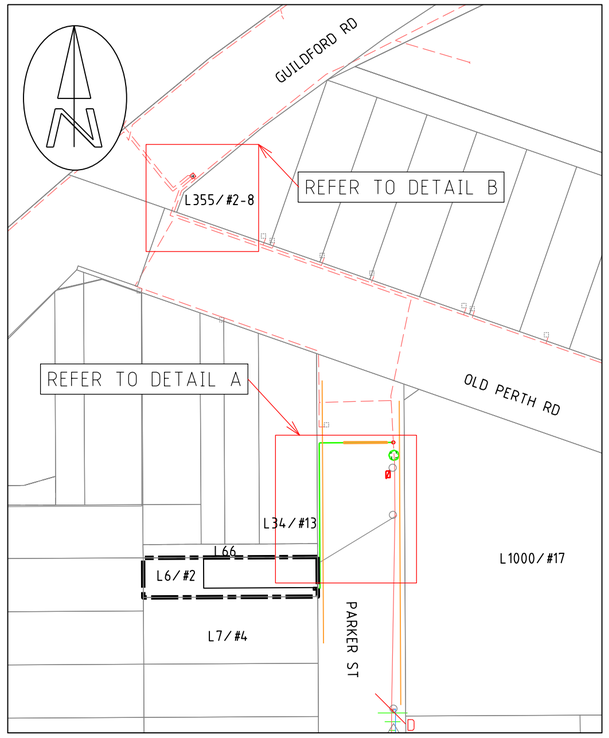
westernpower

DRAWING NUMBER

MP210470

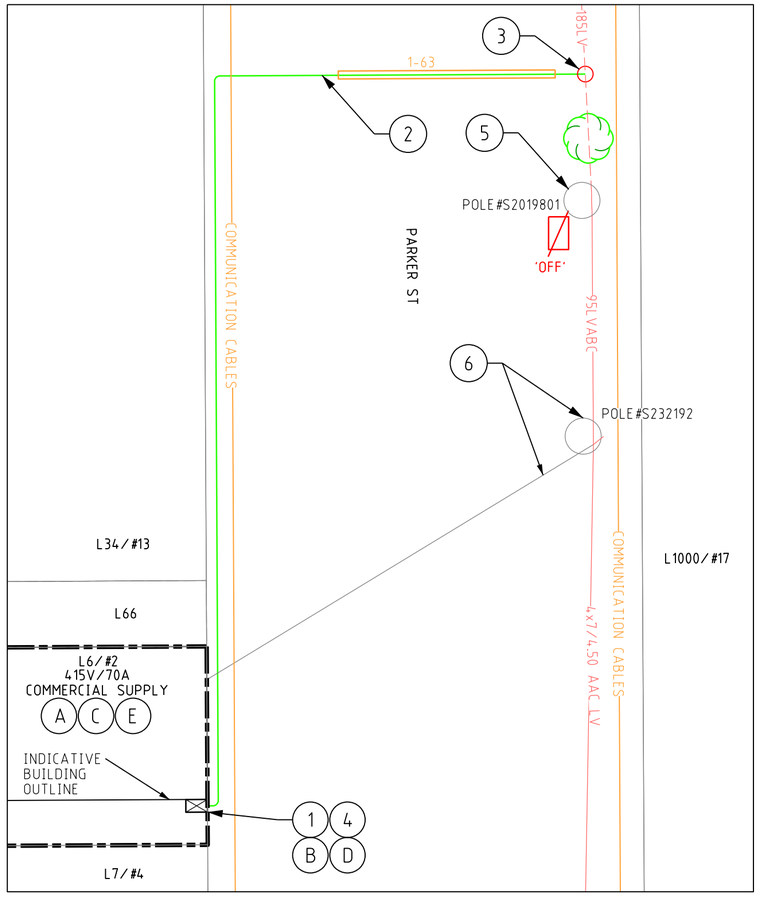
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



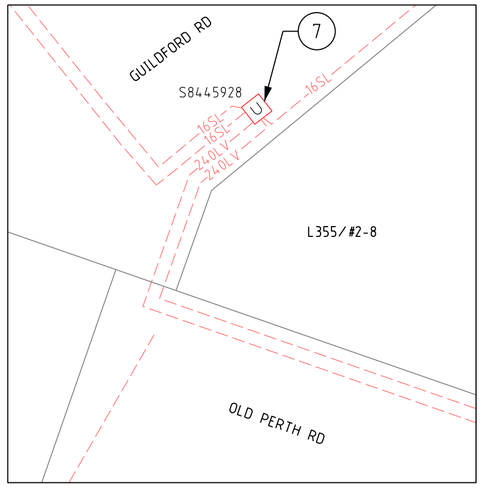
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



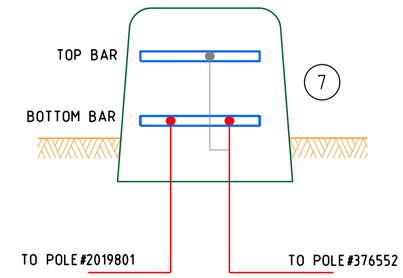
DETAIL A

1:200 — 4 2 0 4 8 12 16



DETAIL B

1:200 — 4 2 0 4 8 12 16



UNI-PILLAR#S8445928 CONNECTION DETAIL

WESTERN POWER SCOPE OF WORK

1. WESTERN POWER TO SUPPLY 100A WALL MOUNTED BOX TO CUSTOMER'S ELECTRICAL CONTRACTOR AND CUSTOMER IS RESPONSIBLE FOR THE INSTALLATION AS PER DCSH U31.
   CU: LU34 x 1
   DCSH: U31, U20
2. INSTALL 25LV CABLE FROM NEW WALL MOUNTED BOX TO 185LV CABLES VIA 63mm POLYPIPE.
   CU: CN42 x 65m, CN73_LV25 x 3, CN54 x 3
3. TEE-JOINT 25LV CABLE WITH EXISITNG 185LV CABLE
   CU: LU4 x 1
4. AT WALL MOUNTED BOX (L6/#2 PARKER ST)
   - TERMINATE NEW 25LV CABLE VIA 100A FUSE.
   - TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   DCSH: U9
5. AT POLE#S2019801,
   - REMOVE EXISTING OVERHEAD FUSE SWITCH
   - INSTALL NEW KRONE BOX
   - RE-TERMINATE EXISTING 185LV CABLE TO LVABC, VIA NEW KRONE BOX (OFF)
   CU: LU37_185O x 1, LU37_185U x 1
6. AT POLE#S232192, REMOVE EXISTING AERIAL SERVICE TO L6/#2 PARKER ST.
7. AT UNI-PILLAR#S8445928, DISCONNECT EXISTING 240LV CABLE FROM TOP BAR AND RE-TERMINATE TO BOTTOM BAR AS PER UNI PILLAR CONNECTION DETAIL
8. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO: BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING. IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 4150794)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. INSTALL THE WESTERN POWER SUPPLIED 100A WALL MOUNTED BOX. WALL MOUNTED BOX TO BE PREPARED AS PER THE DISTRIBUTION CONSTRUCTION STANDARDS HANDBOOK U31.

   1. SHADED AREA MUST BE KEPT CLEAR FOR OPERATIONAL PURPOSES AND WP PERSONNEL (TO A HEIGHT OF 3500).

C. CUSTOMER TO SUPPLY AND INSTAL SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

D. CUSTOMER ELECTRICAL CONTRACTOR TO PROVIDE SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

E. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

LEGEND

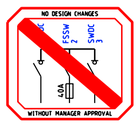
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT
- INTERMEDIATE
- PTS
- DOF
- ISOL HV ISOLATOR
- INLINE STRAIN
- D LV DISCONNECTOR
- PTT
- ISOL TX
- AERIAL STAY
- OUTRIGGER STAY
- R RECLOSER
- GROUND STAY
- L LOAD BREAK SWITCH
- S SECTIONALISER
- C CAPACITOR
- LV ABC KRONE
- LBPTS
- SLB STREETLIGHT BOX



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: PLACE DEVELOPMENT
CONTACT: JOE D×ALESSANDRO TEL: +61 413 761 484

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70102234
HAZARD MANAGEMENT REGISTER EDM#: 70104601

70A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

2 PARKER ST, BASSENDEAN WA 6054

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 10/24	SCALE: AS NOTED @ A2
GEO REF:	Lat. - 31° 54' 16" S Long. - 115° 56' 52" E

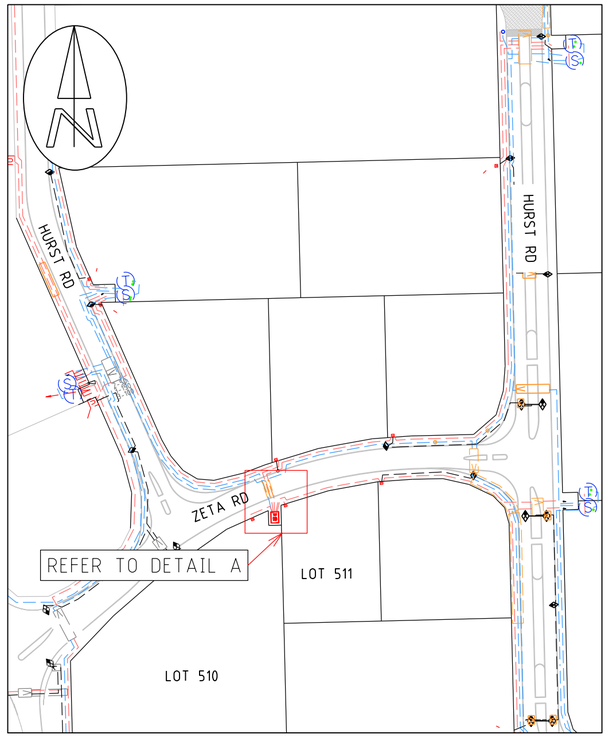
westernpower

DRAWING NUMBER

MP210523

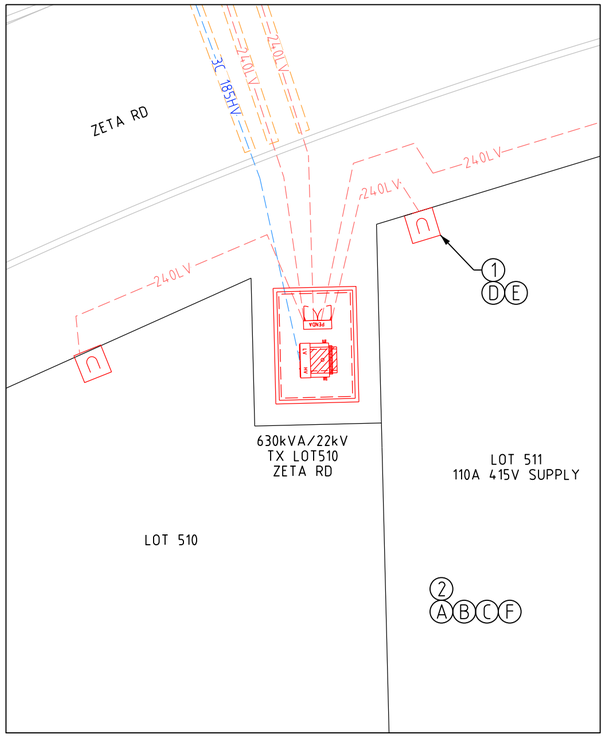
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



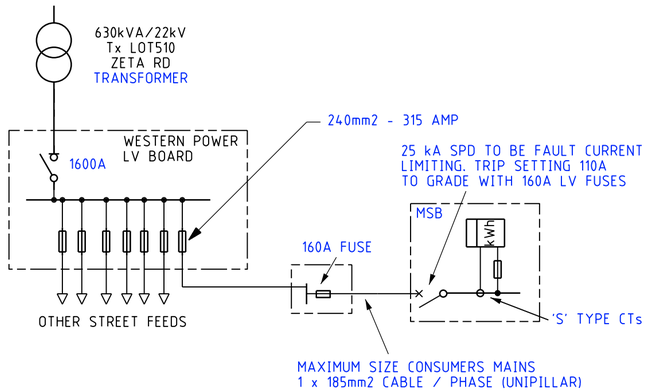
LOCATION PLAN

1:1000 20 10 0 20 40 60 80

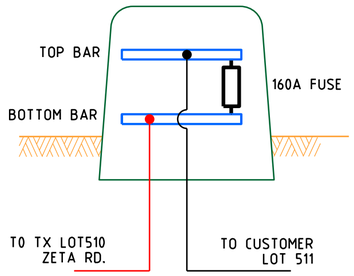


DETAIL A

1:200 4 2 0 4 8 12 16



UNI-PILLAR LOT 511
CONNECTION DETAILS



WESTERN POWER SCOPE OF WORK

i NOTE TO PNO: REFER TO MS018354 FOR LOCAL DISTRIBUTION NETWORK INFORMATION.

1 AT EXISTING UNI-PILLAR (LOT 511 ZETA RD) TERMINATE CUSTOMER'S CABLE TO THE TOP BAR WITH 160A FUSE.
CU: LU36_160A x 1
DCSH: U9

2 INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
CU: LM1_M x 1

CUSTOMER SCOPE OF WORK

A THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B CUSTOMER MUST INSTALL SMSB WITHIN 30m OF THE POINT OF SUPPLY.

C CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

E CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

F CUSTOMER MUST SUPPLY AND INSTALL 'S' TYPE METERING CT's.

LEGEND

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

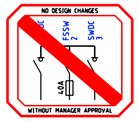
TRANSFORMER (T) EXISTING SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

APPROVED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: ELECTRICAL CONSULTANCY WA
CONTACT: AHMAD AL DAJANI TEL: 0439 988 739

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	07/24	HAZARD MANAGEMENT REGISTER EDM#: 69006346

110A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/ 415V

LOT 511 ZETA ROAD, WATTLEUP, 6166, WA

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 09' 20" S Long. - 115° 47' 37" E

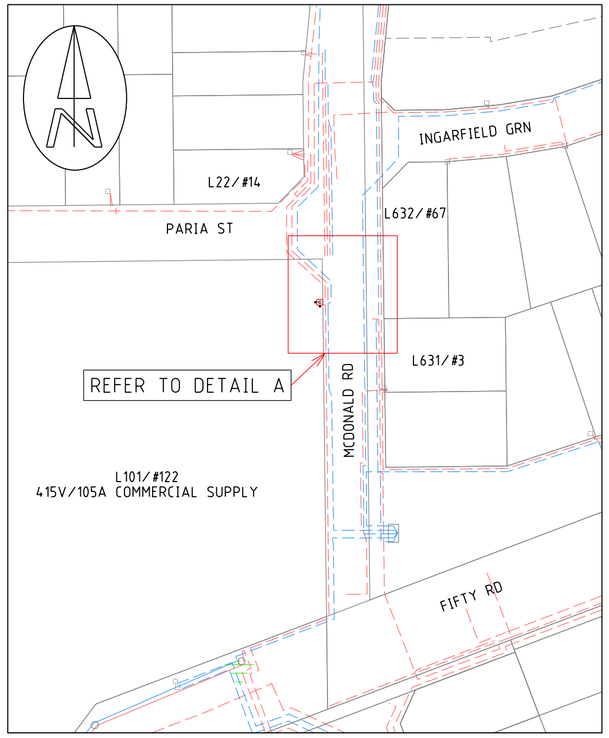
westernpower

DRAWING NUMBER

MP210570

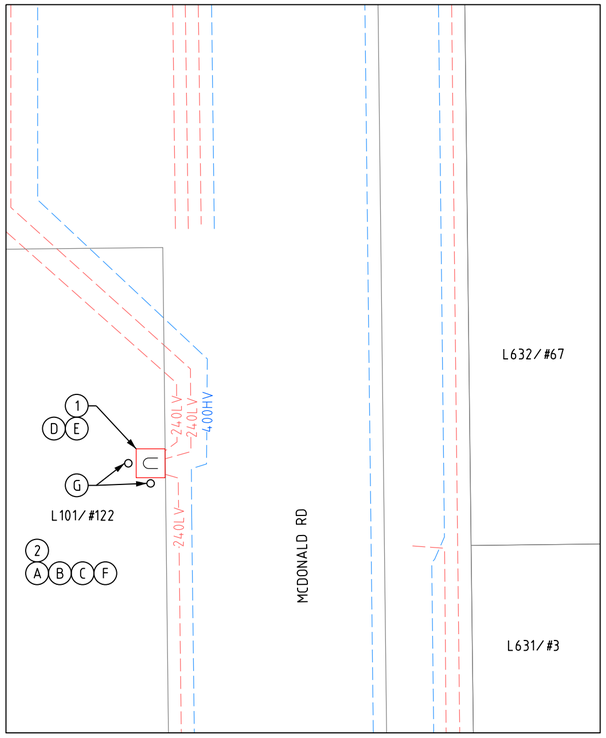
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



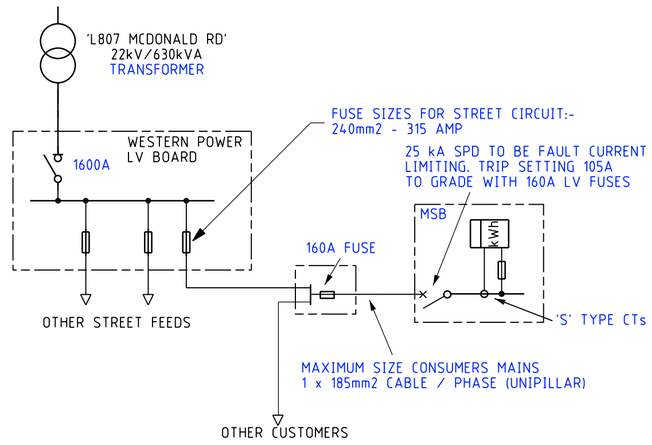
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16



LEGEND

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WESTERN POWER SCOPE OF WORK

1. AT EXISTING UNI-PILLAR (L101/#122 FIFTY RD) TERMINATE CUSTOMER'S CABLE TO THE TOP BAR WITH 160A FUSE IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
   CU: LU36_160A x 1
   DCSH: U09
2. INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1_M x 1

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST INSTALL SMSB WITHIN 30m OF THE POINT OF SUPPLY.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

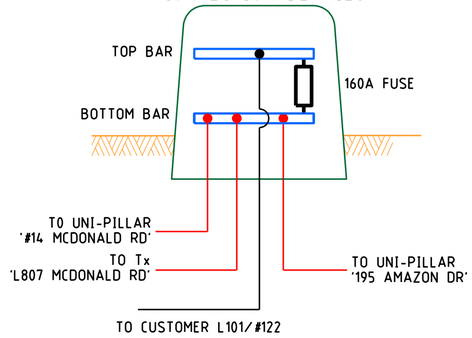
D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR TOP BAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 185mm2/PHASE.

E. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

F. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

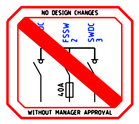
G. CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

UNI-PILLAR L101/#122 CONNECTION DETAILS



ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: ELECTRICAL CONSULTANCY WA
CONTACT: MICHAEL TEL: 0439 988 739

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	07/24	HAZARD MANAGEMENT REGISTER EDM#: 69065126

105A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

122 FIFTY RD, BALDIVIS WA 6171

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 18' 28" S Long. - 115° 48' 31" E

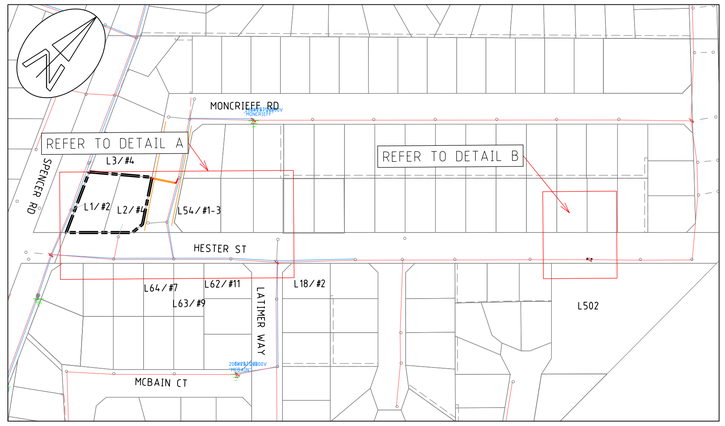
westernpower

DRAWING NUMBER

MP210705

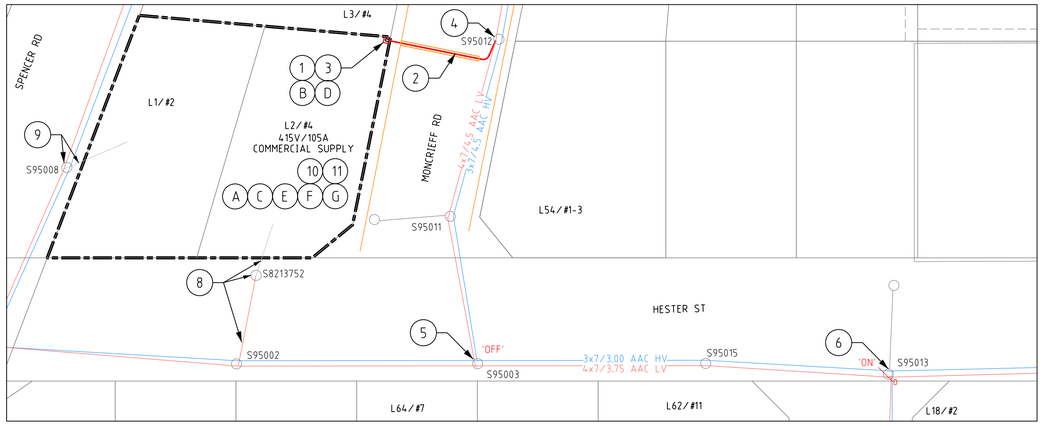
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



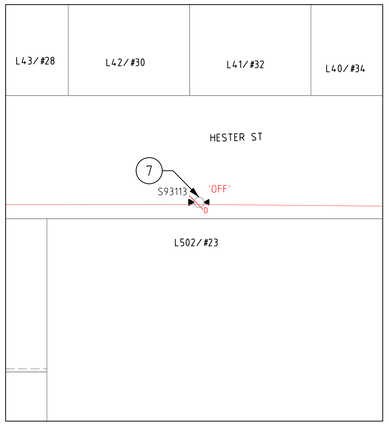
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:500 10 5 0 10 20 30 40



DETAIL B

1:500 10 5 0 10 20 30 40

LEGEND

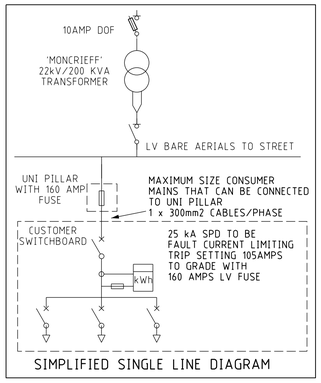
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
CN45 - 240mm2 (1x3c)
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

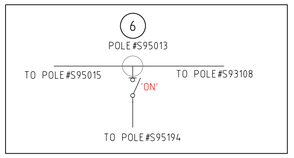
PILLARS / PITS / ETC
U UNIVERSAL
CONDUIT / POLY PIPE

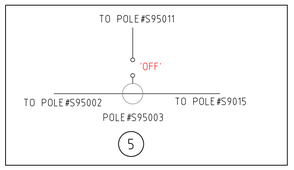
AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

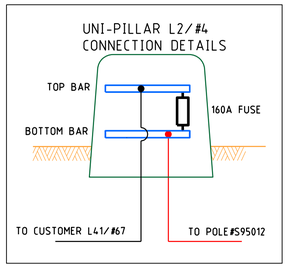
OH EQUIPMENT
INLINE STRAIN
LV DISCONNECTOR



SIMPLIFIED SINGLE LINE DIAGRAM







WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L2/#4 MONCRIEFF RD. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S95012
   CU: CN45 x 35m, CN54 x 3, CN73_LVMAINS x 3
3. AT UNI-PILLAR (L2/#4 MONCRIEFF RD)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR VIA 160A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_100Ax1
4. AT POLE#S95012
   - TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
   CU: LU6_240 x 1, LU6_240U x 1
   DCSH: U16-1
5. AT POLE#S95003
   - REMOVE OH LV TAILS AND ISOLATE LV TOWARDS POLE#S95011.
6. AT POLE#S95013
   - SWITCH DISCONNECTOR TO 'ON' POSITION.
7. AT POLE#S93113
   - REPLACE EXISTING INTERMEDIATE CROSSARM WITH A STRAIN CROSSARM.
   - INSTALL DISCONNECTOR ON NEW LV STRAIN CROSSARM AND ESTABLISH OPEN POINT.
   CU: LV03 x 1, LV06 x 1
   DCSH: L03, L06
8. AT POLE#S8213752
   - REMOVE EXISTING AERIAL SERVICE TO L2/#4.
   - REMOVE EXISTING POLE.
9. AT POLE#S95008
   - REMOVE EXISTING AERIAL SERVICE TO L1/#2.
10. INSTALL CT METER
    CU: LM1_M x 1
11. ENVIRONMENTAL FILTER CHECKED.
    THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

    ACID SULPHATE SOILS (ASS)
    EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO:
    BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
    IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

    UNEXPLODED ORDNANCE
    NO FURTHER ACTIONS REQUIRED FOR THIS CONSTRAINT. HOWEVER, PLEASE BE MINDFUL OF ANY UNEXPECTED FINDS AND MANAGE IN ACCORDANCE WITH EXCAVATION AND TRENCHING PROCEDURE (EDMS# 43740956)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

E. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

G. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

APPROVED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

W.A.P.C. REFERENCE No.

CUSTOMER CONTACT DETAILS
COMPANY NAME: ACCORD PROPERTIES
CONTACT: MATT DAVIES TEL: 0439 988 739

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM#: 70192420
A	ISSUED FOR CONSTRUCTION	I.P.	10/24	HAZARD MANAGEMENT REGISTER - EDM# 70188611

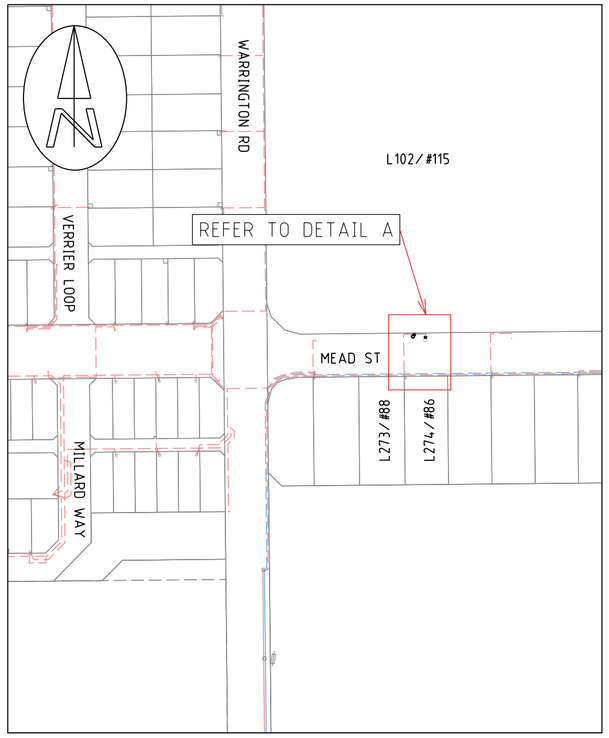
105A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGES: 22kV/415V

2-4 MONCRIEFF RD, LANGFORD WA
DESIGN DRAWING

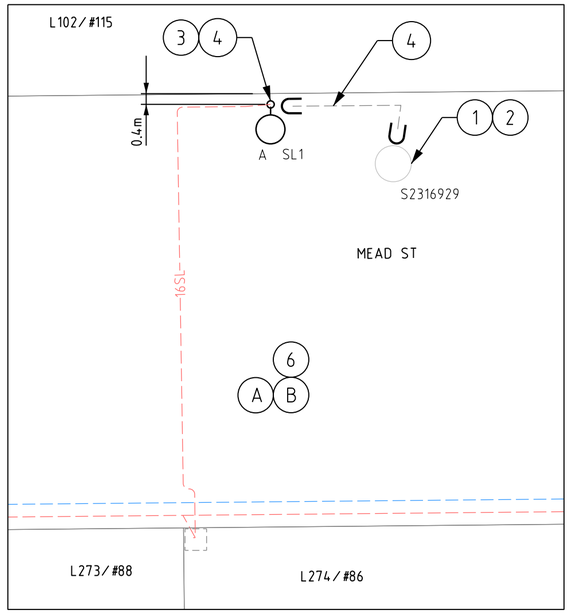
DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 10/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 02' 25" S
Long. - 115° 56' 58" E

westernpower
DRAWING NUMBER
[DQM ID]
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING SL CABLE FROM S2316929.
   CU: CN73_LV10/16 x 1
2. REMOVE EXISTING STREETLIGHT.
3. INSTALL NEW 12.5m STREETLIGHT WITH 165W 3000K LED STREET LIGHT AS PER POLE SCHEDULE TABLE.
   CU: SL8_1E x 1
   DSCH: S08
4. CUT EXISTING 16SL CABLE AND TERMINATE TO NEW STREETLIGHT(SL1). CAP AND ABANDON EXCESS CABLE.
   CU: CN73_10/16x 1
5. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
   IN AREAS WHERE THIS IS A RISK YOU NEED TO:
   BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
   IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

   POTENTAIL WET AREA
   THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

   THREATENED ECOLOGICAL COMMUNITIES
   NATIVE VEGETATION REGS ESAS CLEARING
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- POLE BRACKET — POLE BRACKET
- SINGLE OUTREACH — SINGLE OUTREACH
- DOUBLE OUTREACH — DOUBLE OUTREACH
- 4-WAY — 4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE.
REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

STREET LIGHT SCHEDULE

TYPE	POLE ID	DISCRIPTION	POLE LOCATION		COLOR	CU'S
			EASTING	NORTHING		
A	SL1	NEW 6.5m STANDARD STEEL POLE	405478.605	6434235.831	N/A	SL8_1E

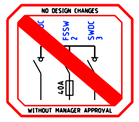
CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE POSITION OF NEW STREETLIGHT IN ACCORDANCE WITH AS1158 AND SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD ADJUSTMENT AUTHORITY AT NO COST TO WESTERN POWER.

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: SHIRE OF SERPENTINE JARRAHDALE
CONTACT: TANJINA RAHMAN TEL: 08 95261581

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	01/25

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 70879496
HAZARD MANAGEMENT REGISTER EDM#: 70879057

INSTALLATION/REMOVAL OF STREET LIGHT

NETWORK VOLTAGE: 22kV/415V

#86 MEAD ST, BYFORD WA 6122

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 01/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 13' 28" S Long. - 115° 59' 48" E

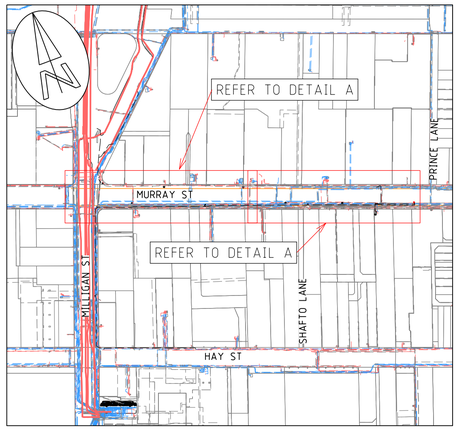
westernpower

DRAWING NUMBER

MP210873

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



SITE PLAN 1

1:2500 50 25 0 50 100 150 200



LOCATION PLAN
NTS

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
CN53 - 400mm2 (3X1c)
CN49 - 240mm2 (3X1c)
CN48 - 185mm2 (3X1c)
CN50 - 95mm2 (3X1c)
CN51 - 50mm2 (3X1c)
CN51 - 50mm2 (1c)
CN46 - 35mm2 (3X1c)
CN47 - 35mm2 (1c)
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

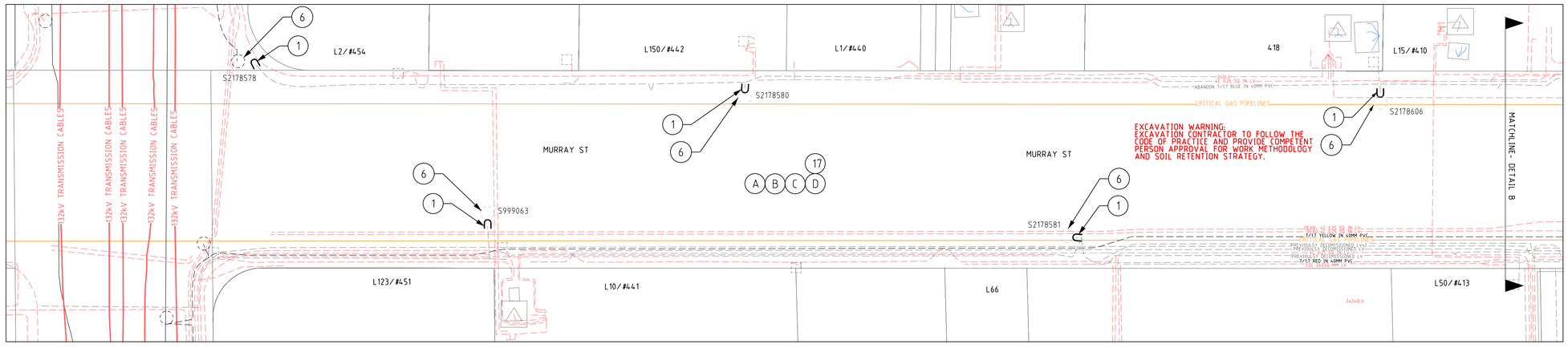
LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
42W CFL · 70W HPS · 70W MH · 1L LED · 1E LED
80W MV · 150W HPS · 150W MH · 2L LED · 2E LED
125W MV · 250W HPS · 250W MH · 3L LED · 3E LED
250W HPS AERODROME · 4L LED · 4E LED
5L LED · 5E LED
6L LED · 6E LED
LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
MINI (WORKING END) · UMS PIT
UNIVERSAL · UTILITIES PIT
MINI · BGSP PIT
LINK · 100/200AMP CUTOUT
CABLE MARKER U/G · CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

TRANSFORMER · EXISTING · SWITCHGEAR

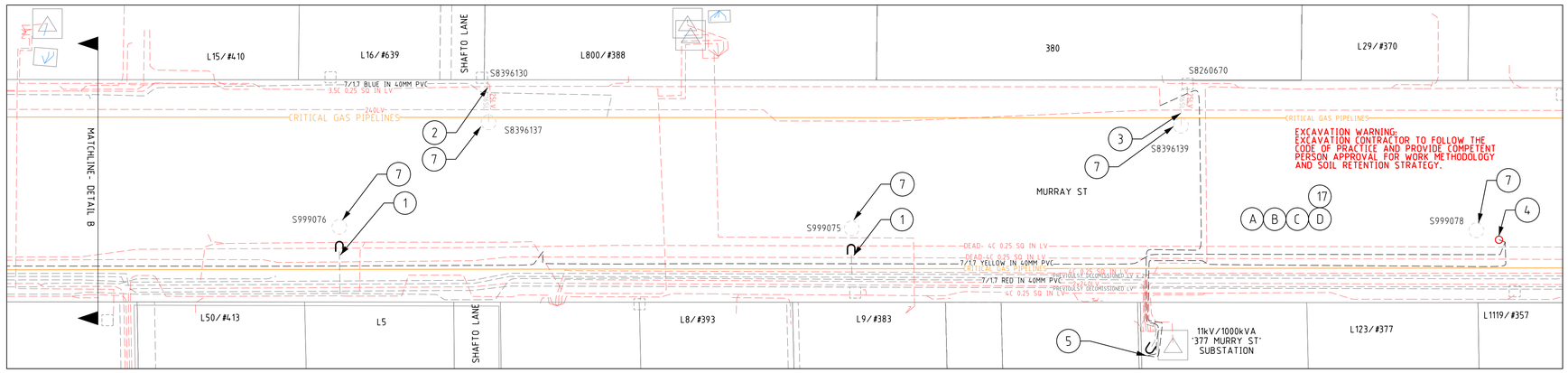
AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV · LV · EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



DETAIL A

1:250 5 2.5 0 5 10 15 20

HV CABLES IN VICINITY OF WORKS NOT SHOWN FOR CLARITY



DETAIL B

1:250 5 2.5 0 5 10 15 20

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

APPROVED FOR CONSTRUCTION

CRITICAL GAS PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF PERTH
CONTACT: ADNAN ABDUL BASET TEL: 0456 938 192

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT EDM#: 70882926
A	UNDER ENGINEERING REVIEW	I.P.	03/25	HAZARD MANAGEMENT REGISTER - EDM# 70882115

STREETLIGHT REMOVAL

NETWORK VOLTAGES: 11kV/415V

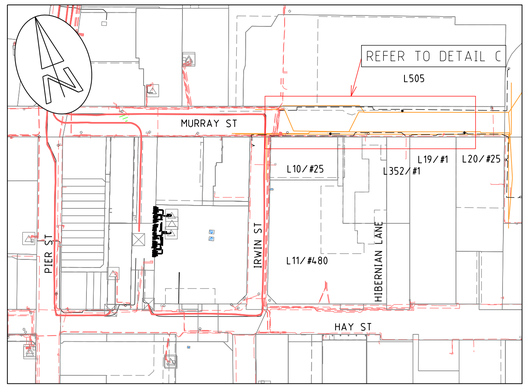
MURRAY ST, PERTH WA 6000
DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 03/25	SCALE: AS NOTED @ A1
GEO REF:	Lat. - 31° 57' 00" S Long. - 115° 51' 01" E

westernpower

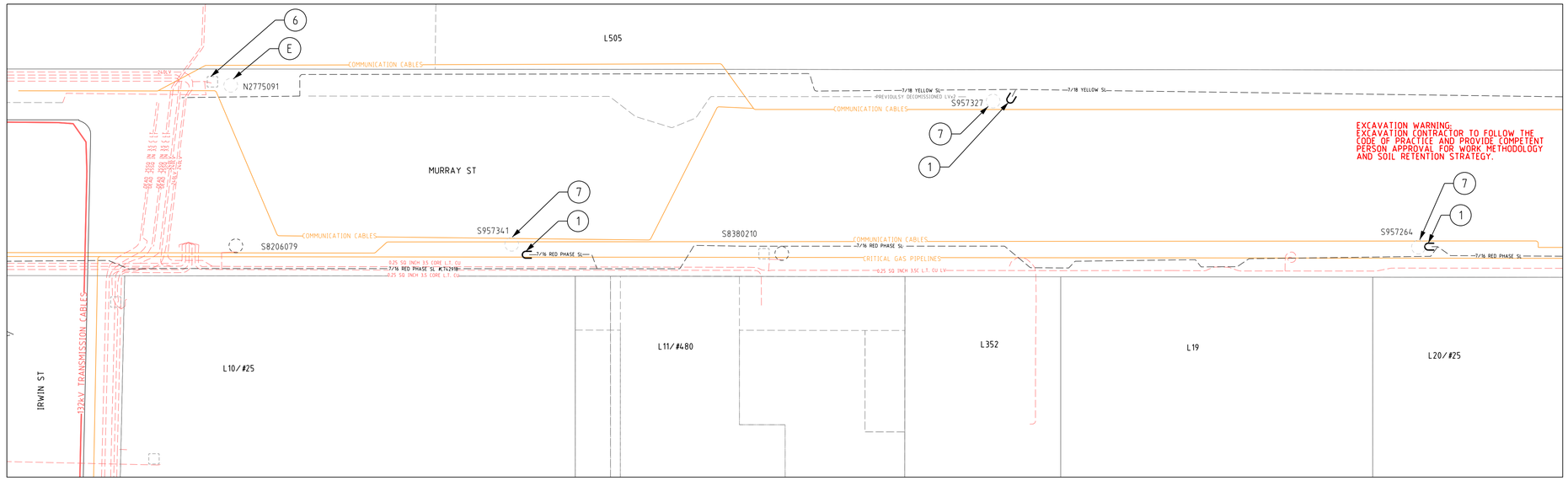
DRAWING NUMBER
MP210935
SHEET 01 OF 02

REV: A ORIGINAL SIZE - A1



SITE PLAN 2

1:2000 — 40 20 0 40 80 120 160



DETAIL C

1:250 — 5 2.5 0 5 10 15 20

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING SL CABLE FROM EXISTING STREETLIGHT. PULL BACK TO CABLE ALIGNMENT. CUT, CAP AND ABANDON EXISTING SL CABLE. CU: CN73/10/16 x 10
2. DISCONNECT EXISTING 16SL CABLE FROM POLE#S8396137. REMOVE SL CABLE TOWARDS U/G PIT#S8396130.
3. DISCONNECT EXISTING 16SL CABLE FROM POLE#S8396139 REMOVE EXISTING 16SL CABLE TOWARDS U/G PIT#S8260670
4. CUT AND LIVE END SEAL EXISTING SL CABLE#C2553058. REMOVE EXCESS. CU: LU62x1
5. AT STREET LIGHT CONTROL BOX IN '377 MURRAY ST' SUBSTATION, DISCONNECT EXISTING 7/1.7 RED SL CABLE#C809721. PULL BACK TO KERB; CUT, CAP AND ABANDON SL CABLE#C809721. CU: CN73/10/16 x 1
6. DISCONNECT EXISTING SL CABLE FROM UMS PIT#N2775091
7. REMOVE EXISTING STREET LIGHTS.
   S2178580
   S2178606
   S8396137
   S8396139
   S999063
   S2178581
   S999076
   S999075
   S999078
   S957327
   S957341
   S957264
8. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   HERITAGE COUNCIL OF WA
   MUNICIPAL INVENTORY
   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 41507941)

   CONTAMINATED SITES:
   THE RISK OF ENCOUNTERING CONTAMINATED SOIL IS LOW DUE TO THE DEPTH OF EXCAVATION FOR THE REMOVAL OF STREETLIGHTS AND CABLES. THE CONTRACTOR UNDERTAKING THE WORKS SHOULD FOLLOW THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM# 41050794)

TRANSMISSION LINE IN THE VICINITY OF WORKS

1. ALL MOBILE PLANT IN PROXIMITY TO THE LINE SHALL OPERATE AT A MINIMUM APPROACH DISTANCE THAT COMPLIES WITH WESTERN POWER WORK PRACTICES AND REQUIREMENTS.
2. WORKS MUST COMPLY WITH WA WORK HEALTH AND SAFETY (GENERAL) REGULATIONS - REGULATION 166A FOR A DUTY OF PERSON WORKING IN VICINITY OF OVERHEAD ELECTRIC LINES.
3. ALL WORKS SHALL COMPLY WITH WESTERN POWER WORK INSTRUCTIONS AND PROCEDURES.
4. CAUTION: INDUCED VOLTAGES IN ISOLATED CONDUCTORS AND APPARATUS DUE TO ELCTRO-MAGNETIC COUPLING FROM ADJACENT ENERGISED TRANSMISSION LINE, INDUCED VOLTAGES MAY BE SIGNIFICANTLY INCREASED DURING FAULT CONDITIONS. ALL WORKS MUST COMPLY WITH WORK INSTRUCTION:
   "INDUCED VOLTAGE IN ISOLATED CONDUCTORS/APPARATUS"

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO REINSTATE GROUND SURFACE AFTER COMPLETION OF WESTERN POWER SCOPE OF WORK.

C. ANY TEMPORARY STREET LIGHTING REQUIRED DURING THE WORKS AFTER REMOVAL OF EXISTING STREET LIGHTS UNTIL NEW STREET LIGHTS HAVE BEEN ENERGISED IS TO BE SUPPLIED AN INSTALLED BY THE CUSTOMER AT NO COST TO WESTERN POWER.

D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT REMOVAL OF STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

E. CUSTOMER IS RESPONSIBLE FOR REMOVAL OF SL#N2775091 AFTER COMPLETION OF WESTERN POWER SCOPE OF WORKS.





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

HV CABLES IN VICINITY OF WORKS NOT SHOWN FOR CLARITY

CRITICAL GAS PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF PERTH
CONTACT: ADNAN ABDUL BASET TEL: 0456 938 192

APPROVED FOR CONSTRUCTION

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

TRANSFORMER EXISTING SWITCHGEAR

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE FILTER ASSESSMENT- SITE 2 EDM#: 71494740
A	UNDER ENGINEERING REVIEW	I.P.	03/25	HAZARD MANAGEMENT REGISTER - EDM# 70882115

STREETLIGHT REMOVAL

NETWORK VOLTAGES: 11kV/415V

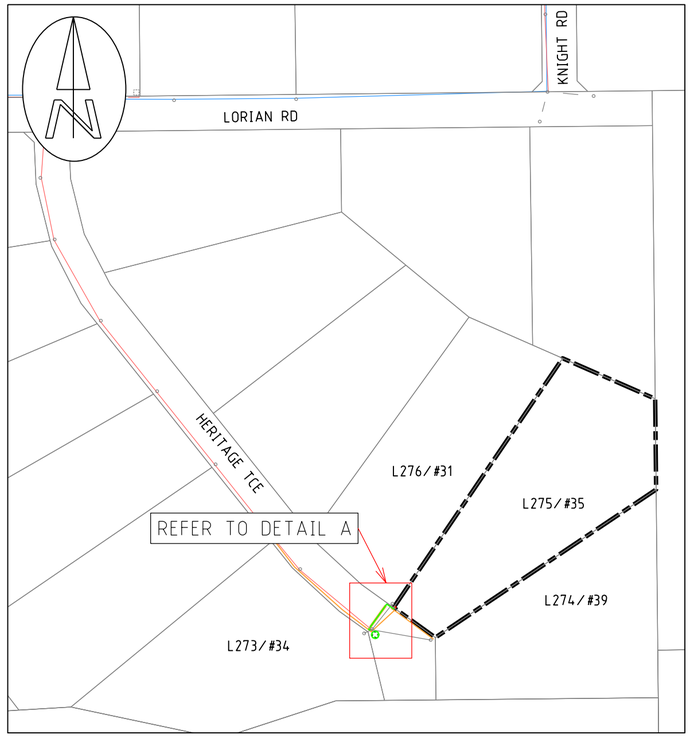
MURRAY ST, PERTH WA 6000
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 03/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 57' 00" S Long. - 115° 51' 01" E

westernpower

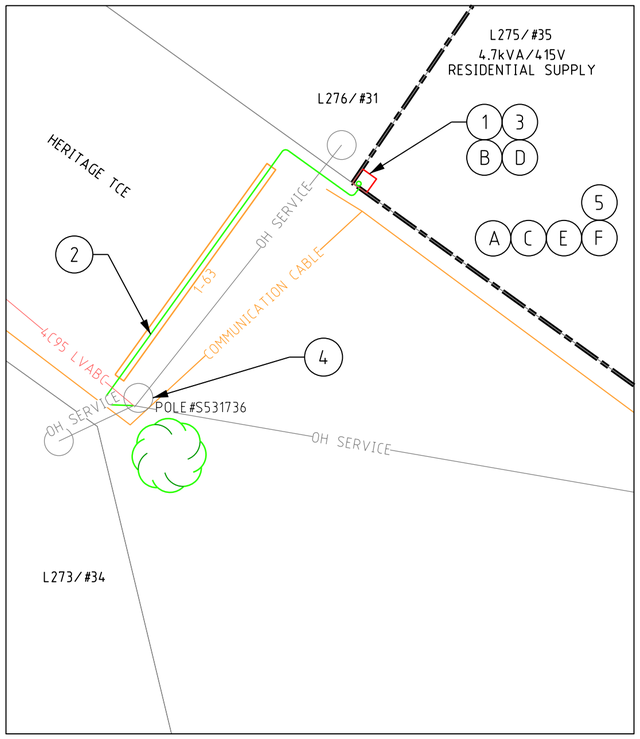
DRAWING NUMBER
MP210935
SHEET 02 OF 02

REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:250 — 5 2.5 0 5 10 15 20

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

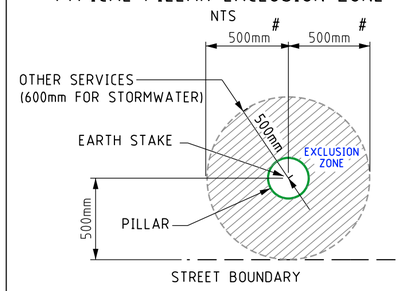
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L275/#35 HERITAGE TCE AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 1
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S531736 VIA 63mm POLYPIPE.
   CU: CN42 x 35m, CN73_LV25 x 3, CN54 x 2
3. AT MINI-PILLAR (L275/#35 HERITAGE TCE)
   - TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
4. AT POLE#S41040
   - TERMINATE NEW 25LV CABLE TO LVABC CONDUCTORS.
   CU: LU50_25/95O x 1, LU50_25/95U x 1
   DCSH: U18
5. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ACID SULPHATE SOILS (ASS)
   EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
   IN AREAS WHERE THIS IS A RISK YOU NEED TO:
   - BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
   - IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

   PUBLIC DRINKING WATER SOURCE AREA
   THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

   DECLARED RARE AND PRIOTIY FLORA
   THREATENED ECOLOGICAL COMMUNITIES
   NATIVE VEGETATION REMAINING
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER MUST INSTALL SMSB WITHIN 30m OF POINT OF SUPPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

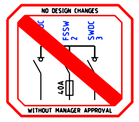
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV / LV — EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: COFFEY RHYS TEL: 0407 200 456

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69930361
HAZARD MANAGEMENT REGISTER EDM#: 69948098

4.7kVA STANDARD RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

35 HERITAGE TERRACE, GNANGARA WA 6077

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 10/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 46' 45" S Long. - 115° 52' 20" E

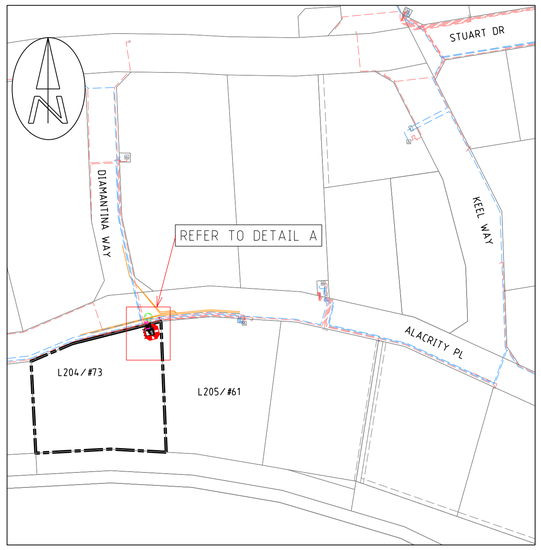
westernpower

DRAWING NUMBER

MP211295

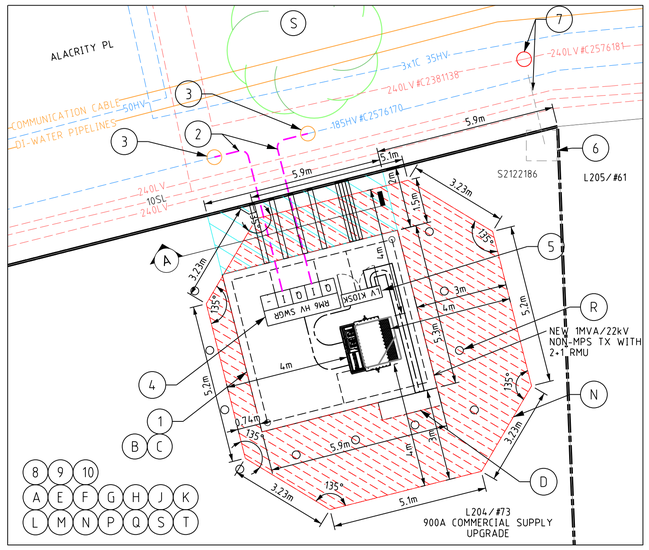
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000

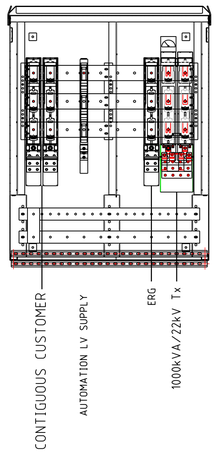


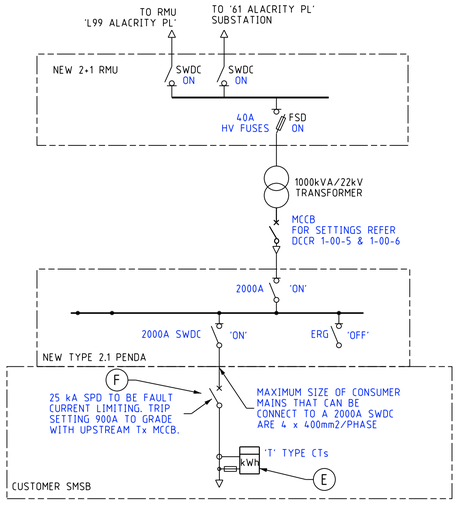
DETAIL A

1:100

EXCAVATION WARNING:
EXCAVATION AND SHORING TO BE CONDUCTED IN CONJUNCTION WITH CIVIL ENGINEERING ADVICE CONTAINED IN EDM VOLT

TYPE 2.1 PENDA ARRANGEMENT





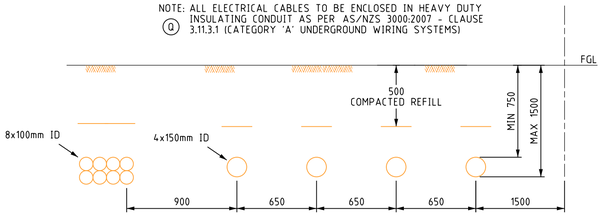
TYPE 2.1 PENDA CIRCUIT TABLE

	LEFT	NAME/LOCATION
Circuit 1 (C1)	LU66_CUST	CONTIGUOUS CUSTOMER
Circuit 2 (C2)		
Circuit 3 (C3)	LU70	FUTURE AUTOMATION SUPPLY
Circuit 4 (C4)		
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)	LU68	ERG
Circuit 8 (C8)	LU66	TO 1000kVA/22kV TRANSFORMER
Circuit 9 (C9)		
Circuit 10 (C10)		

RIGHT

ISSUED FOR CONSTRUCTION

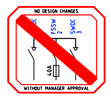
SIMPLIFIED SINGLE LINE DIAGRAM



A CUSTOMER INSTALLED CONDUITS
SECTION DETAIL (NTS)

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☒	☐	☐	☒	☐



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CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING, COMPACTION AND LEVELLING.
2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

(C) CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

(D) CUSTOMER MUST INSTALL MSB CONTIGUOUS TO THE SUBSTATION.

(E) CUSTOMER TO SUPPLY AND INSTALL 'T' TYPE METERING CT's.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(G) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(H) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(J) CUSTOMER TO CONTACT WESTERN POWER PROPERTY SPECIALIST AT conveyancing@westernpower.com.au TO ARRANGE THE PREPARATION AND REGISTRATION OF THE REQUIRED 129BA RESTRICTIVE COVENANT.

RESTRICTIVE COVENANT TO BE REGISTERED AT TIME OF SUBDIVISION (IF APPLICABLE) OR PRIOR TO COMMENCEMENT OF WORKS OR ENERGISATION. (Delete which ever does not apply)

(K) IF THE CUSTOMER PROPOSES TO INSTALL SCREENING, APPROVAL SHALL BE OBTAINED FROM WESTERN POWER PRIOR TO ITS INSTALLATION. THE SCREENING SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE. REFER TO THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSES 14.4.10.3.2, 14.4.11 AND 14.5.4 FOR DETAILS.

(L) CUSTOMER TO ARRANGE FOR AN 'EASEMENT IN GROSS' UNDER THE ENERGY OPERATORS POWERS ACT 1979. CUSTOMER TO SUBMIT THE DEPOSITED PLAN TO WESTERN POWER PRIOR TO THE COMMENCEMENT OF WORKS.
CUSTOMER TO CONTACT WESTERN POWER PROPERTY SPECIALIST AT conveyancing@westernpower.com.au TO ARRANGE THE PREPARATION AND REGISTRATION.

(M) WESTERN POWER WILL REQUIRE OUTGOING RETAILER AGREEMENT (VIA SUBMISSION OF AN ETAF) PRIOR TO SCHEDULING OF THE PROJECT WORKS. CUSTOMER REQUESTED TO CONTACT THE OUTGOING RETAILER AND PROVIDE RELEVANT PROJECT DETAILS TO FACILITATE ETAF.
CUSTOMER TO CONTACT WP PNO PRIOR TO COMMENCEMENT OF METERING CHANGES.

(N) NO BUILDINGS ALLOWED WITHIN 4m OF THE SUBSTATION UNLESS 2 HOUR FIRE RATED. SEE WESTERN POWER DRAWING DSM-5-01 FOR DETAILS.

(P) SUBSTATION TO BE DESIGNED TO MINIMUM DIMENSIONS SHOWN ON THE DRAWING AND IN-CONJUNCTION WITH WESTERN POWER'S DRAWING DSM-3-04

(Q) CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

(R) CUSTOMER TO INSTALL NON-CONDUCTIVE BOLLARDS ADJACENT TO THE SUBSTATION SITE AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6.

(S) THE CUSTOMER IS RESPONSIBLE FOR OBTAINING ALL NECESSARY CLEARING AND THIRD PARTY APPROVALS AND,
- WRITTEN DETAILS OF A RELEVANT CLEARING PERMIT EXEMPTION AND SUPPORTING DOCUMENTS
NOT REQUIRED BY THE CUSTOMER; OR
- A REFERRAL DETERMINATION NOTICE ISSUED BY DWER CONFIRMING A CLEARING PERMIT IS
- A CLEARING PERMIT ISSUED IN THE CUSTOMER'S NAME; OR
COPY OF EITHER;
PRIOR TO WESTERN POWER COMMENCING WORKS, THE CUSTOMER IS REQUIRED TO PROVIDE A (REGULATOR CORRESPONDENCE/LANDOWNER EXEMPTION CLEARING LETTER EDM 46147597).

(T) IF REQUIRED, CUSTOMER TO PROVIDE WESTERN POWER WITH CIVIL DETAILS REGARDING THE SHORING OF THE EXISTING RETAINING WALL SO THE EXISTING WALL CAN BE ADEQUATELY SUPPORTED PRIOR TO THE COMMENCEMENT OF CONSTRUCTION WORKS.

WESTERN POWER SCOPE OF WORK

(1) INSTALL 1000kVA 22kV SUBSTATION WITH 2+1 MOTORISED SWITCHGEAR AND A TYPE 2.1 PENDA TO DSPM-3-04 AS NOTED ON DRAWING.
INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER.
INSTALL OIL CONTAINMENT PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5 AND DRAWING DSPM-4-06 (NON-MPS).
CU: HU76x1, HU59_10002 x1, HU66_22Sx1, HU6_A400x1, HU55_1000/22x1, CN46x15m, CN60_400CU x80m, LU16_400CU16 x 1, LU16_400CU12x1, LU42x1, LU64_0x1, LU66_CUSTx1, LU68x1, LU66_TXx1, LU70x1
DSPM: DSPM-3-04, DSPM-4-06

(2) INSTALL 2x400 HV CABLES FROM NEW HV SWITCHGEAR TO EXISTING 185HV CABLE IN ROAD RESERVE VIA CUSTOMER INSTALLED CONDUITS.
CU: CN53x40m, CN54x2, CN73_240x5

(3) CUT EXISTING 185HV CABLE#C2576170 AND STRAIGHT JOINT TO NEW 400HV CABLES.
CU: HU2_22_400x2, CN54x2

(4) TERMINATE 2x400HV CABLES AND 35HV TO NEW 2+1 SWITCHGEAR AS PER SINGLE LINE DIAGRAM.
DSPM: 4-04

(5) AT NEW PENDA, TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT & CIRCUIT TABLE.
DSPM: 4-05

(6) DISCONNECT EXISTING CUSTOMER MAINS, 240LV CABLES AND REMOVE EXISTING UNIPILLAR#S2122186

(7) SWING EXISTING 240 LV CABLE#C2576181 AND STRAIGHT JOINT TO EXISTING 240LV CABLE#C2381138. REMOVE EXCESS CABLE.
CU: LU2 x1, CN54 x2

(8) INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
CU: LM1_M x 1

(9) WP SCHEDULER TO CONFIRM RECEIPT OF ETAF PRIOR TO SCHEDULING WORKS.
WP PNO TO OBTAIN FINAL METER READ DATE FROM WP METERING.

(10) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

THREATENED ECOLOGICAL COMMUNITIES
HYDROGRAPHIC CATCHMENTS - SUBCATCHMENTS (DWER-030)
VEGETATION COMPLEXES - SWAN COASTAL PLAIN (DBCA-046)

REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT
OPERATIONAL CLEARANCE ZONE

HV CABLES
CN53 - 400mm2 (3X1c)
CN49 - 240mm2 (3X1c)
CN48 - 185mm2 (3X1c)
CN50 - 95mm2 (3X1c)
CN51 - 50mm2 (3X1c)
CN51 - 50mm2 (1c)
CN46 - 35mm2 (3X1c)
CN47 - 35mm2 (1c)
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (3x3c)
CN44 - 185mm2 (3x3c)
CN43 - 120mm2 (3x3c)
CN42 - 25mm2 (3x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS	
COMPANY NAME: ELECTRICAL CONSULTANCY WA	
CONTACT: MICHAEL ANILE	TEL: 0439 988 739

CUSTOMER CONTACT DETAILS	
COMPANY NAME: Kerr Engineering (wa) Pty Ltd	
CONTACT: MARK KINSMAN	TEL: 0439988739

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				ENVIRONMENTAL ASSESSMENT
A	ISSUED FOR CONSTRUCTION	I.P.	08/25	HAZARD MANAGEMENT REGISTER

900A COMMERCIAL SUPPLY UPGRADE

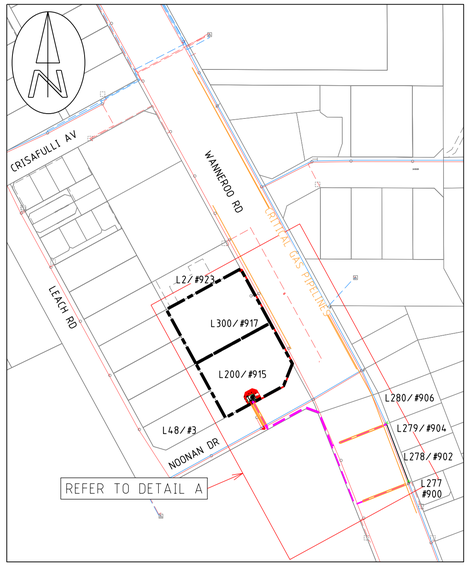
NETWORK VOLTAGES: 22kV/415V

L204/#73 ALACRITY PL, HENDERSON WA 6166
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN	
DESIGN TEAM: CCS COMM. SVM- RASP	
EMAIL: ccs.design.resi.solution@westernpower.com.au	
TEL: 13 10 87	
DATE: 08/25	SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 09' 56" S Long. - 115° 46' 30" E	

westernpower

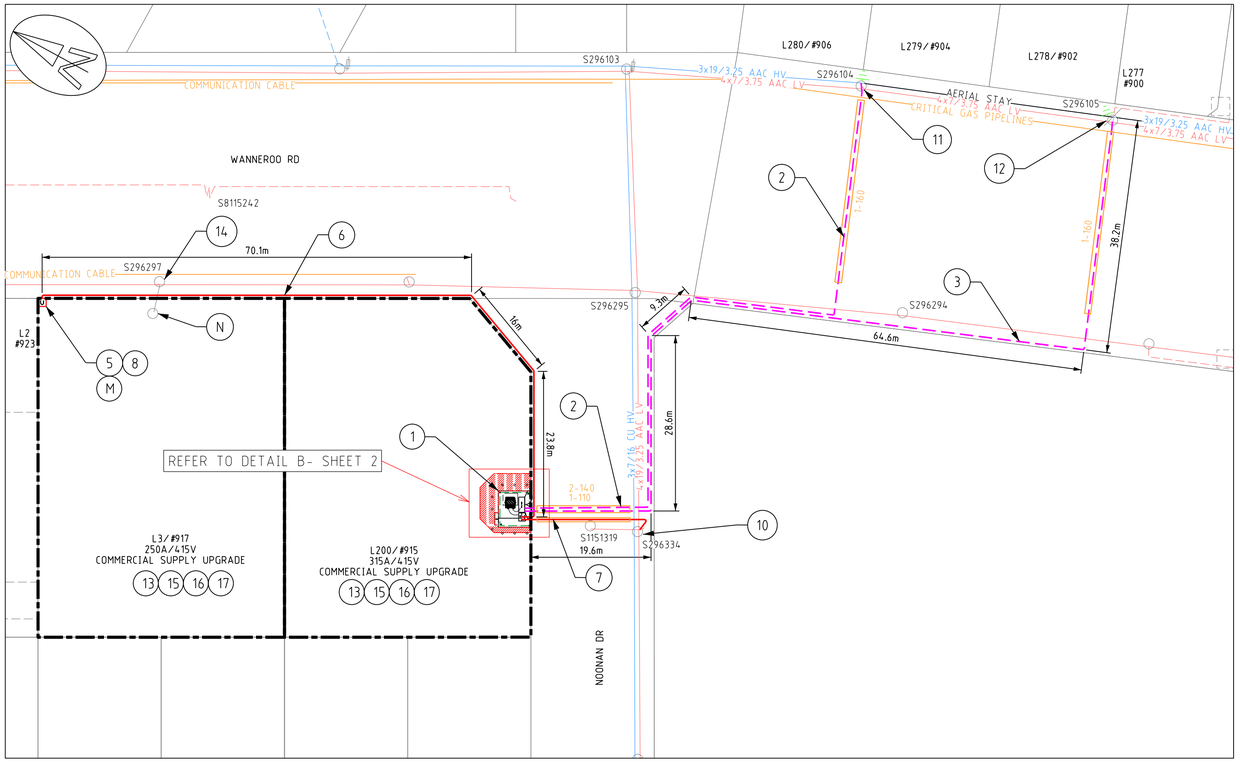
DRAWING NUMBER
MP212323
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

40 20 0 40 80 120 160

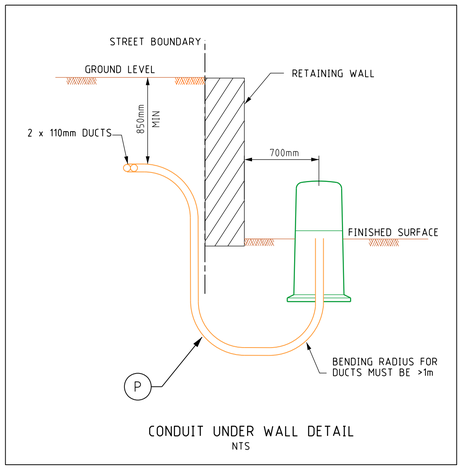
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DETAIL A

10 5 0 10 20 30 40

1:500



CONDUIT UNDER WALL DETAIL

NTS

WESTERN POWER SCOPE OF WORK

1. INSTALL 630kVA 22kV SUBSTATION WITH 2+1 RMU AND A TYPE 2.1 PENDA TO DSPM-3-04 AS NOTED ON DRAWING.
   INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER.
   INSTALL OIL CONTAINMENT PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5 AND DRAWING DSPM-4-06 (NON-MPS).
   CU: HU76x1, HU59_630/22 x1, HU66_22x1, HU6_A400x1, HU55_630/22x1, CN46x15m, CN60_400CU x80m, LU16_400CU16 x 1, LU16_400CU12x1, LU42_Mx1, LU64_0x1, LU67x1, LU68x1, LU69x2, LU70x1, LU14x2
   DSPM: DSPM-3-04, DSPM-4-06
2. INSTALL 1x400 HV CABLES FROM NEW HV SWITCHGEAR TO EXISTING POLE#S296104.
   CU: CN53x135m, CN54x2, CN73_240x3
3. INSTALL 1x400 HV CABLES FROM NEW HV SWITCHGEAR TO EXISTING POLE#S296105.
   CU: CN53x175m, CN54x2, CN73_240x3
4. TERMINATE 2x400HV CABLES AND 35HV TO NEW 2+1 SWITCHGEAR AS PER SINGLE LINE DIAGRAM. REFER SHEET 2.
   DSPM: 4-04
5. INSTALL NEW UNI-PILLAR ON L3/#917 WANNEROO RD WHERE PEGGED BY CUSTOMER AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
6. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO NEW LV PENDA AT L200/#915 WANNEROO RD VIA CUSTOMER INSTALLED DUCT.
   CU: CN45 x 125m, CN54 x 3, CN73_LVMAINS x 3
7. INSTALL 240LV CABLE FROM NEW NEW LV PENDA AT L200/#915 WANNEROO RD TO POLE#S296334.
   CU: CN45 x 35m, CN54 x 3, CN73_LVMAINS x 3
8. AT UNI-PILLAR (L300/#917 WANNEROO RD)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 400A LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_LINKx1
9. AT NEW PENDA, TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT & CIRCUIT TABLE. REFER TO SHEET 2.
   NON CU: GF1096x6
   DSPM: 4-05
10. AT POLE#S296334, TERMINATE NEW 240LV CABLE TO LV BARE.
    CU: LU6_240Ox1, LU6_240Ux1, CN9_19/325AAx4
    DCSH: U16-1
11. AT POLE#S296104,
    - REPLACE EXISTING HV INTERMEDIATE WITH HV TERMINATION CROSSARM AND REMOVE HV CONDUCTOR TOWARD POLE S296105.
    - TERMINATE NEW 400HV CABLE TO HV BARE TOWARDS NORTH WEST
    - INSTALL NEW DOWN EARTH.
    - INSTALL AERIAL STAY TOWARDS POLE S296105.
    CU: HV5_3MNx1, HU16_U_22x1, HU16_0x1, CN1_19/325x3, ST2_19/200Wx1, CN9_T19/3.25x3
    NON CU: CE0058x1, CE0135x6, CE0078x6, CE0201x1, EE1264x15m
    DCSH: H09-1, R14-3
12. AT POLE#S296105,
    - REMOVE EXISTING HV POLE TOP SWITCH.
    - INSTALL HV TERMINATION CROSSARM.
    - TERMINATE NEW 400HV CABLE TO HV BARE TOWARDS SOUTH EAST.
    - INSTALL NEW DOWN EARTH
    - ATTACH AERIAL STAY FROM POLE S296104.
    CU: HV5_3MNx1, HU16_U_22x1, HU16_0x1, CN1_19/325x3, CN9_T19/3.25x3
    NON CU: CE0058x1, CE0135x6, CE0078x6, CE0201x1, EE1264x15m
    DCSH: H09-1, R14-3
13. INSTALL CT METER IN CUSTOMERS SWITCHBOARD.
    CU: LM1x2
14. AT POLE#S296297, REMOVE EXISTING AERIAL SERVICE.
15. WP SCHEDULER TO CONFIRM RECEIPT OF ETAF PRIOR TO SCHEDULING WORKS.
    WP PNO TO OBTAIN FINAL METER READ DATE FROM WP METERING.
16. POSSESSION OF SITE CERTIFICATE IS TO BE REQUESTED. AND MAIN ROADS DOCUMENT D18#604587 'SEEKING APPROVAL TO UNDERTAKE WORKS WITHIN THE ROAD RESERVE' TO BE COMPLETED.

    CM TO COMPLY WITH SIGNED CONDITIONAL AGREEMENT-REFER VOLT
17. ENVIRONMENTAL FILTER CHECKED.
    NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

    THREATENED ECOLOGICAL COMMUNITIES
    MUNICIPAL INVENTORY
    PUBLIC DRINKING WATER SOURCE AREAS
    BLACK COCKATOO ROOSTING SITES - BUFFERED (DBCA-064)
    LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
    REGION SCHEME - ZONES AND RESERVES (DPLH-023)
    INDIGENOUS LAND USE AGREEMENTS (ILUAS)
    NATIVE TITLE BOUNDARIES
    PUBLIC DRINKING WATER SOURCE AREAS (DWER-033)
    RIWI ACT, GROUNDWATER AREAS (DWER-034)
    PRE-EUROPEAN VEGETATION (DPIRD-006)

    REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- S.d. Serviced Unclearad De-energised Lot

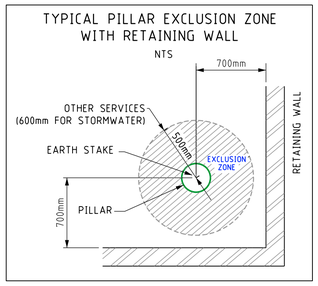
AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

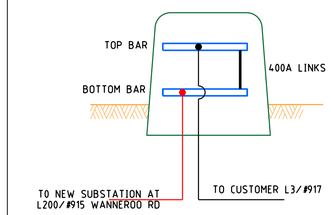
REFER TO SHEET 2

TYPICAL PILLAR EXCLUSION ZONE WITH RETAINING WALL

NTS



UNI-PILLAR
L3/#917 WANNEROO RD
CONNECTION DETAILS



CRITICAL GAS PIPELINES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

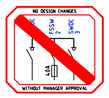
CUSTOMER CONTACT DETAILS
COMPANY NAME: Bayerstevens Pty Ltd
CONTACT: MICHAEL STEVENS TEL: 0439988739

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	☐	☒	☐

SAFETY - MAKE IT YOUR PRIMARY VALUE



NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
A	ISSUED FOR CONSTRUCTION	I.P.	03/26	ENVIRONMENTAL ASSESSMENT
				HAZARD MANAGEMENT REGISTER

315A & 250A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGES: 22kV/415V

L3/#917 & L200/#915 WANNEROO ROAD, WANNEROO WA 6065

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 03/26 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 45' 21" S Long. - 115° 48' 18" E

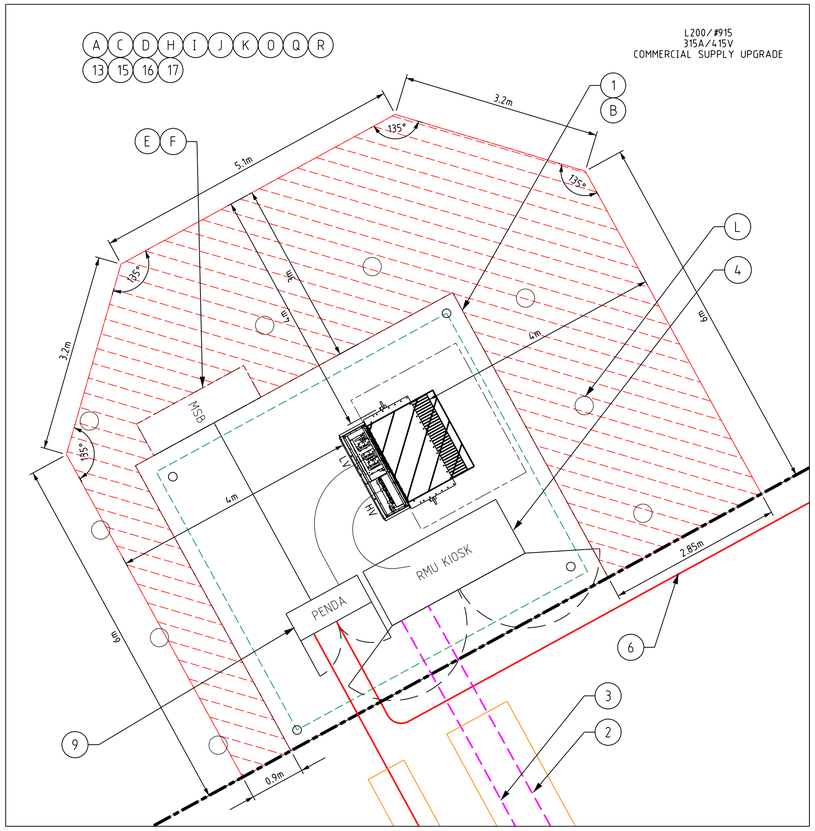
westernpower

DRAWING NUMBER

MP214838

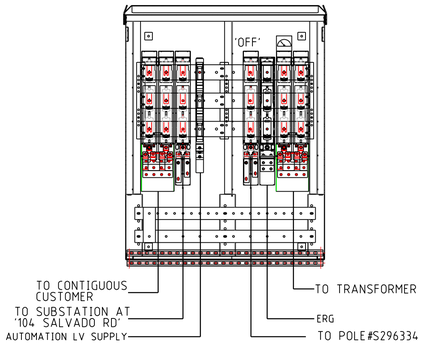
SHEET 01 OF 02

REV: A ORIGINAL SIZE - A1



DETAIL B

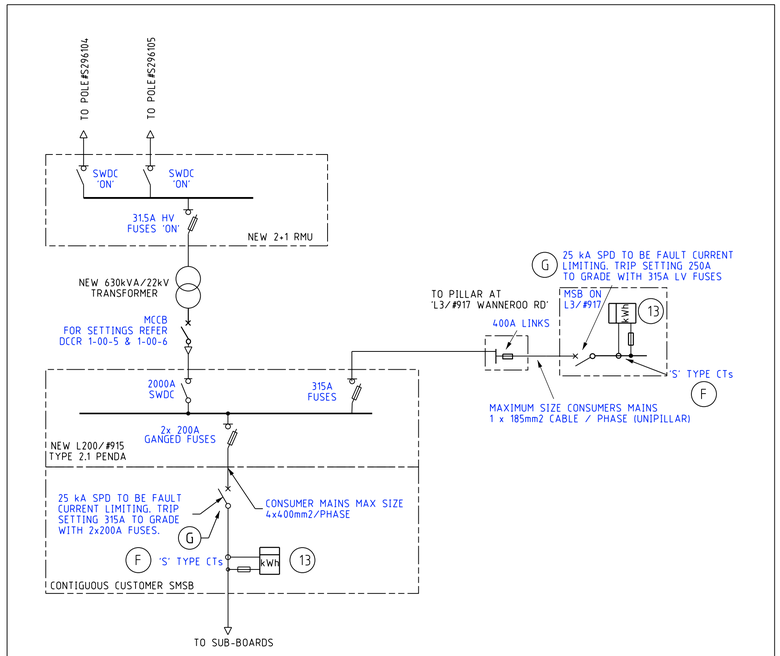
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TYPE 2.1 PENDA LAYOUT

	LEFT	NAME/LOCATION
Circuit 1 (C1)	LU67	CONTIGUOUS CUSTOMER
Circuit 2 (C2)	LU69	TO PILLAR AT L3/#917 WANNEROO RD
Circuit 3 (C3)	LU70	AUTOMATION SUPPLY
Circuit 4 (C4)		
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)	LU69 'OFF'	TO POLE#S296334
Circuit 8 (C8)	LU68	ERG
Circuit 9 (C9)	LU66	TO 630kVA/22kV TRANSFORMER
Circuit 10 (C10)	RIGHT	

TYPE 2.1 PENDA CIRCUIT TABLE



SIMPLIFIED SINGLE LINE DIAGRAM

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) SUBSTATION TO BE DESIGNED TO MINIMUM DIMENSIONS SHOWN ON THE DRAWING AND IN-CONJUNCTION WITH WESTERN POWER'S DRAWING DSM-03-04.

(C) SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING, COMPACTION AND LEVELLING. A COMPACTION CERTIFICATE IS REQUIRED.

2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.

4. CLEAN SAND FILL TO A DEPTH OF 1m. EXCAVATIONS GREATER THAN 1.5m TO BE IN ACCORDANCE WITH THE EXCAVATION CODE OF PRACTICE.

(D) CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

(E) CUSTOMER MUST INSTALL MSB CONTIGUOUS WITH SUBSTATION.

(F) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(G) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(H) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(I) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(J) IF THE CUSTOMER PROPOSES TO INSTALL SCREENING, APPROVAL SHALL BE OBTAINED FROM WESTERN POWER PRIOR TO ITS INSTALLATION. THE SCREENING SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE. REFER TO THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSES 14.4.10.3.2, 14.4.11 AND 14.5.4 FOR DETAILS.

(K) WESTERN POWER WILL REQUIRE OUTGOING RETAILER AGREEMENT (VIA SUBMISSION OF AN ETAF) PRIOR TO SCHEDULING OF THE PROJECT WORKS. CUSTOMER REQUESTED TO CONTACT THE OUTGOING RETAILER AND PROVIDE RELEVANT PROJECT DETAILS TO FACILITATE ETAF. CUSTOMER TO CONTACT WP PNO PRIOR TO COMMENCEMENT OF METERING CHANGES.

(L) CUSTOMER TO INSTALL NON-CONDUCTIVE BOLLARDS ADJACENT TO THE SUBSTATION SITE AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6.

(M) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(N) CUSTOMER RESPONSIBLE TO REMOVE CUSTOMERS POLE AFTER COMPLETION OF WORK.

(O) NO BUILDINGS ALLOWED WITHIN 3m OF THE SUBSTATION UNLESS 2 HOUR FIRE RATED. SEE WESTERN POWER DRAWING DSM-5-01 FOR DETAILS.

(P) CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

(Q) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

(R) CUSTOMER IS RESPONSIBLE FOR COMPLETING THEIR DUE DILIGENCE ASSESSMENT OF POTENTIAL ABORIGINAL HERITAGE IMPACTS AND ADVISING WESTERN POWER OF ANY COMPLIANCE REQUIREMENTS THAT APPLY.

WESTERN POWER SCOPE OF WORK

REFER TO SHEET 1

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
CN53 - 400mm2 (3X1c)
CN49 - 240mm2 (3X1c)
CN48 - 185mm2 (3X1c)
CN50 - 95mm2 (3X1c)
CN51 - 50mm2 (3X1c)
CN51 - 50mm2 (1c)
CN46 - 35mm2 (3X1c)
CN47 - 35mm2 (1c)
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UMS PIT
UNIVERSAL
UTILITIES PIT
MINI
BGSP PIT
LINK
100/200AMP CUTOUT
CABLE MARKER U/G
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CRITICAL RISK MANAGEMENT

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: Bayerstevens Pty Ltd
CONTACT: MICHAEL STEVENS TEL: 0439988739

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	11/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
REFERENCE FILE DETAILS

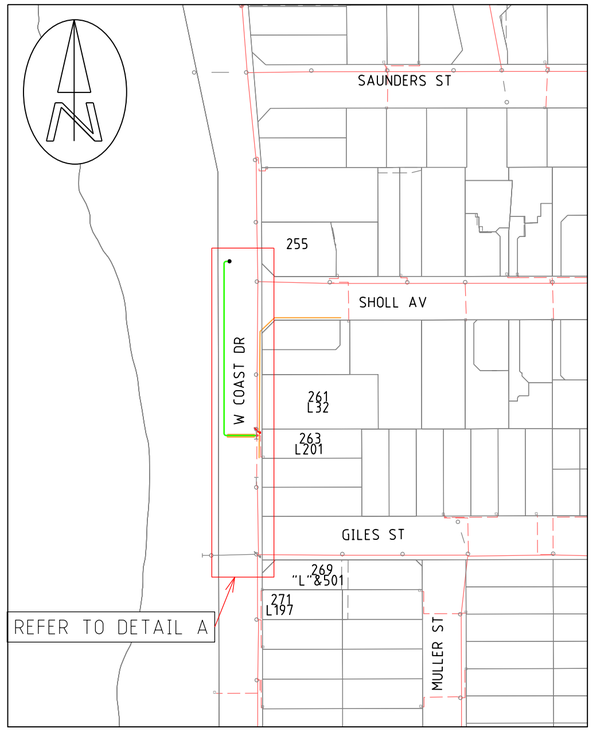
315A & 250A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGES: 22kV/415V

L3/#917 & L200/#915 WANNEROO ROAD, WANNEROO WA 6065
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 45' 21" S Long. - 115° 48' 18" E

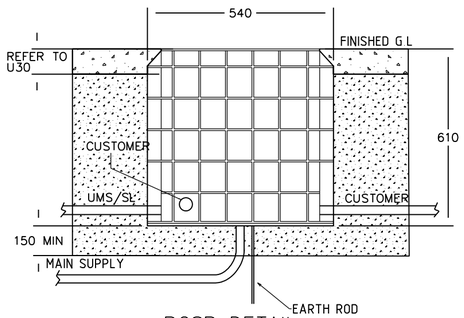
DRAWING NUMBER
MP214838
SHEET 02 OF 02
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

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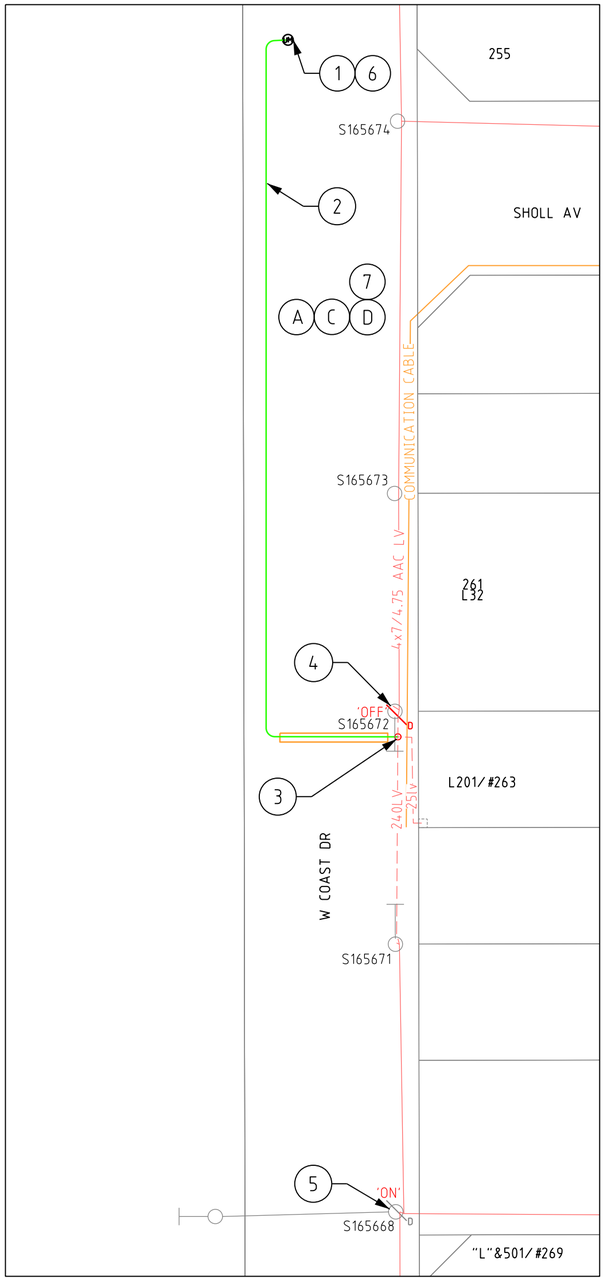
CONSUMER MAINS TO BE CONNECTED BY WESTERN POWER ONLY.



BGSP DETAIL

ISSUED FOR CONSTRUCTION





LOCATION PLAN

1:1000 20 10 0 20 40 60 80

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW TRAFFICABLE BELOW GROUND SERVICE PIT AND N25 (MINIMUM) GREY CONCRETE OVER THE EXCLUSION ZONE AT LOCATION;
   EASTING: 382011.092m,
   NORTHING: 6473726.222m
   CU: UM4x1
   DCSH: U23-1
2. INSTALL 25LV CABLE FROM NEW UMS PIT TO EXISTING 240LV CABLE AT NORTH BEACH RD.
   CU: CN42x110m, CN73_LV25x3, CN54x2
3. DISCONNECT EXISTING 25LV CABLE FROM POLE#S165672.
   CUT AND SWING EXISTING 25LV AT THE BASE OF POLE#S165672.
   T-JOINT EXISTING 25LV AND NEW 25LV TO EXISTING 240LV.
   CU: LU05x1
4. AT POLE#S165672,
   - INSTALL NEW LV ISOLATORSS AND TAP ON TO LV BARE.
   - RE-TERMINATE EXISTING 240LV TO NEW ISOLATOR IN 'OFF' POSITION.
   CU: LU09x1, LU6_240Ox1
   DCSH: U15
5. AT POLE#S165668
   - SWITCH EXSITING ISOLATOR TO 'ON' POSITION.
6. AT NEW UMS PIT.
   - TERMINATE 25LV CABLE.
   - TERMINATE CUSTOMER CABLE VIA RED SPOT FUSE IN CONJUNCTION WITH CUSTOMER ELECTRICAL CONTRACTOR.
   DCSH: U23-1
7. ENVIRONMENTAL FILTER CHECKED
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. BELOW GROUND SERVICE PIT EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE BGSP LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. CLEAR OF ALL SERVICES.

3. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

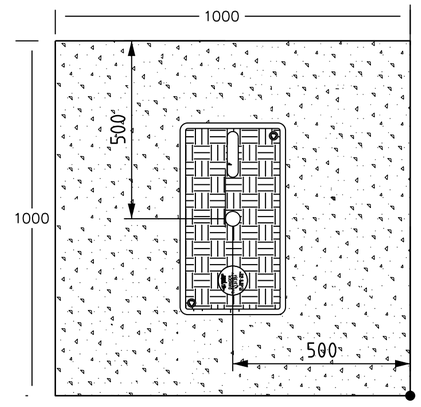
D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT THE CHANGES.
CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

LEGEND

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

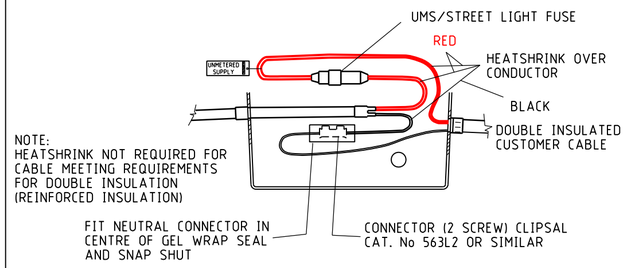
PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDI Serviced Unclearad De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

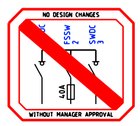


TYPICAL BGSP EXCLUSION ZONE

CONSUMER MAINS TO BE CONNECTED BY WESTERN POWER ONLY.



UMS PIT DETAIL



COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: SAGE CONSULTING ENGINEERS PTY LTD
CONTACT: ANDREEA PUIU TEL: 0431318900

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
FILE REFERENCE DETAILS

UNMETERED SUPPLY

NETWORK VOLTAGE: 22kV/415V

SHOLL AVENUE, NORTH BEACH, 6020, WA

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

DATE: 06/25 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31° 51' 58" S
Long. - 115° 45' 09" E

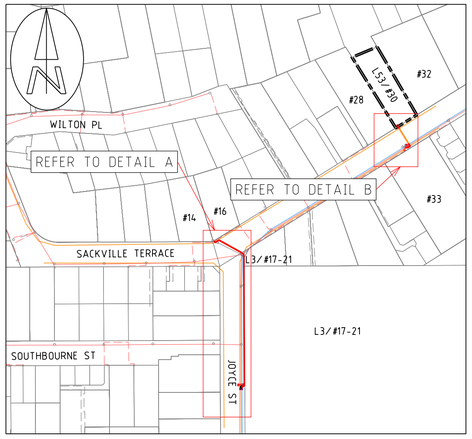
westernpower

DRAWING NUMBER

MP216333

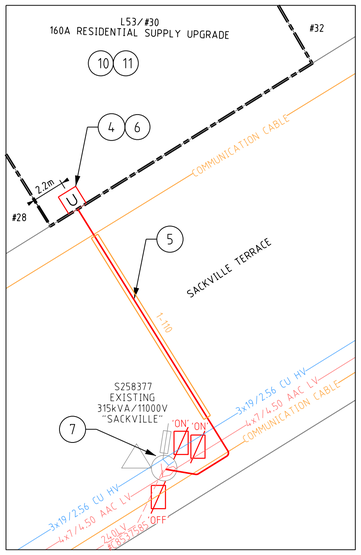
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



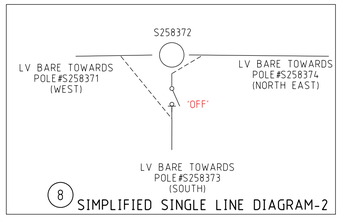
LOCATION PLAN

1:2500 50 25 0 50 100 150 200

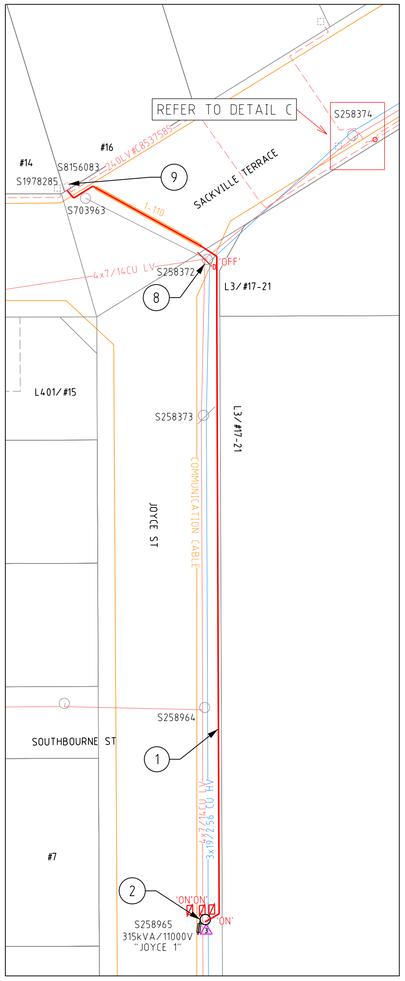


DETAIL B

1:200 4 2 0 4 8 12 16

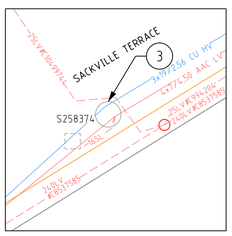


SIMPLIFIED SINGLE LINE DIAGRAM-2



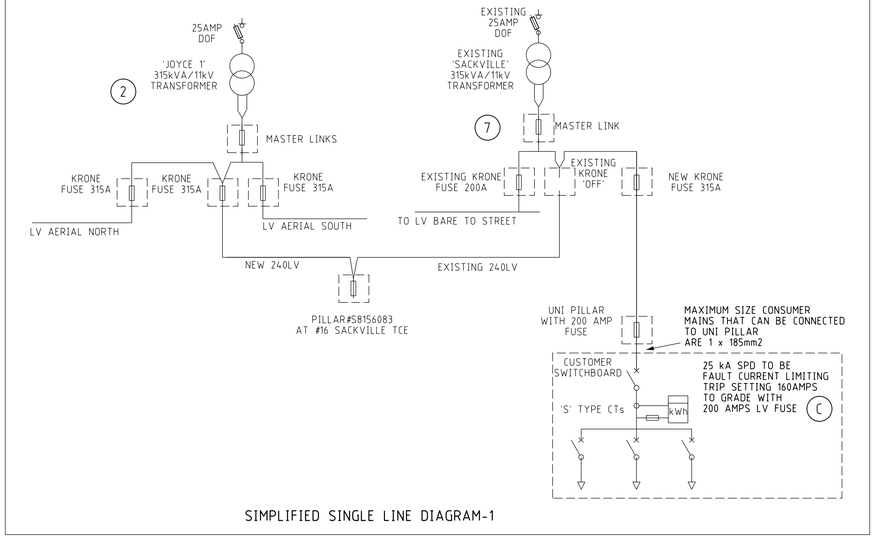
DETAIL A

1:500 10 5 0 10 20 30 40



DETAIL C

1:200 4 2 0 4 8 12 16



SIMPLIFIED SINGLE LINE DIAGRAM-1

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
UTILISE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 240LV CABLE IN THE 0-500mm ALIGNMENT.

(1) INSTALL NEW 240LV CABLE FROM EXISTING POLE#S258965 TO EXISTING PILLAR#S8156083 AT #16 SACKVILLE TCE.
CU: CN45 x 155m, CN54 x 6, CN73_LVMAINS x 3
DCSH: U16-1

(2) AT POLE#S258965
- DISCONNECT LV AERIALS SERVICES.
- REMOVE EXISTING TRANSFORMER AND ASSOCIATED EQUIPMENTS
- REPLACE EXISTING POLE WITH 12.5m/8kN WOOD POLE.
- INSTALL NEW HV INTERMEDIATE CROSSARM.
- INSTALL NEW LV STRAIN CROSSARM.
- INSTALL NEW 315kVA TRANSFORMER AND TAP ONTO HV BARE. VIA NEW 25A DOFs.
- TAP TO LV BARE TOWARDS NORTH AND SOUTH VIA TX BUSHING WITH 2xNEW 315A KRONE FUSES.
- TERMINATE NEW 240LV CABLE VIA NEW 315A KRONE FUSES.

REFER TO SIMPLIFIED LINE DIAGRAM-1.
CU: P09_8x1 CN9_T19/3.25x 3, HV6_3Nx1, HV51_25EDOx3, TX8_1_315/11x1, CN39_150/4Mx3, CN11_150x5, CN9_CUx8, LU6_240Ox1, LU6_240Ux1, CN69_150x3, LV3_W3x1, CN13_7/275CUx8
NON-CU: CB0204x4, GF1101x9, GS0111x4
DCSH: U17, R12-3, R13-3, H10-1, H01-1

(3) CUT EXISTING 25LV CABLE#C1049974 AND #C994204 FROM POLE#S258374. THEN T-JOINT EXISTING 25LV CABLE TO EXISTING 240LV CABLE#C8537585. REMOVE EXCESS CABLE.
CU: LU05x1

(4) INSTALL NEW UNI-PILLAR ON L53/#30 SACKVILLE TCE. WHERE PEGGED ON SITE BY CUSTOMER AS PER PILLAR ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3). REFER TO DETAIL B.
CU: LU11 x 1
DCSH: U9

(5) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S258377. REFER TO DETAIL B.
CU: CN45x40m, CN54x3, CN73_LVMAINSx3

(6) AT NEW UNI-PILLAR (L5/#56 FORREST AV).
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
- TERMINATE CUSTOMER CABLE AT TOP BAR WITH 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR.

CU: LU7_240x1, LU36_200Ax1

(7) AT POLE #S258377.
- INSTALL NEW KRONE FUSE BOX.
- TERMINATE 240LV CABLE TO NEW KRONE BOX.
- REMOVE EXISTING KRONE FUSES AND TURN 'OFF' EXISTING 240LV CABLE #8537585.
- DISCONNECT AND REMOVE EXISTING AERIAL SERVICE TO L53/#30.

REFER TO SIMPLIFIED LINE DIAGRAM-1.
CU: LU37_240x1, LU37_240U x1, CN39_150/4M x2
NON-CU: GF1101 x3 (315A FUSES), CB0204x1, GS0111x1
DCSH: U17

(8) AT POLE#S258372
- REMOVE EXISTING LV TAILS BETWEEN LV BARE TOWARDS NORTH EAST AND WEST.
- INSTALL NEW LV TAILS AND TAP EXISTING BARE LV TOWARDS SOUTH TO LV BARE TOWARDS WEST.
- EXISTING LV DISOS BETWEEN LV BARE SOUTH AND LV BARE NORTH EAST TO REMAIN OFF.

REFER TO SIMPLIFIED LINE DIAGRAM-2
CU: CN39_150/4Mx1, CN9_T70Cx8
DCSH: U16-1

(9) AT EXISTING UNI-PILLAR#S8156083 AT #16 SACKVILLE TCE.
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.

CU: LU7_240x1

(10) INSTALL CT METER IN CUSTOMERS SWITCHBOARD.
CU: LM1x1

(11) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
HYDROGRAPHIC CATCHMENTS - SUBCATCHMENTS (DWER-030)
VEGETATION COMPLEXES - SWAN COASTAL PLAIN (DBCA-046)

REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(E) CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(F) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(G) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

EXCAVATION WARNING:
EXCAVATION CONTRACTOR TO FOLLOW THE CODE OF PRACTICE AND PROVIDE COMPETENT PERSON APPROVAL FOR WORK METHODOLOGY AND SOIL RETENTION STRATEGY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

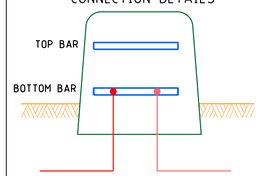
HV CABLES
CN53 - 400mm2 (3X1c)
CN49 - 240mm2 (3X1c)
CN48 - 185mm2 (3X1c)
CN50 - 95mm2 (3X1c)
CN51 - 50mm2 (3X1c)
CN51 - 50mm2 (1c)
CN46 - 35mm2 (3X1c)
CN47 - 35mm2 (1c)
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (3x3c)
CN44 - 185mm2 (3x3c)
CN43 - 120mm2 (3x3c)
CN42 - 25mm2 (3x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

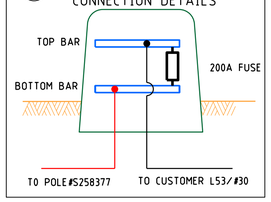
PILLARS / PITS / ETC
MINI (WORKING END)
UMS PIT
UNIVERSAL
UTILITIES PIT
MINI
BGSP PIT
LINK
100/200AMP CUTOUT
CABLE MARKER U/G
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

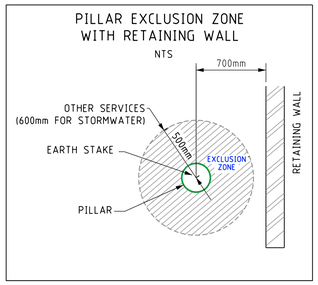
(9) UNI-PILLAR#8156083 #16 SACKVILLE TCE CONNECTION DETAILS



(6) UNI-PILLAR L53/#30 CONNECTION DETAILS



PILLAR EXCLUSION ZONE WITH RETAINING WALL



COMMUNICATION CABLES IN VICINITY OF WORKS

CRITICAL WATER CO. ASSETS IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: The Trustee For Gpg No 16 Unit Trust
CONTACT: LEANNE SHADBOLT TEL: 0892211144

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☐	☐	☒	☐

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				ENVIRONMENTAL ASSESSMENT
A	ISSUED FOR CONSTRUCTION	I.P.	09/25	HAZARD MANAGEMENT REGISTER

160A RESIDENTIAL SUPPLY UPGRADE

NETWORK VOLTAGES: 11kV/415V

30 SACKVILLE TERRACE, SCARBOROUGH WA 6019

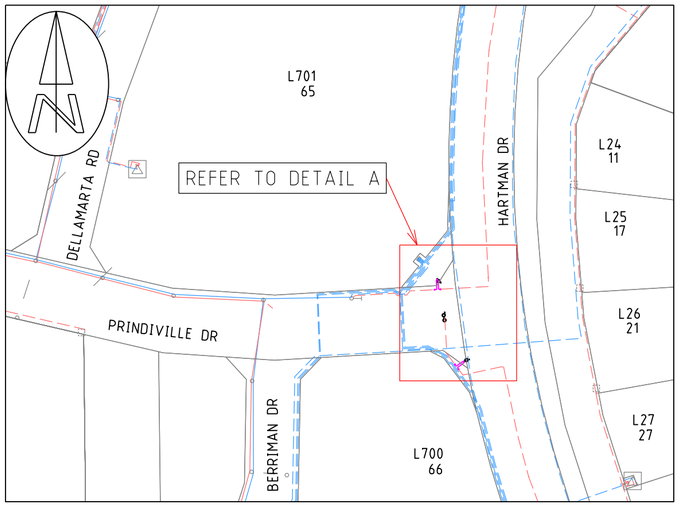
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 53' 17" S Long. - 115° 45' 56" E

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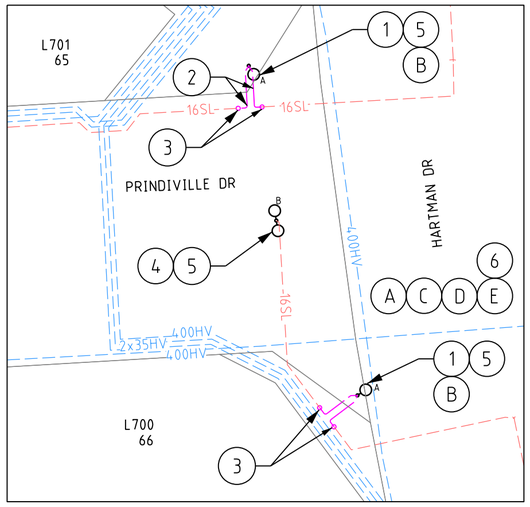
DRAWING NUMBER
MP216762
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:500 10 5 0 10 20 30 40

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 10.5m STEEL STANDARD STREETLIGHT WITH 165W 3000K LED STREETLIGHTS AS PER STREETLIGHT SCHEDULE.
   CU: SL9_6Ex2
   NON CU: FE0112x2
   DCSH: S09
2. INSTALL NEW 2xSL CABLE FROM NEW STREETLIGHT TO EXISTING 16SL CABLE.
   CU: CN41x60m, CN73_LV10/16x10, CN54x4
3. CUT EXISTING 16SL CABLE AND 2xSTRAIGHT JOINT NEW 16SL TO EXISTING 16SL CABLE.
   CU: LU13x2
4. REMOVE EXISTING STREETLIGHT POLE AND ASSOCIATED ASSEMBLIES. INSTALL NEW 10.5m STEEL STANDARD STREETLIGHT WITH 165W DOUBLE OUTREACH 3000K LED STREETLIGHTS AS PER STREETLIGHT SCHEDULE.
   CU: SL12_6Ex1
   DCSH: S12
5. AT NEW STREETLIGHT POLE, TERMINATE NEW 16SL CABLES.
   NON CU: FE0032x3, FE0071x3
6. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO environmentf@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE LOCATION OF ALL NEW STREETLIGHTS IN ACCORDANCE WITH THE CUSTOMERS STREETLIGHTING DESIGN, AS1158, WESTERN POWERS 'UNDERGROUND DISTRIBUTION SCHEMES' MANUAL AND ANY SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER. IT IS THE CUSTOMERS RESPONSIBILITY TO ADHERE TO THE FOLLOWING:
- THE LOCATION OF ALL STREETLIGHTS IS CLEAR OF ANY 3RD PARTY ASSETS
- ALL STREETLIGHTS MAINTAIN A MINIMUM SET BACK FROM THE ROAD IN ACCORDANCE WITH AS1158.1.2 AND WESTERN POWERS UDS MANUAL
- ANY PROPOSED STREETLIGHT LOCATIONS OUTSIDE OF THE WESTERN POWER STANDARD 2.4 - 3.0m POLE ALIGNMENT HAVE THE APPROVAL OF 3RD PARTY UTILITIES IN ACCORDANCE WITH THE UTILITY PROVIDERS CODE OF PRACTICE.
- STREET LIGHTS ARE NOT PERMITTED TO BE INSTALLED WITHIN THE WESTERN POWER STANDARD 0-500mm CABLE ALIGNMENT.
- ALL STREETLIGHTS ARE TO ADHERE TO THE POWER LINE CLEARANCE REQUIREMENTS AS PER AS7000, INCLUDING IN THE EVENT THEY ARE STRUCK BY A VEHICLE AND FALL TOWARDS THE TRANSMISSION LINE.

C. ANY TEMPORARY STREET LIGHTING REQUIRED DURING THE WORKS AFTER REMOVAL OF EXISTING STREET LIGHTS UNTIL NEW STREET LIGHTS HAVE BEEN ENERGISED IS TO BE SUPPLIED AND INSTALLED BY THE CUSTOMER AT NO COST TO WESTERN POWER.

D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT ADDITIONAL STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

E. THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

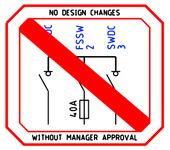
STREET LIGHTS — EXISTING
- POLE BRACKET
- SINGLE OUTREACH
- DOUBLE OUTREACH
- 4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

LUMINAIRE SCHEDULE (DETAIL A)

TYPE	DESCRIPTION	COLOUR	CU'S	QTY
A	INSTALL NEW STANDARD 10.5m POLE WITH 165W 3000K LED	3000K	SL9_6E	2
B	REPLACE EXISTING STREETLIGHT WITH NEW STANDARD 10.5m POLE WITH 165W DOUBLE OUTREACH LED	3000K	SL10_6E	1



ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

GAS PIPES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: CITY OF WANNEROO
CONTACT: KATALIN RICHTER TEL: 0894055895

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	12/25

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT
HAZARD MANAGEMENT REGISTER

STREETLIGHT UPGRADE

INTERSECTION OF HARTMAN DR AND PRINDIVILLE DR

WANGARA

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: Ishan Palikhe
DESIGN TEAM: CCS COMM SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

DATE: 12/25 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31°47'29.7"S Long. - 115°49'40.9"E

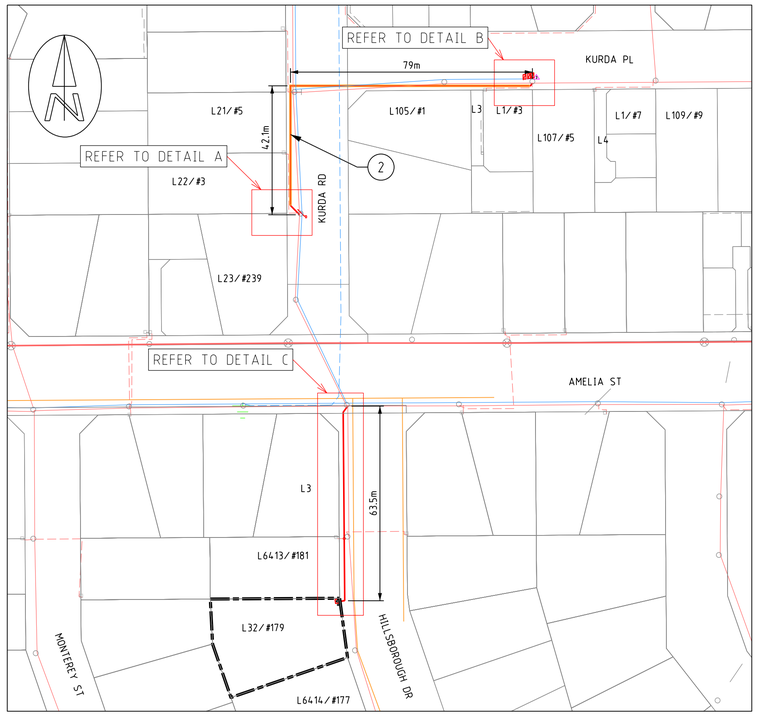
westernpower

DRAWING NUMBER

MP217616

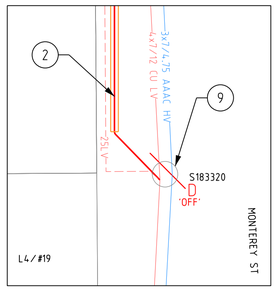
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



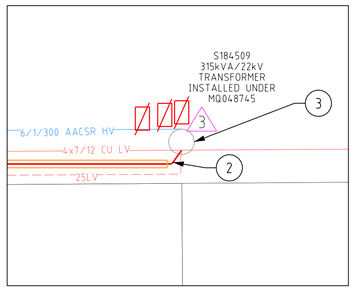
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



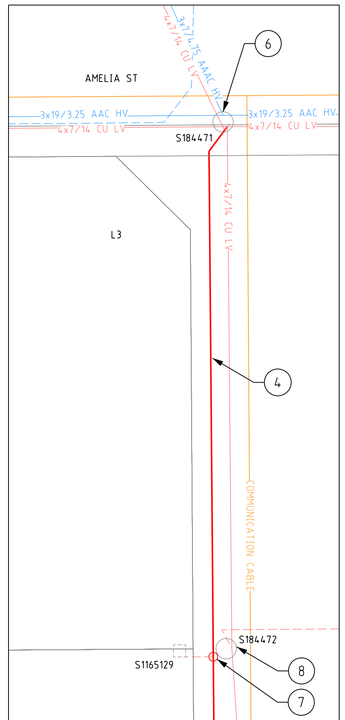
DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16



DETAIL C

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

(i) THIS PROJECT IS DEPENDENT ON THE COMPLETION OF PROJECT MQ048745.

1. INSTALL NEW UNI-PILLAR ON L32/#179 HILLSBOROUGH DR. WHERE PEGGED ON SITE BY CUSTOMER AS PER PILLAR ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3). REFER TO LOCATION PLAN.
   CU: LU11x1
   DCSH: U09
2. INSTALL NEW 240LV CABLE FROM POLE#S183320 TO POLE#S184509 VIA 110mm CONDUIT. UTILIZE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 240LV CABLE IN THE 0-500mm ALIGNMENT.
   CU: CN45x145m, CN54x6, CN73_LVMAINSx3
3. AT POLE#S184509,
   - TERMINATE NEW 240LV CABLE VIA NEW 250A KRONE FUSE.
   REFER TO SIMPLIFIED LINE DIAGRAM-2
   CU: LU61_240_3POx1, LU61_240_3PUx1
   NON CU: GF2129x3, GS0109x1, GS6022x1, GS6021, AB6732x1
   DCSH: U17, R12-3
4. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S184471 IN 2.7m ALIGNMENT.
   REFER TO DETAIL C.
   CU: CN45x80m, CN54x3, CN73_LVMAINSx3
5. AT NEW UNI-PILLAR (L32/#179 HILLSBOROUGH DR)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 160A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_160Ax1
   DCSH: U09
6. AT POLE#S184471,
   - TERMINATE NEW 240LV CABLE TO LV BARE TOWARDS NORTH.
   REFER TO SIMPLIFIED LINE DIAGRAM-1
   CU: LU6_2400x1, LU6_240Ux1
   DCSH: U16-1
7. TEE JOINT EXISTING 25LV TO NEW 240LV CABLE.
   REMOVE EXCESS CABLE.
   CU: L15x1
8. AT POLE#S184472
   - DISCONNECT AND REMOVE EXISTING 25LV UG SERVICE TO PILLAR #S1165129
9. AT POLE#S183320,
   - DISCONNECT EXISITNG 25LV CABLE.
   - REPLACE EXISTING LV INTERMEDIATE CROSSARM WITH NEW LV STRAIN CROSSARM WITH LV ISOLATOR IN 'OFF' POSITION.
   - TERMINATE NEW 240LV CABLE TO LV BARE CONDUCTOR HEADING SOUTH.
   - RE CONNECT EXISTING 25LV CABLE TO LV BARE CONDUCTOR HEADING NORTH.
   REFER TO SIMPLIFIED LINE DIAGRAM-2
   CU: LV3_w3x1, LV6_600Ax1, CN9_CU250x8, LU6_2400x1, LU6_240Ux1, CN39_150/4Mx1, CN13_7/160CUx8, CN11_150x6, CN69_150x8
   DCSH: U16-1, L03, L06
10. ENVIRONMENTAL FILTER CHECKED.
    NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
    CARNABY'S COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
    LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
    REGION SCHEME - ZONES AND RESERVES (DPLH-023)
    INDIGENOUS LAND USE AGREEMENTS (ILUAS)
    NATIVE TITLE BOUNDARIES
    RIWI ACT. GROUNDWATER AREAS (DWER-034)
    PRE-EUROPEAN VEGETATION (DPIRD-006)

    REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

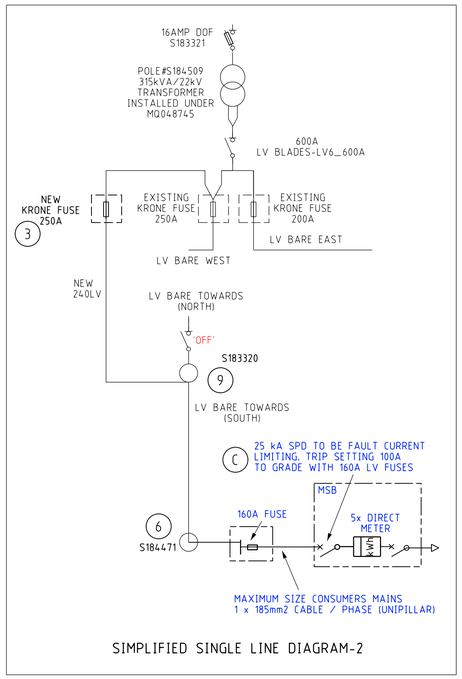
(E) CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL CONTRACTOR SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(G) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(H) CUSTOMER TO INSTALL NON-CORROSIVE BOLLARDS ADJACENT TO THE PILLAR AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3.

(I) CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.



SIMPLIFIED SINGLE LINE DIAGRAM-2

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

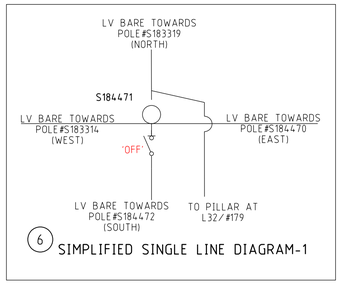
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

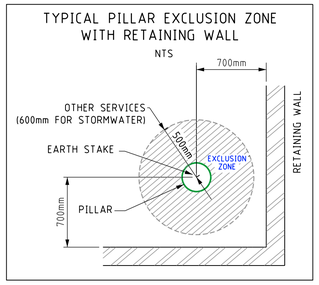
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

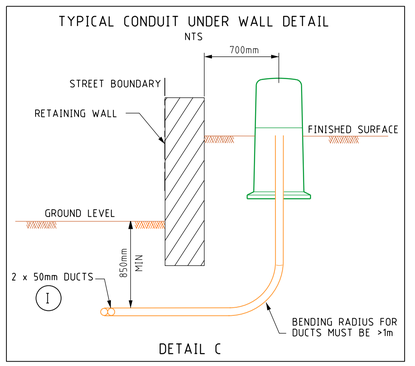
PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Ex Serviced Unclearad De-energised Lot

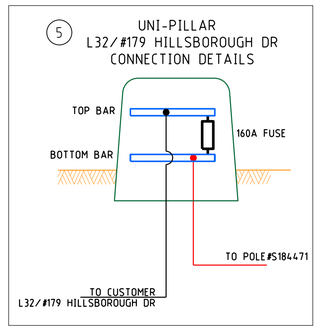
TRANSFORMER (T) EXISTING SWITCHGEAR (S)

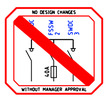
AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)











CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☐	☐	☐	☐	☐

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: Bw Residential Pty Ltd
CONTACT: TONY CUMACE TEL: 0439988739

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				ENVIRONMENTAL ASSESSMENT
A	ISSUED FOR CONSTRUCTION	I.P.	11/25	HAZARD MANAGEMENT REGISTER

5x 3.5kVA STANDARD RESIDENTIAL SUPPLIES

NETWORK VOLTAGES: 22kV/415V

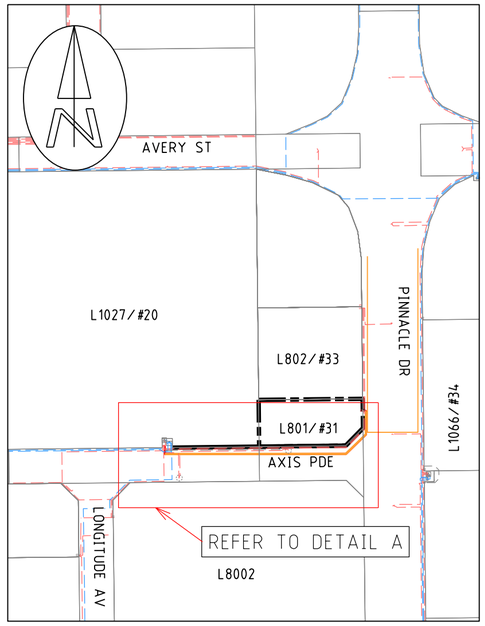
L32/#179 HILLSBOROUGH DR, NOLLAMARA WA 6061

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 52' 27" S Long. - 115° 50' 04" E

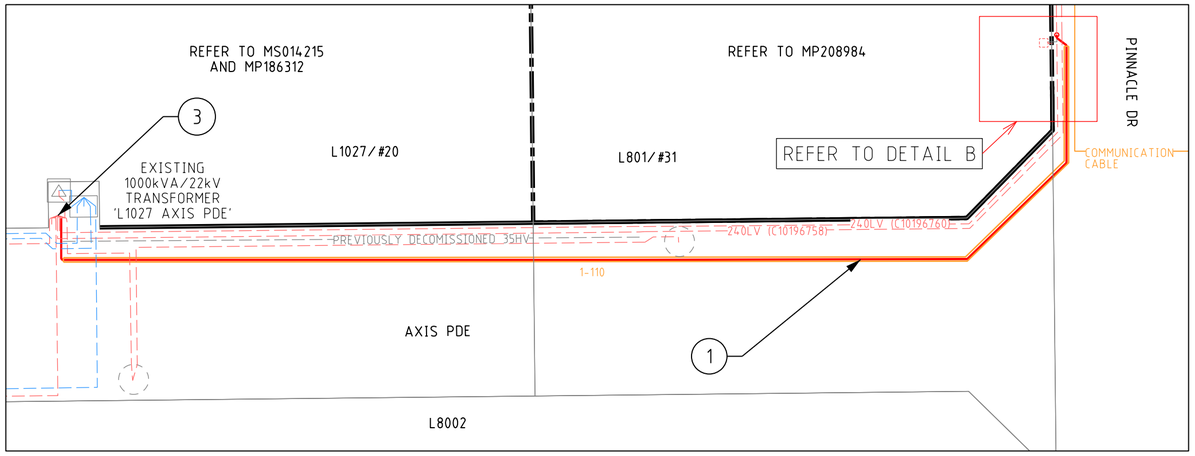
westernpower

DRAWING NUMBER
MP217673
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



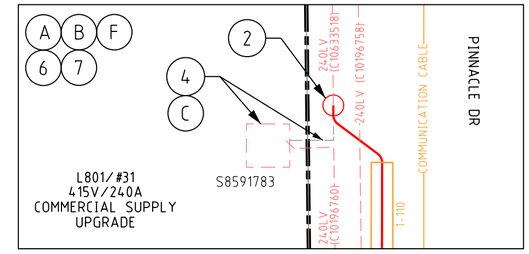
LOCATION PLAN

1:2500 50 25 0 50 100 150 200



DETAIL A

1:500 10 5 0 10 20 30 40



DETAIL B

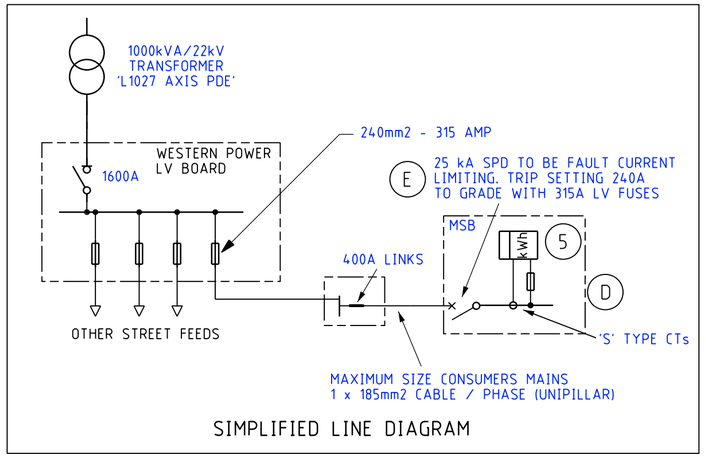
1:100 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 240LV CABLE FROM EXISTING LV KIOSK AT 'L1027 AXIS PDE' SUBSTATION TO EXISTING 240LV AT #31 PINNACLE DR IN 2.7m ALLIGNMENT.
   CU: CN45×145m, CN73_LVMAINSx3, CN54x4
2. CUT EXISTING 240LV CABLE#C10633518 AND STRAIGHT JOINT TO NEW 240LV CABLE.
   CU: LU2×1
3. AT EXISTING LV KIOSK AT 'L1027 AXIS PDE'.
   - INSTALL NEW 315A FUSE IN EXISITING FUSE HOLDER AND TERMINATE NEW 240LV CABLE TO SPARE FUSEWAY 7.
   - UPDATE LV KIOSK LABELS.
   CU: LU14_240x1
   NON CU: GF1101x3
   DCSH: U27-1
4. AT UNI-PILLAR (L801/#31 PINNACLE DR)
   - DISCONNECT AND REMOVE EXISTING 240LV CABLE#C10715375 TOWARDS NORTH.
   - DISCONNECT EXISTING CONSUMER MAINS FROM BACK BLOCKS AND TERMINATE NEW CUSTOMER CABLE AT TOP BAR VIA 400A LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU36_LINK x 1
   DCSH: U09
5. INSTALL CT METER IN CUSTOMERS SWITCHBOARD.
   CU: LM1x1
6. WP SCHEDULER TO CONFIRM RECEIPT OF ETAF PRIOR TO SCHEDULING WORKS.
   WP PNO TO OBTAIN FINAL METER READ DATE FROM WP METERING.
7. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   THREATENED ECOLOGICAL COMMUNITIES
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   PRE-EUROPEAN VEGETATION (DPIRD-006)
   TUART WOODLANDS (DBCA-048)

   REFER TO environment@westernpower.com.au



SIMPLIFIED LINE DIAGRAM

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

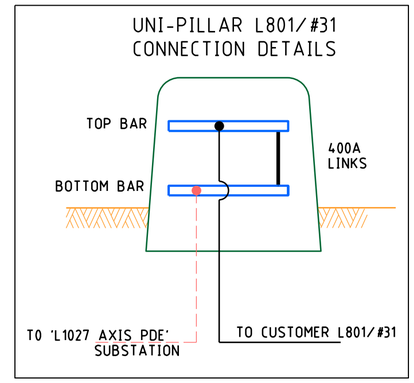
C. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

D. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

E. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

F. WESTERN POWER WILL REQUIRE OUTGOING RETAILER AGREEMENT (VIA SUBMISSION OF AN ETAF) PRIOR TO SCHEDULING OF THE PROJECT WORKS.
CUSTOMER REQUESTED TO CONTACT THE OUTGOING RETAILER AND PROVIDE RELEVANT PROJECT DETAILS TO FACILITATE ETAF.
CUSTOMER TO CONTACT WP PNO PRIOR TO COMMENCEMENT OF METERING CHANGES.

C. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGIZATION WITHIN 30m OF THE POINT OF SUPPLY.



LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

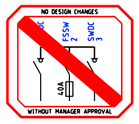
HV CABLES
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDU Serviced Uncleared De-energised Lot

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☐	☐	☐	☒	☐



SAFETY - MAKE IT YOUR PRIMARY VALUE

ISSUED FOR CONSTRUCTION

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: The Trustee For Rivo Family Trust
CONTACT: DANIELE BARBI TEL: 0439988739

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	11/25

FILE REFERENCE DETAILS
- ENVIRONMENTAL ASSESSMENT
- HAZARD MANAGEMENT REGISTER

240A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

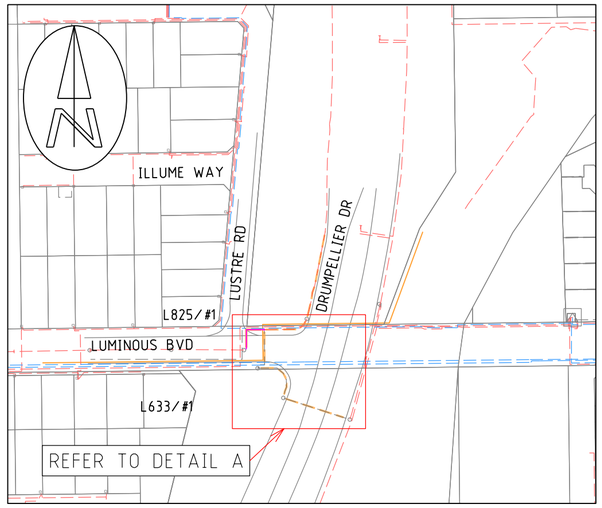
L801/#31 PINNACLE DR, NEERABUP WA 6031

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 11/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31° 40' 59" S Long. - 115° 47' 01" E

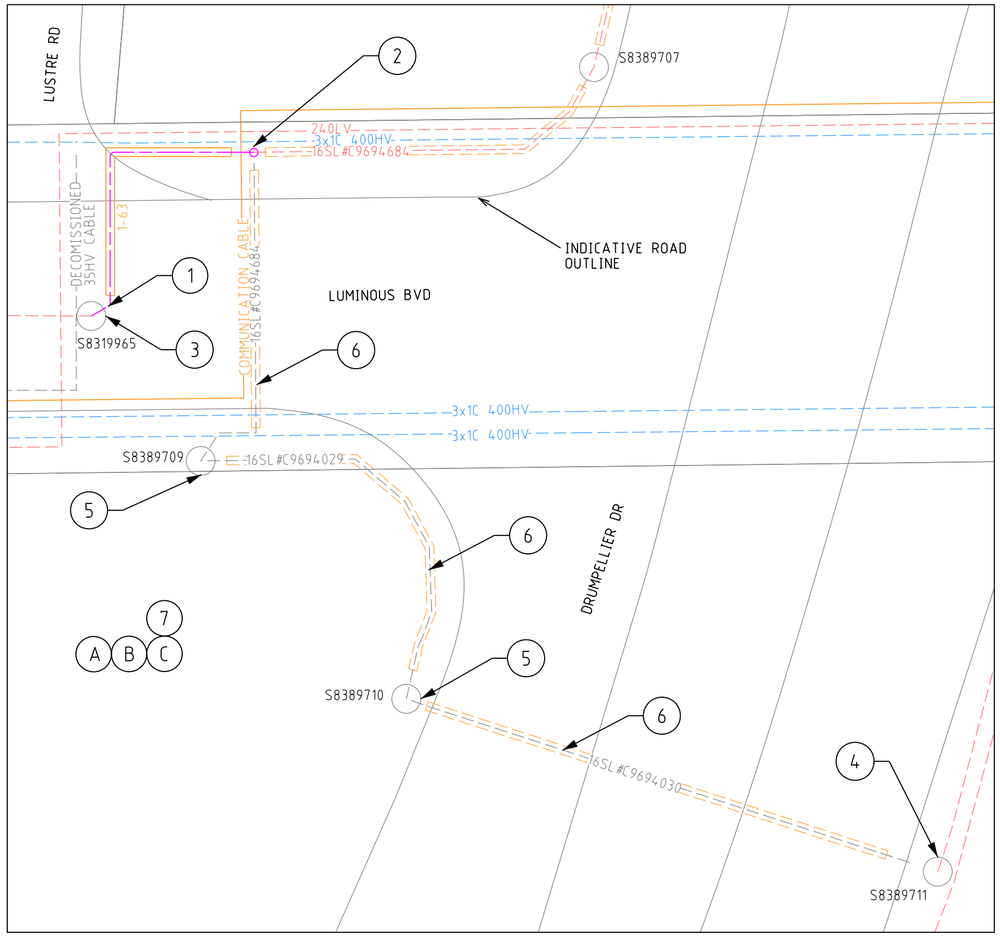
westernpower

DRAWING NUMBER
MP218332
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:250 — 5 2.5 0 5 10 15 20

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW 16SL CABLE FROM EXISTING STREETLIGHT#S8319965 TO EXISTING 16SL CABLE#C9694684 VIA 63mm CONDUIT AS INDICATED IN DETAIL A.
   CU: CN41x30m, CN73_LV10/16x3, CN54x1
   DCSH: R27
2. STRAIGHT JOINT NEW 16SL CABLE TO EXISTING 16SL CABLE#C9694684.
   CU: LU13x1
3. TERMINATE NEW 16SL CABLE TO EXISTING STREETLIGHT#S8319965.
   NON CU: FE0032 X 1, FE0071 X 1
   DCSH: R26, R27
4. DISCONNECT EXISTING 16SL CABLE #C9694030 FROM THE STREETLIGHT#S8389711.
5. DISCONNECT EXISTING 16SL CABLES FROM STREETLIGHTS #S8389709 AND #S8389710. REMOVE STREETLIGHTS.
6. REMOVE EXISTING 16SL CABLES #C9694684, #C9694029, #C9694030 FROM CONDUIT.
7. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ABORIGINAL HERITAGE SITES
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   POTENTIALLY WET AREAS
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   GEOMORPHIC WETLANDS, SWAN COASTAL PLAIN (DBCA-019)
   RIWI ACT, SURFACE WATER AREAS AND IRRIGATION DISTRICTS (DWER-037)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   OFFSETS REGISTER - PROJECTS (DWER-079)
   NATIVE TITLE BOUNDARIES
   PRE-EUROPEAN VEGETATION (DPIRD-006)
   CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT ADDITIONAL STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

C. POSSESSION OF SITE CERTIFICATE IS TO BE REQUESTED, AND MAIN ROADS DOCUMENT D18#604587 'SEEKING APPROVAL TO UNDERTAKE WORKS WITHIN THE ROAD RESERVE' TO BE COMPLETED.
   AS THIS IS A MRWA INITIATED PROJECT, NO CONDITIONAL AGREEMENT FOR THE PROPOSED WORKS IS REQUIRED. PLEASE LIAISE WITH THE MRWA REPRESENTATIVE FOR ANY FURTHER REQUIREMENTS REGARDING ACCESS TO THE MRWA ROAD RESERVE.

LEGEND

HV CABLES
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS

	EXISTING
# POLE BRACKET	POLE BRACKET
# SINGLE OUTREACH	SINGLE OUTREACH
# DOUBLE OUTREACH	DOUBLE OUTREACH
# 4-WAY	4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- SUDL Serviced Uncleared De-energised Lot
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☐	☐	☐	☐

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

WATER PIPES IN VICINITY OF WORKS

ATCO GAS PIPE IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: GEORGIOU GROUP PTY LTD
CONTACT: STEFAN DRASKO TEL: 0482805438

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	12/25

FILE REFERENCE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

MRWA STREETLIGHT REMOVAL

INTERSECTION OF LUMINOUS BVD & DRUMPELLIER DR

BENNETT SPRINGS WA 6063
NETWORK VOLTAGE: 22kV/415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

DATE: 12/25 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31° 51' 28" S
Long. - 115° 57' 58" E

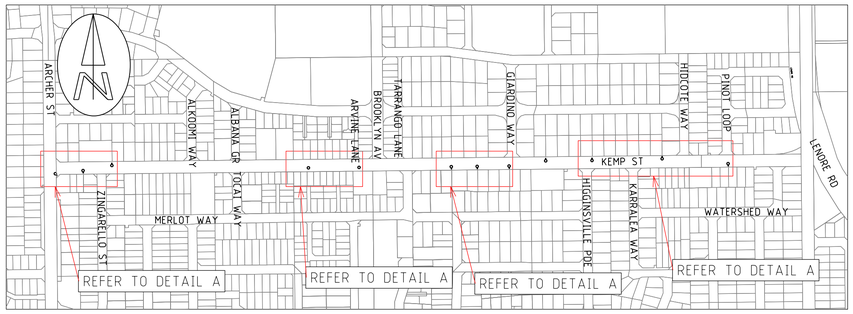
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DRAWING NUMBER

MP218667

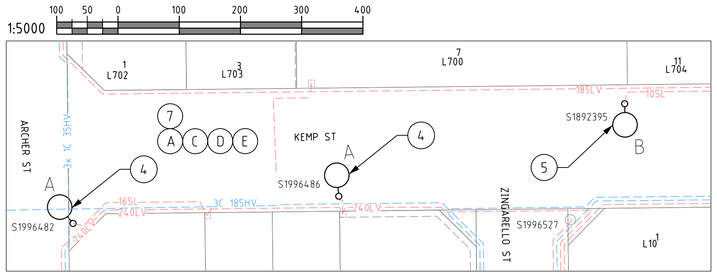
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



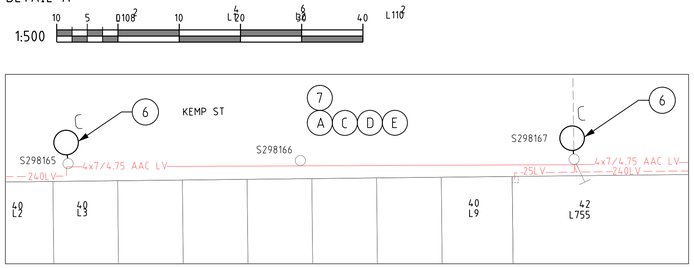
LOCATION PLAN

1:5000



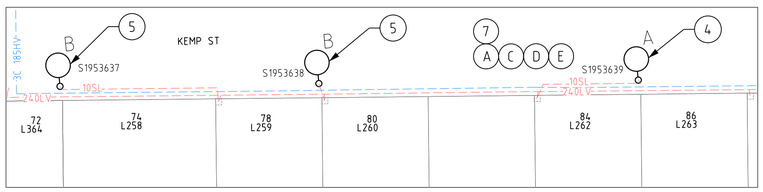
DETAIL A

1:500



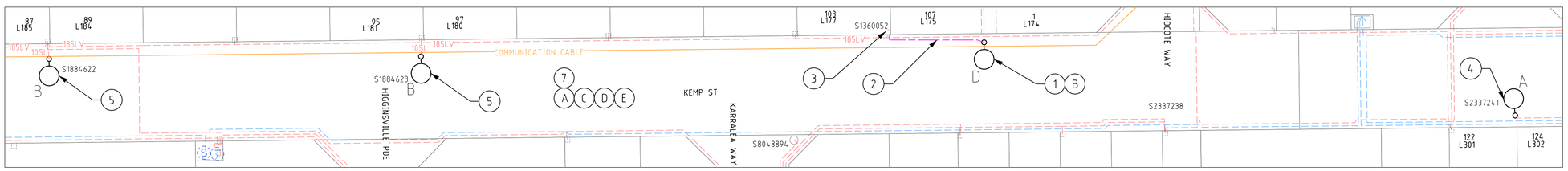
DETAIL B

1:500



DETAIL C

1:500



DETAIL D

1:500

DESIGNED TO BE CONSTRUCTED DE-ENERGISED

SAFETY - MAKE IT YOUR PRIMARY VALUE

WESTERN POWER SCOPE OF WORK

1. INSTALL A NEW 6.5m STEEL STANDARD STREETLIGHT WITH 16W 3000K LED STREETLIGHTS AS PER STREETLIGHT SCHEDULE.
   CU: SL8_1Ex1
   NON CU: FE0112x1
   DCSH: S08
2. INSTALL NEW 16SL CABLE FROM EXISTING MINI-PILLAR AT L177/#103 KEMP ST TO NEW STREETLIGHT.
   CU: CN41x30m, CN73_LV10/16x3, CN54x2
   DCSH: R27
3. AT EXISTING MINI-PILLAR (S1360052)
   - TERMINATE NEW 16SL CABLE VIA RED SPOT FUSE.
   CU: UM3_Rx1
   DCSH: U08-04, U23-1
4. REPLACE EXISTING LUMINARE WITH 43W 3000K LED STREETLIGHT AND REPLACE EXISTING CUT OUTS WITH NEW BILL TYPE.
   CU: SL18_FLAIRx4
   NON CU: HL6604x4, HL5557x4, HZ0131x4, GF1300x4, HZ0135x4
   DCSH: S08, R26-7-1
5. REPLACE EXISTING LUMINARE WITH 16W 3000K LED STREETLIGHT. AND REPLACE EXISTING CUT OUTS WITH NEW BILL TYPE.
   CU: SL18_FLAIRx5
   NON CU: HL6601x5, HL5557x5, HZ0131x5, GF1300x5, HZ0135x5
   DCSH: S08, R26-7-1
6. REPLACE EXISTING STREETLIGHT ASSEMBLY WITH NEW 43W 3000K LED STREETLIGHT ON MINOR ROAD OUTREAC FOR POLE#S298165 & #S298167.
   CU: SL02_2Ex2
   DCSH: S02
7. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, GROUNDWATER AREAS (DWER-034
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CUSTOMER TO PEG THE LOCATION OF ALL NEW STREETLIGHTS IN ACCORDANCE WITH THE CUSTOMERS STREETLIGHTING DESIGN. AS1158, WESTERN POWERS 'UNDERGROUND DISTRIBUTION SCHEMES' MANUAL AND ANY SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER. IT IS THE CUSTOMERS RESPONSIBILITY TO ADHERE TO THE FOLLOWING:
   - THE LOCATION OF ALL STREETLIGHTS IS CLEAR OF ANY 3RD PARTY ASSETS.
   - ALL STREETLIGHTS MAINTAIN A MINIMUM SET BACK FROM THE ROAD IN ACCORDANCE WITH AS1158.1.2 AND WESTERN POWERS UDS MANUAL.
   - ANY PROPOSED STREETLIGHT LOCATIONS OUTSIDE OF THE WESTERN POWER STANDARD 2.4 - 3.0m POLE ALIGNMENT HAVE THE APPROVAL OF 3RD PARTY UTILITIES IN ACCORDANCE WITH THE UTILITY PROVIDERS CODE OF PRACTICE.
   - STREET LIGHTS ARE NOT PERMITTED TO BE INSTALLED WITHIN THE WESTERN POWER STANDARD 0-500mm CABLE ALIGNMENT.
   - ALL STREETLIGHTS ARE TO ADHERE TO THE POWER LINE CLEARANCE REQUIREMENTS AS PER AS7000, INCLUDING IN THE EVENT THEY ARE STRUCK BY A VEHICLE AND FALL TOWARDS THE TRANSMISSION LINE.

C. ANY TEMPORARY STREET LIGHTING REQUIRED DURING THE WORKS AFTER REMOVAL OF EXISTING STREET LIGHTS UNTIL NEW STREET LIGHTS HAVE BEEN ENERGISED IS TO BE SUPPLIED AND INSTALLED BY THE CUSTOMER AT NO COST TO WESTERN POWER.

D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT ADDITIONAL STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

E. CUSTOMER IS RESPONSIBLE FOR COMPLETING THEIR DUE DILIGENCE ASSESSMENT OF POTENTIAL ABORIGINAL HERITAGE IMPACTS AND ADVISING WESTERN POWER OF ANY COMPLIANCE REQUIREMENTS THAT APPLY.

STREETLIGHT SCHEDULE

TYPE	POLE ID	DESCRIPTION	POLE LOCATION MGA 94-50 EASTING	POLE LOCATION MGA 94-50 NORTHING	COLOR	CU
A	S1996482 (2)	REPLACE EXISTING LUMINARE WITH NEW 43W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
A	S1996486 (3)	REPLACE EXISTING LUMINARE WITH NEW 43W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
B	S1892395 (4)	REPLACE EXISTING LUMINARE WITH NEW 16W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
C	S298165 (15)	REPLACE EXISTING OUTREACH WITH NEW 43W LED ON MINOR ROAD OUTREACH	EXISTING	EXISTING	3000K	SL18_FLAIR
C	S298167 (16)	REPLACE EXISTING OUTREACH WITH NEW 43W LED ON MINOR ROAD OUTREACH	EXISTING	EXISTING	3000K	SL18_FLAIR
B	S1953637 (21)	REPLACE EXISTING LUMINARE WITH NEW 16W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
B	S1953638 (22)	REPLACE EXISTING LUMINARE WITH NEW 16W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
A	S1953639 (23)	REPLACE EXISTING LUMINARE WITH NEW 43W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
B	S1884622 (24)	REPLACE EXISTING LUMINARE WITH NEW 16W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
B	S1884623 (25)	REPLACE EXISTING LUMINARE WITH NEW 16W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
A	S2337241 (28)	REPLACE EXISTING LUMINARE WITH NEW 43W LED ON EXISTING POLE	EXISTING	EXISTING	3000K	SL18_FLAIR
D	NEW	NEW STANDARD 6.5m POLE WITH 16W LED	388674.107	6483122.52	3000K	SL8_1E

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1X3c)
- CN44 - 185mm2 (1X3c)
- CN43 - 120mm2 (1X3c)
- CN42 - 25mm2 (1X3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- POLE BRACKET
- SINGLE OUTREACH
- DOUBLE OUTREACH
- 4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE.
REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- MS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WATER PIPES IN VICINITY OF WORKS

GAS PIPES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: City of Wanneroo
CONTACT: BREE ABBASI TEL: 0894055686

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	☐	☐	☒

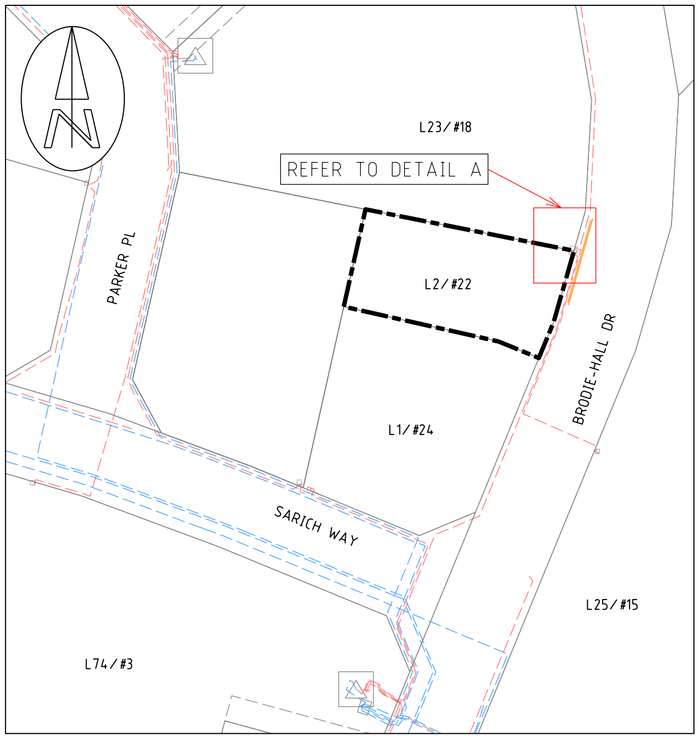
Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	12/25	HAZARD MANAGEMENT REGISTER

STREETLIGHT UPGRADE
KEMP STREET
PEARSALL, WA 6065
NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS COMM SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

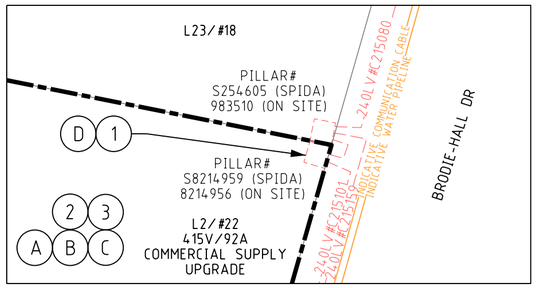
DATE: 12/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31°46'55" S Long. - 115°48'52" E

DRAWING NUMBER
MP218674
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:200 — 4 2 0 4 8 12 16

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY.
CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(E) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
THERE IS DISREPENCY BETWEEN PILLAR# IN SPIDA AND ON SITE.

(1) AT UNI-PILLAR (L2/#22 BRODIE-HALL DR.)
- TERMINATE CUSTOMER CABLE TO THE TOP BAR VIA 160A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU36_160A x 1
DCSH: U09

(2) INSTALL CT METER IN CUSTOMER SWITCHBOARD.
CU: LM1_M x 1

(3) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

MUNICIPAL INVENTORY
ACID SULFATE SOILS
ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
BLACK COCKATOO ROOSTING SITES - BUFFERED (DBCA-064)
CARNABYS COCKATOO CONFIRMED ROOST SITES (DBCA-050)
LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
REGION SCHEME - ZONES AND RESERVES (DPLH-023)
INDIGENOUS LAND USE AGREEMENTS (ILUAS)
NATIVE TITLE BOUNDARIES
RIWI ACT, GROUNDWATER AREAS (DWER-034)
PRE-EUROPEAN VEGETATION (DPIRD-006)

REFER TO environment@westernpower.com.au

LEGEND

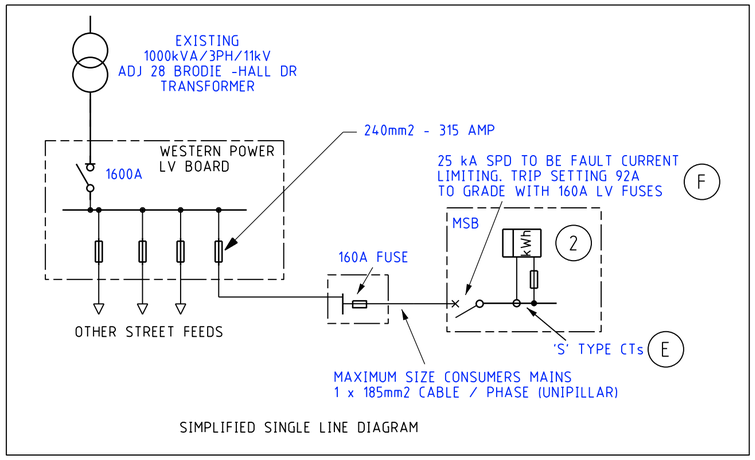
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

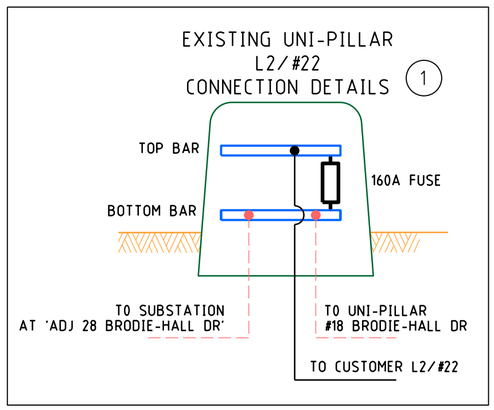
LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

TRANSFORMER (T) EXISTING SWITCHGEAR (S)



SIMPLIFIED SINGLE LINE DIAGRAM

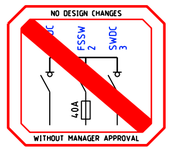


ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☐	☐	☒	☐	☐	☐	☐	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: CAMCODEV PTY LTD
CONTACT: DEON KEET TEL: 0499992191

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	02/26	HAZARD MANAGEMENT REGISTER

92A COMMERCIAL SUPPLY UPGRADE

RESEARCH & DEVELOPMENT CHILD CARE CENTRE

22 BRODIE HALL DRIVE, BENTLEY WA 6102

NETWORK VOLTAGE: 11kV / 415V

DESIGNER: Ishan Palikhe
DESIGN TEAM: CCS.COMM.SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

DATE: 02/26 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31°59'45.9"S
Long. - 115°53'33.9"E

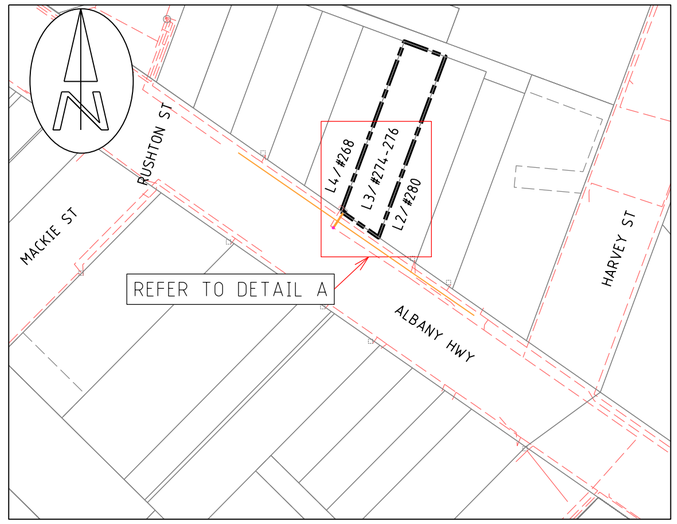
westernpower

DRAWING NUMBER

MP218826

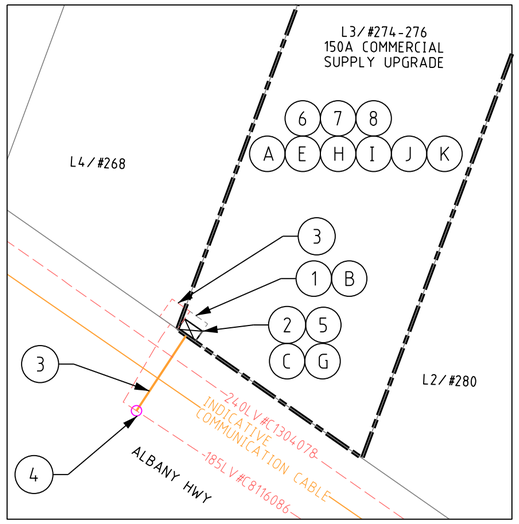
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



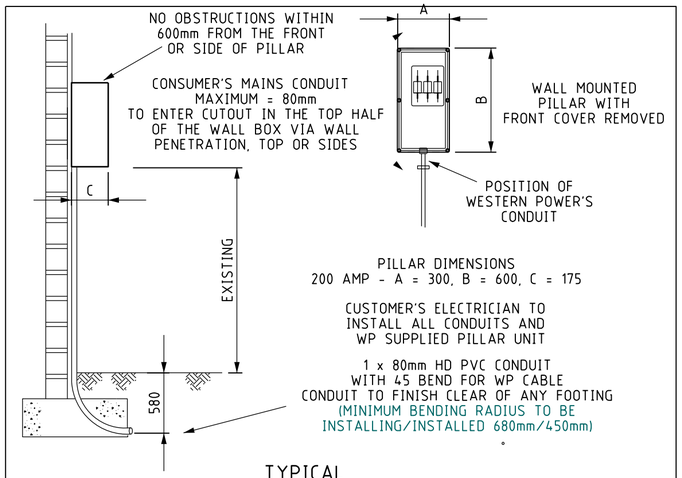
LOCATION PLAN

1:1000

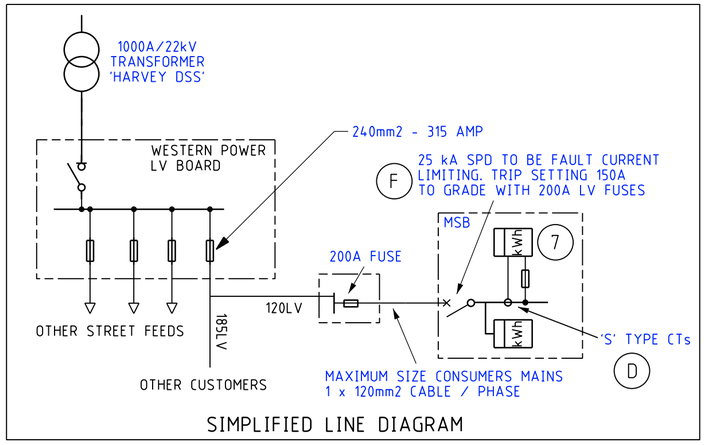


DETAIL A

1:200



TYPICAL
WALL MOUNTED BOX DETAILS



SIMPLIFIED LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING CONSUMER MAINS.
DISCONNECT AND REMOVE CABLE BETWEEN EXISTING 200A WALL PILLAR AND EXISTING 100A WALL MOUNTED BOX TO ALLOW CUSTOMER ELECTRICAL CONTRACTOR TO REMOVE THE 100A WALL MOUNTED BOX SAFELY.
2. PROVIDE NEW 200A WALL MOUNTED BOX TO CUSTOMER.
CUSTOMER RESPONSIBLE TO INSTALL NEW 200A WALL MOUNTED BOX IN APPROXIMATELY THE SAME LOCATION TO DCSH U31.
CU: LU35 x 1
DCSH: U31, U21
3. INSTALL NEW 120LV CABLE FROM NEW WALL MOUNTED BOX TO EXISTING 185LV CABLE VIA CUSTOMER INSTALLED DUCT.
CU: CN43 x 15m, CN54 x 2, CN73_LVMAINS x 3
4. BREACH JOINT NEW 120LV TO EXISTING 185LV CABLE.
CU: LU1 x 1
5. AT NEW WALL MOUNTED BOX,
- TERMINATE NEW 120LV CABLE.
- TERMINATE CUSTOMER CABLE VIA 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU7_240 x 1
DCSH: U32
6. WP SCHEDULER TO CONFIRM RECEIPT OF ETAF PRIOR TO SCHEDULING WORKS.
WP PNO TO OBTAIN FINAL METER READ DATE FROM WP METERING.
7. INSTALL CT METER IN CUSTOMER SWITCHBOARD.
CU: LM1_M x 1
8. ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
ACID SULFATE SOILS
ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
REGION SCHEME - ZONES AND RESERVES (DPLH-023)
INDIGENOUS LAND USE AGREEMENTS (ILUAS)
NATIVE TITLE BOUNDARIES
RIWI ACT, GROUNDWATER AREAS (DWER-034)
PRE-EUROPEAN VEGETATION (DPIRD-006)
REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. CONFIRM WITH WESTERN POWER CM THAT EXISTING 100A WALL PILLAR IS DE-ENERGISED AND REMOVE AS REQUIRED.

C. INSTALL THE WESTERN POWER SUPPLIED 200A WALL MOUNTED BOX IN SAME LOCATION. WALL MOUNTED BOX TO BE PREPARED AS PER THE DISTRIBUTION CONSTRUCTION STANDARDS HANDBOOK U32.
1. SHADED AREA MUST BE KEPT CLEAR FOR OPERATIONAL PURPOSES AND WP PERSONNEL (TO A HEIGHT OF 2500).
2. CUSTOMER TO INSTALL 80mm CONDUIT FROM NEW 200A WALLPILLAR, THROUGH EAVES, TO THE GROUND, WITH 45 DEGREE BEND INTO ROAD RESERVE. REFER TO DCSH U31 FOR DETAILS.

D. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL CONTRACTOR SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

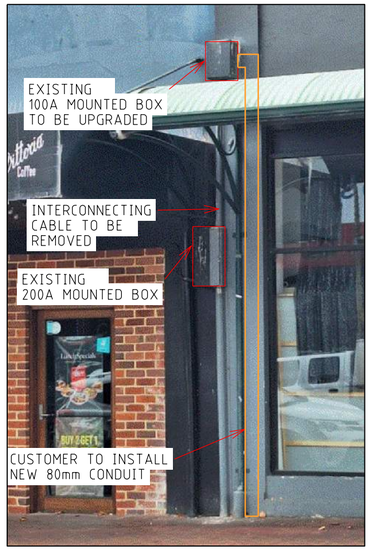
G. CUSTOMER ELECTRICAL CONTRACTOR TO PROVIDE SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

H. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

I. WESTERN POWER WILL REQUIRE OUTGOING RETAILER AGREEMENT (VIA SUBMISSION OF AN ETAF) PRIOR TO SCHEDULING OF THE PROJECT WORKS. CUSTOMER REQUESTED TO CONTACT THE OUTGOING RETAILER AND PROVIDE RELEVANT PROJECT DETAILS TO FACILITATE ETAF. CUSTOMER TO CONTACT WP PNO PRIOR TO COMMENCEMENT OF METERING CHANGES.

J. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF WITHIN 30m OF THE POINT OF SUPPLY

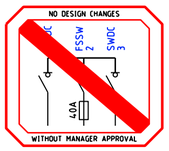
K. CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR WALL MOUNTED BOX 24/7 ACCESS.



PILLAR LOCATION

CRITICAL RISK MANAGEMENT

■	■	□	■	□	■	□	□	□



WATER PIPELINES
IN VICINITY OF WORKS

COMMUNICATION CABLES
IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: JEDCO ELECTRICAL & COMMUNICATION PTY LTD
CONTACT: SIMON JEDRZEIEWSKI TEL: 0409589956

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	03/26	HAZARD MANAGEMENT REGISTER

150A COMMERCIAL SUPPLY UPGRADE

274-276 ALBANY HWY

VICTORIA PARK WA 6100

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS COMM SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 03/26 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31°58'18.9"S Long. - 115°53'41.0"E

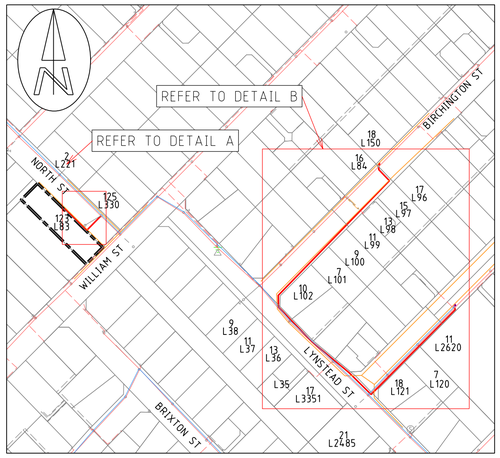
westernpower

DRAWING NUMBER

MP218916

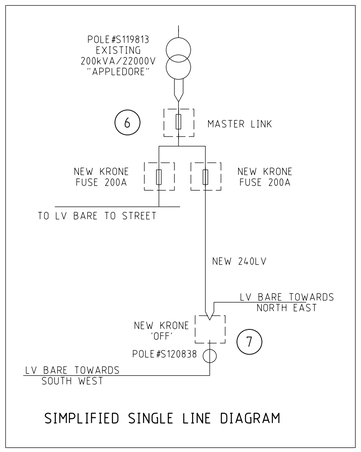
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2

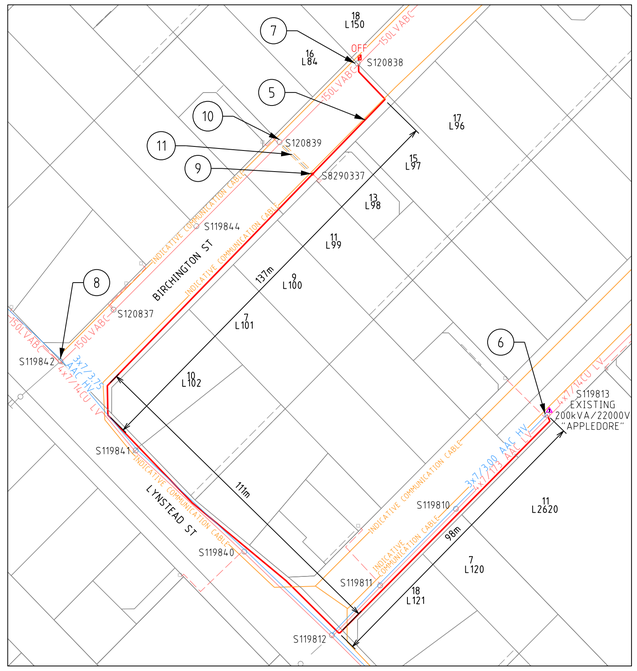


LOCATION PLAN

1:1000 20 10 0 20 40 60 80

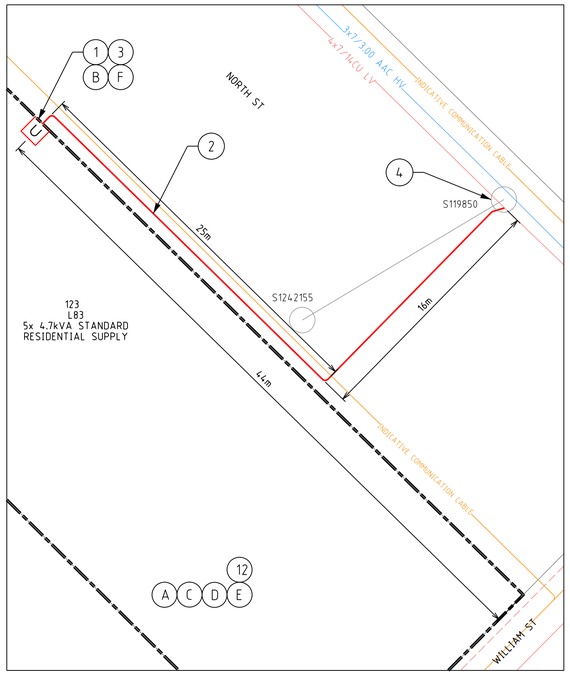


SIMPLIFIED SINGLE LINE DIAGRAM



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L83/#123 WILLIAM ST. WHERE PEGGED ON SITE BY CUSTOMER AS PER PILLAR ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3). REFER TO DETAIL B.
   CU: LU11 x 1
   DCSH: U9
2. INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S119850. REFER TO DETAIL B.
   CU: CN45x60m, CN54x3, CN73_LVMAINSx3
3. AT NEW UNI-PILLAR (L83/#123 WILLIAM ST).
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   CU: LU7_240x1
4. AT POLE#S119850.
   - TAP ONTO LV BARE AND TERMINATE NEW 240LV CABLE
   - REMOVE EXISTING AERIAL SERVICE TO L5/#56
   CU: LU6_240 x 1, LU6_240U x 1
   DCSH: U16-1
5. INSTALL NEW 240LV CABLE FROM POLE#S119813 TO THE POLE#S120838. REFER TO DETAIL B.
   CU: CN45x370m, CN54x6, CN73_LVMAINSx3
6. AT POLE #S119813.
   -TAP EXISTING LV BARE VIA TX BUSHING WITH 2xNEW 200A KRONE FUSES.
   -TERMINATE NEW 240LV CABLE VIA NEW 200A KRONE FUSES.
   REFER TO SIMPLIFIED LINE DIAGRAM.
   CU: LU37_2400x1, LU37_240U x1, CN39_150/4M x2
   NON-CU: QF1096 x6 (200A FUSES), EB020442, QS0111x2
   DCSH: U17
7. AT POLE #S120838.
   - REPLACE LVABC INTERMEDIATE TO LVABC STRAIN WITH ISOLATION 'OFF'
   - TERMINATE 240LV CABLE TO NEW KRONE BOX AS INDICATED IN SIMPLIFIED LINE DIAGRAM.
   CU: AB04x1, LU37_2400x1, LU37_240U x1, CN39_150/4M x2
   NON-CU: EB020441, QS0111x1
   DCSH: A04-1, U17
8. AT POLE#S119842.
   RE-TAP EXISTING AERIAL SERVICES TOWARDS NORTHWEST.
   CU: CN75x2, CN65_150/35x2
9. TEE JOINT EXISTING 25LV TO NEW 240LV CABLE. REMOVE EXCESS CABLE.
   CU: LU5x1
10. AT POLE#S120839, DISCONNECT AND REMOVE EXISTING 25LV CABLE.
11. REMOVE EXISTING 25LV CABLE FROM CONDUIT.
12. ENVIRONMENTAL FILTER CHECKED.
    NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
    ACID SULFATE SOILS
    ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
    NATIVE VEGETATION REGS ESAS CLEARING
    UNEXPLODED ORDNANCE
    CARNABYS COCKATOO CONFIRMED BREEDING AREAS (DBCA-054)
    LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
    REGION SCHEME - ZONES AND RESERVES (DPLH-023)
    INDIGENOUS LAND USE AGREEMENTS (ILUAS)
    NATIVE TITLE BOUNDARIES
    RIWI ACT, GROUNDWATER AREAS (DWER-034)
    PRE-EUROPEAN VEGETATION (DPIRD-006)

    REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

G. THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION

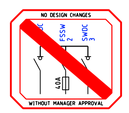
CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

WATER PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS



W.A.P.C. REFERENCE No. 392-23

CUSTOMER CONTACT DETAILS
COMPANY NAME:
CONTACT: ANTHONY BUNTIC TEL: 0414.619101

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	02/26
	HAZARD MANAGEMENT REGISTER		
	SHE ASSESSMENT REPORT		

REFERENCE FILE DETAILS

5x DADMD (4.7kVA) STANDARD RESIDENTIAL SUPPLY

L83/#123 WILLIAM ST

BECKENHAM WA 6107

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 02/26 SCALE: 1:1000 @ A1
GEO REF: Lat. - 32°01'03.2" S Long. - 115°57'35.9" E

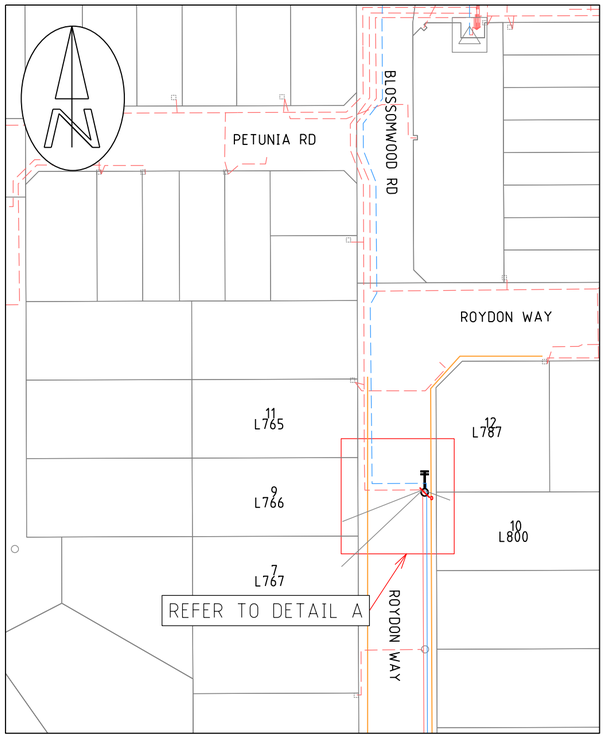
westernpower

DRAWING NUMBER

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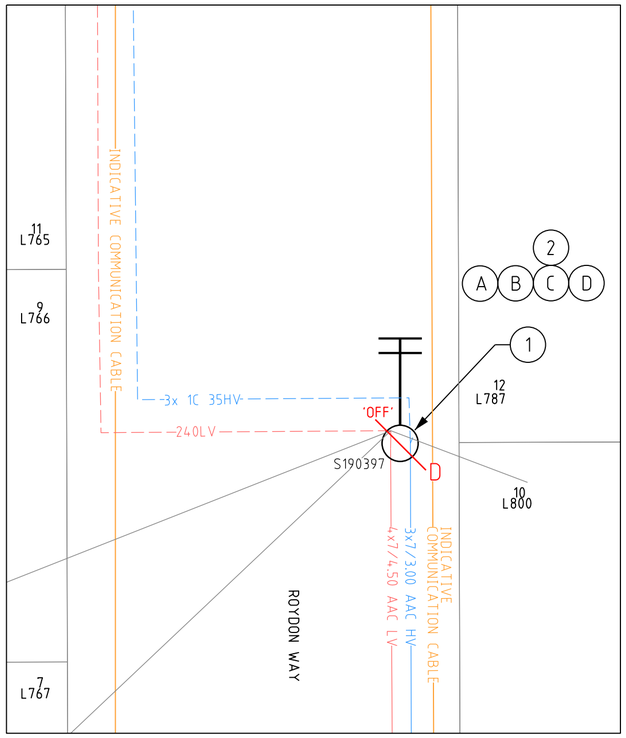
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:1000



DETAIL A

1:200

WESTERN POWER SCOPE OF WORK

1. AT POLE#S190397.
   - DISCONNECT EXISTING HV CABLE, LV CABLE AND AERIAL SERVICES.
   - REPLACE THE POLE WITH 12.5m/8kN SSP2 POLE.
   - INSTALL NEW LV TERMINATION CROSSARM.
   - INSTALL NEW HV TERMINATION CROSSARM.
   - RE-TERMINATE EXISTING HV CABLE TO HV BARE.
   - RE-TERMINATE EXISTING LV CABLE TO LV BARE VIA NEW ISOLATOR 'OFF'.
   - INSTALL NEW OUTRIGGER STAY
   - REATTACH/ REPLACE EXISTING AERIAL SERVICES.

   CU: P018_12.5Mx1, LV05_W3x 1, LV6_600Ax1, LU9_2400x1, LU9_240Ux1, CN75x3, HV5_3MNx1, HU19_Ox1, HU19_U_22x1, ST4_19/275Wx1, CN9_7/475AAx7

   DCSH: L05, L06, U15, H04-1, H09-1, R13-2, R13-3 R16-3

2. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   PUBLIC DRINKING WATER SOURCE AREAS
   PUBLIC DRINKING WATER SOURCE AREAS (DWER-033)
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

(C) CUSTOMER IS RESPONSIBLE FOR COMPLETING THEIR DUE DILIGENCE ASSESSMENT OF POTENTIAL ABORIGINAL HERITAGE IMPACTS AND ADVISING WESTERN POWER OF ANY COMPLIANCE REQUIREMENTS THAT APPLY.

(D) CUSTOMER TO ENSURE MINIMUM 1M SEPARATION IS MAINTAINED BETWEEN EDGE OF PROPOSED DRIVEWAY AND NEW STAY ANCHOR.

LEGEND

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT

- INTERMEDIATE
- DOF
- INLINE STRAIN
- PTT
- AERIAL STAY
- RECLOSER
- LOAD BREAK SWITCH
- CAPACITOR
- LBPTS
- PTS
- ISOL HV ISOLATOR
- LV DISCONNECTOR
- ISOL TX
- OUTRIGGER STAY
- GROUND STAY
- SECTIONALISER
- LV ABC KRONE
- STREETLIGHT BOX

POLE SCHEDULE

POLE ID	POLE DETAILS	INSTRUCTIONS	POLE LOCATION		STRUCTURE MATERIALS / CU'S	DCSH REFERENCE	AS CON CHECK
			EASTING	NORTHING			
S190397	12.5m/8kN SSP2 POLE	-	EXISTING	EXISTING	P018_12.5M	R13-2, R13-3	-

COMMUNICATION CABLES IN VICINITY OF WORKS

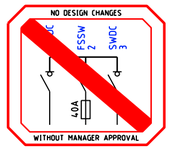
CRITICAL WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



CUSTOMER CONTACT DETAILS
COMPANY NAME:
CONTACT: SHANE HOLLAND TEL: 0487049700

CONSULTANT CONTACT DETAILS
COMPANY NAME: DEVPLEX PTY LTD
CONTACT: FRANK FERRALORO TEL: 0411831183

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	02/26	HAZARD MANAGEMENT REGISTER

STAY REPLACEMENT

L787/#12 ROYDON WAY

GIRRAWHEEN WA 6064

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: Ishan Palikhe
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87

DATE: 02/26 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31°50'39.1" S
Long. - 115°50'37.2" E

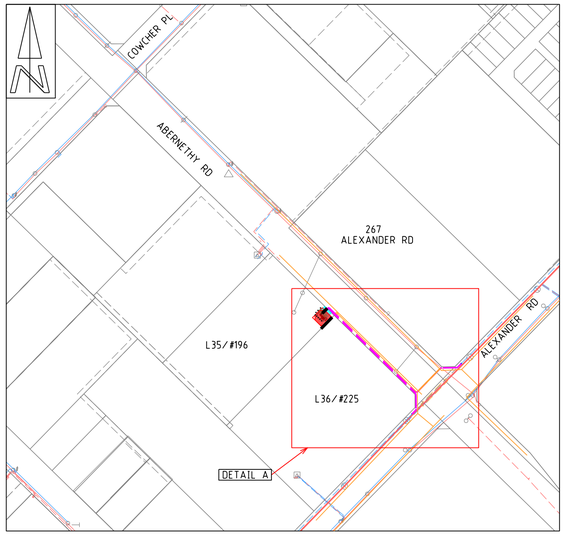
westernpower

DRAWING NUMBER

MP219407

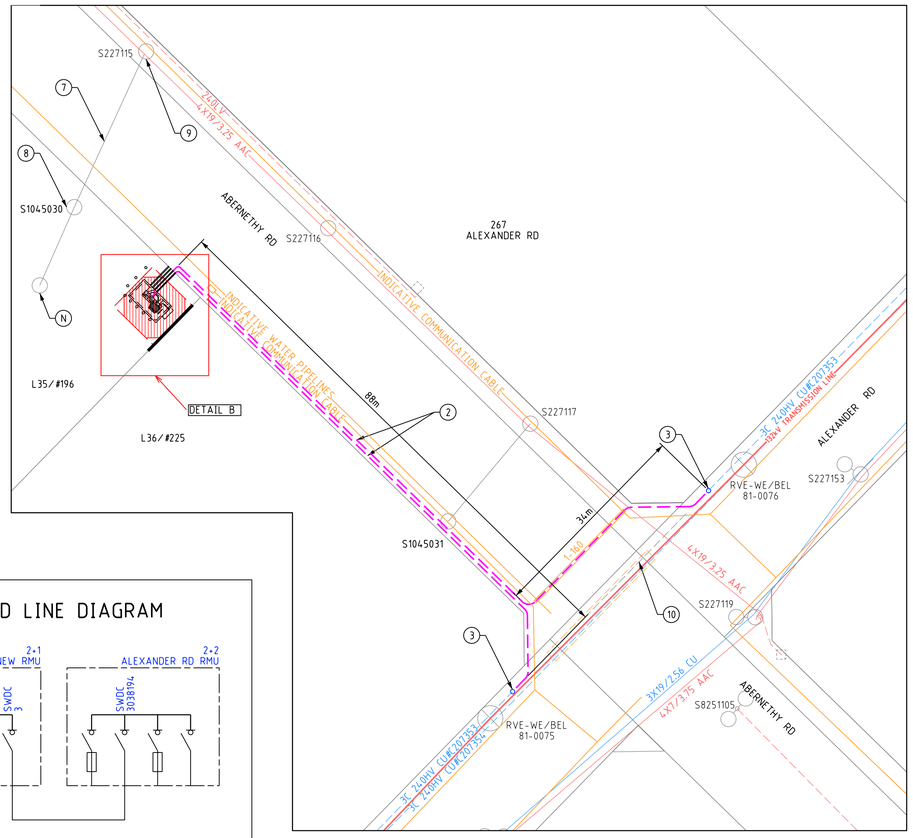
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



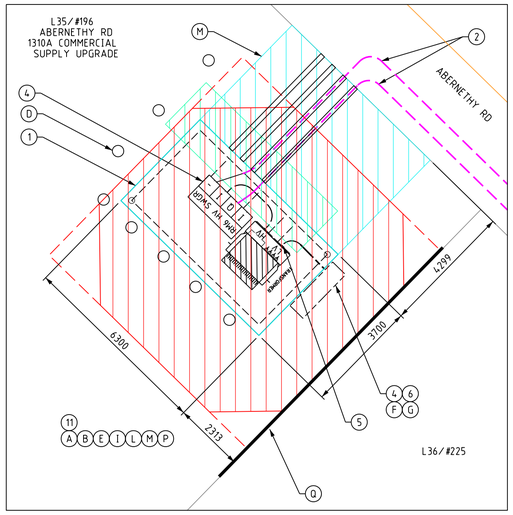
LOCATION PLAN

1:2000 — 40 20 0 40 80 120 160



1:500 — 10 5 0 10 20 30 40

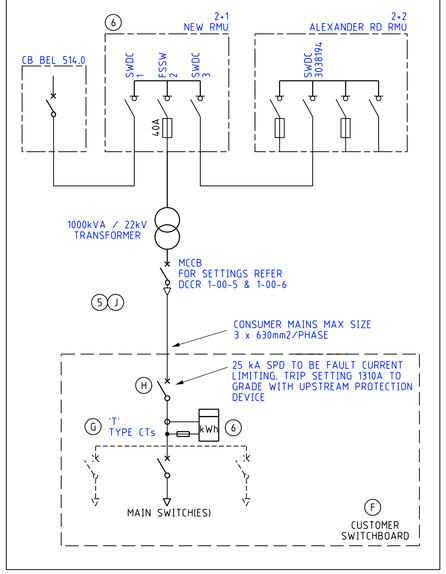
DETAIL A - CABLE ROUTE DETAIL



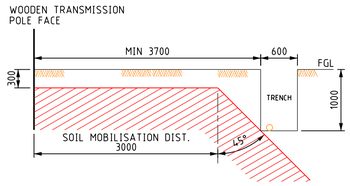
DETAIL B - SUBSTATION DETAIL

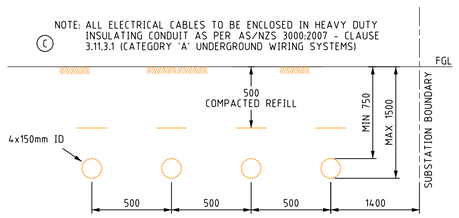
1:100 — 2 1 0 2 4 6 8

SIMPLIFIED LINE DIAGRAM



STANDARD EXCAVATION LIMITS IN PROXIMITY TO WOODEN TRANSMISSION POLE





A CUSTOMER INSTALLED CONDUITS
SECTION DETAIL (NTS)

TRANSMISSION LINE IN THE VICINITY OF WORKS

i ALL MOBILE PLANT IN THE PROXIMITY TO THE LINE SHALL OPERATE AT A MINIMUM APPROACH DISTANCE THAT COMPLIES WITH WESTERN POWER WORKS PRACTICES AND REQUIREMENTS.

ii THE CONTRACTOR SHALL COMPLY WITH WA OCCUPATIONAL SAFETY & HEALTH REGULATION 3.64 FOR THE RESTRICTION ON WORKING IN THE VICINITY OF OVERHEAD POWER LINES.

iii THE CONTRACTOR SHALL ADHERE TO REQUIREMENTS OF 'EXCAVATION IN PROXIMITY TO OVERHEAD LINE POLES AND STEEL LATTICE TOWERS' GUIDELINE. REFER EDM#34165792.

WESTERN POWER SCOPE OF WORK

1. INSTALL 1000kVA / 22kV SOLE USE SUBSTATION TO DSM-3-14 AS NOTED ON THE DRAWING WITH 2+1 HV SWITCHGEAR.
   INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER.
   INSTALL PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5, AND DRAWING DSPM-4-06 (NON-MPS).
   CU: HU59__100022x1, HU66__22x1, HU76x1, HU6__A400x1, HU55__1000/22, HU69x2, CN46x15
   DSPM: DSPM-3-14, DSPM-4-06
2. INSTALL 2x400 HV CABLES THROUGH CUSTOMER INSTALLED CONDUIT FROM NEW HV SWITCHGEAR TO EXISTING 400HV CABLE ON ALEXANDER RD.
   CU: CN53x245m, CN54x8, CN73x6
3. CUT INTO EXISTING 240 SQ MM 3C CU 22kV CABLE#C207353 AND STRAIGHT JOINT TO NEW 400HV CABLES.
   CU: HU29x2
4. TERMINATE 2x 400HV CABLE AND 1x 35HV CABLE TO NEW HV SWITCHGEAR AS PER SIMPLIFIED LINE DIAGRAM.
5. TERMINATE CUSTOMER'S CABLE TO TX MCCB IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
6. INSTALL CT METER IN CUSTOMERS SWITCHBOARD.
   CU: LM1x1
7. REMOVE BARE LV CONDUCTOR BETWEEN POLE#S1045030 TO POLE#S227115.
8. REMOVE POLE#S1045030.
9. AT POLE#S227115:
   - REMOVE LV TERMINATION CROSS ARM.
   - REMOVE TAPS.
   - REMOVE OUTRIGGER STAY.
10. REMOVE 240 SQ MM 3C CU 22KV FROM ROAD RESERVE IF REASONABLY PRACTICAL TO DO SO. ELSE CAP AND DECOMISSION. ABANDON CONDUIT.
    CU: CN73_240x3
11. ENVIRONMENTAL FILTER CHECKED
    NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE VICINITY OF WORKS:

    ACID SULFATE SOILS
    ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
    CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
    LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
    REGION SCHEME - ZONES AND RESERVES (DPLH-023)
    INDIGENOUS LAND USE AGREEMENTS (ILUAS)
    NATIVE TITLE BOUNDARIES
    RIWI ACT, GROUNDWATER AREAS (DWER-034)
    PRE-EUROPEAN VEGETATION (DPIRD-006)
    CONTAMINATED SOILS - REMEDIATED FOR RESTRICTED USE - 267 ALEXANDER RD
    CONCENTRATIONS EXCEEDING ECOLOGICAL INVESTIGATION LEVELS, BUT BELOW HEALTH-BASED INVESTIGATION LEVELS FOR COMMERCIAL AND INDUSTRIAL SITES, AS PUBLISHED IN 'ASSESSMENT LEVELS FOR SOIL, SEDIMENT AND WATER' (DEPARTMENT OF ENVIRONMENT, 2003).

    LOW RISK - REFER TO EDM# 68761355 & 68760274

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:
   1. PEGGING, COMPACTION AND LEVELING.
   2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
   3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
   4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

C. CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

D. CUSTOMER TO INSTALL NON-CONDUCTIVE BOLLARDS ADJACENT TO THE SUBSTATION SITE AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6.

E. CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

F. CUSTOMER MUST INSTALL MSB CONTIGUOUS WITH SUBSTATION.

G. CUSTOMER TO SUPPLY AND INSTALL 'T' TYPE METERING CT's.

H. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

I. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

J. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

K. CUSTOMER TO PROVIDE RETAILER CONSENT TO METERING CHANGES. PROJECT WILL NOT BE SCHEDULED UNTIL THE ETAF IS RECEIVED BY WESTERN POWER.

L. CUSTOMER TO PROVIDE A 129BA RESTRICTIVE COVENANT BURDENING THE LOT/S AROUND THE SUBSTATION SITE FOR FIRE PROTECTION CLEARANCE REQUIREMENTS TO FUTURE BUILDINGS AS PER UNDERGROUND DISTRIBUTION SCHEMES MANUAL CLAUSE 5.3.18.1. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS.

M. CUSTOMER TO ARRANGE FOR AN 'EASEMENT IN GROSS' UNDER THE ENERGY OPERATORS POWERS ACT 1979 COVERING THE ENTIRE SUBSTATION SITE AND CABLE ROUTE WITHIN CUSTOMER PROPERTY. CUSTOMER TO SUBMIT THE DEPOSITED PLAN TO WESTERN POWER PRIOR TO THE COMMENCEMENT OF WORKS. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS

N. CUSTOMER RESPONSIBLE TO REMOVE CUSTOMERS POLE AFTER COMPLETION OF WORK.

P. THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

Q. THE CUSTOMER IS RESPONSIBLE TO CONSTRUCT 2HR FIRE RATED WALL / FENCE IN ACCORDANCE WITH DSMP SECTION 5 ON SHARED PROPERTY BOUNDARY TO AVOID FIRE RESTRICTIVE COVENANT IMPACTING NEIGHBOURING PROPERTY. IT IS THE CUSTOMER'S RESPONSIBILITY TO ENSURE THAT THE WALL FOOTING IS CONSTRUCTED TO ALLOW SAFE EXCAVATION ANYWHERE WITHIN THE SUBSTATION SITE TO A DEPTH OF 1.5m AND DOES NOT ENCROACH INTO THE WESTERN POWER SUBSTATION SITE. THE FIRE WALL SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS / EXISTING
- POLE BRACKET / POLE BRACKET
- SINGLE OUTREACH / SINGLE OUTREACH
- DOUBLE OUTREACH / DOUBLE OUTREACH
- 4-WAY / 4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINIRE DETAILS.

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SLD Serviced Unclearad De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	□	☒	☒

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINE IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: COSSILL & WEBLEY
CONTACT: JEY SHIVAKUMAR TEL: 0403 314 136

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE ASSESSMENT REPORT
1	ISSUED FOR CONSTRUCTION	I.P.	03/26	HAZARD MANAGEMENT REGISTER

1310A COMMERCIAL SUPPLY UPGRADE

L35/#196 ABERNETHY RD,

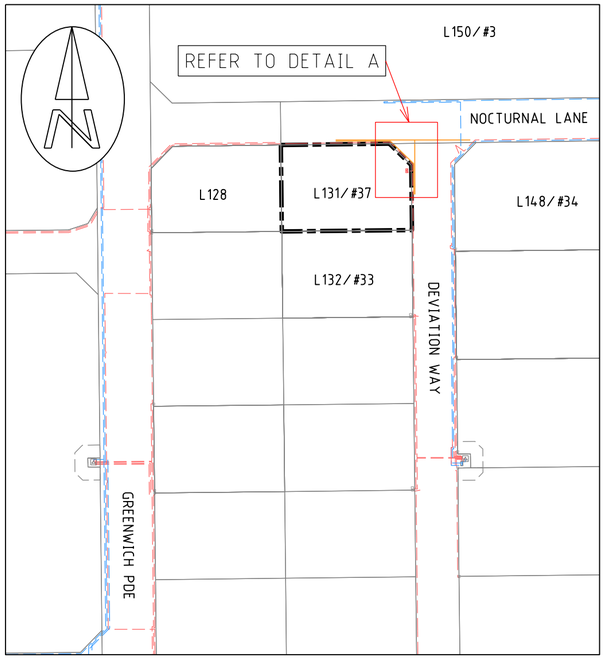
BELMONT, WA, 6104

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS COMM.SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 03/26 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31°57'28.4"S Long. - 115°55'50.4"E

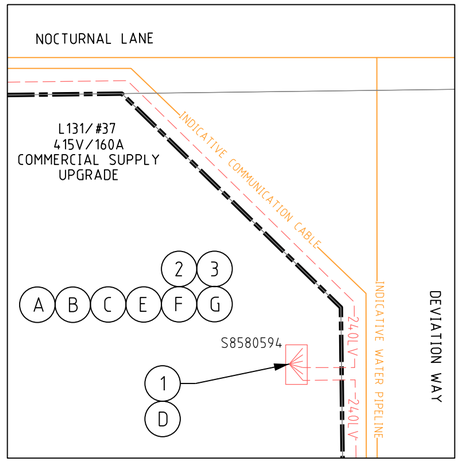
westernpower

DRAWING NUMBER
MP219426
SHEET 01 OF 01
REV: 1 ORIGINAL SIZE - A1



LOCATION PLAN

1:2000 — 40 20 0 40 80 120 160



DETAIL A

1:200 — 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. AT LV FRAME (L131 DEVIATION WAY)
   - TERMINATE CUSTOMER CABLE VIA 200A FUSE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   NON CU: GF1096x3
   DCSH: U27-1
2. INSTALL CT METER IN CUSTOMER'S SWITCHBOARD.
   CU: LM1_M x 1
3. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   THREATENED ECOLOGICAL COMMUNITIES
   PRE-EUROPEAN VEGETATION (DPIRD-006)
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, GROUNDWATER AREAS (DWER-034)

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

- (A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.
- (B) CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.
- (C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.
- (D) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.
- (E) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.
- (F) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.
- (G) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

LEGEND

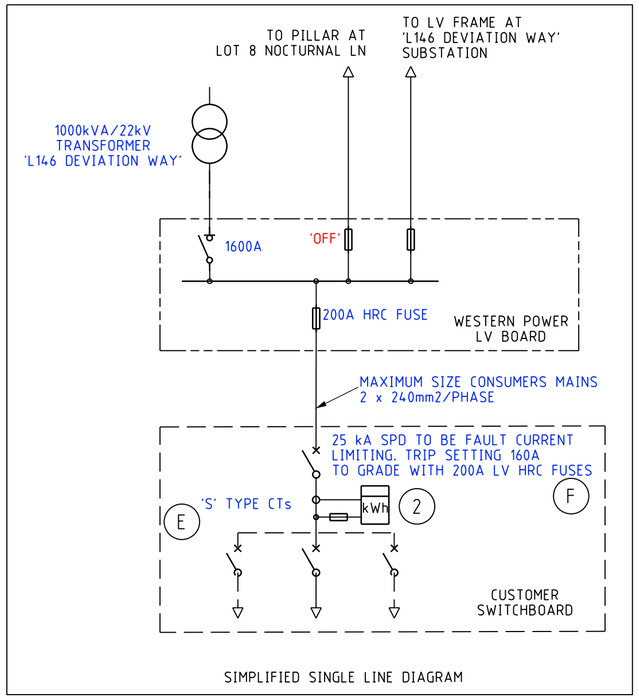
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)



SIMPLIFIED SINGLE LINE DIAGRAM

COMMUNICATION CABLES IN VICINITY OF WORKS

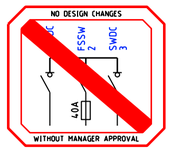
WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☐	☐	☒	☐	☐	☐	☐	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



CONSULTANT CONTACT DETAILS
COMPANY NAME: ECWA
CONTACT: MICHAEL ANILE TEL: 0863235838

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: MATTHEW JAMES TEL: 0439988739

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	02/26	HAZARD MANAGEMENT REGISTER

160A COMMERCIAL SUPPLY UPGRADE

L131/#37 DEVIATION WY

NEERABUP, WA 6031

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS COMM SVM - RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87

DATE: 02/26 | SCALE: AS NOTED @ A2

GEO REF: Lat. - 31°40'42.0" S
Long. - 115°46'55.0" E

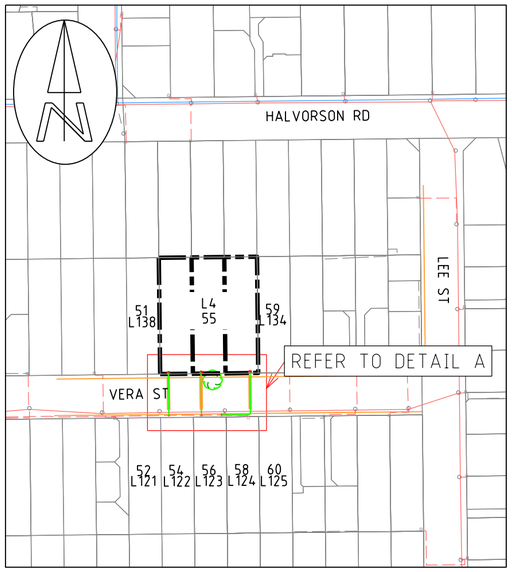
westernpower

DRAWING NUMBER

MP219499

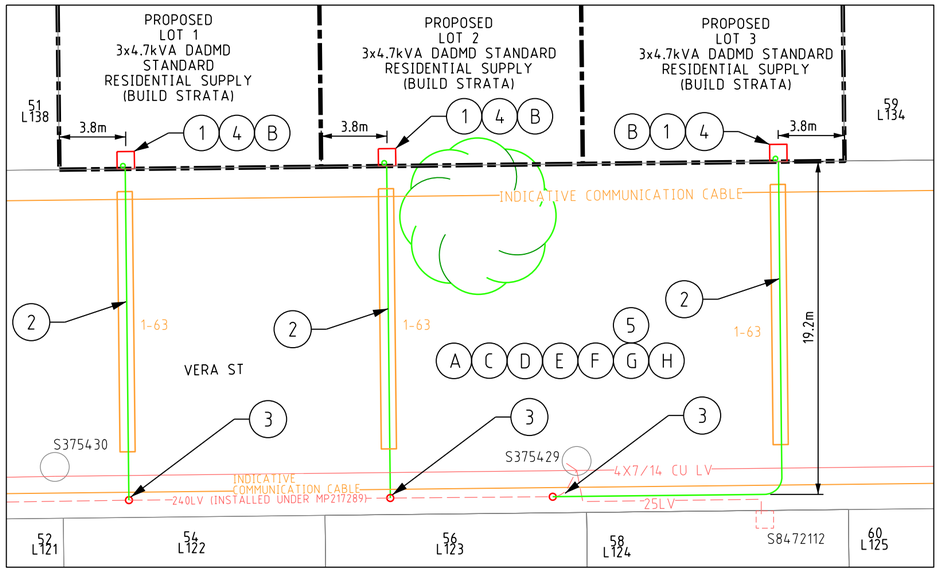
SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
CN53 - 400mm2 (3X1c)
CN49 - 240mm2 (3X1c)
CN48 - 185mm2 (3X1c)
CN50 - 95mm2 (3X1c)
CN51 - 50mm2 (3X1c)
CN51 - 50mm2 (1c)
CN46 - 35mm2 (3X1c)
CN47 - 35mm2 (1c)
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

W.A.P.C. REFERENCE No. 202128

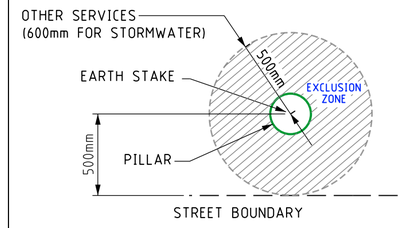
CONSULTANT CONTACT DETAILS
COMPANY NAME: TSA SURVEYS PTY LTD
CONTACT: KELLY SKINNER TEL: 0436397522

ISSUED FOR CONSTRUCTION

COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

PILLAR EXCLUSION ZONE
NTS

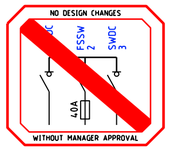


NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☒	☐	☐	☒	☐



TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
THESE WORKS ARE DEPENDENT ON THE COMPLETION OF MP217289 AND MM012039.

(1) INSTALL NEW MINI-PILLAR ON PROPOSED LOT1, LOT 2 AND LOT 3 WHERE PEGGED ON SITE BY CUSTOMER AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU10x3
DCSH: U08-04

(2) INSTALL 25LV CABLE FROM NEW MINI-PILLAR TO EXISTING 240LV CABLES INSTALLED UNDER MP217289.
CU: CN42x100m, CN73_LV25x9, CN54x6

(3) CUT EXISTING 240LV CABLE INSTALLED UNDER MP217289 AND TEE-JOINT 25LV CABLE WITH EXISITNG 240LV CABLE
CU: LU5x3

(4) AT MINI-PILLAR (PROPOSED LOT 1, LOT 2 AND LOT 3)
- TERMINATE NEW 25LV CABLE.
DCSH: U8-04

(5) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
ACID SULFATE SOILS
ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED
CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
REGION SCHEME - ZONES AND RESERVES (DPLH-023)
INDIGENOUS LAND USE AGREEMENTS (ILUAS)
NATIVE TITLE BOUNDARIES
RIWI ACT, GROUNDWATER AREAS (DWER-034)
PRE-EUROPEAN VEGETATION (DPIRD-006)

REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS AND UNDERGROUND DISTRIBUTION SCHEMES MANUAL.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(E) CUSTOMER TO SUPPLY AND INSTALL A STRATA DEVELOPMENT SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(F) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

(G) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(H) SHOULD ANY VEGETATION CLEARING BE REQUIRED TO COMPLETE THE SCOPE OF WORKS. CUSTOMER SHALL CLEAR AS REQUIRED AND OBTAIN RELEVANT APPROVALS.

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	04/26	HAZARD MANAGEMENT REGISTER

3 x GREEN TITLE (9x4.7KVA DADMD) SUBDIVISION

L4 #55 VERA ST

MORLEY WA 6062

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 04/26 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31°53'28.3"S Long. - 115°53'53.4"E

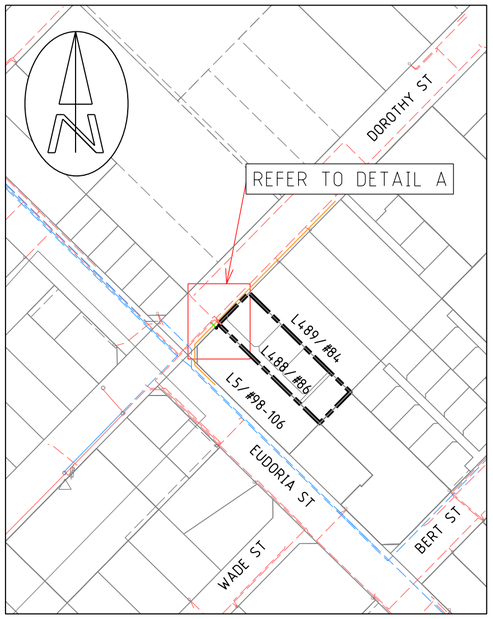
westernpower

DRAWING NUMBER

MP219649

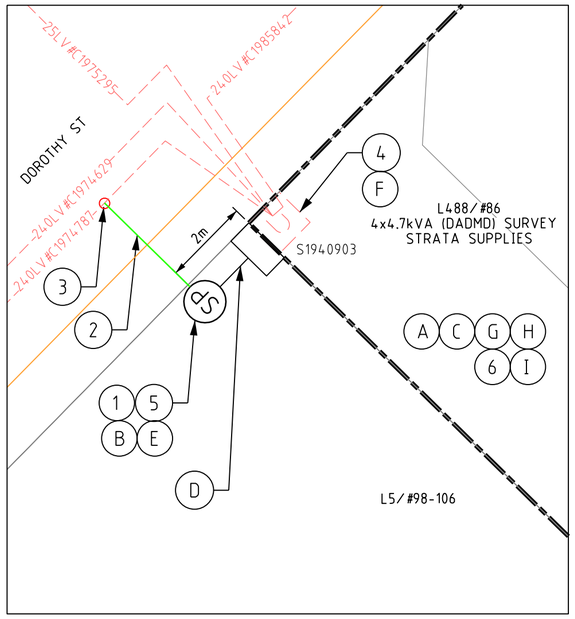
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

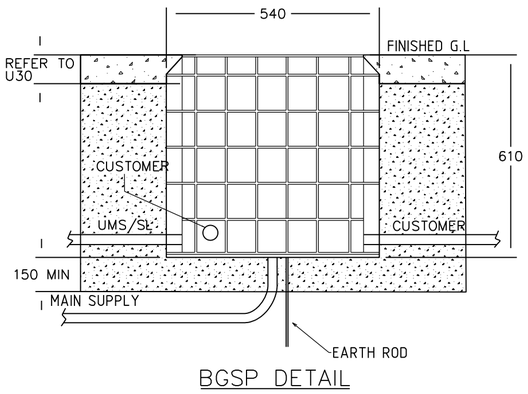
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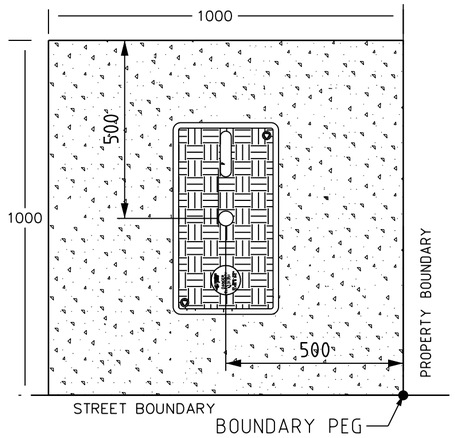
DETAIL A

1:200 4 2 0 4 8 12 16

CONSUMER MAINS TO BE CONNECTED BY WESTERN POWER ONLY.



BGSP DETAIL



TYPICAL BGSP EXCLUSION ZONE

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
- NEW BGSP WILL HAVE 5x DWELLINGS CONNECTED TO BACKBLOCK.
- EXISTING UNIPILLAR#S1940903 WILL HAVE 4x DWELLINGS CONNECTED TO BACKBLOCK.

(1) INSTALL NEW BELOW GROUND SERVICE PIT AND N25 (MINIMUM) GREY CONCRETE OVER THE EXCLUSION ZONE WHERE PEGGED BY CUSTOMER AT L5/#98-106 DOROTHY ST.
CU: LU52 x 1
DCSH: U30

(2) INSTALL 25LV CABLE FROM NEW BGSP TO EXISTING 240LV CABLE.
CU: CN42 x 15m, CN73_LV25 x 3, CN54 x 2

(3) TEE-JOINT NEW 25LV CABLE WITH EXISTING 240LV CABLE#C1974787
CU: LU5 x 1

(4) AT EXISTING UNI-PILLAR#S1940903
-DISCONNECT CONSUMER MAINS CABLE TO NEIGHBOURING PROPERTY #98-106 FROM BACK BLOCKS

(5) AT NEW BELOW GROUND SERVICE PIT.
-TERMINATE NEW 25LV CABLE.
-TERMINATE CONSUMER CABLE FROM L5/#98-106 DOROTHY ST IN CONJUNCTION WITH CUSTOMER ELECTRICAL CONTRACTOR.
DCSH: U30

(6) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
ACID SULFATE SOILS
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID. IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED
UNEXPLODED ORDNANCE
ACID SULFATE SOIL RISK MAP, SWAN COASTAL PLAIN (DWER-055)
LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
REGION SCHEME - ZONES AND RESERVES (DPLH-023)
RIWI ACT, GROUNDWATER AREAS (DWER-034)
PRE-EUROPEAN VEGETATION (DPIRD-006)

REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS AND UNDERGROUND DISTRIBUTION SCHEME MANUAL.

(B) BELOW GROUND SERVICE PIT EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE BGSP LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. CLEAR OF ALL SERVICES.

3. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO INSTALL NEW CONSUMER MAINS FROM CUSTOMER'S SWITCHBOARD AT L5/#98-106 DOROTHY ST TO NEW BGSP.

(E) CUSTOMER ELECTRICAL CONTRACTOR TO BE AVAILABLE ON SITE WHEN WESTERN POWER CARRIES OUT TERMINATION OF CONSUMER MAINS CABLE TO THE BGSP.

(F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(G) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

(H) THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGIZED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME.

(I) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL CONTRACTOR SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

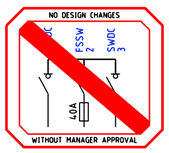
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUN Serviced Uncleared De-energised Lot

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☐	☐	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



COMMUNICATION CABLES IN VICINITY OF WORKS

W.A.P.C. REFERENCE No. 958-25

CUSTOMER CONTACT DETAILS
COMPANY NAME: JAGHORI GROUP PTY LTD
CONTACT: ALI JAFFARY TEL: 0408917244

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	06/26	HAZARD MANAGEMENT REGISTER

4x 4.7kVA (DADMD)SURVEY STRATA SUPPLY

86 DOROTHY ST

GOSNELLS WA 6110

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 06/26 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32°04'40.4" S Long. - 115°59'44.1" E

westernpower

DRAWING NUMBER

MP219698

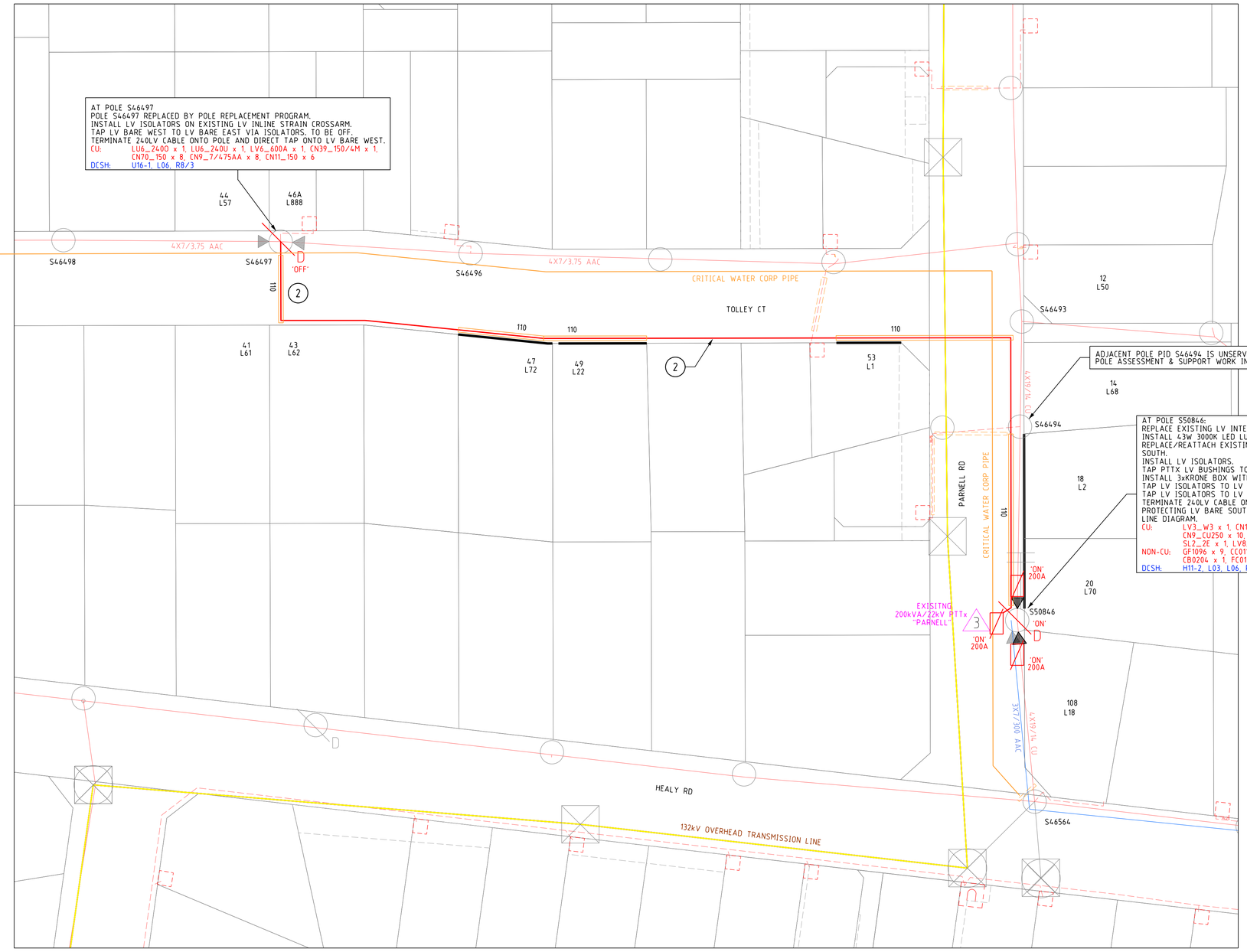
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2

SAFETY - MAKE IT YOUR PRIMARY VALUE

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL



CONSTRUCTION DETAIL

1:500 (10 5 0 10 20 30 40)

SCOPE OF WORK (THIS SHEET ONLY)

1. REFER TO INDIVIDUAL POLES FOR CONSTRUCTION NOTES.
2. INSTALL 240LV CABLE FROM POLE S50846 TO POLE S46497 AS PER CONSTRUCTION DETAIL.
   CU: CN45 x 275m, LU2 x 1, CN73_LVMAINS x 4, CN54 x 12

ALL MOBILE PLANT IN PROXIMITY TO THE TRANSMISSION LINE SHALL OPERATE AND A MINIMUM APPROACH DISTANCE THAT COMPLIES WITH WESTERN POWER'S PRACTICES AND REQUIREMENTS.

LEGEND

SCHEME BDY.
RETAINING WALL

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

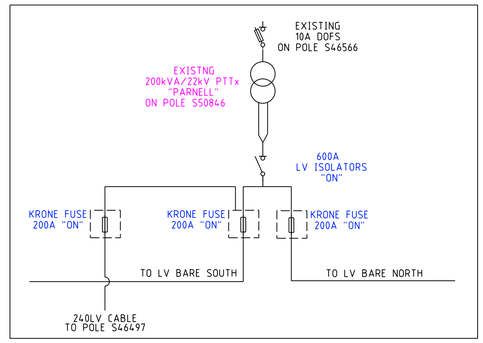
STREET LIGHTS — EXISTING
- 42w CFL
- 80w MV
- 125w MV
- 70w HPS
- 150w HPS
- 250w HPS
- 70/75w MH
- 150w MH
- 250w MH

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

TRANSFORMER — EXISTING — SWITCHGEAR

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



SINGLE LINE DIAGRAM - PARNELL

POLE SCHEDULE (THIS SHEET ONLY)						
POLE ID	POLE DETAILS	POLE LOCATION		POLE UNSERVICEABLE?	ASCON CHECK	WIND REGION
		EASTING	NORTHING			
S46497	EXISTING	EXISTING	EXISTING	NO		AM
S50846	EXISTING	EXISTING	EXISTING	NO		AM

DESIGN NOTES

THE OBJECTIVE OF THESE WORKS IS TO EXPAND OR RECONFIGURE THE NETWORK TO INCREASE ITS OVERALL CAPACITY. THE PRIMARY DRIVER IS TO INCREASE THE RATING OR CAPACITY OF THE NETWORK TO MEET THE NEEDS OF CURRENT AND FUTURE CUSTOMERS. THE SCOPE OF WORKS MAY INCLUDE ADDITIONAL WORKS TO SUPPORT THIS OVERALL OBJECTIVE OR TO EFFICIENTLY DELIVER WORK IN PROXIMITY.

FOR QUERIES REGARDING THE DESIGN SCOPE CONTACT THE PROJECT MANAGER

THIS DRAWING TO BE READ IN CONJUNCTION WITH SCOPE OF SERVICES DOCUMENT EDM# 64295918

APPROVED FOR CONSTRUCTION

CRITICAL WATER CORP PIPES IN VICINITY OF WORKS

132kVA TRANSMISSION LINES IN VICINITY OF WORKS

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SCOPE OF SERVICES EDM: #64295918
A	APPROVED FOR CONSTRUCTION	R.P.	07/23	HAZARD MANAGEMENT REGISTER EDM#64198804

N0551596 - RESPONSIVE PACKAGE 91

NETWORK VOLTAGE: 22kV/415V

TOLLEY ST & PARNELL RD, HAMILTON HILL WA 6163

DESIGN DRAWING

PROJECT MANAGER: MARTIN FORMATO
CONTACT: 9326 6173
LEAD DESIGNER: DINITH JAYASUNDERA
DESIGNER: ROBERT PEACOCK
DESIGN TEAM: E&D RD GHD

DATE: 07/23 | SCALE: 1:1000 @ A1

GEO REF: Lat. - 32°04'38" S, Long. - 115°46'22" E

westernpower

DRAWING NUMBER

MQ045765

SHEET 01 OF 01

REV: A | ORIGINAL SIZE - A1

Appendix to Design Drawing

Scope of Services for MQ045765

General

This scope of Services* summarises the underground and overhead service requirements for Hamilton Hill, serviced by the South West Interconnected System (SWIS*), in the Perth metropolitan area. This project includes for the installation of 275m of 240LV cable and pole replacement necessary for LV network reconfiguration.

Maintenance Zone

APM/Z014

Disclaimer

All work shall be undertaken in accordance with Western Power's Work Instructions, Guidelines and Procedures, Network Standards (e.g. Distribution Construction Standards Handbook) and Western Power's Safety, Health and Environment (SEQT) requirements.

Important Project Documents

Description	EDM#	Guardian Permit ID
Hazard Management Register	64198804	N/A
Environmental Filter	60966347	N/A
Scope of Work (Basis of Design)	60969973	N/A

Preliminaries (Specific Notes)

Pole S46497 is replaced under the pole replacement program. Refer to Ellipse Project number N0588480.

CM to ensure all extra materials are returned to depot.

Constructor is responsible for removal and reinstatement.

CM to ensure all excavations are done as per WP Work Instructions, Guidelines and Procedures.

High value communications cable, critical water pipe, and critical sewer main in the vicinity of works. Refer to BYDA drawings for location information, specific control measures and relevant utility contacts.

Carry out necessary switching to isolate the network prior to commencement of work.

Poles and Conductors

Replace aerial service / overhead customer service connection (OCSC) if PVC insulated or TWISTY type or too short to reconnect (if applicable), reconnect all others.

All poles, conductors and associated hardware marked for removal are to be completely removed from site inclusive of pole butts.

Foundations

Pole foundations have been designed for "Good" soil conditions. If it is found during construction that foundations are "Medium", Construction must follow the "Enhanced" foundation details in R13.

Cable Installation

For clarity not all underground services are shown. Contractor must contact Dial Before You Dig prior to commencement of work.

No cables to be installed in 2.7m alignment unless specified by Designer / Project Manager. Locations of cable joints are indicative only, exact location to be determined on site.

Footpaths and roadways/road reserves affected by cable installation are to be reinstated.

Work in Vicinity of Transmission Line

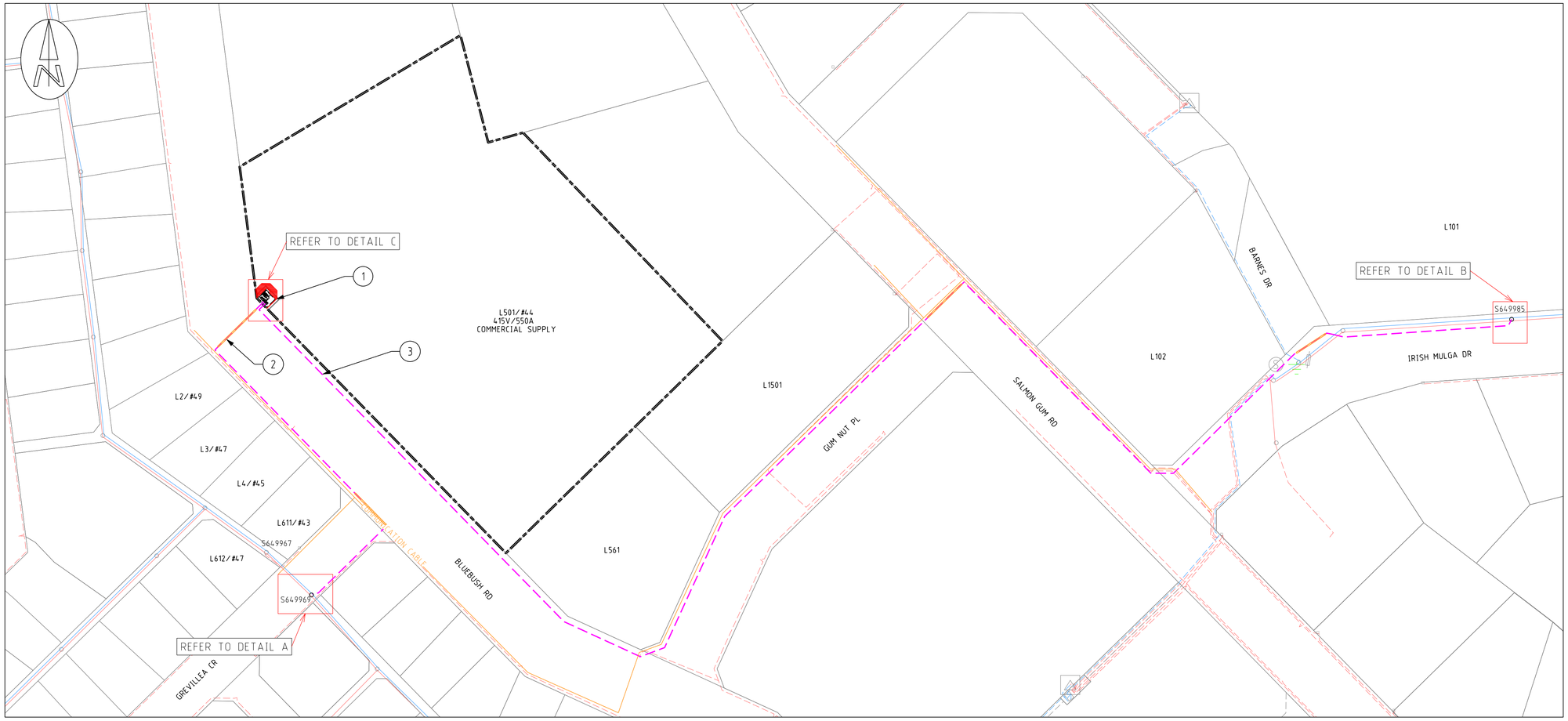
All mobile plant in proximity to the transmission line shall operate at a minimum approach distance that complies with Western Power's practices and requirements.

Environmental Requirements

There are no environmental issues.

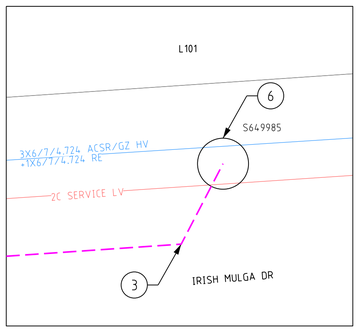
Surface Gradient

The project area poses a possible risk for setting up of equipment. Before construction commences the constructor is to assess the surface gradient of the site and ensure appropriate steps are taken.



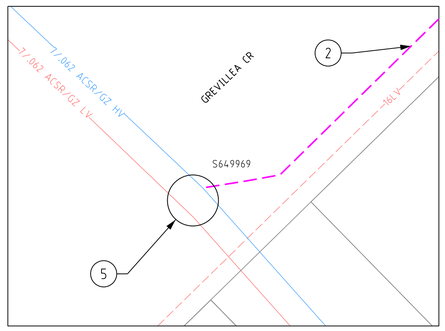
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL B

1:100 2 1 0 2 4 6 8



DETAIL A

1:100 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

REFER TO SHEET 2 AND 3

CUSTOMER SCOPE OF WORK

REFER TO SHEET 2 AND 3

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Unclearad De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES 15kV 33kV WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: THE SHIRE OF COOLGARDIE
CONTACT: AARON COOK TEL: (08) 9080 2111

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/25

REFERENCE FILE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

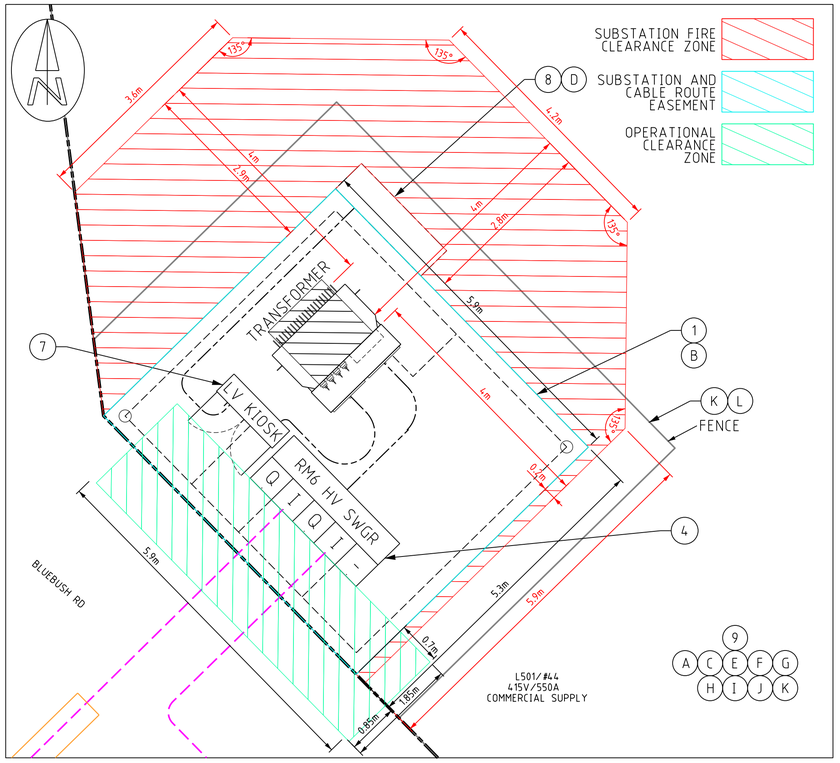
500A COMMERCIAL SUPPLY

NETWORK VOLTAGES: 11kV/415V

#44 BLUEBUSH RD, KAMBALDA WEST, WESTERN AUSTRALIA, 6442
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 06/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 12' 32" S Long. - 121° 37' 04" E

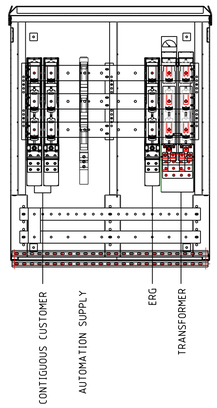
western power
DRAWING NUMBER
NP031662
SHEET 01 OF 03
REV: A ORIGINAL SIZE - A1



DETAIL C

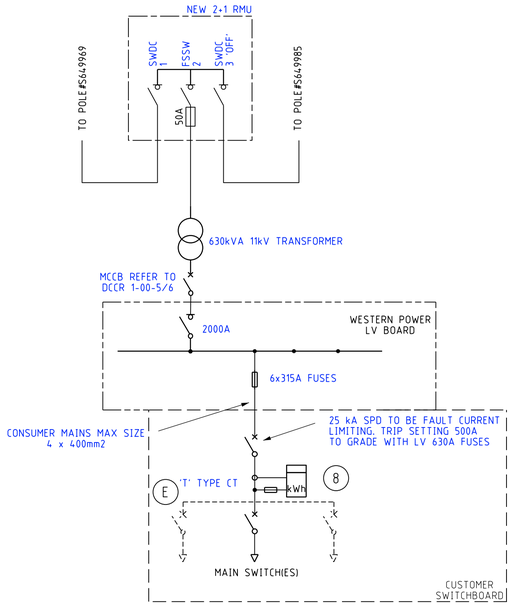
1:250 5 2.5 0 5 10 15 20

TYPE 2.1 PENDA ARRANGEMENT



	LEFT	NAME/LOCATION
Circuit 1 (C1)	LU67	CONTIGUOUS CONNECTION
Circuit 2 (C2)		
Circuit 3 (C3)		
Circuit 4 (C4)	LU70	FUTURE AUTOMATION SUPPLY
Circuit 5 (C5)		
Circuit 6 (C6)		
Circuit 7 (C7)		
Circuit 8 (C8)	LU68	ERG
Circuit 9 (C9)	LU66_TX	TRANSFORMER CONNECTION
Circuit 10 (C10)	RIGHT	

DETAIL F



SIMPLIFIED LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. INSTALL 630kVA 11kV SUBSTATION WITH 2+1 MOTORISED SWITCHGEAR AND A TYPE 2.1 PENDA TO DSPM-3-04 AS NOTED ON DRAWING.
   INSTALL 35HV CABLE BETWEEN HV SWITCHGEAR AND TRANSFORMER.
   INSTALL OIL CONTAINMENT PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5 AND DRAWING DSPM-4-06 (NON-MPS).
   HU6_A400x1, HU59_630/11x1, HU55_630/11x1, HU66_22x1, HU76x1, CN46x15m CN60_400CUx80m, CN73_35/185x3, LU16_400CU16x1, LU16_400CU12x1, LU42_Mx1, LU64_0x1, LU66_TXx1, LU67x1, LU68x1, LU70x1
   NON-CU: GF1101x6 (315A FUSE CARTRIDGE)
   DSPM: DSPM-3-04, DSPM-4-06
2. INSTALL 400 HV CABLE FROM NEW 2+1 HV SWITCHGEAR TO THE EXISTING POLE S649969.
   UTILISE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 400HV CABLE IN THE 0-500mm ALIGNMENT.
   CU: CN53x195m, CN54x6, CN73_240x3
3. INSTALL 400 HV CABLE FROM NEW 2+1 HV SWITCHGEAR TO THE EXISTING POLE S649985.
   UTILISE THE 2.7m ALIGNMENT IN AREAS WHERE ITS NOT PRACTICAL TO INSTALL THE 400HV CABLE IN THE 0-500mm ALIGNMENT.
   CU: CN53x660m, CN54x12, CN73_240x6, HU2_22_400x2
4. AT NEW 2+1 HV SWITCHGEAR.
   - TERMINATE NEW 400HV CABLE FROM POLE#S649985 IN 'OFF' POSITION
   - TERMINATE NEW 400HV CABLE FROM POLE#S649969 IN 'ON' POSITION
   - TERMIANTE TRANSFORMER CABLE
   AS PER SIMPLIFIED LINE DIAGRAM.
5. AT POLE#S649969.
   - DETTACH EXISTING SERVICE
   - REPLACE EXISTING POLE WITH NEW 12.5m/6kN POLE.
   - INSTALL HV INTERMEDIATE CROSSARM.
   - INSTALL LV INTERMEDIATE CROSSARM.
   - TERMINATE NEW 400HV CABLE TO HV BARE.
   - RE-ATTACH EXISTING AERIAL SERVICE TO NEW POLE.
   CU: P08_R×1, RE6×1, HV6_3N×1, LV1_W3×1, HU16_U_11×1, HU16_0×1, CN27_3/275x3, CN9_T6/475x3, LV08_3Px1
   DCSH: H08-3, L01, R13-3, R11-1
6. AT POLE#S649985.
   - REPLACE EXISTING POLE WITH NEW 12.5m/6kNPOLE.
   - INSTALL NEW HV INTERMEDIATE CROSSARM.
   - TERMINATE NEW 400HV CABLE TO HV BARE VIA .
   - REPLACE EXISTING STREETLIGHT WITH NEW 170W LED STREETLIGHT.
   CU: P08_R×1, RE6x1, HV6_3N×1, HU16_U_11x1, HU16_0×1 CN27x3, CN9_T6/475x3, RE6x1, RE8_IWx1, SL3_6L x1
   DCSH: H08-3, R13-3, S03
7. AT NEW TYPE 2.1 PENDA.
   - TERMINATE CUSTOMER'S CABLE TO THE TYPE 2.1 PENDA IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR. REFER TO PENDA ARRANGEMENT AND CIRCUIT TABLE.
   DSPM-4-05
8. INSTALL 1x 'T' TYPE CT METER IN CUSTOMERS SWITCHBOARD.
   CU: LM1x1
13. ENVIRONMENTAL FILTER CHECKED

THE FOLLOWING ENVIRONMENTAL ISSUES HAVE BEEN IDENTIFIED FOR THIS PROJECT. ALL RELEVANT PERMITS / PROCEDURES MUST BE OBTAINED / ADHERED TO:
ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

LAND POTENTIALLY SUBJECT TO NATIVE TITLE
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

NATIVE VEGETATION REGS ESAS CLEARING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL
FROM SEQT.

REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:
   1. PEGGING. COMPACTION AND LEVELLING.
   2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
   3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
   4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

C. CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

D. CUSTOMER MUST INSTALL MSB CONTIGUOUS WITH SUBSTATION.

E. CUSTOMER TO SUPPLY AND INSTALL 'T' TYPE METERING CT's.

F. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

G. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

H. CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

I. NO BUILDINGS ALLOWED WITHIN 4m OF THE SUBSTATION UNLESS 2 HOUR FIRE RATED. SEE WESTERN POWER DRAWING DSM-5-01 FOR DETAILS.

J. CUSTOMER TO PROVIDE A 129BA RESTRICTIVE COVENANT BURDENING THE LOT/S AROUND THE SUBSTATION SITE FOR FIRE PROTECTION CLEARANCE REQUIREMENTS TO FUTURE BUILDINGS AS PER UNDERGROUND DISTRIBUTION SCHEMES MANUAL CLAUSE 5.3.18.1. CONTACT THE WESTERN POWER PROPERTY SPECIALIST ON 9326 6308 FOR DETAILS.

K. CUSTOMER TO ARRANGE FOR MODIFICATION OF THE EXISTING FENCE TO ACCOMMODATE THE SUBSTATION SITE. THE FENCE SURROUNDING THE SUBSTATION MUST BE CONSTRUCTED FROM NON-METALLIC MATERIALS.

L. IF METALLIC FENCE IS TO BE USED IT IS TO BE LOCATED A MINIMUM OF 1.25m AWAY FROM THE EDGE OF THE SUBSTATION SITE BOUNDARY AND ALLOW FOR SAFE ACCESS TO THE CONTIGUOUS SMSB.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: THE SHIRE OF COOLGARDIE
CONTACT: AARON COOK TEL: (08) 9080 2111

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				ENVIRONMENTAL ASSESSMENT
A	ISSUED FOR CONSTRUCTION	I.P.	06/25	HAZARD MANAGEMENT REGISTER

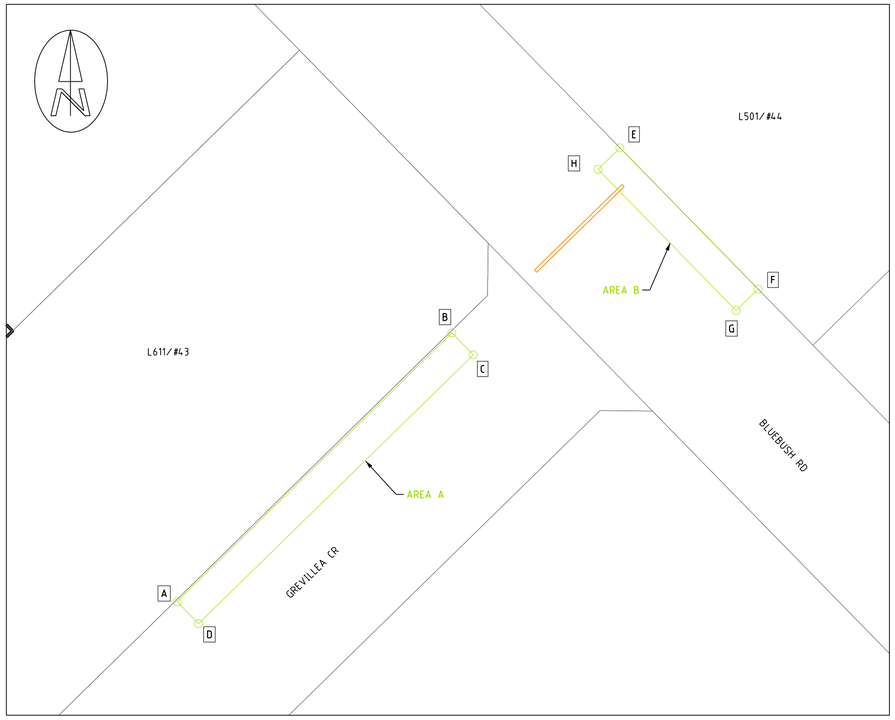
500A COMMERCIAL SUPPLY

NETWORK VOLTAGES: 11kV/415V

#44 BLUEBUSH RD, KAMBALDA WEST, WESTERN AUSTRALIA, 6442
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 06/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 12' 32" S Long. - 121° 37' 04" E

westernpower
DRAWING NUMBER
NP031662
SHEET 02 OF 03
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:500 10 5 0 10 20 30 40

AREA A- CLEARING ENVELOPE		
POINT ID	NORTHING	EASTING
A	940319.16	6537984
B	940364.02	6538028
C	940367.52	6538024
D	940322.66	6537980

AREA B- CLEARING ENVELOPE		
POINT ID	NORTHING	EASTING
E	940391.45	6538058
F	940414	6538035
G	940410.42	6538031
H	940387.88	6538054

CLEARING PERMIT WORK INSTRUCTION

With regard to your requested works, it has been identified by Western Power that clearing or potential clearing is required to facilitate the connection. Being a customer funded project, the customer is responsible for obtaining all necessary clearing approvals prior to the commencement of works, you can engage an environmental consultancy to assist with this matter.

The Department of Water and Environmental Regulation (DWER) administers the clearing provisions of the Environmental Protection Act 1986 (EP Act). Under the EP Act, a clearing permit is required to authorise clearing of native vegetation unless a valid exemption applies or DWER identifies that a clearing permit is not required via the formal clearing referral process.

Refer to https://www.wa.gov.au/service/environment/environment-information-services/native-vegetation-clearing-permits for further information on clearing approval requirements. Advice on relevant exemptions can be obtained via DWER's "Guide 1 Exemptions and regulations for clearing native vegetation" and clarification can be sought via emailing DWER directly via info@dwer.wa.gov.au.

Detail A identifies the area for which clearing approvals will need to be obtained and in which you may have to arrange to clear/remove vegetation.

Please note that DWER may identify that the clearing has significant impacts and there may be additional requirements before or after DWER issue a clearing permit that may impact the delivery date. These may include flora or fauna survey requirements and hygiene management plans.

Please ensure you provide the Western Power Scoping Specialist / Construction Manager with a copy of one of the following, as soon as practicable, for review and suitability confirmation:

- A clearing permit; or
- A referral determination notice issued by DWER confirming a clearing permit is not required; or
- Written details of a relevant clearing permit exemption and supporting documents (including any Regulator correspondence).

Further relevant and supporting information is available via DWER's website via https://www.wa.gov.au/organisation/department-of-water-and-environmental-regulation. If you have any questions about Western Power's requirements, please contact the Western Power Quality Assurance team via ccs.sbm.qat@westernpower.com.au.



ENVIRONMENT ISSUES AND REQUIREMENTS

This page outlines environmental issues and requirements related to the scope of work, as well as any concerns in the vicinity that the Construction Manager must consider to prevent environmental damage. Particular attention is given to issues that could lead to prosecution. Project-specific information is referenced on this page and must be followed.

CUSTOMER TO OBTAIN PERMIT AND CLEAR VEGETATION TO THE SATISFACTION OF THE WESTERN POWER SCOPER/PNO
AS PER THE FOLLOWING.

NATIVE VEGETATION REMAINING
SHOULD CLEARING BE REQUIRED;
THE CUSTOMER IS RESPONSIBLE FOR OBTAINING ALL NECESSARY CLEARING APPROVALS AND, PRIOR TO WESTERN POWER COMMENCING WORKS, THE CUSTOMER IS REQUIRED TO PROVIDE A COPY OF EITHER:
- A CLEARING PERMIT ISSUED IN THE CUSTOMER'S NAME; OR
- A REFERRAL DETERMINATION NOTICE ISSUED BY DWER CONFIRMING A CLEARING PERMIT IS NOT REQUIRED BY THE CUSTOMER; OR
- WRITTEN DETAILS OF A RELEVANT CLEARING PERMIT EXEMPTION AND SUPPORTING DOCUMENTS (REGULATOR CORRESPONDENCE/LANDOWNER EXEMPTION CLEARING LETTER EDM 46147597).

THIS ALSO INCLUDES RELEVANT DEVELOPMENT APPROVALS FOR THE WORKS (I.E. BUILDING APPROVALS FOR A STRUCTURE THAT REQUIRES CLEARING TO CONSTRUCT). APPROVALS / EXEMPTIONS CAN BE SENT TO THE SEQT ENVIRONMENT TEAM VIA ENVIRONMENT@WESTERNPOWER.COM.AU FOR REVIEW AND SUITABILITY CONFIRMATION. ALLOW 10 BUSINESS DAYS FOR ENVIRONMENT TEAM TO REVIEW.
TO ENABLE THE CUSTOMER TO OBTAIN CLEARING APPROVALS, WESTERN POWER MUST PROVIDE THE CUSTOMER WITH A DRAWING SHEET OF THE VEGETATION CLEARING REQUIRED (VEGETATION CLEARING REQUIREMENTS TEMPLATE EDM 50997884) AND ANY PHOTOS OR GIS SHAPEFILES, SHOWING THE EXACT LOCATION(S) WHERE CLEARING IS REQUIRED.
REFER TO CLEARING PERMIT WORK INSTRUCTION (EDM 43938228) FOR FURTHER DETAILS.

LAND POTENTIALLY SUBJECT TO NATIVE TITLE
- 1020210
- 792669
IT IS THE CUSTOMER'S RESPONSIBILITY TO COMPLETE THEIR OWN DUE DILIGENCE ASSESSMENT AND PROVIDE WESTERN POWER WITH THE EVIDENCE OF SURVEYS/CONSULTATIONS AND APPROVALS (WHERE APPLICABLE) PRIOR TO THE COMMENCEMENT OF WORKS.

IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 41507941).

REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT. IF IN DOUBT PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL OPERATION TO ARRANGE A PERMIT.
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: THE SHIRE OF COOLGARDIE
CONTACT: AARON COOK TEL: (08) 9080 2111

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/25

REFERENCE FILE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

500A COMMERCIAL SUPPLY

NETWORK VOLTAGES: 11kV/415V

#44 BLUEBUSH RD, KAMBALDA WEST, WESTERN AUSTRALIA, 6442

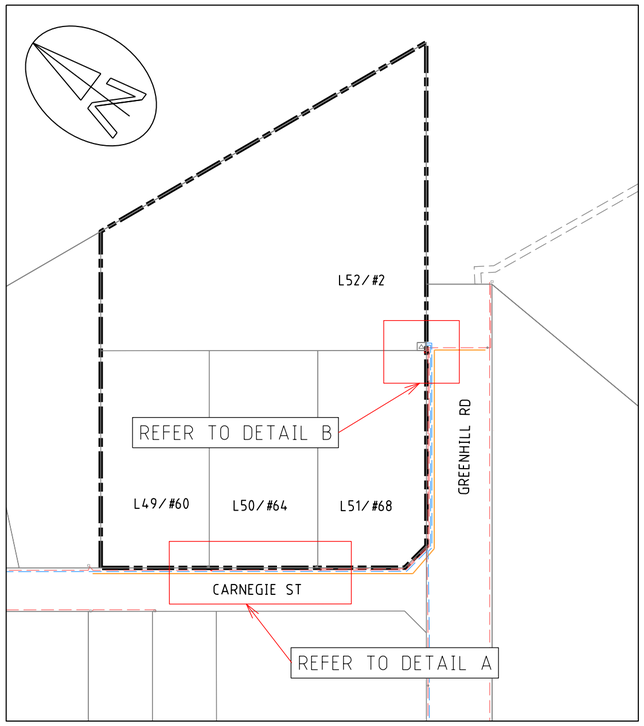
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 06/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 12' 32" S Long. - 121° 37' 04" E

westernpower

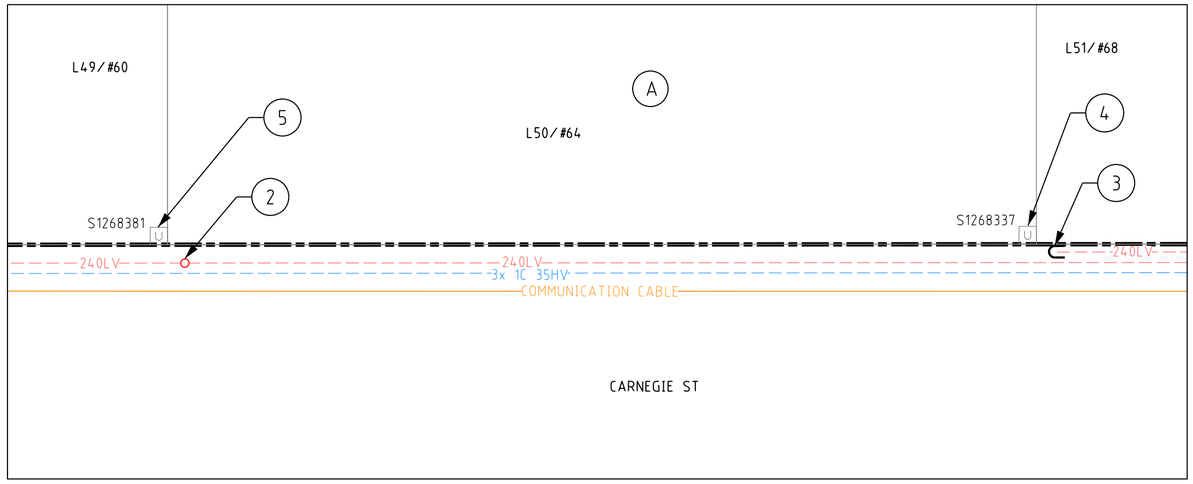
DRAWING NUMBER
NP031662
SHEET 01 OF 03

REV: A ORIGINAL SIZE - A1



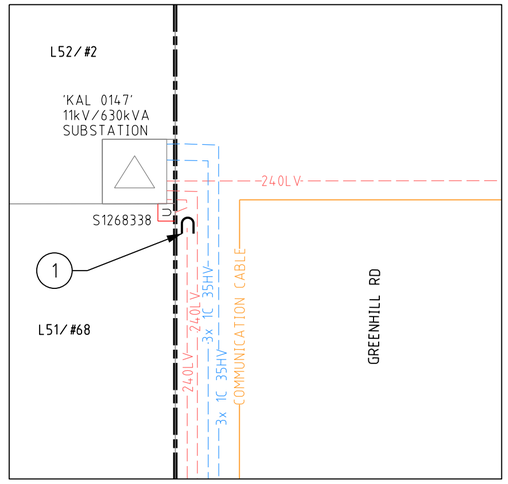
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:250 5 2.5 0 5 10 15 20



DETAIL B

1:250 5 2.5 0 5 10 15 20

WESTERN POWER SCOPE OF WORK

1. DISCONNECT EXISTING 240LV CABLE FROM EXISTING PILLAR (S1268338) AND PULL BACK TO WP ALIGNMENT. CUT THEN CAP AND DE-COMMISSION THE EXISTING 240LV CABLE. REMOVE EXCESS CABLE.
   CU: CN73_LVMAINS X 1
2. DISCONNECT EXISTING 2 X 240LV CABLES FROM EXISTING PILLLAR (S1268381) AND PULL BACK TO WP ALIGNMENT. CUT AND STRAIGHT JOINT EXISTING 240LV CABLES. REMOVE EXCESS CABLES
   CU: LU2 x 1
3. DISCONNECT EXISTING 240LV CABLE FROM EXISTING PILLAR (S1268337) AND PULL BACK TO WP ALIGNMENT. CUT THEN CAP AND DE-COMMISSION THE EXISTING 240LV CABLE. REMOVE EXCESS CABLE.
   CU: CN73_LVMAINS X 1
4. REMOVE PILLAR#S1268337 AT L50/#64.
5. REMOVE PILLAR#S1268381 AT L49/#60.
6. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   NATIVE VEGETATION REMAINING
   DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- L LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUD Serviced Uncleared De-energised Lot

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

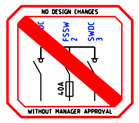
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

W.A.P.C. REFERENCE No. 163247

ISSUED FOR CONSTRUCTION



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: SPECTRUM SURVEYS
CONTACT: BRENDAN BUCHANAN TEL: 0890218343

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 69886453
HAZARD MANAGEMENT REGISTER EDM#: 69886668

UNDERGROUND ASSET REMOVAL

NETWORK VOLTAGE: 11kV/415V

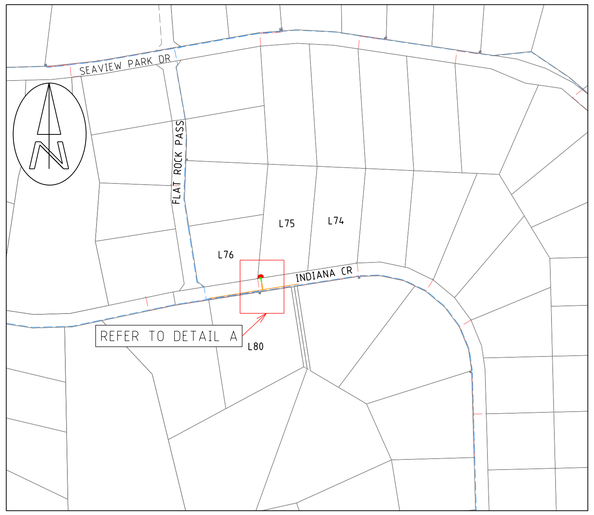
2 GREENHILL RD, BROADWOOD WA 6430

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 10/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 30° 47' 22" S Long. - 121° 26' 52" E

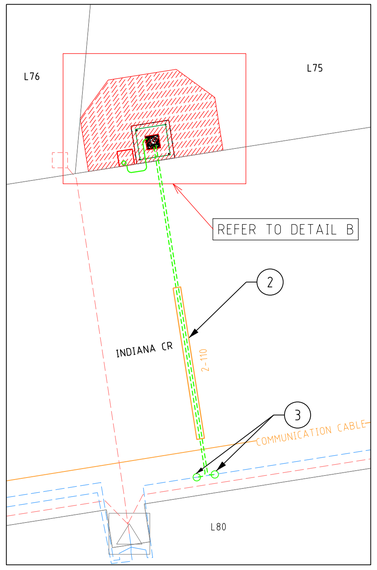
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DRAWING NUMBER
NP031850
SHEET 01 OF 01
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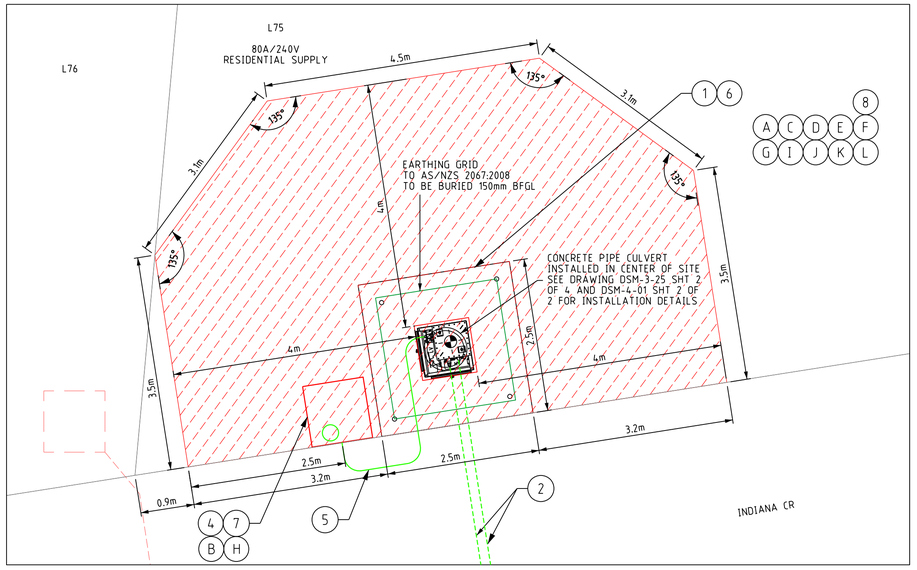
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:50 1 0.5 0 1 2 3 4

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES

- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES

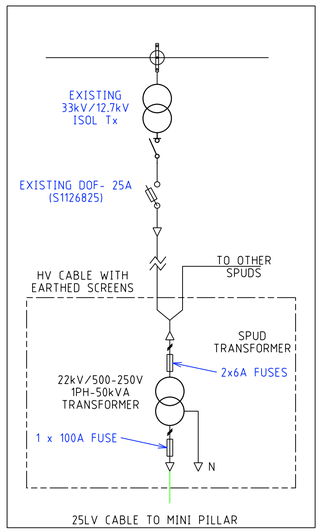
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Unclearged De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)



SIMPLIFIED LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. INSTALL 25kVA 22kV 3-BUSHING SPUD TRANSFORMER TO DSPM-3-25 AS NOTED ON DRAWING. INSTALL OIL CONTAINMENT PANEL AND POST WITH RAILWAY BALLAST/FLAME TRAP IN ACCORDANCE WITH WASIR CLAUSE 14.6.5.
   CU: HU34_R25/22 x 1, HU75 x 1
2. INSTALL 2x NEW 1C 35mm HV CABLES FROM NEW INSTALLED 25kVA SPUD TRANSFORMER AT LOT 75 INDIANA CR TO EXISTING 1C 35HV CABLES.
   CU: CN47 x 60m, CN73_35/185 x 5, CN54 x 6
3. CUT EXISTING 1C 35HV CABLE AND STRAIGHT JOINT TO 2x NEW 35HV CABLES
   CU: CN73_35/185 x 2, CN54 x 2, HU72 x 2
4. INSTALL NEW MINI-PILLAR ON LOT 75 INDIANA CR WHERE PEGGED BY CUSTOMER ON SITE AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 1
5. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO NEW TRANSFOMER.
   CU: CN42 x 10m, CN73_LV25 x 3, CN54 x 2
6. AT NEW 25kVA SPUD TRANSFORMER.
   - TERMINATE 2x35mm HV CABLE TO HV TERMINALS.
   - TERMINATE THE 25LV CABLES TO THE LV DISTRIBUTION BOARD AND CAP SPARE CONDUCTORS.
   CU: LU10_CABTERM x 1
   NON_CU: FL1448 x 3
   DCSH: R29
7. AT NEW MINI PILLAR.
   - TERMINATE NEW INSTALLED 25LV CABLE.
   DCSH: U08-04 (1PH NETWORK)
8. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   FIRE - SLIP PUBLIC SERVICE
   ABORIGINAL HERITAGE SITES
   THREATENED ECOLOGICAL COMMUNITIES
   HYDROGRAPHIC CATCHMENTS - SUBCATCHMENTS (DWER-030)
   VEGETATION COMPLEXES - SWAN COASTAL PLAIN (DBCA-046)

   REFER TO ENVIRONMENTAL DRAWING (SHEET 2)

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES AND OBSTRUCTIONS INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. SUBSTATION SITE WILL BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 14.4.6. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING, COMPACTION AND LEVELING.
2. LOCATED 500mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
3. CLEAR OF ALL SERVICES INCLUDING ENCROACHMENT.
4. CLEAN SAND FILL TO A DEPTH OF 1m IN SANDY SOILS TO A DEPTH OF 1.6m IN CLAY OR ROCK.

D. IF THE CUSTOMER PROPOSES TO INSTALL SCREENING, APPROVAL SHALL BE OBTAINED FROM WESTERN POWER PRIOR TO ITS INSTALLATION. THE SCREENING SHALL BE INSTALLED ONLY AFTER WESTERN POWER HAS COMPLETED ALL OF THE WORK ON THE SUBSTATION SITE. REFER TO THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSES 14.4.10.3.2, 14.4.11 AND 14.5.4 FOR DETAILS.

E. CUSTOMER TO CONTACT WESTERN POWER PROPERTY SPECIALIST AT conveyancing@westernpower.com.au TO ARRANGE THE PREPARATION AND REGISTRATION OF THE REQUIRED 129BA RESTRICTIVE COVENANT.

F. THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME.

G. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

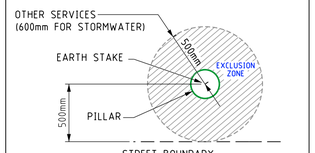
H. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

I. CUSTOMER METER BOARD TO BE INSTALLED WITHIN 30m OF 'POINT OF SUPPLY' (MINI PILLAR).

J. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL CONTRACTOR SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

K. CUSTOMER TO PROVIDE WESTERN POWER WITH ACCESS KEY AND SECURITY INFORMATION FOR SUBSTATION 24/7 ACCESS.

PILLAR EXCLUSION ZONE
NTS



COMMUNICATION CABLES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: THE RURAL BUILDING CO.
CONTACT: CHRISTOPHER NOEL KANE TEL: 0417932501

CUSTOMER CONTACT DETAILS
COMPANY NAME:
CONTACT: LEAR PAPAS TEL: 0416739864

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☐	☐	☐	☒	☐

NO DESIGN CHANGES WITHOUT MANAGER APPROVAL

SAFETY - MAKE IT YOUR PRIMARY VALUE

THIS DRAWING TO BE READ IN CONJUNCTION ENVIRONMENTAL DRAWING IN SHEET 2

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/25

REFERENCE FILE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

NEW 80A RESIDENTIAL SUPPLY

NETWORK VOLTAGES: 12.7kV/240V

LOT 75 INDIANA CRESCENT KARAKIN WA 6044

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 09/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 02' 31" S Long. - 115° 23' 36" E

westernpower

DRAWING NUMBER

NP032685

SHEET 01 OF 02

REV: A ORIGINAL SIZE - A1



DETAIL A

1:100 2 1 0 2 4 6 8

AREA A- CLEARING ENVELOPE		
POINT ID	NORTHING	EASTING
A	346693	6564639
B	346687	6564639
C	346693	6564631
D	346688	6564631

AREA B- CLEARING ENVELOPE		
POINT ID	NORTHING	EASTING
E	346690	6564619
F	346695	6564620
G	346696	6564615
H	346691	6564614

CLEARING PERMIT WORK INSTRUCTION

With regard to your requested works, it has been identified by Western Power that clearing or potential clearing is required to facilitate the connection. Being a customer funded project, the customer is responsible for obtaining all necessary clearing approvals prior to the commencement of works, you can engage an environmental consultancy to assist with this matter.

The Department of Water and Environmental Regulation (DWER) administers the clearing provisions of the Environmental Protection Act 1986 (EP Act). Under the EP Act, a clearing permit is required to authorise clearing of native vegetation unless a valid exemption applies or DWER identifies that a clearing permit is not required via the formal clearing referral process.

Refer to https://www.wa.gov.au/service/environment/environment-information-services/native-vegetation-clearing-permits for further information on clearing approval requirements. Advice on relevant exemptions can be obtained via DWER's "Guide 1 Exemptions and regulations for clearing native vegetation" and clarification can be sought via emailing DWER directly via info@dwer.wa.gov.au.

Detail A identifies the area for which clearing approvals will need to be obtained and in which you may have to arrange to clear/remove vegetation.

Please note that DWER may identify that the clearing has significant impacts and there may be additional requirements before or after DWER issue a clearing permit that may impact the delivery date. These may include flora or fauna survey requirements and hygiene management plans.

Please ensure you provide the Western Power Scoping Specialist / Construction Manager with a copy of one of the following, as soon as practicable, for review and suitability confirmation:

- A clearing permit; or
- A referral determination notice issued by DWER confirming a clearing permit is not required; or
- Written details of a relevant clearing permit exemption and supporting documents (including any Regulator correspondence).

Further relevant and supporting information is available via DWER's website via https://www.wa.gov.au/organisation/department-of-water-and-environmental-regulation. If you have any questions about Western Power's requirements, please contact the Western Power Quality Assurance team via ccs.sbm.qat@westernpower.com.au.



ENVIRONMENT ISSUES AND REQUIREMENTS

This page outlines environmental issues and requirements related to the scope of work, as well as any concerns in the vicinity that the Construction Manager must consider to prevent environmental damage. Particular attention is given to issues that could lead to prosecution. Project-specific information is referenced on this page and must be followed.

CUSTOMER TO OBTAIN PERMIT AND CLEAR VEGETATION TO THE SATISFACTION OF THE WESTERN POWER SCOPER/PNO
AS PER THE FOLLOWING.

NATIVE VEGETATION REGS ESAS CLEARING
25m² OF NATIVE VEGETATION CLEARING MAY BE REQUIRED TO FACILITATE PROJECT WORKS.

1. IF NATIVE VEGETATION CLEARING IS REQUIRED:
BEING A CUSTOMER FUNDED PROJECT, THE CUSTOMER IS RESPONSIBLE FOR OBTAINING ALL NECESSARY CLEARING APPROVALS AND, PRIOR TO WESTERN POWER COMMENCING WORKS, THE CUSTOMER IS REQUIRED TO PROVIDE A COPY OF EITHER:
- A CLEARING PERMIT ISSUED IN THE CUSTOMER×S NAME; OR
- A REFERRAL DETERMINATION NOTICE ISSUED BY DWER CONFIRMING A CLEARING PERMIT IS NOT REQUIRED BY THE CUSTOMER; OR
- WRITTEN DETAILS OF A RELEVANT CLEARING PERMIT EXEMPTION AND SUPPORTING DOCUMENTS (REGULATOR CORRESPONDENCE/LANDOWNER EXEMPTION CLEARING LETTER EDM 46147597)

THIS ALSO INCLUDES RELEVANT DEVELOPMENT APPROVALS FOR THE WORKS (I.E. BUILDING APPROVALS FOR A STRUCTURE THAT REQUIRES CLEARING TO CONSTRUCT). APPROVALS / EXEMPTIONS CAN BE SENT TO THE SEQT ENVIRONMENT TEAM VIA ENVIRONMENT@WESTERNPOWER.COM.AU FOR REVIEW AND SUITABILITY CONFIRMATION. ALLOW 10 BUSINESS DAYS FOR ENVIRONMENT TEAM TO REVIEW.

TO ENABLE THE CUSTOMER TO OBTAIN CLEARING APPROVALS, WESTERN POWER MUST PROVIDE THE CUSTOMER WITH A DRAWING SHEET OF THE VEGETATION CLEARING REQUIRED (VEGETATION CLEARING REQUIREMENTS TEMPLATE EDM 50997884) AND ANY PHOTOS OR GIS SHAPEFILES, SHOWING THE EXACT LOCATION(S) WHERE CLEARING IS REQUIRED.

REFER TO CLEARING PERMIT WORK INSTRUCTION (EDM 43938228) FOR FURTHER DETAILS.

2. IF PLANTED/NON-NATIVE VEGETATION CLEARING IS REQUIRED:
NO FORMAL CLEARING APPROVAL REQUIRED AS ONLY PLANTED NON-NATIVE VEGETATION IS PROPOSED TO BE CLEARED. THE RELEVANT LANDOWNER (I.E. LOCAL GOVERNMENT IN ROAD RESERVE OR PRIVATE LANDOWNER) SHOULD BE CONSULTED IN RELATION TO ANY VEGETATION PROPOSED TO BE REMOVED.

3. IF NO VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

FLORA
1. IF NATIVE VEGETATION CLEARING IS REQUIRED:
THE CUSTOMER SHALL UNDERTAKE ANY NECESSARY FLORA SURVEYS AND IF REQUIRED OBTAIN RELEVANT APPROVALS TO CLEAR THREATENED OR PRIORITY FLORA. ALL SURVEYS AND APPROVALS SHALL BE PROVIDED TO WESTERN POWER FOR RECORDKEEPING AND REVIEW.

2. IF PLANTED/NON-NATIVE VEGETATION CLEARING IS REQUIRED:
NO FORMAL CLEARING APPROVAL REQUIRED AS ONLY PLANTED NON-NATIVE VEGETATION IS PROPOSED TO BE CLEARED. THE RELEVANT LANDOWNER (I.E. LOCAL GOVERNMENT IN ROAD RESERVE OR PRIVATE LANDOWNER) SHOULD BE CONSULTED IN RELATION TO ANY VEGETATION PROPOSED TO BE REMOVED.

3. IF NO VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

FAUNA
1. IF NATIVE VEGETATION CLEARING IS REQUIRED:
THE CUSTOMER SHALL UNDERTAKE ANY NECESSARY FAUNA SURVEYS AND IF REQUIRED OBTAIN RELEVANT STATE/COMMONWEALTH APPROVALS FOR IMPACTS TO THIS SPECIES &/OR ITS HABITAT. ALL SURVEYS, APPROVALS AND RELEVANT MANAGEMENT ACTIONS SHALL BE PROVIDED TO WESTERN POWER FOR RECORDKEEPING AND REVIEW.

2. IF PLANTED/NON-NATIVE VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

3. IF NO VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

THREATENED-PRIORITY ECOLOGICAL COMMUNITIES
1. IF NATIVE VEGETATION CLEARING IS REQUIRED:
THE CUSTOMER IS RESPONSIBLE FOR UNDERTAKING ALL SURVEYS REQUIRED TO CONFIRM THE PRESENCE OR ABSENCE OF THREATENED OR PRIORITY ECOLOGICAL COMMUNITIES, AND FOR OBTAINING THE RELEVANT APPROVALS TO CLEAR TECS/PECS IF REQUIRED. ALL SURVEYS AND APPROVALS SHALL BE PROVIDED TO WESTERN POWER PRIOR TO THE COMMENCEMENT OF WORKS.

2. IF PLANTED/NON-NATIVE VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

3. IF NO VEGETATION CLEARING IS REQUIRED:
NO FURTHER ACTION IS REQUIRED FOR THIS CONSTRAINT.

CONSTRUCTION NOISE
IF WORKS WILL BE UNDERTAKEN OUTSIDE OF 07:00-19:00 MON-SAT; OR WORKS ARE LOCATED NEAR MAJOR SENSITIVE RECEIVERS, CONSULTATION WITH THE LGA WILL BE REQUIRED TO DETERMINE MITIGATION MEASURES AND THE REQUIREMENTS FOR A NOISE MANAGEMENT PLAN.

ABORIGINAL HERITAGE SITES
THE WORKS ARE LOCATED WITHIN A HISTORIC ABORIGINAL HERITAGE PLACE: PLACE 3237 - LEDGE POINT.
IT IS THE CUSTOMER'S RESPONSIBILITY TO COMPLETE THEIR OWN DUE DILIGENCE ASSESSMENT AND PROVIDE WESTERN POWER WITH THE EVIDENCE OF SURVEYS/CONSULTATIONS AND APPROVALS (WHERE APPLICABLE) PRIOR TO THE COMMENCEMENT OF WORKS.

LAND USE PLANNING
THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LOCAL GOVERNMENT AUTHORITY (LGA) AND CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT MAY APPLY.

INFRASTRUCTURE - ROADS, GAS, RAIL
THE WORKS ARE OCCURRING IN A LOCAL GOVERNMENT ROAD RESERVE.
FORMAL APPROVAL IS NOT REQUIRED HOWEVER A NOTIFICATION OF WORKS (EDM 34174076) SHALL BE SUBMITTED TO THE RELEVANT LOCAL GOVERNMENT AUTHORITY AT LEAST 10 DAYS BEFORE WORKS COMMENCE IN A LOCAL ROAD RESERVE (EXCLUDING INSPECTIONS).
REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT. IF IN DOUBT PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL OPERATION TO ARRANGE A PERMIT.
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING
- 42W CFL
- 80W MV
- 125W MV
- 70W HPS
- 150W HPS
- 250W HPS
- 250W HPS AERODROME
- 70W MH
- 150W MH
- 250W MH
- 1L LED
- 2L LED
- 3L LED
- 4L LED
- 5L LED
- 6L LED
- 1E LED
- 2E LED
- 3E LED
- 4E LED
- 5E LED
- 6E LED

LEGEND DEFINES TYPE & WATTAGE - NOT COLOUR

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- Serviced Uncleared De-energised Lot
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE

TRANSFORMER (T) EXISTING SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☐	☒	☐	☐	☐	☒	☐



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CONSULTANT CONTACT DETAILS
COMPANY NAME: THE RURAL BUILDING CO.
CONTACT: CHRISTOPHER NOEL KANE TEL: 0417932501

CUSTOMER CONTACT DETAILS
COMPANY NAME:
CONTACT: LEAR PAPAS TEL: 0416739864

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				ENVIRONMENTAL ASSESSMENT
A	ISSUED FOR CONSTRUCTION	I.P.	09/25	HAZARD MANAGEMENT REGISTER

NEW 80A RESIDENTIAL SUPPLY

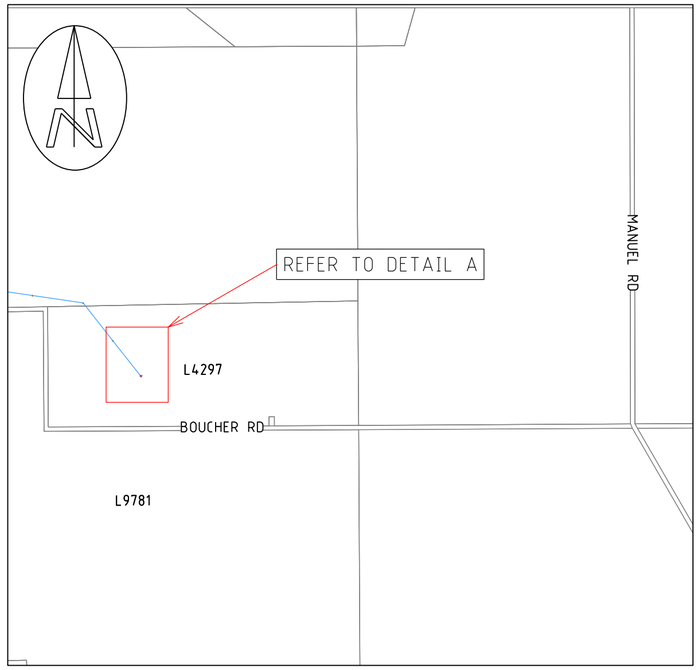
NETWORK VOLTAGES: 12.7kV/240V

LOT 75 INDIANA CRESCENT KARAKIN WA 6044
DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 09/25 SCALE: AS NOTED @ A1
GEO REF: Lat. - 31° 02' 31" S Long. - 115° 23' 36" E

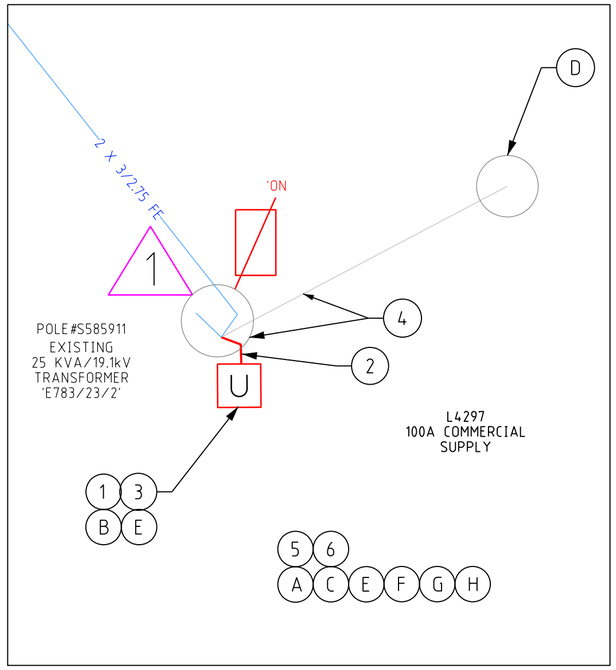
westernpower

DRAWING NUMBER
NP032685
SHEET 02 OF 02
REV: A ORIGINAL SIZE - A1



LOCATION PLAN

1:20,000 — 400 200 0 400 800 1200 1600



DETAIL A

1:100 — 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

1. REMOVE EXISTING MINI-PILLAR AND INSTALL NEW UNI-PILLAR 1m FROM THE POLE#S585911 ON LOT4297. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU11 x 1
   DCSH: U9
2. REMOVE EXISTING 25LV AND INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S585911
   CU: CN45 x 20m, CN54 x 2, CN73_LVMAINS x 3
3. AT NEW UNI-PILLAR (L5/#56 FORREST AV)
   - TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
   - TERMINATE CUSTOMER CABLE AT TOP BAR WITH 400A LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
   CU: LU7_240x1, LU36_LINKx1
4. AT POLE#SS585911
   - TERMINATE NEW 240LV CABLE TO EXISTING Tx VIA 160A KRONE FUSE.
   - REMOVE EXISTING AERIAL SERVICE TO CUSTOMER POLE.
   CU: LU61_240_3POx 1, LU61_240_3PU x 1
   NON CU: GF1100 x 3, GS0111 x 1, GS0109 x 1, AB6732 x 2
   DCSH: U19-3 MODIFIED EDM#70400904
5. INSTALL CT METER.
   CU: LM1_M x 1
6. ENVIRONMENTAL FILTER CHECKED
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER RESPONSIBLE TO REMOVE CUSTOMER'S POLES AS REQUIRED AFTER COMPLETION OF WORK.

E. CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

F. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

G. CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

H. CUSTOMER TO SUPPLY AND INSTALL SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY.
CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

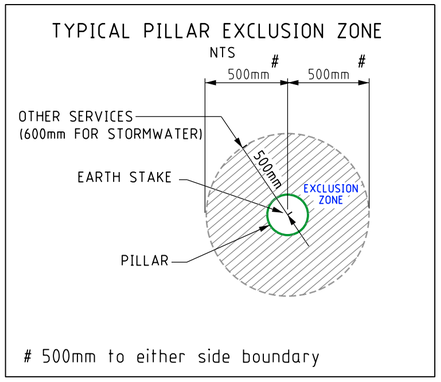
HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

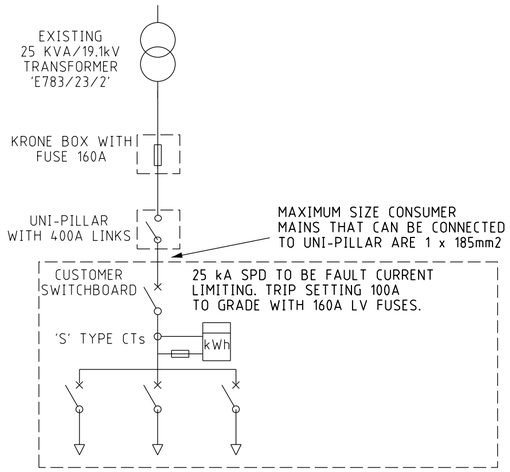
PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

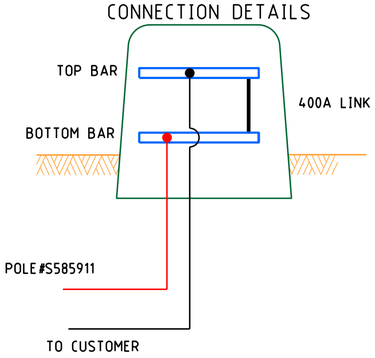
TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary

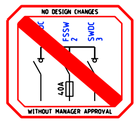


UNI-PILLAR LOT 4297
CONNECTION DETAILS



ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: DC ELECTRICAL
CONTACT: DARAN HUMFREY TEL: 0488 254 500

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS
- DCSH U19-3 MODIFIED EDM#70400904
- SHE FILTER ASSESSMENT EDM#: 69520218
- HAZARD MANAGEMENT REGISTER EDM#: 69520217

100A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 19.1kV/250V

LOT 4297, BOUCHER RD, BUNTINE, 6613, WA

DESIGN DRAWING

DESIGNER:	CCD
DESIGN TEAM:	CCS SBM QAT - RASP
EMAIL:	ccs.sbm.qat@westernpower.com.au
TEL:	13 10 87
DATE: 10/24	SCALE: AS NOTED @ A2
GEO REF:	Lat. - 30° 02' 11" S Long. - 116° 38' 14" E

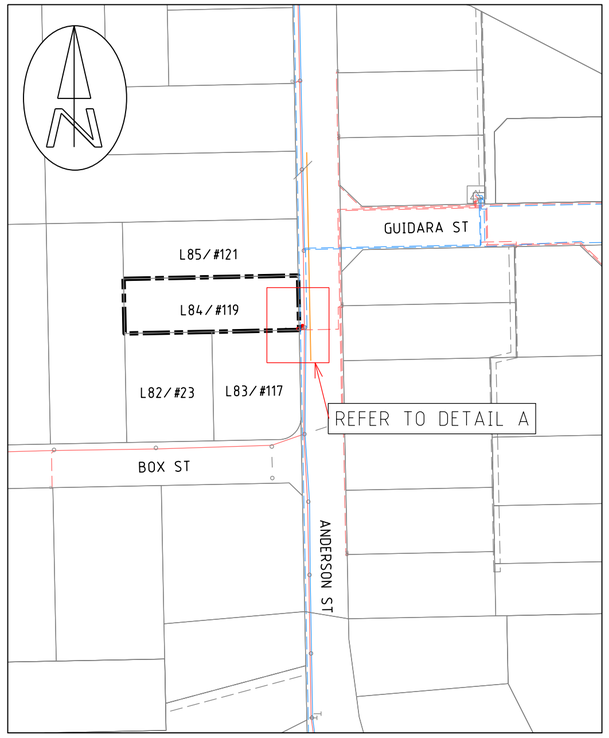
westernpower

DRAWING NUMBER

NP031984

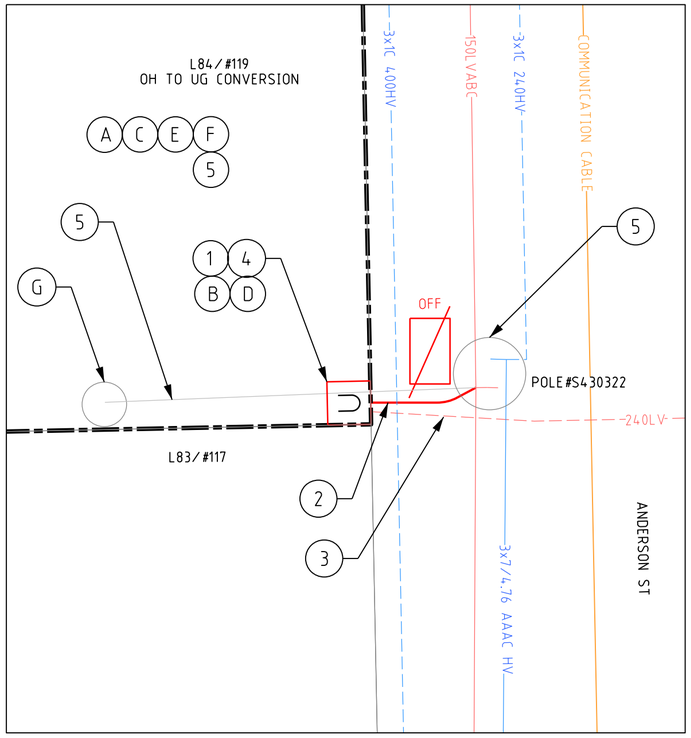
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 (40 20 0 40 80 120 160)



DETAIL A

1:100 (2 1 0 2 4 6 8)

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW UNI-PILLAR ON L84/#119 ANDERSON ST AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU11 x 1
   DCSH: U9
2. INSTALL NEW 240LV CABLE FROM NEW UN-PILLAR TO POLE#S430322
   CU: CN45 x 20m, CN54 x 3, CN73_LVMAINS x 3
3. DISCONNECT EXISTING 240LV CABLE FROM POLE#S430322. CUT AND SWING EXISTING 240LV CABLE FROM POLE S430322 TO NEW UNI-PILLAR. REMOVE THE EXCESS CABLE
4. AT UNI-PILLAR (L84/#119 ANDERSON ST), TERMINATE 2 x 240LV CABLE.
   CU: LU7_240 x 2
   DCSH: U9
5. AT POLE#S430322
   - REMOVE AERIAL SERVICE TO #119 AND ASSOCIATED MCBs.
   - TERMINATE NEW 240LV CABLE TO EXISTING KRONE.
   NON CU: CB3055 x 3, CC0101 x 3, AS1440 x 6, FC0499 x 1, HT6320 x 1, GC7195 x 3
   DCSH: U17
6. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   UNEXPLODED ORDNANCE
   NO FURTHER ACTIONS REQIUIRED FOR THIS CONSTRAINT. HOWEVER PLEASE BE MINDFUL OF ANY UNEXPECTED FINDS AND MANAGE IN ACCORDANCE WITH EXCAVATION AND TRENCHING PROCEDURE (EDMS# 43740956)

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

F. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

G. CUSTOMER RESPONSIBLE TO REMOVE CUSTOMER'S POLES AS REQUIRED AFTER COMPLETION OF WORK.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- U UNIVERSAL
- UTILITIES PIT
- MINI
- SP BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDL Serviced Uncleared De-energised Lot

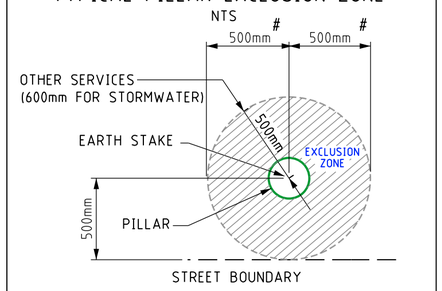
AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

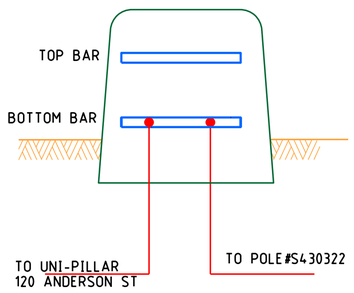
TYPICAL PILLAR EXCLUSION ZONE

NTS

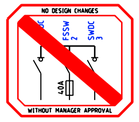


500mm to either side boundary

UNI-PILLAR L84/#119
CONNECTION DETAILS







SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: FORRESTER ELECTRICAL
CONTACT: DARREN FORRESTER TEL: 0400809199

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/24

SHE FILTER ASSESSMENT EDM#: 69502700
HAZARD MANAGEMENT REGISTER EDM#: 69509673
FILE REFERENCE DETAILS

OVERHEAD TO UNDERGROUND CONVERSION

NETWORK VOLTAGE: 22kV/415V

119 ANDERSON ST, WEBBERTON 6530

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 28° 44' 54" S Long. - 114° 38' 00" E

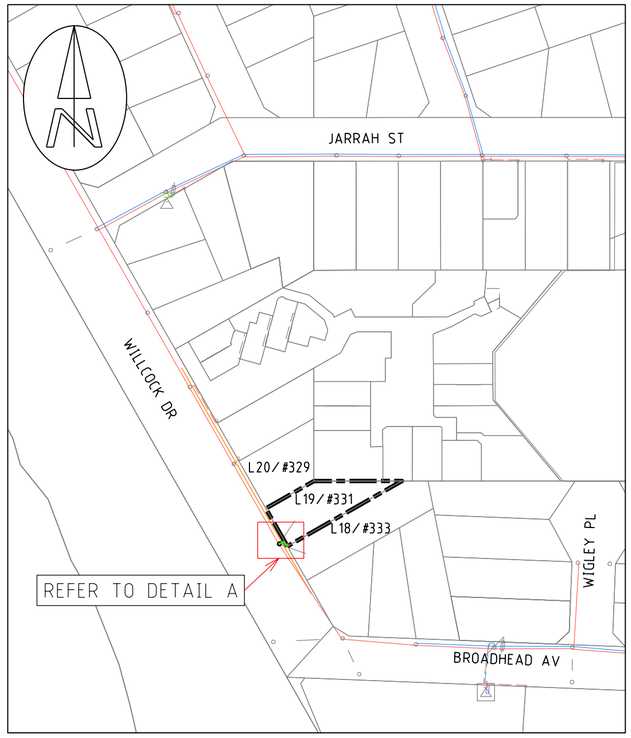
westernpower

DRAWING NUMBER

NP032006

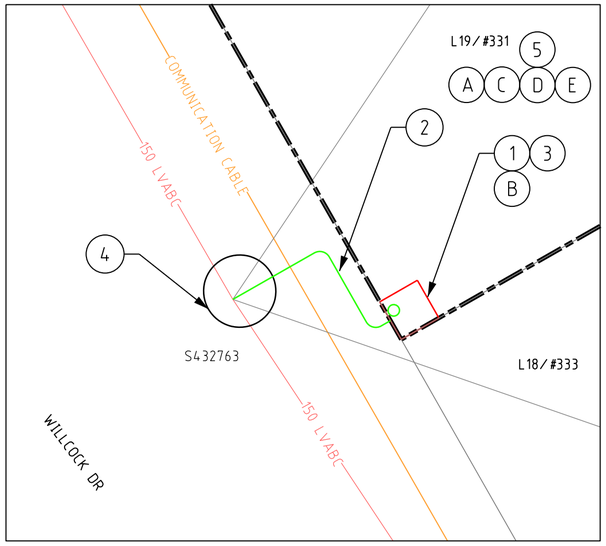
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:100 2 1 0 2 4 6 8

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L19/#331 WILLCOCK DR WHERE PEGGED BY CUSTOMER. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 1
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S432763.
   CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 2
3. AT NEW MINI-PILLAR, TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
4. AT POLE#S432763
   - REMOVE EXISTING AERIAL SERVICE TO L19/#331.
   - DETTACH EXISTING SERVICE TO L18/#333.
   - REPLACE THE POLE WITH NEW 9.5m/5kN WOOD POLE.
   - INSTALL NEW LVABC INTERMEDIATE.
   - REATTACH/REPLACE EXISTING SERVICE TO L18/#333.
   - TERMINATE NEW 25LV CABLE TO LVABC CONDUCTORS.
   CU: P02_Rx1, AB1_150x1, CN65_150/35x2, LV8_1Px1, LU50_25/150Ox1, LU50_25/150Ux1, CN75x1
   DCSH: U18, R13-3, A01
5. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   UNEXPLODED ORDNANCE
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

POLE SCHEDULE

POLE ID	POLE DETAILS	INSTRUCTIONS	POLE LOCATION		STRUCTURE MATERIALS / CU'S	DCSH REFERENCE	AS CON CHECK
			EASTING	NORTHING			
S432763	9.5m/5kN WOOD POLE	-	EXISTING	EXISTING	P02_R	R13-3	-

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

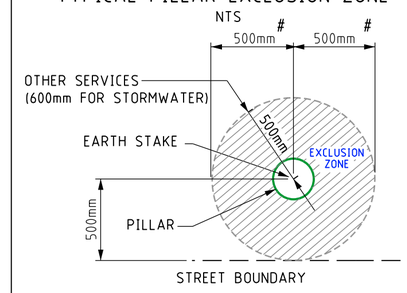
1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER TO SUPPLY AND INSTALL A SMSB WITHIN 30m AND PRIOR TO THE ENERGISATION OF THE POINT OF SUPPLY.

TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary

ISSUED FOR CONSTRUCTION

COMMUNICATION CABLES IN VICINITY OF WORKS

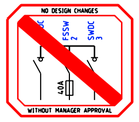
WATER PIPELINES IN VICINITY OF WORKS

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

SAFETY - MAKE IT YOUR PRIMARY VALUE

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐



CUSTOMER CONTACT DETAILS
COMPANY NAME: The Trustee For Mcm Trust
CONTACT: MCAULLAY BUILDERS TEL: 0428903034

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	11/25

FILE REFERENCE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

OVERHEAD TO UNDERGROUND CONVERSION

L19/#331 WILLCOCK DR

TARCOOLA BEACH WA 6530

NETWORK VOLTAGE: 22kV/415V

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 11/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 28° 48' 06" S Long. - 114° 37' 05" E

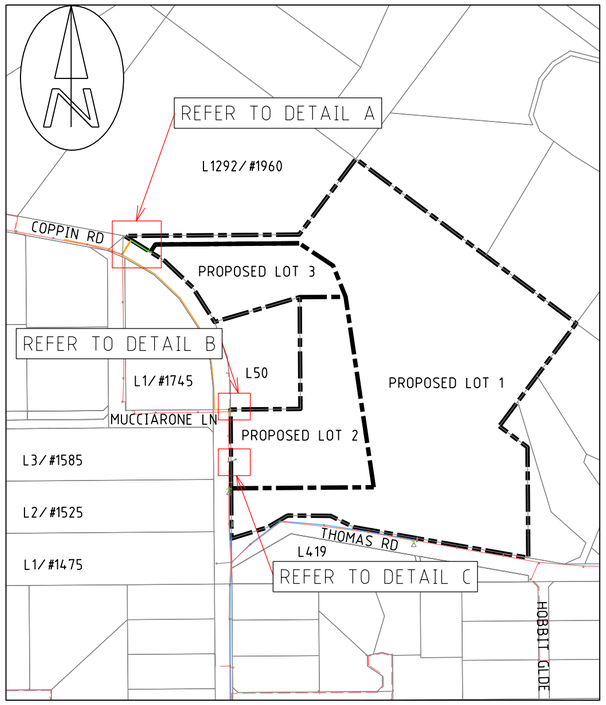
westernpower

DRAWING NUMBER

NP032932

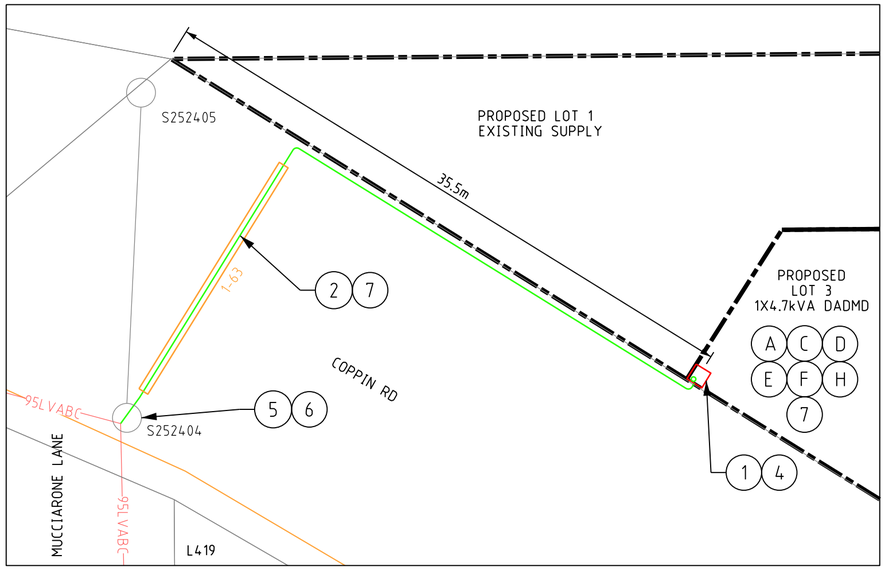
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



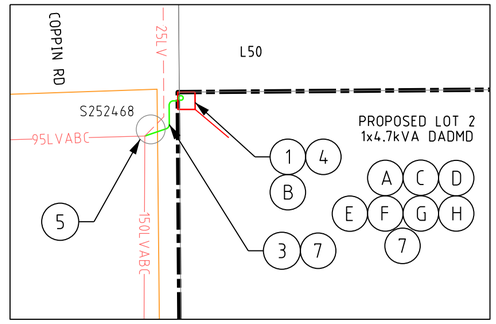
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:250 — 5 2.5 0 5 10 15 20



DETAIL B

1:250 — 5 2.5 0 5 10 15 20

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

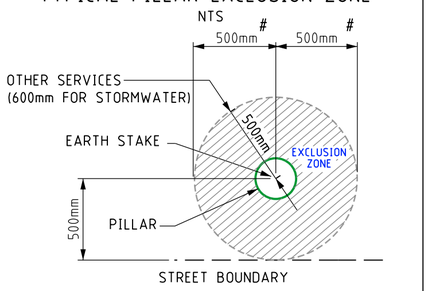
TRANSFORMER (T) — EXISTING — SWITCHGEAR (S)

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV — EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

ISSUED FOR CONSTRUCTION

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON CUSTOMER'S PROPERTY WHERE PEGGED BY CUSTOMER. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 2
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S252404
   CU: CN42 x 60m, CN73_LV25 x 3, CN54 x 2
3. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S252468
   CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 1
4. AT NEW MINI-PILLARS, TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
5. AT POLE#S252404 & S252468,
   - TERMINATE NEW 25LV CABLES TO 95LVABC CONDUCTORS.
   CU: LU50_25/95O x 2, LU50_25/95U x 2
   DCSH: U18
6. AT POLE#S252470
   - REMOVE EXISTING AERIAL SERVICE WIRE
7. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   ABORIGINAL HERITAGE SITES
   CARNABYS COCKATOO CONFIRMED BREEDING AREAS (DBCA-054)
   CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, SURFACE WATER AREAS AND IRRIGATION DISTRICTS (DWER-037)
   BUSH FIRE PRONE AREAS - CURRENT (OBRM-001)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO ENVIRONMENTAL DRAWING (SHEET 2)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS. UNDERGROUND DISTRIBUTION SCHEMES MANUAL.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

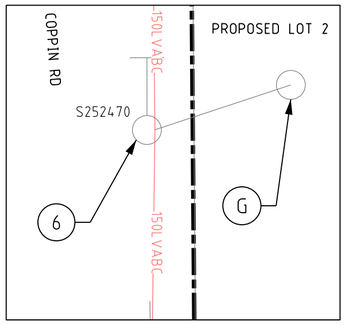
D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

E. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

F. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO TERMINATE ALL CONSUMER MAINS TO THE PILLAR BACK BLOCKS . MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35MM2/PHASE CABLE

G. CUSTOMER RESPONSIBLE TO REMOVE CUSTOMER SERVICE POLE AFTER COMPLETION OF WORK.

H. CUSTOMER RESPONSIBLE TO OBTAIN RELEVANT PERMITS AND REMOVE ANY VEGETATION REQUIRED TO FACILITATE WESTERN POWER WORKS. REFER TO ENVIRONMENTAL DRAWING (SHEET 2)



DETAIL C

1:250 — 5 2.5 0 5 10 15 20

COMMUNICATION CABLES IN VICINITY OF WORKS

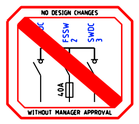
WATER PIPELINES IN VICINITY OF WORKS

W.A.P.C. REFERENCE No. 201772

CONSULTANT CONTACT DETAILS
COMPANY NAME: STATE WEST PLANNING
CONTACT: JASON HUNT TEL: 0408599187

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE
1	ISSUED FOR CONSTRUCTION	I.P.	12/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
FILE REFERENCE DETAILS

RESIDENTIAL SUBDIVISION

1 x NEW+ 2 x EXISTING DADMD

LOT 51/#1740 COPPING RD, PARKERVILLE WA 6081

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 12/25 | SCALE: 1:1000 @ A2
GEO REF: Lat. - 31°53'31" S Long. - 116°09'01" E

westernpower

DRAWING NUMBER

NP032946

SHEET 01 OF 02

REV: 1 | ORIGINAL SIZE - A2



DETAIL A

1:250 5 2.5 0 5 10 15 20

CLEARING PERMIT WORK INSTRUCTION

With regard to your requested works, it has been identified by Western Power that clearing or potential clearing is required to facilitate the connection. Being a customer funded project, the customer is responsible for obtaining all necessary clearing approvals prior to the commencement of works, you can engage an environmental consultancy to assist with this matter.

The Department of Water and Environmental Regulation (DWER) administers the clearing provisions of the Environmental Protection Act 1986 (EP Act). Under the EP Act, a clearing permit is required to authorise clearing of native vegetation unless a valid exemption applies or DWER identifies that a clearing permit is not required via the formal clearing referral process.

Refer to https://www.wa.gov.au/service/environment/environment-information-services/native-vegetation-clearing-permits for further information on clearing approval requirements. Advice on relevant exemptions can be obtained via DWER's "Guide 1 Exemptions and regulations for clearing native vegetation" and clarification can be sought via emailing DWER directly via info@dwer.wa.gov.au.

Detail A identifies the area for which clearing approvals will need to be obtained and in which you may have to arrange to clear/remove vegetation.

Please note that DWER may identify that the clearing has significant impacts and there may be additional requirements before or after DWER issue a clearing permit that may impact the delivery date. These may include flora or fauna survey requirements and hygiene management plans.

Please ensure you provide the Western Power Scoping Specialist / Construction Manager with a copy of one of the following, as soon as practicable, for review and suitability confirmation:

- A clearing permit; or
- A referral determination notice issued by DWER confirming a clearing permit is not required; or
- Written details of a relevant clearing permit exemption and supporting documents (including any Regulator correspondence).

Further relevant and supporting information is available via DWER's website via https://www.wa.gov.au/organisation/department-of-water-and-environmental-regulation. If you have any questions about Western Power's requirements, please contact the Western Power Quality Assurance team via ccs.sbm.qat@westernpower.com.au.

ENVIRONMENT ISSUES AND REQUIREMENTS

This page outlines environmental issues and requirements related to the scope of work, as well as any concerns in the vicinity that the Construction Manager must consider to prevent environmental damage. Particular attention is given to issues that could lead to prosecution. Project-specific information is referenced on this page and must be followed.

CUSTOMER TO OBTAIN PERMIT AND CLEAR VEGETATION TO THE SATISFACTION OF THE WESTERN POWER SCOPER/PNO
AS PER THE FOLLOWING.

Vegetation
Being a customer funded project, the customer is responsible for obtaining all necessary clearing approvals and,prior to Western Power commencing works, the customer is required to provide a copy of either:
A clearing permit issued in the customer×s name; or
A referral determination notice issued by Department of Water and Environmental Regulation (DWER) confirming a clearing permit is not required by the customer; or Written details of a relevant clearing permit exemption and supporting documents (Regulator correspondence/Landowner Exemption Clearing letter)
The above must include relevant development approvals for the works (i.e. building approvals for a structure that requires clearing to construct), if required. Approvals / Exemptions can be sent to the Safety, Health and Environment (SHE) Environment team via environmentf@westernpower.com.au for review and suitability confirmation. Allow 10 business days for Environment team to review.
To enable the customer to obtain clearing approvals, Western Power must provide the customer with a drawing sheet of the vegetation clearing required Vegetation Clearing Template (See CAD Conform Seed Files), any photos and/or shapefiles showing the exact location(s) where clearing is required. Refer to Clearing Permit Work Instruction for further details.

Fauna
The customer shall undertake any necessary fauna surveys and, if required, obtain relevant State / Commonwealth approvals for impacts to this species and/or its habitat. All surveys, approvals, and relevant management actions shall be provided to Western Power for record keeping and review.

Dangerous Goods - Hazardous Chemicals
The proposed works involves the removal or installation of assets associated with hazardous substances or dangerous goods. (e.g. oil filled assets). All crews should follow the Western Power Hazardous substances and dangerous goods work instruction which outlines the minimum safety, health and environmental requirements associated with handling, storing, transporting, and disposing of hazardous substances and dangerous goods.

Construction Noise
Standard construction noise management measures apply. If the proposed works are to be undertaken outside of 07:00 - 19:00 on any other day which is not a Sunday or a public holiday and/or are located near major sensitive receivers, consultation with the Local Government Authority will be required to determine mitigation measures and requirements for a noise management plan.

Aboriginal Heritage
The proposed works intersect with the Aboriginal Heritage site (Place ID:3188 - Darling Range). It is the Customer's responsibility to complete their own due diligence assessment and provide Western Power with the evidence of surveys/consultations and approvals (where applicable) prior to the commencement of works.

Land Use Planning
The Customer is responsible for consulting with the relevant local government (e.g. Shire of Mundaring) and confirming and satisfying any Development Application (DA) requirements providing evidence to Western Power.

Infrastructure - roads, gas, rail
In the event of works being undertaken within a LGA Road reserve, Formal approval is not required however a Notification of works (EDM 34174076) shall be submitted to the relevant local government authority at least 10 days before works commence in a local road reserve (excluding inspections).



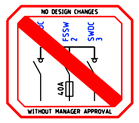
AREA A- CLEARING ENVELOPE		
POINT ID	NORTHING	EASTING
A	419661	6471223
B	419664	6471221
C	419653	6471203
D	419650	6471205

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT. IF IN DOUBT PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL OPERATION TO ARRANGE A PERMIT.
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: JASON HUNT TEL: 0408599187

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				ENVIRONMENTAL ASSESSMENT
1	ISSUED FOR CONSTRUCTION	I.P.	12/25	HAZARD MANAGEMENT REGISTER

ENVIRONMENTAL DRAWING

LOT 51/#1740 COPPIN RD

PARKERVILLE WA 6081

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: RASP
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 12/25 SCALE: 1:1000 @ A2
GEO REF: Lat. - 31°53'31" S Long. - 116°09'01" E

westernpower

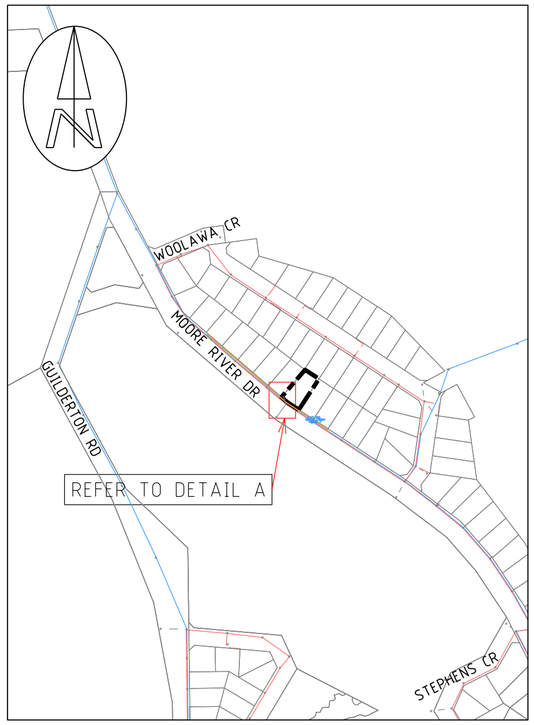
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NP032946

SHEET 02 OF 02

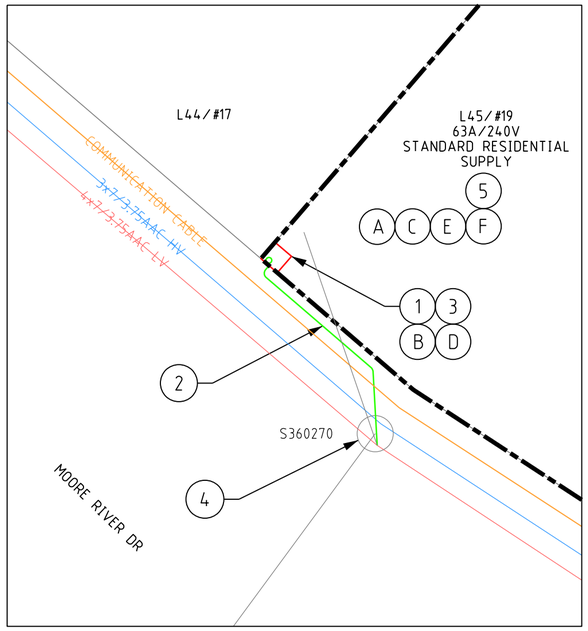
REV: 1 ORIGINAL SIZE - A2

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LOCATION PLAN

1:5000 100 50 0 100 200 300 400



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

(i) THIS PROJECT IS DEPENDENT ON COMPLETION OF PROJECT ND010180

1. INSTALL NEW MINI-PILLAR ON L45/#19 MOORE RIVER DR WHERE PEGGED BY CUSTOMER. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 1
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S360270.
   CU: CN42 x 25m, CN73_LV25 x 3, CN54 x 2
3. AT NEW MINI-PILLAR, TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
4. AT POLE#S360270,
   - TERMINATE NEW 25LV CABLE TO LV BARE CONDUCTORS.
   - REMOVE EXISTING AERIAL SERVICE TO L45/#19.
   CU: LU51_25ALOx1, LU51_25ALUx1
   DCSH: U19-1
5. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   THREATENED ECOLOGICAL COMMUNITIES
   ABORIGINAL HERITAGE SITES
   NATIVE TITLE BOUNDARIES
   UNEXPLODED ORDNANCE
   CARNABYS COCKATOO CONFIRMED BREEDING AREAS (DBCA-054)
   CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
   BLACK COCKATOO ROOSTING SITES - BUFFERED (DBCA-064)
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   RIWI ACT, GROUNDWATER AREAS (DWER-034)
   BUSH FIRE PRONE AREAS - CURRENT (OBRM-001)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

REFER TO environment@westernpower.com.au

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- U UNIVERSAL
- MINI
- L LINK
- CABLE MARKER U/G
- UM UMS PIT
- UTILITIES PIT
- SP BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

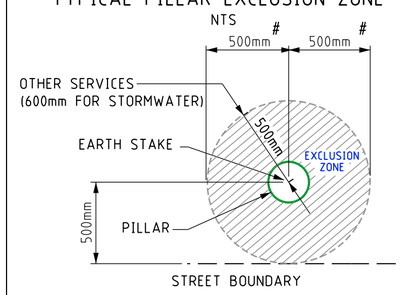
(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE FOR COMPLETING THEIR DUE DILIGENCE ASSESSMENT OF POTENTIAL ABORIGINAL HERITAGE IMPACTS AND ADVISING WESTERN POWER OF ANY COMPLIANCE REQUIREMENTS THAT APPLY.

(F) THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

CRITICAL WATER PIPE IN VICINITY OF WORKS

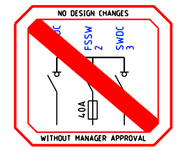
COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: BRAD GIBLETT TEL: 0428220075

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	12/25	HAZARD MANAGEMENT REGISTER

OVERHEAD TO UNDERGROUND CONVERSION

19 MOORE RIVER DR

GUILDERTON WA, 6041

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: ISHAN PALIKHE
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87

DATE: 12/25 SCALE: AS NOTED @ A2

GEO REF: Lat. - 31°20'46.1"S
Long. - 115°30'32.6"E

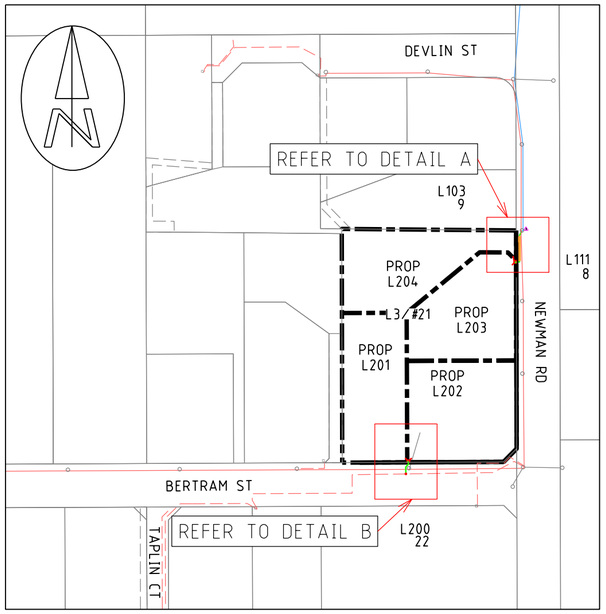
westernpower

DRAWING NUMBER

NP032951

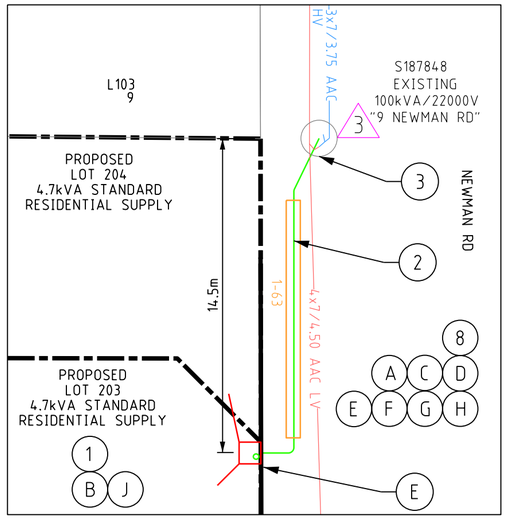
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



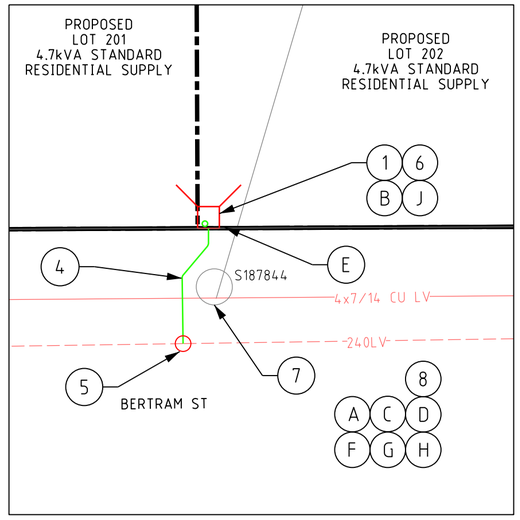
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL A

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON CUSTOMER'S PROPERTY WHERE PEGGED BY CUSTOMER. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10 x 2
   DCSH: U08-04
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S187848 IN 2.7m ALLIGNMENT VIA 63mm CONDUIT.
   CU: CN42 x 30m, CN73_LV25 x 3, CN54 x 2
3. AT POLE#S187848
   - TERMINATE NEW 25LV CABLE TO LV BARE CONDUCTORS.
   CU: LU51_25ALOx1, LU51_25ALUx1,
   DCSH: U19-1
4. INSTALL 25LV CABLE FROM NEW MINI-PILLAR TO EXISTING 250LV CABLE.
   CU: CN42 x 15m, CN73_LV25 x 3, CN54 x 1
5. TEE-JOINT 25LV CABLE WITH EXISTNG 240LV CABLE
   CU: L15 x 1
6. AT NEW MINI-PILLAR, TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
7. AT POLE#S187844, REMOVE EXISTING OVERHEAD SUPPLY TO L3/#21
8. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   ABORIGINAL HERITAGE SITES
   CARNABYS COCKATOO CONFIRMED BREEDING AREAS (DBCA-054)
   CARNABYS COCKATOO UNCONFIRMED BREEDING AREAS (DBCA-055)
   BLACK COCKATOO ROOSTING SITES - BUFFERED (DBCA-064)
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, SURFACE WATER AREAS AND IRRIGATION DISTRICTS (DWER-037)
   BUSH FIRE PRONE AREAS - CURRENT (OBRM-001)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO environment@westernpower.com.au

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

E. CUSTOMER TO REMOVE SECTION OF STONE WALL INFORNT OF THE PILLAR LOCATION.

F. CUSTOMER TO SUPPLY AND INSTALL A SMSB PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY.
   CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

G. CUSTOMER IS RESPONSIBLE FOR COMPLETING THEIR DUE DILIGENCE ASSESSMENT OF POTENTIAL ABORIGINAL HERITAGE IMPACTS AND ADVISING WESTERN POWER OF ANY COMPLIANCE REQUIREMENTS THAT APPLY.

H. THE CUSTOMER IS RESPONSIBLE FOR CONSULTING WITH THE RELEVANT LGA CONFIRMING AND SATISFYING ANY DEVELOPMENT APPROVAL REQUIREMENTS THAT APPLY.

J. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER ELECTRICAL SERVICE REQUEST OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

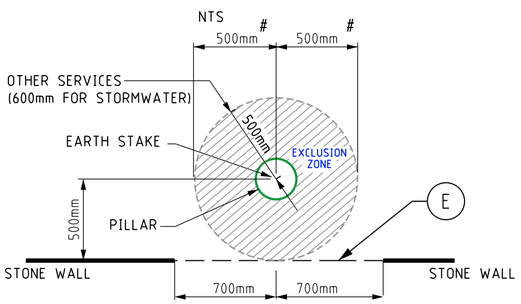
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDN Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT
- INTERMEDIATE
- DOF
- INLINE STRAIN
- PTT
- AERIAL STAY
- RECLOSER
- LOAD BREAK SWITCH
- CAPACITOR
- LBPTS
- PTS
- HV ISOLATOR
- LV DISCONNECTOR
- ISOL TX
- OUTRIGGER STAY
- GROUND STAY
- SECTIONALISER
- LV ABC KRONE
- STREETLIGHT BOX

PILLAR EXCLUSION ZONE



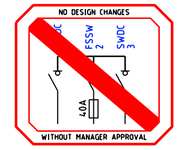
500mm to either side boundary

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☐	☒	☐	☒	☐

TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



W.A.P.C. REFERENCE No. 201711

WATER PIPELINES IN VICINITY OF WORKS

CONSULTANT CONTACT DETAILS
COMPANY NAME: PROWEST SURVEYING PTY LTD
CONTACT: RYAN PARKER TEL: 0439828950

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT
A	ISSUED FOR CONSTRUCTION	I.P.	12/25	HAZARD MANAGEMENT REGISTER

4x 4.7kVA(DADMD) STANDARD RESIDENTIAL SUPPLY

21 BERTRAM ST

DARLINGTON, WA, 6070

NETWORK VOLTAGE: 22kV / 415V

DESIGNER: Ishan Palikhe
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 12/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 31°54'39.2"S Long. - 116°05'12.0"E

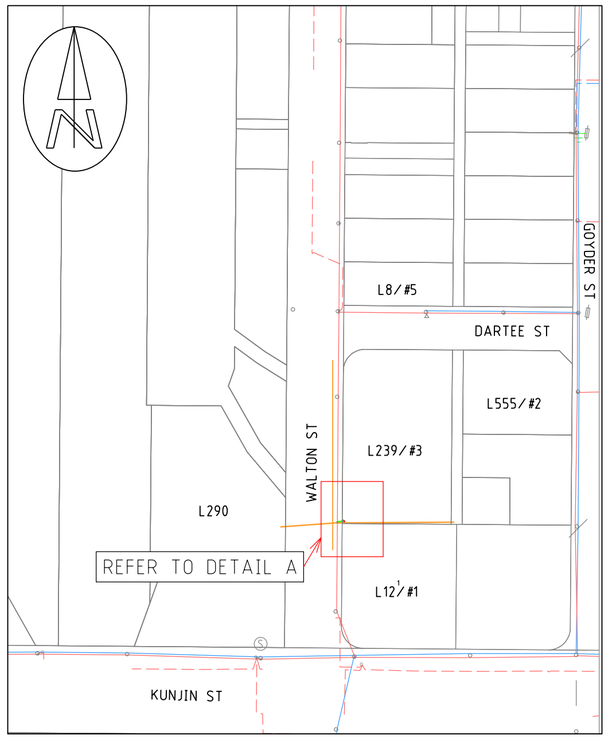
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DRAWING NUMBER

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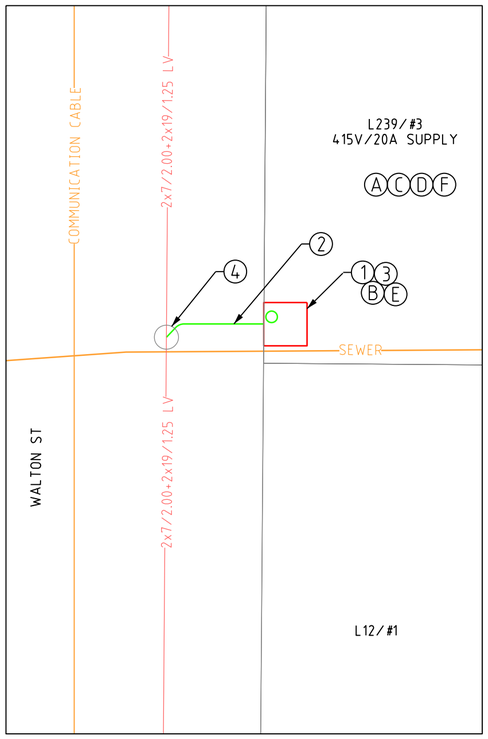
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) THIS DESIGN IS INTENDED TO BE CONSTRUCTED DE-ENERGISED. YOUR ELECTRICITY SUPPLY MAY NOT BE AVAILABLE DURING THIS TIME

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(F) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WP, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENGERGISED.

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW MINI-PILLAR ON L239/#3 WALTON ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10

(2) INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#872136.
CU: CN42 x 20m, CN73_LV25 x 2, CN54 x 2

(3) AT MINI-PILLAR (L6/#128 CRYSTAL BROOK RD)
- TERMINATE NEW 25LV CABLE.
- TERMINATE CUSTOMER'S CABLE TO THE BACK BLOCKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
DCSH: U8-4

(4) AT POLE#872136
- TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
CU: LU51_25ALO x 1, LU51_25ALU x 1
DCSH: U19-1

(5) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

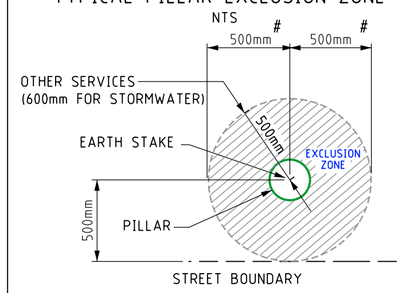
HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE ADN REPORT THE INCIDENTT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERTAIONS WORK INSTRUCTION (EDM 4150794)

COMMUNICATION CABLES IN VICINITY OF WORKS

SEWER PIPE AND PIT IN VICINITY OF WORKS

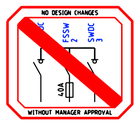
TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary





SAFETY - MAKE IT YOUR PRIMARY VALUE

APPROVED FOR CONSTRUCTION

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: HUTNORSALES
CONTACT: JOHN TRUNFIO TEL: 0429 411 077

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	06/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68699526
HAZARD MANAGEMENT REGISTER EDM#: 68715204

MINI-PILLAR INSTALLATION

NETWORK VOLTAGE: 415V

3 WALTON ST, CORRIGIN WA 6375

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 32° 19' 57" S Long. - 117° 52' 25" E

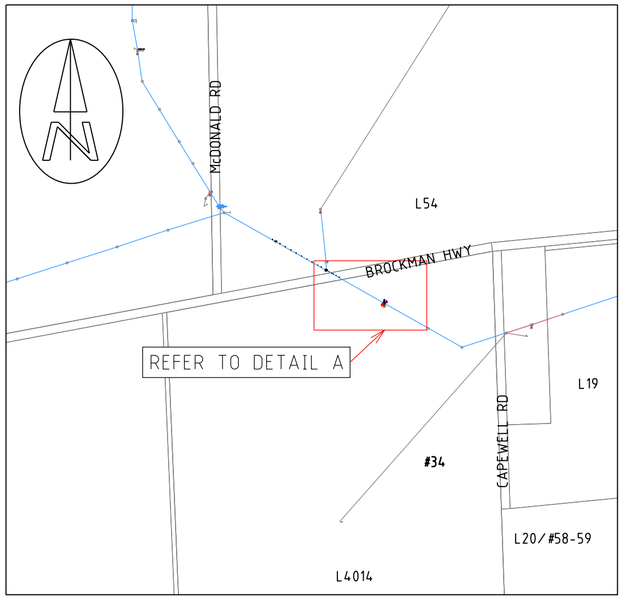
westernpower

DRAWING NUMBER

SP050747

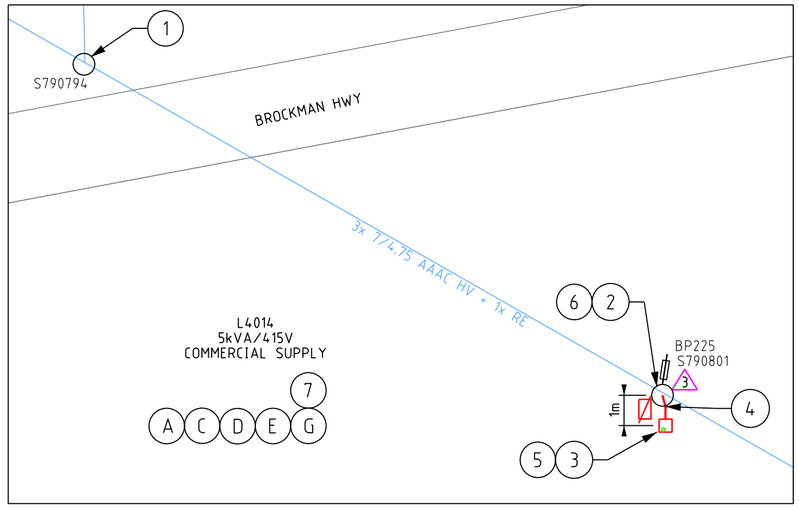
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



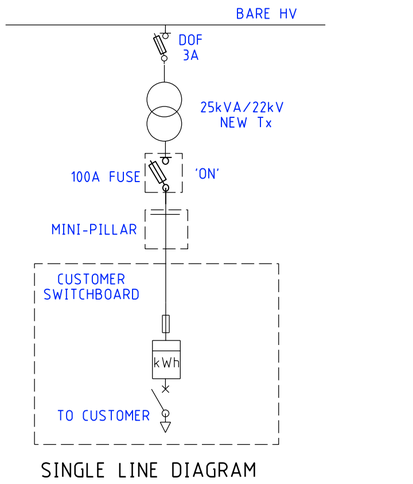
LOCATION PLAN

1:10,000 — 200 100 0 200 400 600 800



DETAIL A

1:500 — 10 5 0 10 20 30 40



SINGLE LINE DIAGRAM

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WP, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENGERGISED.

F. CUSTOMER MUST INSTALL SMSB WITHIN 30m OF POINT OF SUPPLY.

G. INSTALLATION SHALL COMPLY WITH THE WAER CLAUSE 3.6.7 FOR MULTIPLE POINTS OF SUPPLY FOR RURAL / SEMI-RURAL LOTS. ALL REQUIREMENTS OF THE WAER CONDITIONS, WHICH INCLUDES 200m SEPARATION FROM EACH POINT OF SUPPLY AND 50m SEPARATION FROM EACH SUPPLIES ELECTRICAL INSTALLATION SHALL BE UNDERTAKEN PRIOR TO REQUESTING ENERGISATION. PLEASE ENSURE THE ELECTRICAL CONTRACTOR IS AWARE OF THE REQUIREMENTS.

WESTERN POWER SCOPE OF WORK

1. AT POLE#S790794.
- REPLACE THE POLE WITH 14m/8kN POLE.
- INSTALL HV ANTI-SWAN INTERMEDIATE X-ARM.
- INSTALL NEW 1PH HV T-OFF
- INSTALL RUNNING EARTH INTERMEDIATE.
- INSTALL RUNNING EARTH TERMINATION.

CU: P011_Rx1, HV43_3NASx1, HV30x1, RE1x1, RE3_7/250AAx1, CN9_7/475AAx4, CN1_7/475AAx2, CN18_7/475AAx4
DCSH: H02-1, R12-2, R13-3, R3-3

2. AT POLE#S790801.
- REPLACE THE POLE WITH 14m/8kN POLE.
- INSTALL HV ANTI-SWAN INTERMEDIATE X-ARM.
- INSTALL 3A FAULT TAMER DOF AND STANDOFF BRACKET.
- TAP DROP OUT FUSE TO HV BARE.
- INSTALL RUNNING EARTH INTERMEDIATE.
- INSTALL NEW 22kV/25kVA POLE TOP TRANSFORMER.

CU: P011_Rx1, HV43_3NASx1, HV20x1, RE1x1, TX4_25/22 x 1, CN9_7/4.75x3, HV52_3FT x3, CN18_7/457AAx4
DCSH: H10-1, R12-2, R13-3

3. INSTALL NEW MINI-PILLAR WITHIN 1m OF POLE#S790801 AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10 x 1
DCSH: U08-04, U19-3

4. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S790801.
CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 1

5. AT NEW MINI-PILLAR, TERMINATE NEW 25LV CABLE.
DCSH: U08-04

6. AT POLE#S790801, TERMINATE NEW 25LV CABLE
CU: LU51_25_3PHU x 1, LU51_25_3PHO x 1
DCSH: U19-1

7. POSSESSION OF SITE CERTIFICATE IS TO BE REQUESTED, AND MAIN ROADS DOCUMENT D18#604587 'SEEKING APPROVAL TO UNDERTAKE WORKS WITHIN THE ROAD RESERVE' TO BE COMPLETED.

CONSTRUCTOR TO ENSURE THAT ALL WORKS IN MAIN ROADS RESERVE COMPLIES WITH MRWA CONDITIONAL APPROAL DOCUMENT AVAILABLE IN VOLT.

7. ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

ABORIGINAL HERITAGE SITES
IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT AND LAND ACCESS IN OPERATIONS WORK INSTRUCTION (EDM 4150794)

NATIVE VEGETATION REGS ESAS CLEARING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
- CN53 - 400mm2 (3X1c)
- CN49 - 240mm2 (3X1c)
- CN48 - 185mm2 (3X1c)
- CN50 - 95mm2 (3X1c)
- CN51 - 50mm2 (3X1c)
- CN51 - 50mm2 (1c)
- CN46 - 35mm2 (3X1c)
- CN47 - 35mm2 (1c)
- EXISTING HV CABLES (SIZE AND TYPE INDICATED)

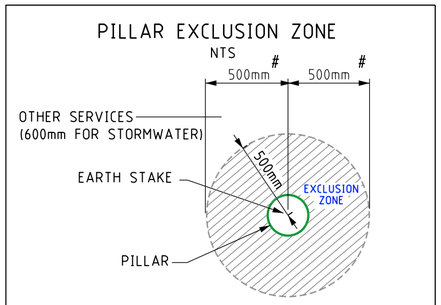
LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

TRANSFORMER — EXISTING — SWITCHGEAR

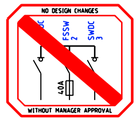
AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

PILLAR EXCLUSION ZONE



ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: Mrces (WA) Pty Ltd
CONTACT: Ian & Nathan TEL: 0418932847

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P	06/25

FILE REFERENCE DETAILS
ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER

5kVA NEW RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

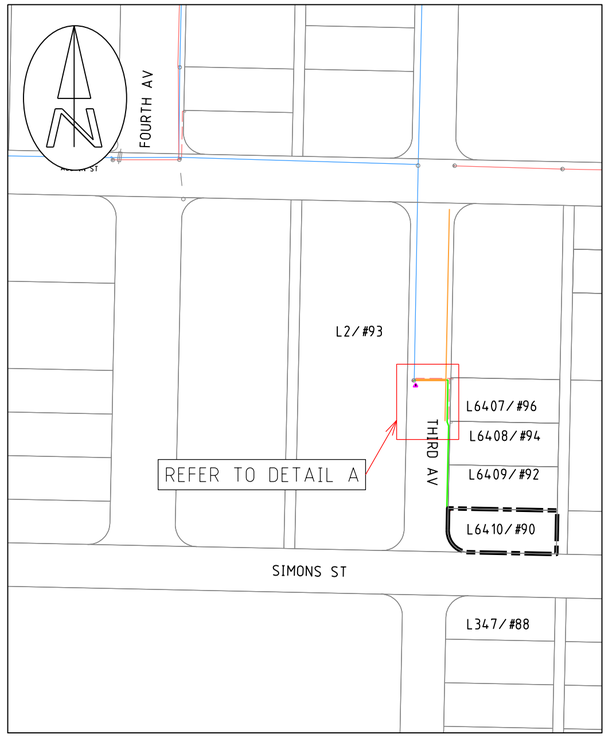
34 CAPEWELL RD, KARRIDALE WA 6288

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 06/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 34° 11' 32" S Long. - 115° 07' 27" E

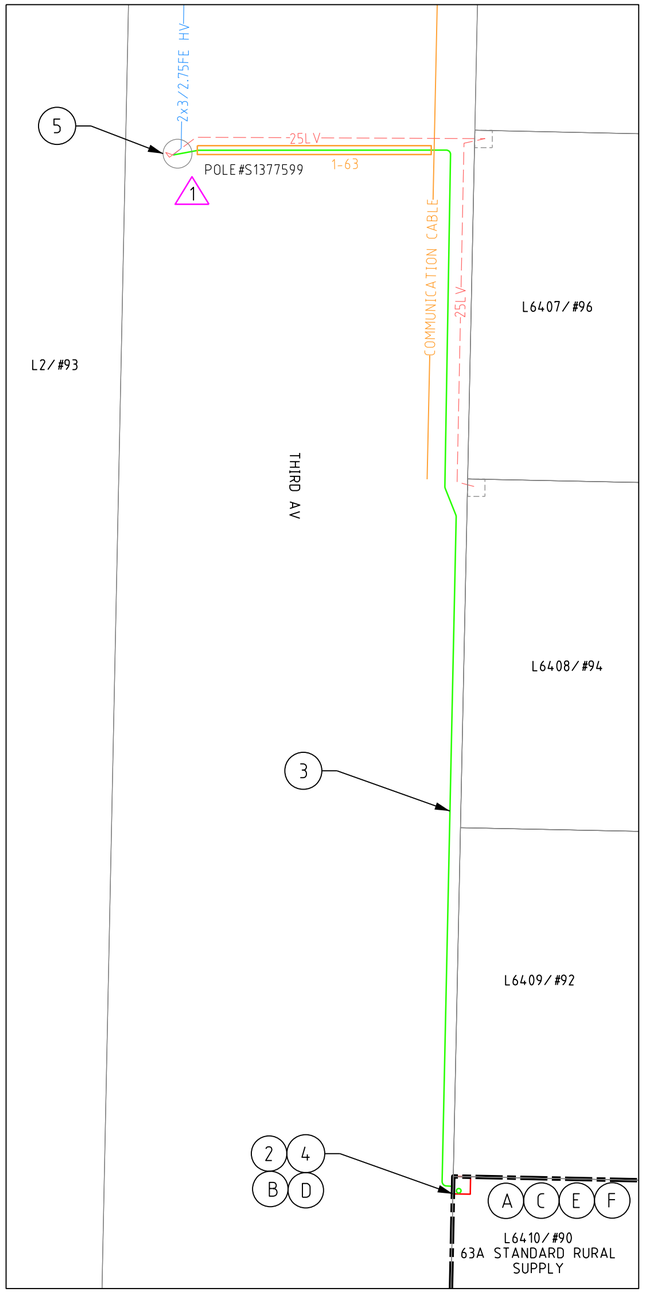
westernpower

DRAWING NUMBER
SP050956
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:250 5 2.5 0 5 10 15 20

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO: VERIFY THE NEIGHBOURS CONNECTION ARRANGEMENT AT PROPERTY #94 & #96 TO CONFIRM 240V OR 480V CONFIGURATION AND SET UP THE NEW TRANSFORMER TO MATCH.

1. MODIFY THE NEIGHBOURS CONNECTION AT RESPECTIVE PILLARS ACCORDINGLY IF THE CONNECTION ARRAGEMENT IS MODIFIED.
2. INSTALL NEW MINI-PILLAR ON NORTHERN CORNER OF L6410/#90 THIRD AV, AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
   CU: LU10 x 1
3. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S1377599 VIA 63mm POLYPIPE.
   CU: CN42 x 90m, CN73_LV25 x 3, CN54 x 2
4. AT MINI-PILLAR (L6410/#90 THIRD AV)
   - TERMINATE NEW 25LV CABLE FOR 480V.
   DCSH: U08-04
5. AT POLE#S1377599
   - REPLACE EXISTING 10kVA Tx WITH NEW 25kVA/12.7kV Tx
   - TERMINATE NEW 25LV CABLE TO Tx VIA 80A FUSE.
   - RE-TERMINATE EXISTING 25LV CABLE TO Tx VIA 80A FUSE.
   CU: LU61_25_240VO x 1, LU61_25_240VU x 1 TX5_25/12.7P x 1, CN35_POLE x 3, CN35_80 x 4
   NON CU: GF1947 x 4, GF1944 x 4
   DCSH: H47-1, H49, U19-3, R12-1
6. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(F) CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

LEGEND

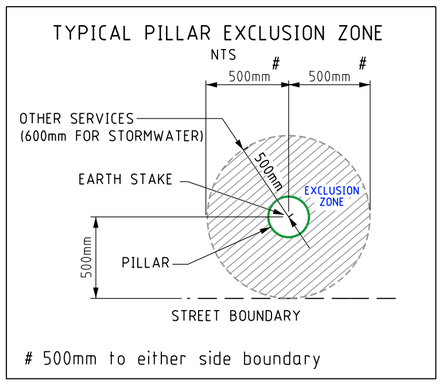
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

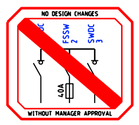
AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

SINGLE LINE DIAGRAM



ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: DANIEL WOODRUFFE TEL: 0437309414

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	08/24

SHE FILTER ASSESSMENT EDM#: 69431824
HAZARD MANAGEMENT REGISTER EDM#: 69431765
FILE REFERENCE DETAILS

5kVA STANDARD RURAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

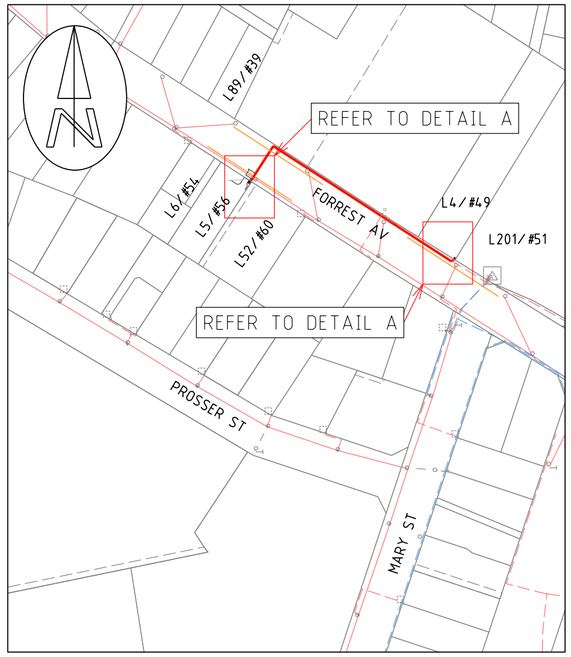
90 THIRD AV, KENDENUP 6323 WA

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 06/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 34° 29' 04" S Long. - 117° 37' 27" E

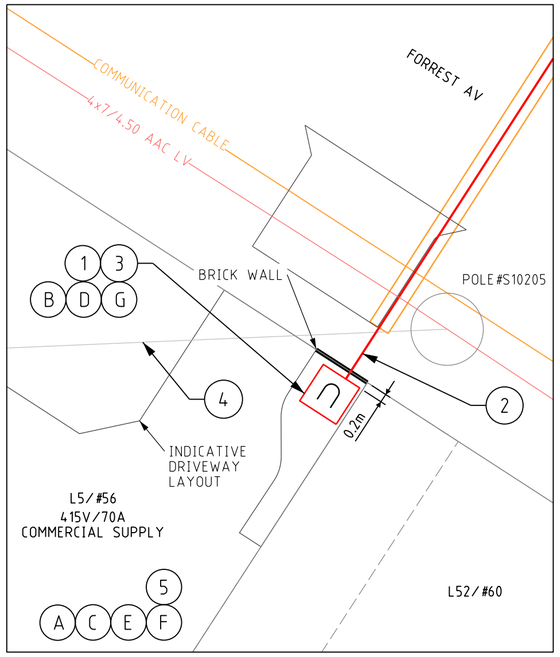
westernpower

DRAWING NUMBER
SP050979
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



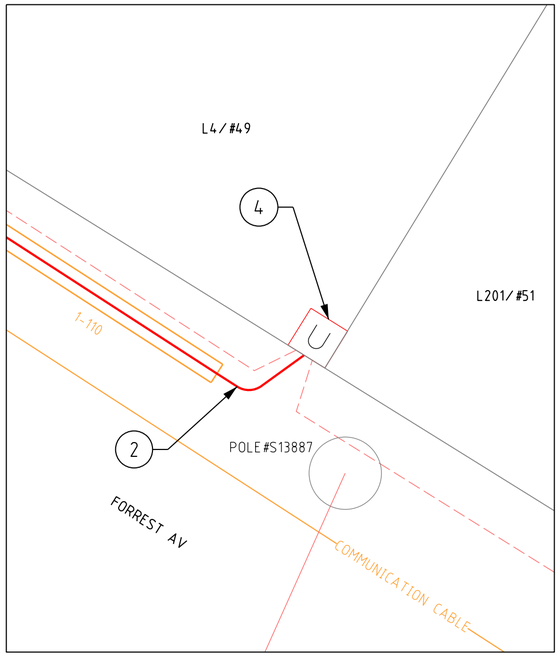
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:100 — 2 1 0 2 4 6 8



DETAIL B

1:100 — 2 1 0 2 4 6 8

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

(G) CUSTOMER TO SUPPLY AND INSTALL DUCTING AS NOTED ON THE DESIGN DRAWING.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW UNI-PILLAR ON L5/#56 FORREST AV. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11 x 1
DCSH: U9

(2) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO EXISTING UNI-PILLAR AT L4/#49 FORREST AV.
CU: CN45 x 130m, CN54 x 3, CN73_LVMAINS x 3

(3) AT NEW UNI-PILLAR (L5/#56 FORREST AV)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
CU: LU7_240x1

(4) AT EXISTING UNI-PILLAR (L4/#49 FORREST AV)
- TERMINATE NEW 240LV TO TOP BAR VIA LINKS.
CU: LU7_240 x 1, LU36_Link x 1
DCSH: U9

(5) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
- BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
- IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

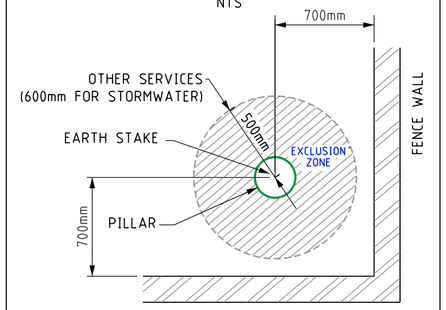
PUBLIC DRINKING WATER SOURCE AREA
THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

LEGEND

- SCHEME BDY.
- RETAINING WALL
- LV-SL CABLES
  - CN45 - 240mm2 (1x3c)
  - EXISTING LV CABLES (SIZE AND TYPE INDICATED)
- PILLARS / PITS / ETC
  - U UNIVERSAL
  - CONDUIT / POLY PIPE
- AERIAL CONDUCTORS AND POLES
  - NEW POLE (ANY TYPE AS INDICATED)
  - EXISTING POLE (ANY TYPE AS INDICATED)
  - HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

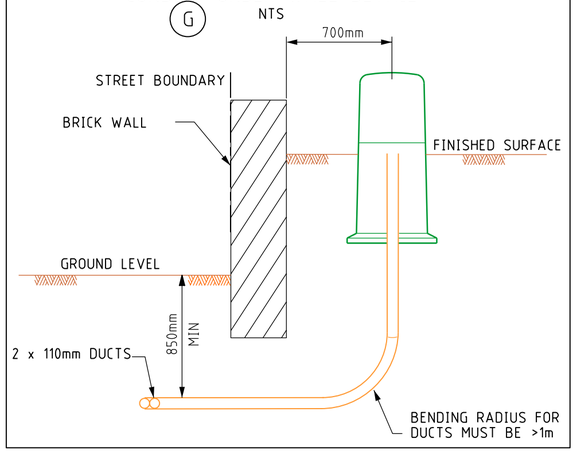
PILLAR EXCLUSION ZONE WITH BRICK WALL

NTS

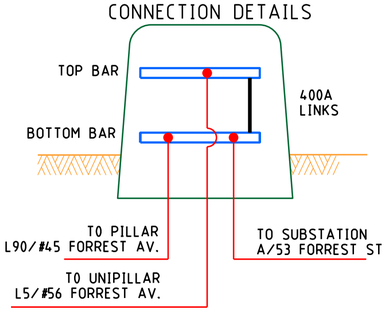


CONDUIT UNDER WALL DETAIL

(G) NTS

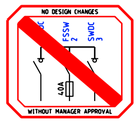


UNI-PILLAR L4/#49 CONNECTION DETAILS



ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
NAME: TERRY PANTLIN
EMAIL: Terry@pantlin.com.au TEL: 08) 9791-8188

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	08/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69236342
HAZARD MANAGEMENT REGISTER EDM#: 69236028

415V/70A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

56 FORREST AVE, SOUTH BUNBURY WA 6230

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 08/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 33° 20' 23" S
Long. - 115° 38' 40" E

westernpower

DRAWING NUMBER

SP051021

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2

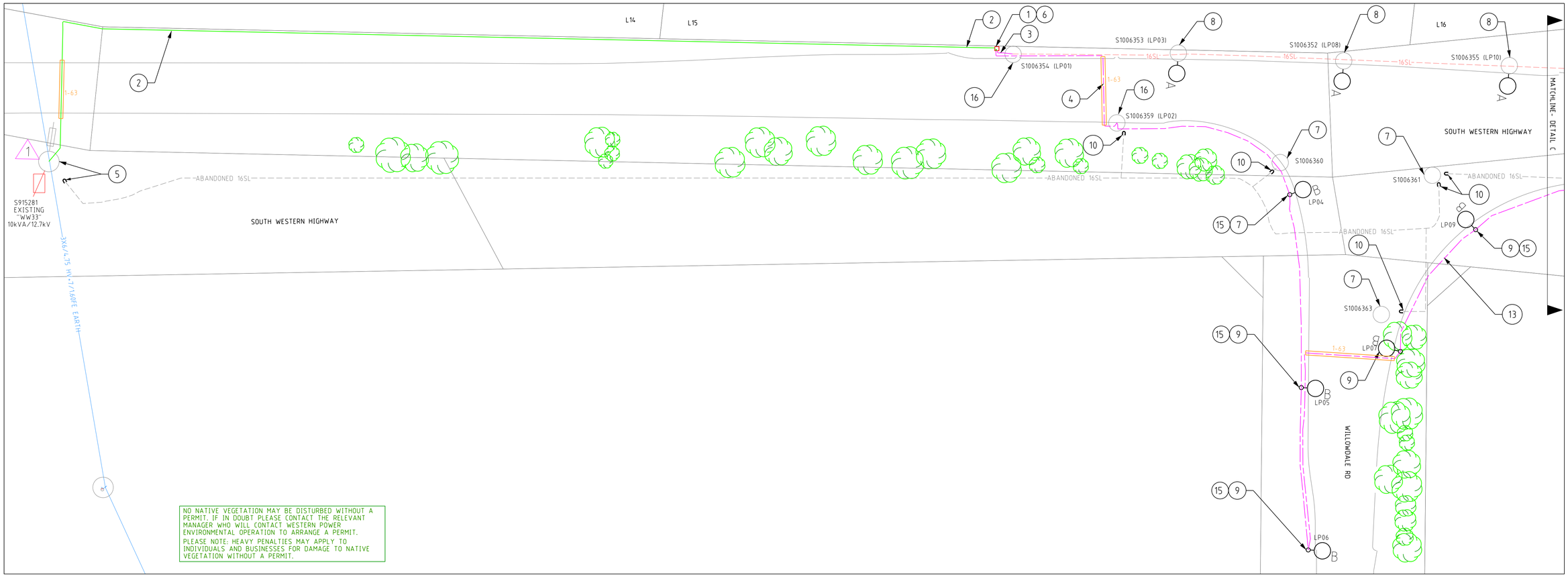
THIS SHEET IS TO BE READ INCONJUNCTION WITH SHEET 02

GOLDEN SAFETY RULES ARE INDICATIVE ONLY

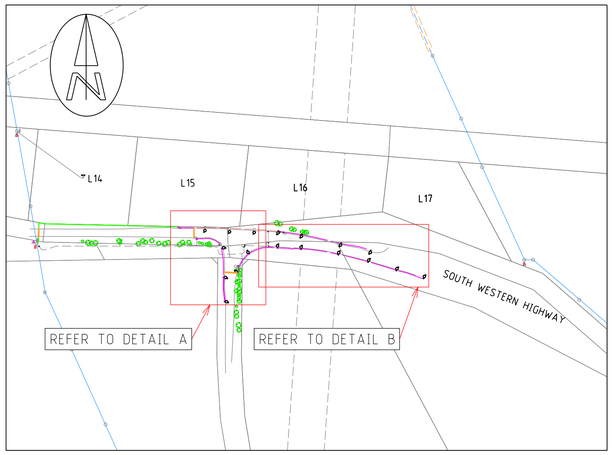
HV ☒
LV ☒
☒
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ALWAYS PERFORM A DETAILED HAZARD ANALYSIS

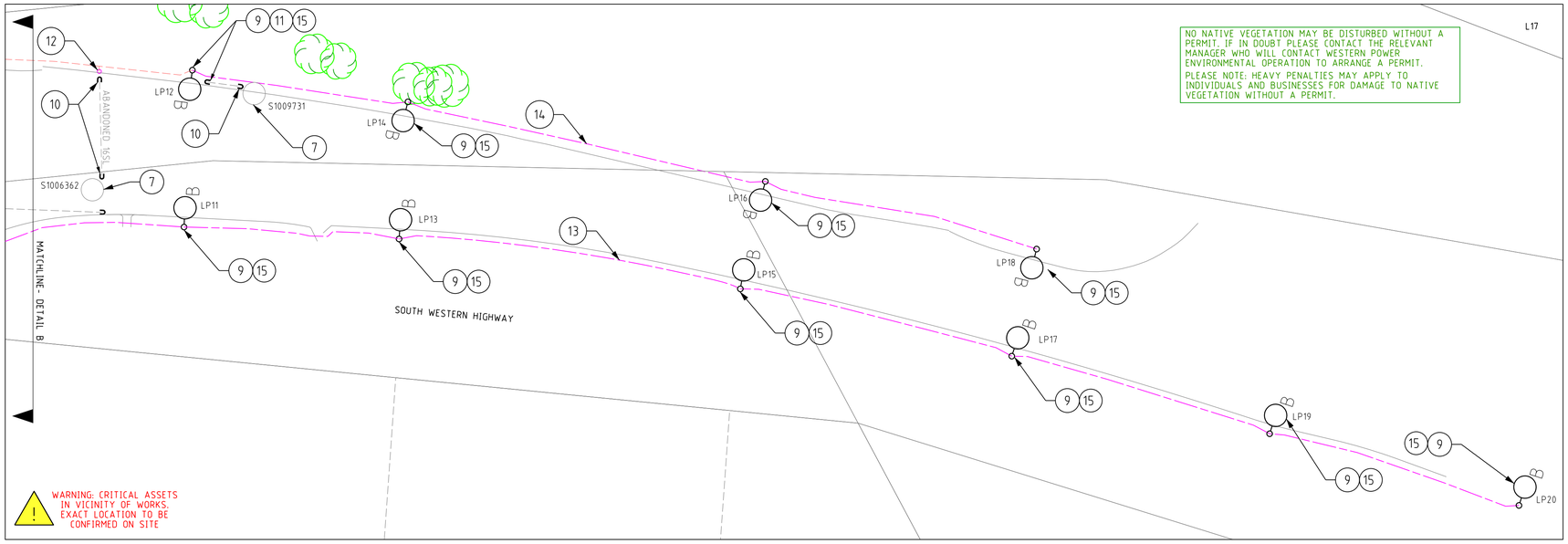
NO DESIGN CHANGES
WITHOUT MANAGER APPROVAL



DETAIL A
1:500 10 5 0 10 20 30 40



LOCATION PLAN
1:5000 100 50 0 100 200 300 400



DETAIL B
1:500 10 5 0 10 20 30 40

ISSUED FOR CONSTRUCTION

CUSTOMER CONTACT DETAILS
COMPANY NAME: MRWA
CONTACT: Louieji Estrella TEL: 0419 359 960

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

Rev.	REVISIONS	DRN	DATE
A	UNDER ENGINEERING REVIEW	XX	MM/YY

REFERENCE FILE DETAILS
SHE ASSESSMENT REPORT EDM#: 70799010
HAZARD MANAGEMENT REGISTER - EDM# 70799654

MRWA STREETLIGHT UPGRADE

NETWORK VOLTAGES: 12.7kV/250V

WILLOWDALE RD, WAGERUP
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 53' 47" S Long. - 115° 54' 20" E

DRAWING NUMBER
SP051023
SHEET 01 OF 02
REV: A ORIGINAL SIZE - A1

THIS SHEET IS TO BE READ INCONJUNCTION WITH SHEET 1

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

STREET LIGHTS — EXISTING

# POLE BRACKET	POLE BRACKET
# SINGLE OUTREACH	SINGLE OUTREACH
# DOUBLE OUTREACH	DOUBLE OUTREACH
# 4-WAY	4-WAY

- ALPHA CHARACTER DENOTES STREET LIGHT TYPE. REFER STREET LIGHT SCHEDULE FOR POLE & LUMINAIRE DETAILS.

PILLARS / PITS / ETC

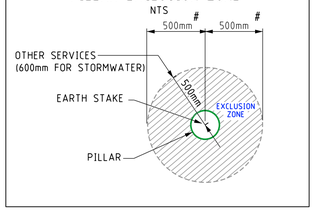
MINI (WORKING END)	UMS PIT
U UNIVERSAL	UTILITIES PIT
MINI	BGSP PIT
L LINK	100/200AMP CUTOUT
CABLE MARKER U/G	CONDUIT / POLY PIPE
SUG Serviced Unclearad De-energised Lot	

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

PILLAR EXCLUSION ZONE

NTS



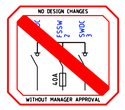
STREET LIGHT SCHEDULE

TYPE	POLE ID	DISCRIPTION	POLE LOCATION		COLOR	CU'S
			EASTING	NORTHING		
A	S1006353 (LP03)	REPLACE LUMINAIRE WITH NEW 165W 6E LED	EXISTING	EXISTING	N/A	
B	LP04	12.5m STANDARD STEEL POLE	6359590.973	397649.987	N/A	SL11_6E
B	LP05	12.5m STANDARD STEEL POLE	6359571.827	397692.929	N/A	SL11_6E
B	LP06	12.5m STANDARD STEEL POLE	6359555.068	397728.733	N/A	SL11_6E
B	LP07	12.5m STANDARD STEEL POLE	6359597.272	397696.298	N/A	SL11_6E
A	S1006352(LP08)	REPLACE LUMINAIRE WITH NEW 165W 6E LED	EXISTING	EXISTING	N/A	
B	LP09	12.5m STANDARD STEEL POLE	6359627.19	397678.486	N/A	SL11_6E
A	S1006355(LP10)	REPLACE LUMINAIRE WITH NEW 165W 6E LED	EXISTING	EXISTING	N/A	
B	LP11	12.5m STANDARD STEEL POLE	6359672.059	397691.101	N/A	SL11_6E
B	LP12	12.5m STANDARD STEEL POLE	6359686.348	397666.918	N/A	SL11_6E
B	LP13	12.5m STANDARD STEEL POLE	6359705. 179	397710.774	N/A	SL11_6E
B	LP14	12.5m STANDARD STEEL POLE	6359717.903	397689.774	N/A	SL11_6E
B	LP15	12.5m STANDARD STEEL POLE	6359755.179	397746.909	N/A	SL11_6E
B	LP16	12.5m STANDARD STEEL POLE	6359768.022	397731.936	N/A	SL11_6E
B	LP17	12.5m STANDARD STEEL POLE	6359792.66	397780.017	N/A	SL11_6E
B	LP18	12.5m STANDARD STEEL POLE	6359805.456	397765.093	N/A	SL11_6E
B	LP19	12.5m STANDARD STEEL POLE	6359827.142	397813.635	N/A	SL11_6E
B	LP20	12.5m STANDARD STEEL POLE	6359860.871	397845.58	N/A	SL11_6E
N/A	S1006360	REMOVED	EXISTING	EXISTING	N/A	
N/A	S1006361	REMOVED	EXISTING	EXISTING	N/A	
N/A	S1006362	REMOVED	EXISTING	EXISTING	N/A	
N/A	S1006363	REMOVED	EXISTING	EXISTING	N/A	
N/A	S1009731	REMOVED	EXISTING	EXISTING	N/A	

ISSUED FOR CONSTRUCTION

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT. IF IN DOUBT PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL OPERATION TO ARRANGE A PERMIT.
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

WARNING: CRITICAL ASSETS IN VICINITY OF WORKS. EXACT LOCATION TO BE CONFIRMED ON SITE





GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: MRWA
CONTACT: Louieji Estrella TEL: 0419 359 960

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

B. CUSTOMER TO PEG THE LOCATION OF ALL NEW STREET LIGHTS IN ACCORDANCE WITH THE CUSTOMERS STREET LIGHTING DESIGN. AS1158. WESTERN POWERS 'UNDERGROUND DISTRIBUTION SCHEMES' MANUAL AND ANY SPECIFIC REQUIREMENTS FROM LGA AND/OR ROAD AUTHORITY AT NO COST TO WESTERN POWER. IT IS THE CUSTOMERS RESPONSIBILITY TO ADHERE TO THE FOLLOWING:
- THE LOCATION OF ALL STREET LIGHTS IS CLEAR OF ANY 3RD PARTY ASSETS
- ALL STREET LIGHTS MAINTAIN A MINIMUM SET BACK FROM THE ROAD IN ACCORDANCE WITH AS1158.1.2 AND WESTERN POWERS UDS MANUAL.
- ANY PROPOSED STREET LIGHT LOCATIONS OUTSIDE OF THE WESTERN POWER STANDARD 2.4 - 3.0m POLE ALIGNMENT HAVE THE APPROVAL OF 3RD PARTY UTILITIES IN ACCORDANCE WITH THE UTILITY PROVIDERS CODE OF PRACTICE.
- TREET LIGHTS ARE NOT PERMITTED TO BE INSTALLED WITHIN THE WESTERN POWER STANDARD 0-500mm CABLE ALIGNMENT.
- ALL STREET LIGHTS ARE TO ADHERE TO THE POWER LINE CLEARANCE REQUIREMENTS AS PER AS7000. INCLUDING IN THE EVENT THEY ARE STRUCK BY A VEHICLE AND FALL TOWARDS THE TRANSMISSION LINE.

C. ANY TEMPORARY STREET LIGHTING REQUIRED DURING THE WORKS AFTER REMOVAL OF EXISTING STREET LIGHTS UNTIL NEW STREET LIGHTS HAVE BEEN ENERGISED IS TO BE SUPPLIED AN INSTALLED BY THE CUSTOMER AT NO COST TO WESTERN POWER.

D. CUSTOMER TO ENSURE UNMETERED SUPPLY ACCOUNT IS CREATED / UPDATED TO REFLECT ADDITIONAL STREET LIGHTING. CUSTOMER TO COMPLY WITH THE UNMETERED SUPPLY STANDARD AVAILABLE THE WESTERN POWER WEBSITE.

E. IF THE BACK OF KERB CONSTRUCTION HAS NOT YET BEEN COMPLETED, THE CUSTOMER TO MARK THE BACK OF THE KERB ON-SITE FOR REFERENCE.

WESTERN POWER SCOPE OF WORK

i. NOTE TO PND:
ALL NEW 16SL CABLES ARE TO BE INSTALLED AT ALLIGNMENT 1m OFFSET FROM BACK OF KERB.

1. INSTALL NEW MINI-PILLAR 1m FROM EXISTING POLE#S1006354 AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10 x 1

2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S915281
CU: CN42 x 275m, CN73_LV25 x 3, CN54 x 3

3. INSTALL NEW 16SL CABLE FROM NEW MINI-PILLAR TO EXISTING POLE#S1006354.
CU: CN41 x 15m, CN73_LV10/16 x 3, CN54 x 2

4. INSTALL NEW 16SL CABLE FROM NEW MINI-PILLAR TO EXISTING POLE#S1006359.
CU: CN41 x 60m, CN73_LV10/16 x 2, CN54 x 2

5. AT POLE#S41040
- DISCONNECT AND REMOVE EXSITING SL CABLE FROM THE POLE. THEN, CUT, CAP AND ABANDON EXSITING SL CABLE.
- TERMINATE NEW 25LV CABLE TO EXISTING Tx VIA 100A FLOWLINE FUSE.
CU: LU61_25_240Ox1, LU61_25_240Ux1
DCSH: U19-1

6. AT NEW MINI-PILLAR
- TERMINATE NEW 25LV CABLE.
- TERMINATE 2x 16SL CABLES VIA RED SPOT FUSES.
CU: LM3_R x 2
DCSH: U08-04

7. REMOVE EXISTING STREETLIGHTS S1006360, S1006361, S1006362, S1006363, S1009731.

8. REPLACE EXISTING LUMINAIRE WITH 165W 3000K LED STREET LIGHT.
NON CU: HL6616 x 3

9. INSTALL 15x 12.5m STANDARD POLE WITH 165W 3000K LED STREETLIGHTS AS PER POLE SCHEDULE TABLE.
CU: SL11_6E x 15
DCSH: S11

10. DISCONNECT EXISTING SL CABLE FROM EXISTING STREETLIGHT. PULL BACK TO CABLE ALLIGNMENT, CUT CAP AND ABANDON EXISTING SL CABLE.
CU: CN73/10/16x7

11. CUT EXISTING SL CABLE, PULL BACK AND TERMINATE TO NEW STREETLIGHT LP12. CAP AND ABANDON EXCESS CABLE.
CU: CN73/10/16x1

12. CUT EXISTING SL CABLE AND LIVE END SEAL AS INDICATED IN DETAIL B. CAP AND ABANDON EXCESS CABLE.
CU: LU62x1, CN73/10/16x1

13. INSTALL NEW 16SL CABLE FROM EXISTING STREETLIGHT S1006354 AND LOOP IN AND OUT FROM:
EXISTING STEETLIGHT S1006359
NEW STREETLIGHT LP04
NEW STREETLIGHT LP05
NEW STREETLIGHT LP06
NEW STREETLIGHT LP07
NEW STREETLIGHT LP08
NEW STREETLIGHT LP09
NEW STREETLIGHT LP10
NEW STREETLIGHT LP11
NEW STREETLIGHT LP13
NEW STREETLIGHT LP15
NEW STREETLIGHT LP17
NEW STREETLIGHT LP19
AND TERMINATE TO NEW STREETLIGHT LP20.
CU: CN41x730m, CN73/10/16x29, CN54x14
DCSH: R27

14. INSTALL NEW 16SL CABLE FROM NEW STREETLIGHT LP12 AND LOOP IN AND OUT FROM:
NEW STREETLIGHT LP14
NEW STREETLIGHT LP16
AND TERMINATE TO NEW STREETLIGHT LP18.
CU: CN41x185m, CN73/10/16x7, CN54x4
DCSH: R27

15. AT NEW STREETLIGHT, TERMINATE NEW 16SL CABLE.
DCSH: R27

16. AT EXISTING STREETLIGHT, TERMINATE NEW 16SL CABLE.
DCSH: R27

17. ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

PUBLIC DRINKING WATER SOURCE AREA
POTENTIAL WET AREAS
THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

DECLARED RARE FLORA
DECLARED RARE AND PRORITY FLORA
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

TRANSMISSION LINE IN THE VICINITY OF WORKS

1. ALL MOBILE PLANT IN PROXIMITY TO THE LINE SHALL OPERATE AT A MINIMUM APPROACH DISTANCE THAT COMPLIES WITH WESTERN POWER WORK PRACTICES AND REQUIREMENTS.
2. WORKS MUST COMPLY WITH WA WORK HEALTH AND SAFETY (GENERAL) REGULATIONS - REGULATION 166A FOR A DUTY OF PERSON WORKING IN VICINITY OF OVERHEAD ELECTRIC LINES.
3. ALL WORKS SHALL COMPLY WITH WESTERN POWER WORK INSTRUCTIONS AND PROCEDURES.
4. CAUTION: INDUCED VOLTAGES IN ISOLATED CONDUCTORS AND APPARATUS DUE TO ELCTRO-MAGNETIC COUPLING FROM ADJACENT ENERGISED TRANSMISSION LINE. INDUCED VOLTAGES MAY BE SIGNIFICANTLY INCREASED DURING FAULT CONDITIONS. ALL WORKS MUST COMPLY WITH WORK INSTRUCTION:
"INDUCED VOLTAGE IN ISOLATED CONDUCTORS/APPARATUS"

Rev.	REVISIONS	DRN	DATE	REFERENCE FILE DETAILS
				SHE ASSESSMENT REPORT EDM#: 70799010
A	UNDER ENGINEERING REVIEW	XX	MM/YY	HAZARD MANAGEMENT REGISTER - EDM# 70799654

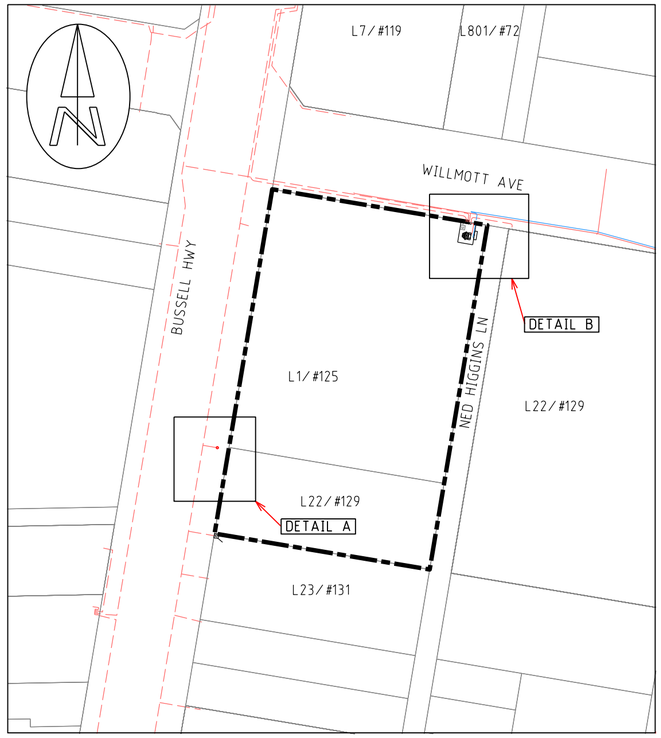
MRWA STREETLIGHT UPGRADE

NETWORK VOLTAGES: 12.7kV/250V

WILLOWDALE RD, WAGERUP
DESIGN DRAWING

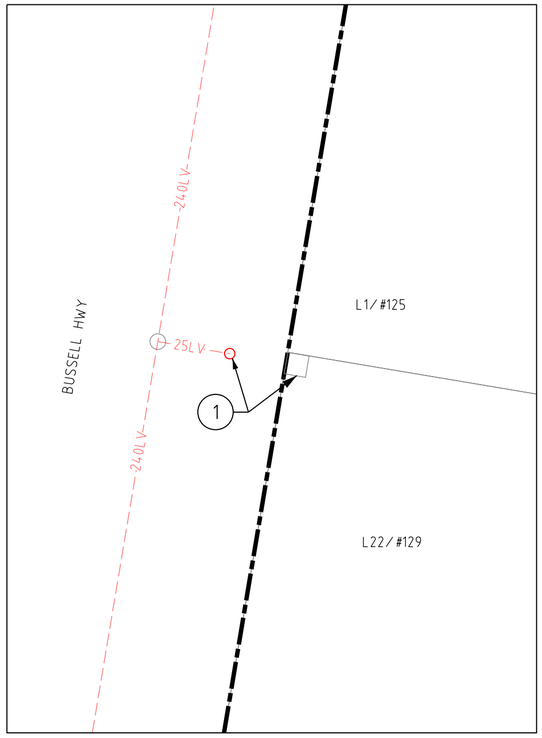
DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A1
GEO REF: Lat. - 32° 53' 47" S
Long. - 115° 54' 20" E

DRAWING NUMBER
SP051023
SHEET 02 OF 02
REV: A ORIGINAL SIZE - A1



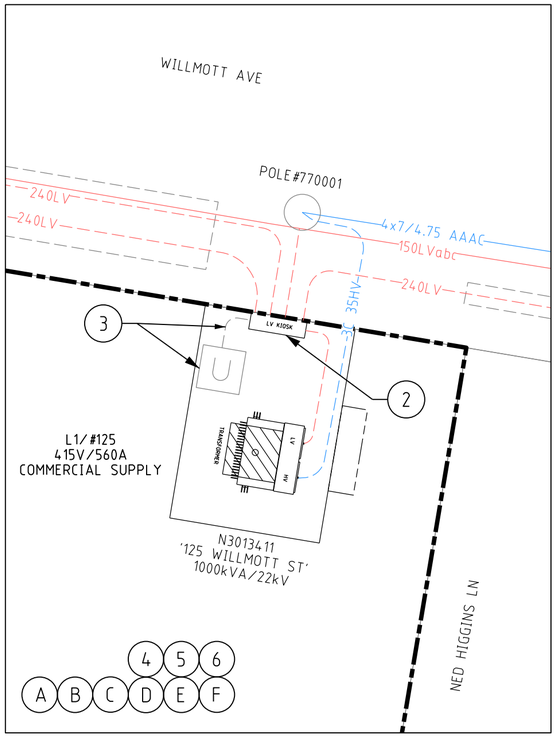
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

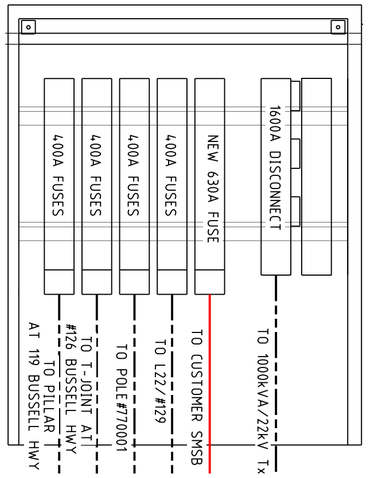
1:200 4 2 0 4 8 12 16



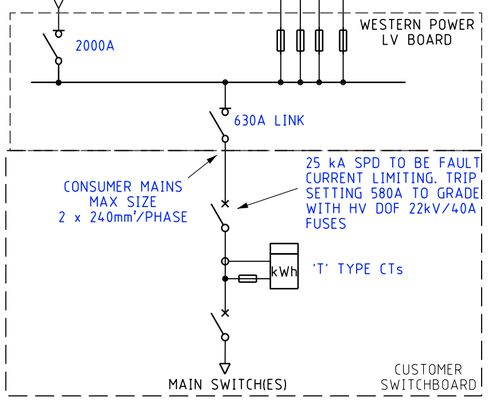
DETAIL B

1:200 4 2 0 4 8 12 16

SIMPLIFIED LINE DIAGRAM



TYPE 2 LV KIOSK
125 BUSSELL HWY SUBSTATION



SIMPLIFIED SINGLE LINE DIAGRAM

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

HV CABLES
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UMS PIT
UNIVERSAL
UTILITIES PIT
MINI
BGSP PIT
LINK
100/200AMP CUTOUT
CABLE MARKER U/G
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST INSTALL MSB CONTIGUOUS WITH SUBSTATION.

(C) CUSTOMER TO SUPPLY AND INSTALL 'T' TYPE METERING CT's.

(D) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(E) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/NZS TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(F) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

WESTERN POWER SCOPE OF WORK

(1) AT L22/#129 EXISTING MINI-PILLAR
- DISCONNECT EXISITNG 240LV.
- CABLE, PULL AND LIVE END SEAL IN WP ALIGNMENT. REMOVE EXISTING MINI-PILLAR.
CU: LU62_50_240 × 1

(2) AT L1/#125 WP LV KIOSK
- REMOVE EXISTING FUSES TOWARDS UNIPILLAR INSIDE SUBSTATION COMPOUND.
- TERMINATE NEW CUSTOMER MAINS VIA 630A FUSE SWITCH.
CU: LU45 × 1

(3) AT EXISTING UNI-PILLAR INSIDE SUBSTATION COMPOUND, DISCONNECT EXISTING CUSTOMER MAINS AND REMOVE CABLE TOWARD LV KIOSK AND REMOVE EXISTING UNI-PILLAR.

(4) AT L22/#129, DISCONNECT CONSUMER MAINS AND REMOVE METER.

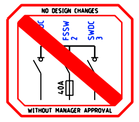
(5) AT L1/#125 SMSB, INSTALL NEW CT METER
CU: LM1_M × 1

(6) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THE PROJECT.

APPROVED FOR CONSTRUCTION

CRITICAL INFRASTRUCTURE IN VICINITY OF WORKS





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

W.A.P.C. REFERENCE No. 164175

CUSTOMER CONTACT DETAILS
COMPANY NAME: REBECCA FOSTER
CONTACT: rebecca@heytesbury.com.au TEL: 0477200848

Rev.	REVISIONS	DRN	DATE
A	APPROVED FOR CONSTRUCTION	I.P.	12/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 68766174
HAZARD MANAGEMENT REGISTER EDM#: 68765075

560A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 415V

125 BUSSELL HWY - MARGARET RIVER WA 6285

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 12/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 32° 57' 04" S
Long. - 115° 04' 27" E

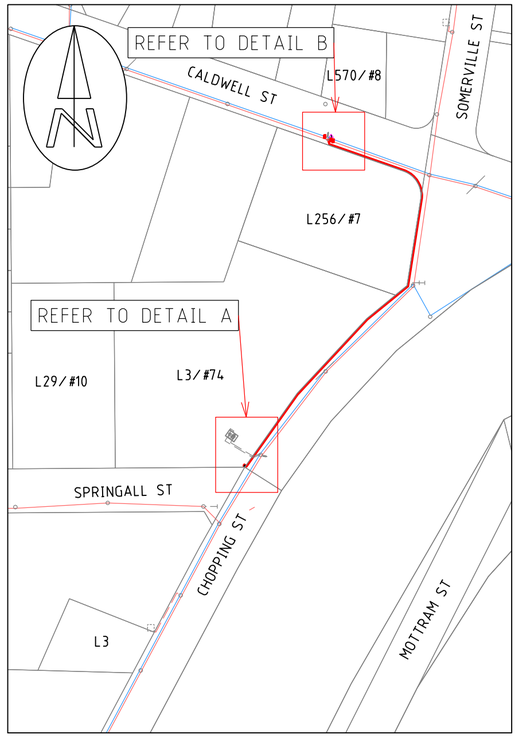
westernpower

DRAWING NUMBER

SP051030

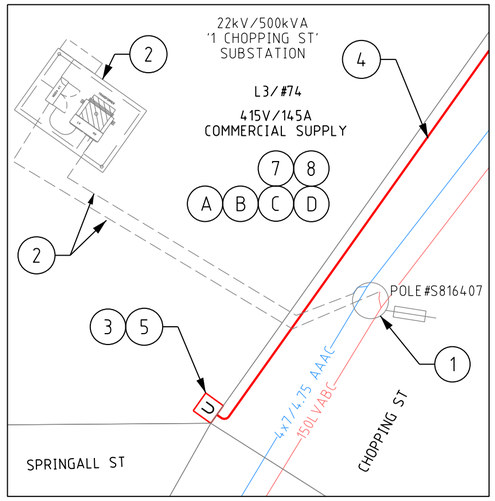
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



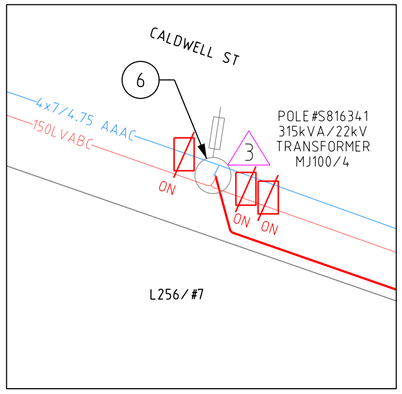
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



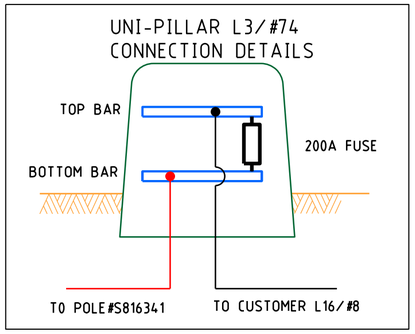
DETAIL A

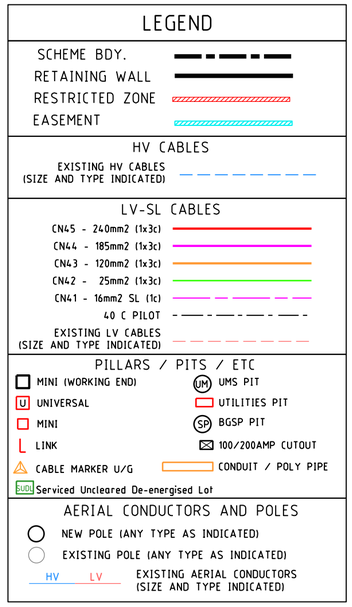
1:200 4 2 0 4 8 12 16



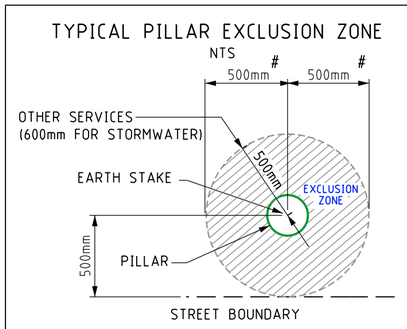
DETAIL B

1:200 4 2 0 4 8 12 16





TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(E) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(F) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

(G) CUSTOMER TO PROVIDE RETAILER CONSENT TO METERING CHANGES. PROJECT WILL NOT BE SCHEDULED UNTIL THE ETAF IS RECEIVED BY WESTERN POWER.

(H) NOTE: SUBSTATION STRUCTURE IS NOT THE PROPERTY OF WESTERN POWER. CUSTOMER IS RESPONSIBLE FOR REMOVAL OF SUBSTATION STRUCTURE IF REQUIRED.

WESTERN POWER SCOPE OF WORK

(1) AT POLE#S816407
- REMOVE EXISTING HV FUSES AND DISCONNECT EXISTING HV CABLE TOWARDS '1 CHOPPING ST' SUBSTATION
- REMOVE EXISTING LV ISOLATORS AND DISCONNECT EXISTING LV CABLE TOWARDS '1 CHOPPING ST' SUBSTATION.

(2) DECOMMISSION EXISTING '1 CHOPPING ST' SUBSTATION. REMOVE EXISTING HV AND LV CABLES WHERE PRACTICABLE. IF REMOVAL IS NOT FEASIBLE, CABLES ARE TO BE CAPPED AND DECOMMISSIONED.

(3) INSTALL NEW UNI-PILLAR ON L3/#74 CHOPPING ST. AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11x1
DCSH: U9

(4) INSTALL 240LV CABLE FROM NEW UNI-PILLAR TO THE POLE#S816341
CU: CN45x225m, CN54x3, CN73_LVMAINSx3

(5) AT UNI-PILLAR (L3/#74 CHOPPING ST)
- TERMINATE NEW 240LV CABLE AT BOTTOM BAR.
- TERMINATE CUSTOMER CABLE AT TOP BAR WITH LINKS IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU7_240x1, LU36_200Ax1

(6) AT POLE#S705723
- INSTALL MASTER KRONE FUSE BOX WITH 630A LINKS AS PER DCSH R12-3.
- TAP PTTX LV BUSHING ONTO MASTER KRONE AS PER SLD - TO BE 'ON'.
- TAP LVABC NORTH WEST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TAP LVABC SOUTH EAST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TERMINATE NEW 240LV ONTO POLE VIA NEW SLAVE KRONE AS PER SLD.
CU: CN39_150/4Mx4, CN11_150x8, LU61_2403PO×1, LU61_2403U×1
NON CU: GF1101×3, GF1508×3, GS0109 × 2
DCSH: U17, R12-3

(7) INSTALL CT METER.
CU: LM1_M x 1

(8) CM TO CONFIRM RETAILER CONSENT HAS BEEN OBTAINED AS PER CUSTOMER SCOPE OF WORK F. METERING POINT CHANGES CANNOT PROCEED UNTIL CONFIRMED.

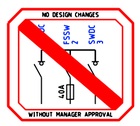
(9) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

CONTAMINATED SOIL
PUBLIC DRINKING WATER SOURCE AREA
THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

REFER TO environment@westernpower.com.au

SIMPLIFIED SINGLE LINE DIAGRAM





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

ISSUED FOR CONSTRUCTION

CONSULTANT CONTACT DETAILS
COMPANY NAME: ECWA
CONTACT: MATT SCANTLEBURY TEL: 0427 760 146

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	05/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
FILE REFERENCE DETAILS

145A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

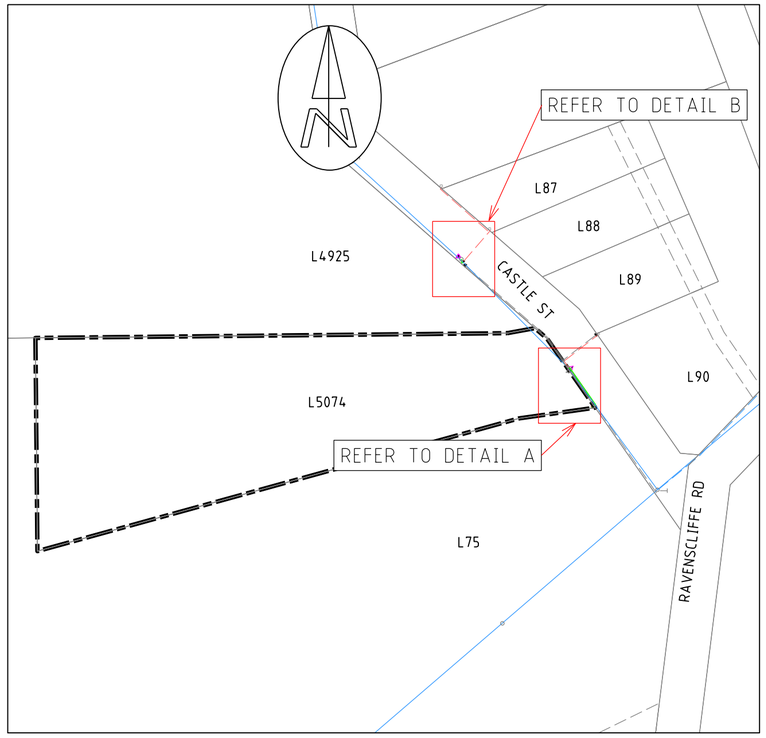
74 CHOPPING ST, MANJIMUP WA 6258

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS COMM. SVM- RASP
EMAIL: ccs.design.commercial.svm@westernpower.com.au
TEL: 13 10 87
DATE: 05/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 34° 15' 09" S
Long. - 116° 08' 08" E

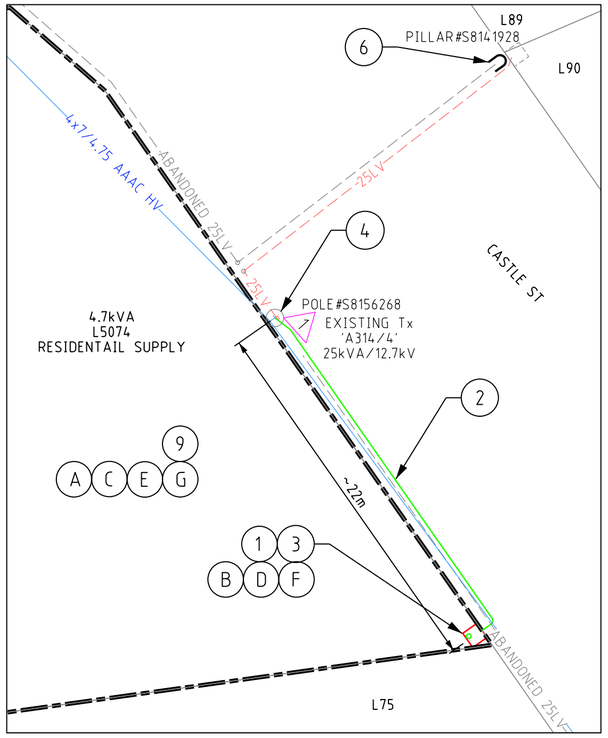
westernpower

DRAWING NUMBER
SP051182
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



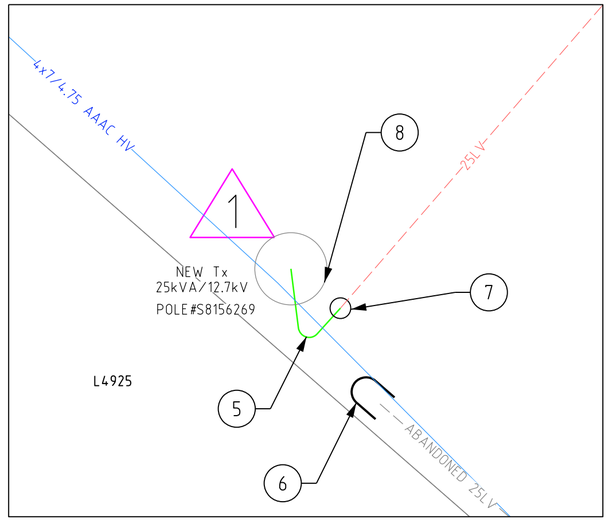
LOCATION PLAN

1:1000 — 20 10 0 20 40 60 80



DETAIL A

1:250 — 5 2.5 0 5 10 15 20



DETAIL B

1:100 — 2 1 0 2 4 6 8

LEGEND

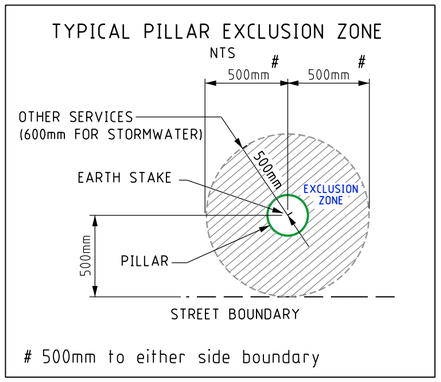
SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

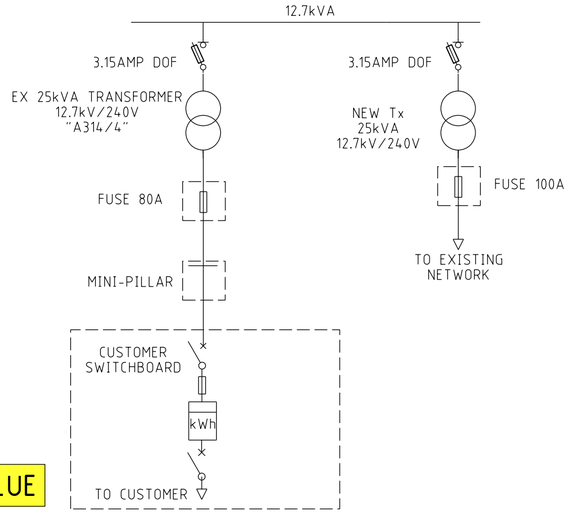
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE



500mm to either side boundary



CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR BACK BLOCK. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED. CUSTOMER TO ENSURE SPD GRADES WITH 80A FUSE.

(F) CUSTOMER ELECTRICAL CONTRACTOR TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMER'S CABLE.

(G) CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW MINI-PILLAR ON SOUTH EAST CORNER OF L5074 CASTLE ST AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU10

(2) INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S8156268
CU: CN42 x 40m, CN73_LV25 x 3, CN54 x 3

(3) AT MINI-PILLAR (L5074 CASTLE ST,)
- TERMINATE NEW 25LV CABLE.
- TERMINATE CUSTOMER'S CABLE TO THE BACK BLOCKS IN CONJUNCTION WITH THE CUSTOMER'S ELECTRICAL CONTRACTOR.
DCSH: U08-04

(4) AT POLE#S8156268
- TERMINATE NEW 25LV CABLE TO EXISTING Tx VIA FUSE.
CU: CN35_80 x 1, CN35_POLE x 1, LU61_25_240O x1, LU61_25_240U x 1
DCSH: U19-3

(5) INSTALL NEW 25LV CABLE FROM EXISTING 25LV TO POLE#S8156269
CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 3

(6) AT PILLAR LOCATED ON L90 (PID:S8141928):
- DISCONNECT EXISTING 25LV AND PULL BACK 25LV
- THEN CUT, CAP AND ABANDON THE EXISTING 25LV CABLE BETWEEN POLE#S8156268 AND THE MINI PILLAR LOCATED ON L90.
CU: CN73_LV25 x 2

(7) STRAIGHT JOINT NEW 25LV CABLE WITH EXISTING 25LV CABLE TOWARDS PILLAR AT LOT 87.
CU: LU8 x 1

(8) AT POLE#S8156268
- INSTALL NEW 25kVA/12.7kV Tx.
- TERMINATE NEW 25LV CABLE TO NEW Tx.
CU: HV52_3FT x 1, TX5_25/12.7 x 1, CN9_7/475AA x 1, LU61_25_240O x1, LU61_25_240U x 1, CN35_100 x 1
DCSH: H46, U19-3, R12-1

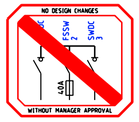
(9) ENIVRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
NAME: SAMUEL BORGAS
E-MAIL: borgassam@outlook.com TEL: 0528590029

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#:69766949
HAZARD MANAGEMENT REGISTER EDM#: 69767442

4.7kVA DADMD RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

LOT 5074 CASTLE ST, KIRUP, 6251, WA

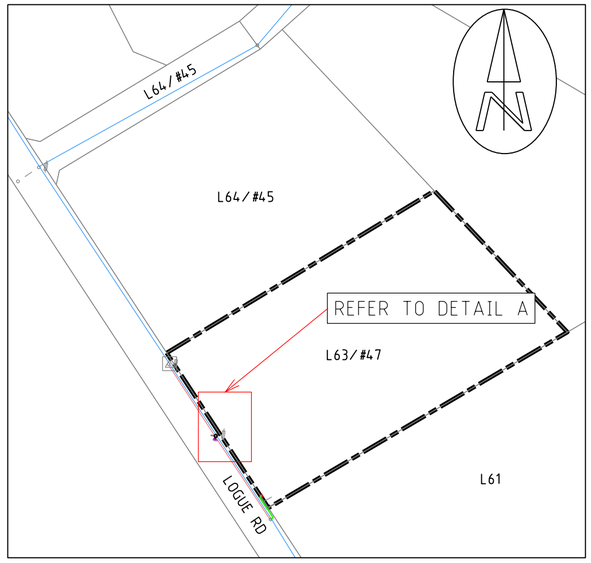
DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 09/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 33° 43' 00" S Long. - 115° 53' 36" E

westernpower

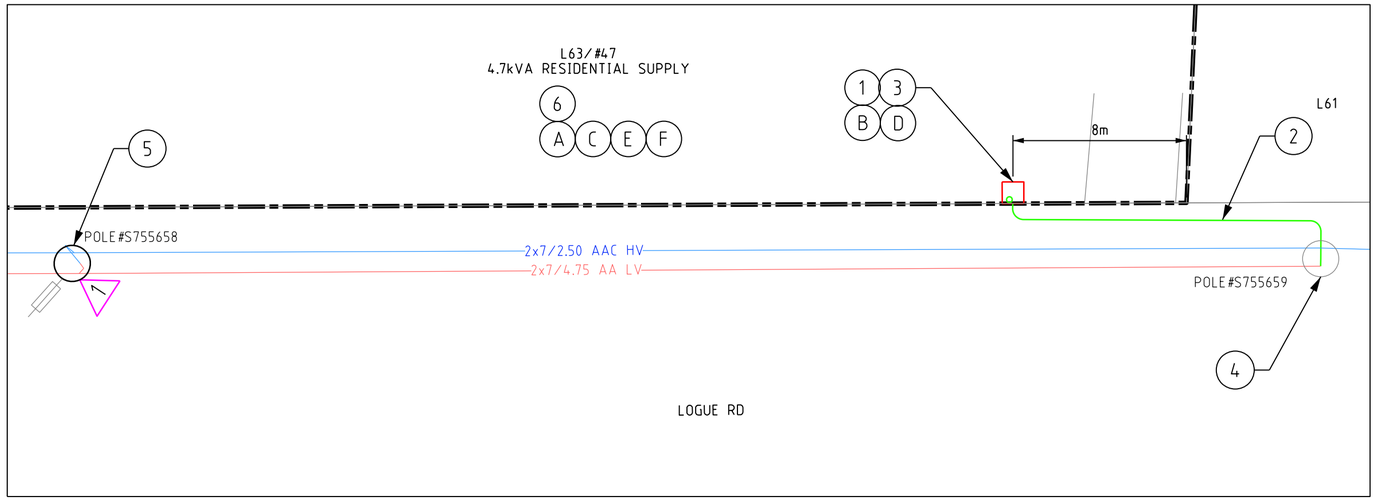
DRAWING NUMBER
SP051323
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

LEGEND

SCHEME BDY.

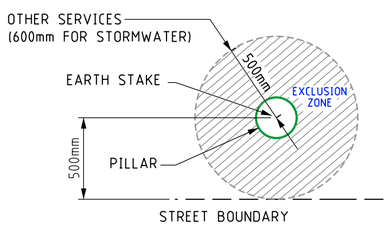
LV-SL CABLES
CN42 - 25mm2 (1x3c)
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI
CONDUIT / POLY PIPE

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT
INTERMEDIATE
PTT
DOF

TYPICAL PILLAR EXCLUSION ZONE
NTS



500mm to either side boundary

COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON THE SOUTH EAST CORNER OF L63/#47 LOGUE RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S755659.
   CU: CN42 x 30m, CN73_LV25 x 3, CN54 x 4
3. AT MINI-PILLAR (L3/#47 LOGUE RD)
   - TERMINATE NEW 25LV CABLE.
   DCSH: U08-04
4. AT POLE#S755659.
   - TERMINATE NEW 25LV CABLE TO BARE LV CONDUCTORS.
   CU: LU51_25ALO x 1, LU51_25ALU x 1,
   DCSH: U19-1
5. AT POLE#S755658.
   - REPLACE THE POLE WITH 12.5m/6kN POLE.
   - INSTALL 1PH HV INTERMEDIATE.
   - INSTALL 1PH 3A FAULT TAMER DOF AND STANDOFF BRACKET.
   - TAP DROP OUT FUSE TO HV BARE.
   - INSTALL RUNNING EARTH INTERMEDIATE.
   - INSTALL NEW 12.7kV/25kVA POLE TOP TRANSFORMER FOR 240V
   - INSTALL LV INTERMEDIATE CROSSARM.
   CU: P08_R x 1, HV29_N x 1, RE1 x 1, TX5_25/12.7 x 1, LV1_W1 x 1 CN30_7/250 x 1, CN9_7/4.75 x 2, HV52_3FT x1, CN26_7/250 x2, CN26_7/475 x2, CN39_150/4M, CN70_150 x2
   DCSH: H46, U19-2, R12-1, R13-3
6. ENIVRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED FOR THIS PROJECT.

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

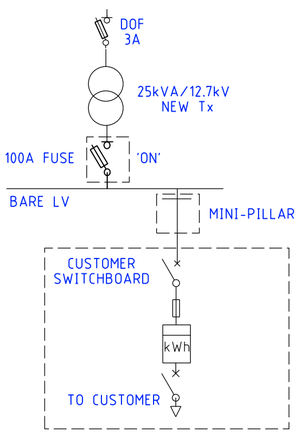
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WP, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENGERGISED.

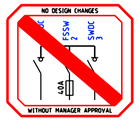
F. CUSTOMER MUST INSTALL SMSB WITHIN 30m OF POINT OF SUPPLY.



SINGLE LINE DIAGRAM



GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
NAME: ANTON FIMMANO
EMAIL: afimmano@hotmail.com TEL: 0408 796 732

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	09/24

FILE REFERENCE DETAILS
HAZARD MANAGEMENT REGISTER EDM#: 69294050
SHE FILTER ASSESSMENT EDM#: 69293450

4.7kVA STANDARD RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

47 LOGUE RD, HARVEY WA 6220

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 09/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 33° 06' 19" S
Long. - 115° 54' 48" E

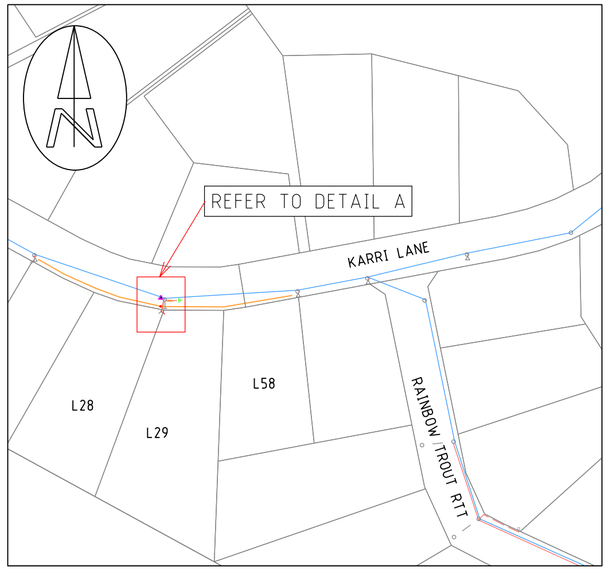
westernpower

DRAWING NUMBER

SP051398

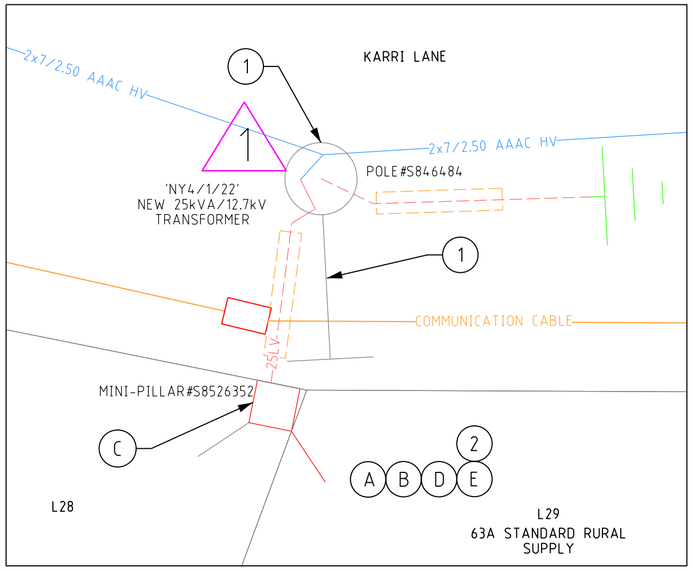
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



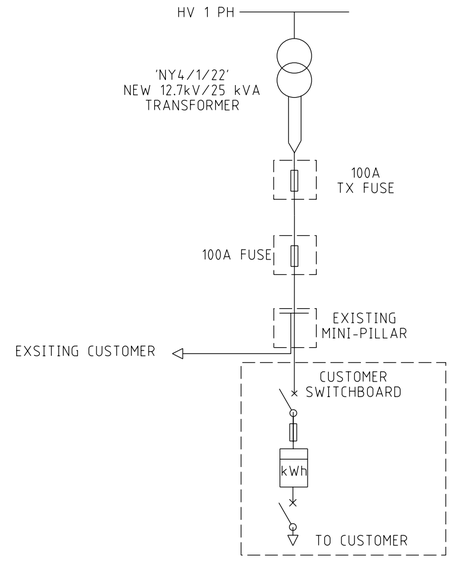
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:100 2 1 0 2 4 6 8



SINGLE LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

(1) AT POLE#S846484
- REPLACE EXISTING 10kVA Tx WITH NEW 25kVA/12.7kV Tx AND TAP ONTO HV BARE.
- RE-ATTACH NEW Tx TO EXISTING REMOTE EARTH
- RE-TERMINATE EXISTING 25LV CABLE TO NEW Tx VIA 100A FUSE.
- RE-TENSION EXISTING STAY.

CU: TX5_25/12.7 x 1
NON CU: GF1947 x 1
DCSH: H49, R12-1, H47-1 NON-STANDARD DCSH

(2) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

NATIVE VEGETATION REMAINING
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(C) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

(E) CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

LEGEND

SCHEME BDY.
RETAINING WALL
RESTRICTED ZONE
EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

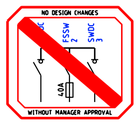
PILLARS / PITS / ETC
- MINI (WORKING END)
- UMS PIT
- UNIVERSAL
- UTILITIES PIT
- MINI
- BGSP PIT
- LINK
- 100/200AMP CUTOUT
- CABLE MARKER U/G
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT
- INTERMEDIATE
- PTS
- DOF
- HV ISOLATOR
- INLINE STRAIN
- LV DISCONNECTOR
- PTT
- ISOL TX

ISSUED FOR CONSTRUCTION





SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

COMMUNICATION CABLES IN VICINITY OF WORKS

CUSTOMER CONTACT DETAILS
COMPANY NAME: N/A
CONTACT: BRIAN HARDISTY TEL: 0458 595 918

Rev.	REVISIONS	DRN	DATE	FILE REFERENCE DETAILS
				SHE ASSESSMENT REPORT EDM#: 70424387
				H47-1 MODIFIED EDM#: 70390388
A	ISSUED FOR CONSTRUCTION	I.P.	11/24	HAZARD MANAGEMENT REGISTER EDM#: 70389949

63A STANDARD RURAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

LOT 29 KARRI LANE, QUINNINUP, 6258, WA

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 11/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 34° 26' 26" S Long. - 116° 15' 28" E

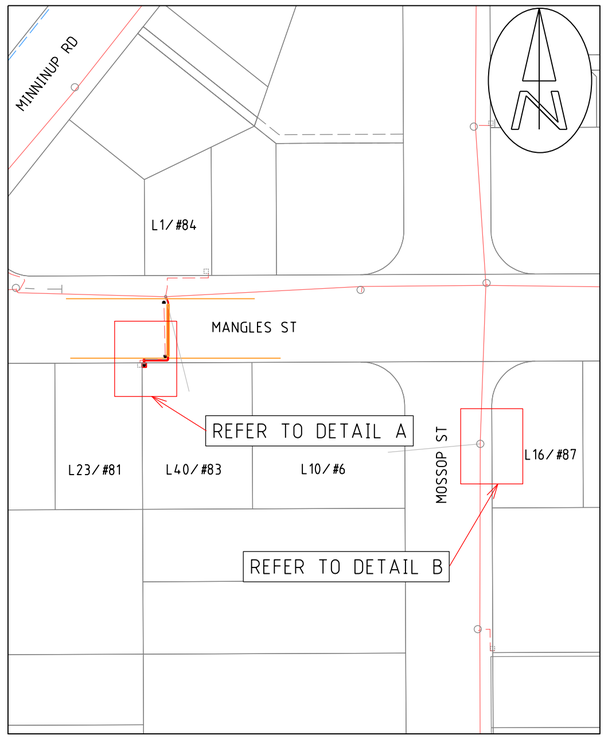
westernpower

DRAWING NUMBER

SP051406

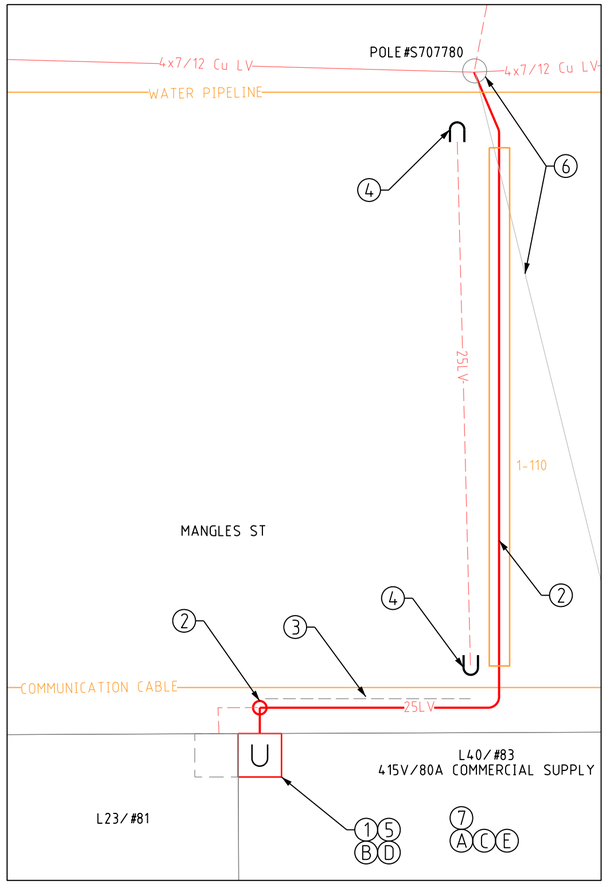
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



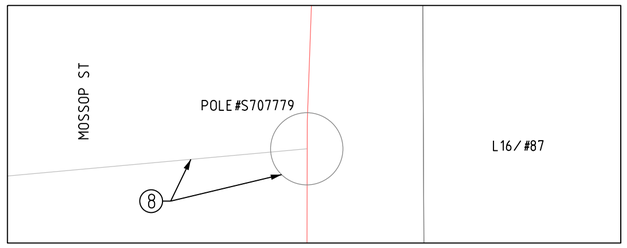
LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16

WESTERN POWER SCOPE OF WORK

(i) NOTE TO PNO:
ASSESS THE CONDITION OF POLE#S707780
PRIOR TO COMMENCEMENT OF OH WORKS.

(1) INSTALL NEW UNI-PILLAR ON L40/#83 MANGLES ST.
AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3).
CU: LU11 x 1
DCSH: U09

(2) INSTALL NEW 240LV CABLE FROM POLE#S707780 TO NEW UNI-PILLAR VIA 110mm POLYPILE AND T-JOINT EXISTING 25LV CABLE TO NEW 240LV CABLE.
CU: CN45 x 35m, CN54 x 4, CN73_LVMAINS x 3, LU5 x 1
DCSH: U16-1

(3) REMOVE EXISTING 25LV CABLE ALONG THE TRENCH AS INDICATED.

(4) CUT, CAP AND ABANDON EXISTING 25LV CABLE
CU: CN73_LV25 x 2

(5) AT UNI-PILLAR (L40/#83 MANGLES ST)
- TERMINATE NEW 240LV CABLE.
- TERMINATE CUSTOMER CABLE IN CONJUNCTION WITH CUSTOMER'S ELECTRICAL CONTRACTOR
CU: LU07_240 x 1
DCSH: U9

(6) AT POLE#S707780
- REMOVE OH SERVICE TO L40/#83
- DISCONNECT AND REMOVE EXISTING 25LV CABLE
- TERMINATE NEW 240LV CABLE
CU: LU6_2400 x 1, LU6_240U x 1
DCSH: U16-1

(7) INSTALL DIRECT METER.
CU: LM1_M x 1

(8) AT POLE#S707779
- REMOVE OH SERVICE TO L10/#6

(9) ENVIRONMENTAL FILTER CHECKED.
NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

THREATENED ECOLOGICAL COMMUNITIES
NATIVE VEGETATION REGS ESAS CLEARING
PRIORITY FLORA
DO NOT IMPACT ON NATIVE VEGETATION WITHOUT PRIOR APPROVAL FROM SEQT.

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
- BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
- IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

PUBLIC DRINKING WATER SOURCE AREA
THERE SHOULD BE NO REFUELLING, REPAIR OR MAINTENANCE OF MOTOR VEHICLES OR EQUIPMENT ON SITE, NOR ANY ON-SITE USE OR STORAGE OF CHEMICALS. A CONTINGENCY PLAN FOR MANAGING AND RESPONDING TO OIL SPILLS SHOULD BE IN PLACE.

REFER TO environment@westernpower.com.au

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

(C) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(D) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE UNI-PILLAR BACK BLOCKS. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

(E) CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENERGISED.

LEGEND

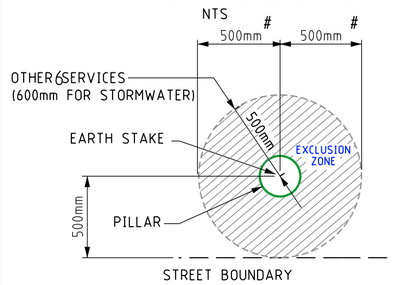
HV CABLES
EXISTING HV CABLES (SIZE AND TYPE INDICATED)

LV-SL CABLES
CN45 - 240mm2 (1x3c)
CN44 - 185mm2 (1x3c)
CN43 - 120mm2 (1x3c)
CN42 - 25mm2 (1x3c)
CN41 - 16mm2 SL (1c)
40 C PILOT
EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC
MINI (WORKING END)
UNIVERSAL
MINI
LINK
CABLE MARKER U/G
UMS PIT
UTILITIES PIT
BGSP PIT
100/200AMP CUTOUT
CONDUIT / POLY PIPE
Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
NEW POLE (ANY TYPE AS INDICATED)
EXISTING POLE (ANY TYPE AS INDICATED)
HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

TYPICAL PILLAR EXCLUSION ZONE
NTS



500mm to either side boundary

ISSUED FOR CONSTRUCTION

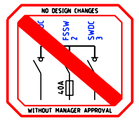
WATER PIPELINE IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED





W.A.P.C. REFERENCE No. 164322

CUSTOMER CONTACT DETAILS
COMPANY NAME: CWC CONSULTANCY PTY LTD
CONTACT: KEVIN KELLY TEL: 0439952378

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	07/24

FILE REFERENCE DETAILS
SHE ASSESSMENT REPORT EDM#: 69065785
HAZARD MANAGEMENT REGISTER EDM#: 69066428

80A COMMERCIAL SUPPLY

NETWORK VOLTAGE: 22kV/415V

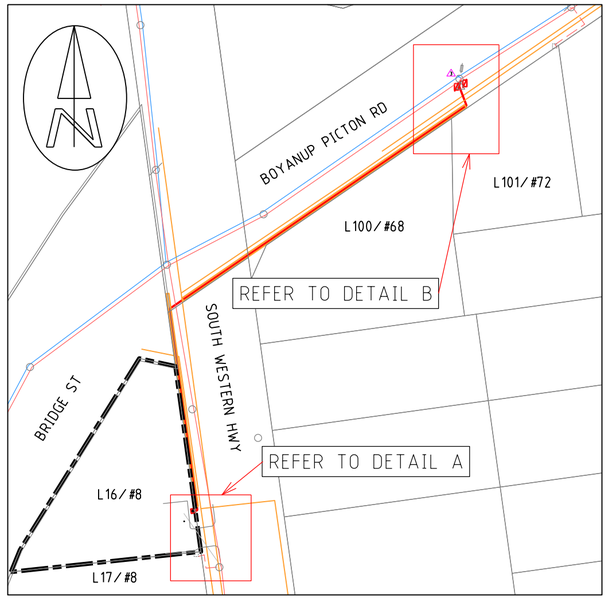
83 MANGLES ST, SOUTH BUNBURY WA 6230

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87
DATE: 07/24 SCALE: AS NOTED @ A2
GEO REF: Lat. - 33° 20' 58" S Long. - 115° 37' 59" E

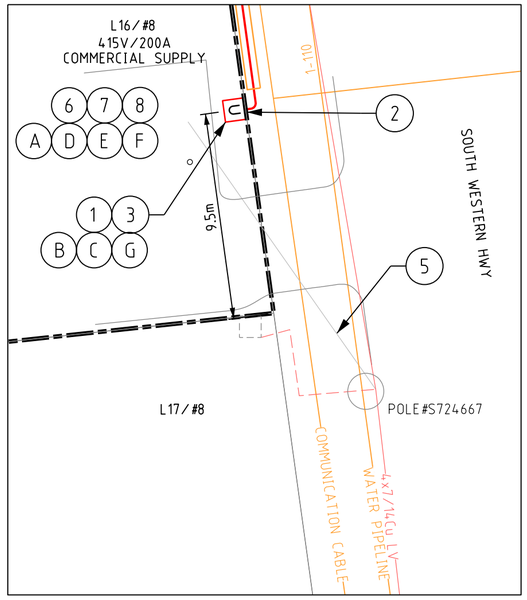
westernpower

DRAWING NUMBER
SP051457
SHEET 01 OF 01
REV: A ORIGINAL SIZE - A2



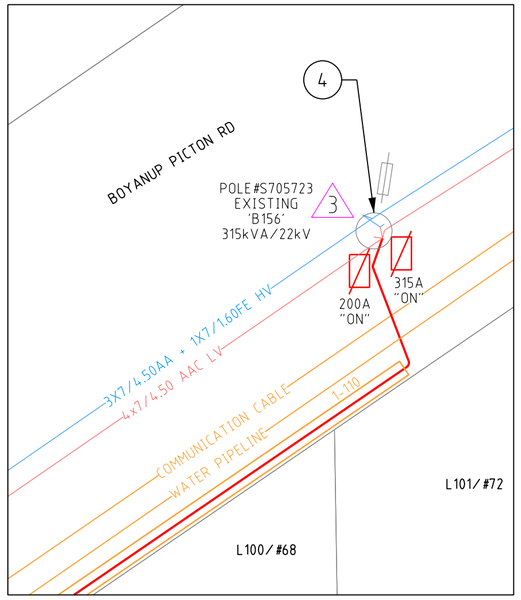
LOCATION PLAN

1:2000 40 20 0 40 80 120 160



DETAIL A

1:200 4 2 0 4 8 12 16



DETAIL B

1:200 4 2 0 4 8 12 16

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

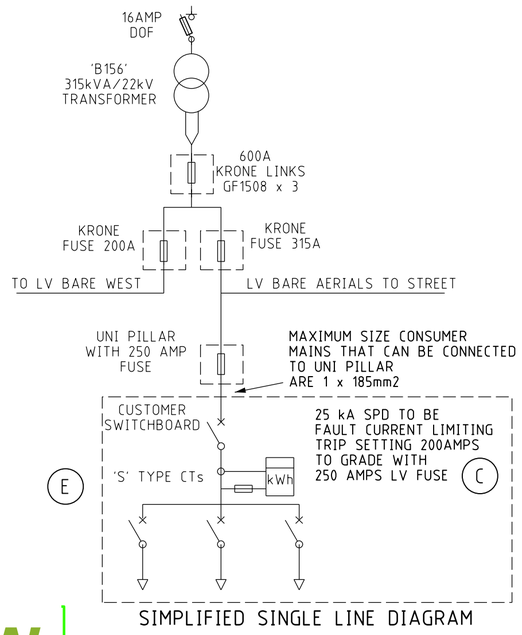
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

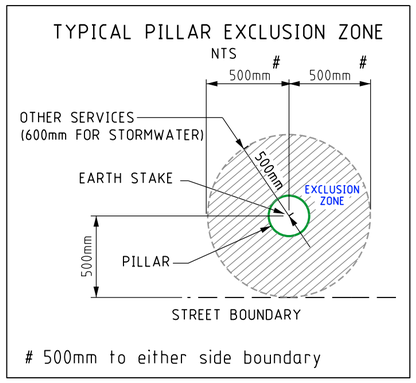
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

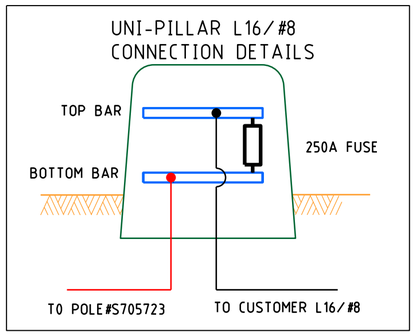
OH EQUIPMENT

- DOF
- PTT
- LV ABC KRONE
- STREETLIGHT BOX



SIMPLIFIED SINGLE LINE DIAGRAM





WATER CO. PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

8 ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
BOYANUP GENERAL STORE

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
- BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
- IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(E) CUSTOMER TO SUPPLY AND INSTALL SMSB, PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY.

(F) CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

(G) CUSTOMER ELECTRICAL CONTRACTOR IS TO SUPPLY SUITABLE LUGS AND CRIMPING TOOLS FOR TERMINATION OF CUSTOMERS CABLE.

WESTERN POWER SCOPE OF WORK

(1) INSTALL NEW UNI-PILLAR ON L16/#8 SOUTH WESTERN HWY AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU11
DCSH: U09

(2) INSTALL NEW 240LV CABLE FROM POLE#S705723 AT L101/#72 BOYANUP PICTON RD TO NEW UNI-PILLAR VIA 110mm POLYPIPE AT 0.5M ALLIGNMENT AND MINIMUM 1.2m DEPTH.
CU: CN45 x 170m, CN54 x 6, CN73_LVMAINS x 3
DCSH: U16-1

(3) AT UNI-PILLAR (L16/#8 SOUTH WESTERN HWY)
- TERMINATE NEW 240LV STREET MAINS CABLE
- TERMINATE CUSTOMER'S CABLE TO THE TOP BAR VIA 250A FUSES.
CU: LU7 x 1
NON-CU: GF0260 x 3, GF2129 x 3
DCSH: U09

(4) AT POLE#S705723
- REMOVE 600A DISCONNECTORS AND ASSOCIATED TAPPING FROM EXISTING IN-LINE STRAIN XARM.
- INSTALL MASTER KRONE FUSE BOX WITH 600A LINKS AS PER DCSH R12-3
- INSTALL 1 x NEW KRONE FUSE BOX WITH 315A FUSES.
- INSTALL 1 x NEW KRONE FUSE BOX WITH 200A FUSES.
- TAP PTTX LV BUSHING ONTO MASTER KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE EAST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE WEST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TERMINATE NEW 240LV ONTO POLE AND TAP TO LV BARE EAST
CU: CN39_150/4M x 5, CN11_150 x 4, CN70_150 x 8, CN9_7/475AA x 8, LU6_240O x 1, LU6_240U x 1
NON CU: GF1508 x 3, OZ0012 x 1, OZ0017 x 1, GF1101 x 3, GF1096 x 3, GS0111 x 2, GS0109 x 2
DCSH: U17, R12-3

(5) AT POLE#S724667, REMOVE EXISITNG OVERHEAD SERVICE TO L16/#8

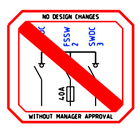
(6) INSTALL CT METER.
CU: LM1_M x 1

(7) POSSESSION OF SITE CERTIFICATE IS TO BE REQUESTED, AND MAIN ROADS DOCUMENT D18#604587 'SEEKING APPROVAL TO UNDERTAKE WORKS WITHIN THE ROAD RESERVE' TO BE COMPLETED.

CM TO COMPLY WITH SIGNED CONDITIONAL AGREEMENT-REFER EDM#_____

For Review
10/12/2024 8:26:30 AM

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS
COMPANY NAME: SCOPE ELECTRICAL
CONTACT: SCOTT ROWE TEL: 0427 345 377

A		I.P.	08/24
Re	VISIONS	DRN	DATE

SHE FILTER ASSESSMENT EDM#: 69329479
HAZARD MANAGEMENT REGISTER EDM#: 69330152
FILE REFERENCE DETAILS

For Review
10/12/2024 8:26:36 AM

200A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

8A WESTERN HWY, BOYANUP WA 6237

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 08/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 33° 29' 00" S
Long. - 115° 43' 40" E

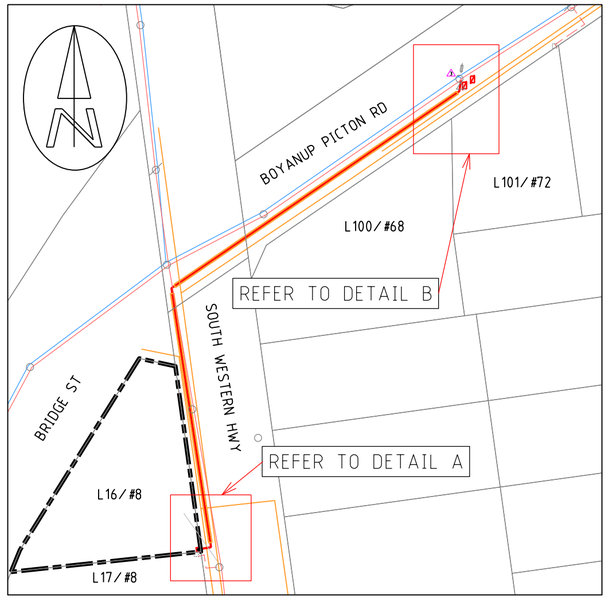
westernpower

DRAWING NUMBER

SP051462

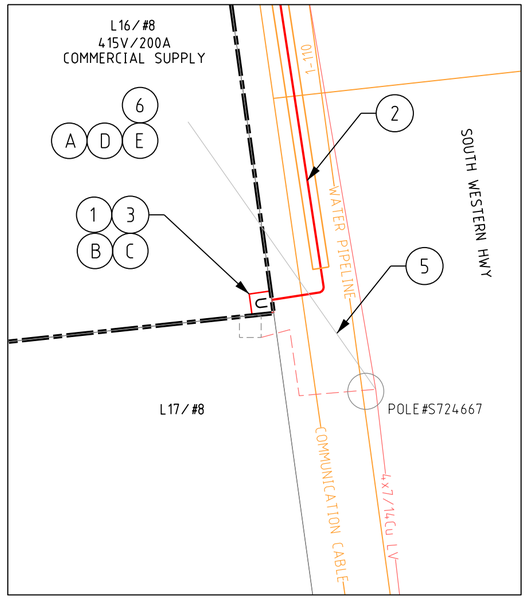
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



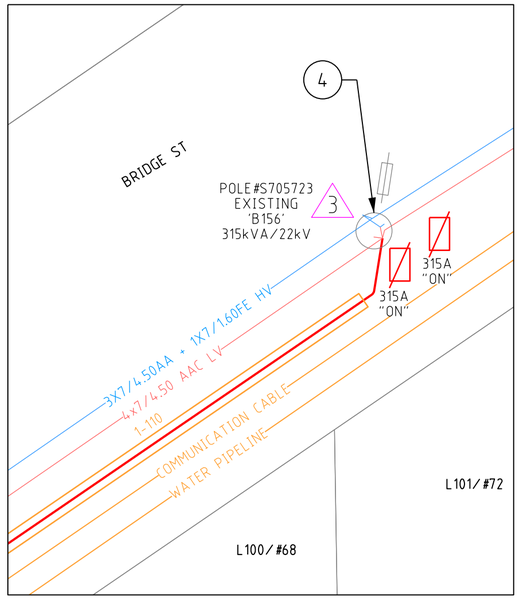
LOCATION PLAN

1:2000 — 40 20 0 40 80 120 160



DETAIL A

1:200 — 4 2 0 4 8 12 16



DETAIL B

1:200 — 4 2 0 4 8 12 16

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

LV-SL CABLES
- CN45 - 240mm2 (1x3c)
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

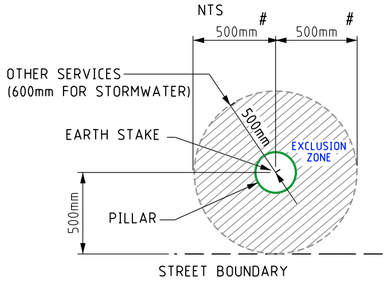
PILLARS / PITS / ETC
- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES
- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

OH EQUIPMENT
- DOF
- LV ABC KRONE
- PTT
- STREETLIGHT BOX (SLB)

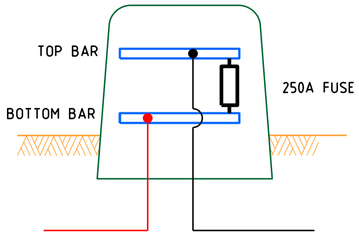
TYPICAL PILLAR EXCLUSION ZONE

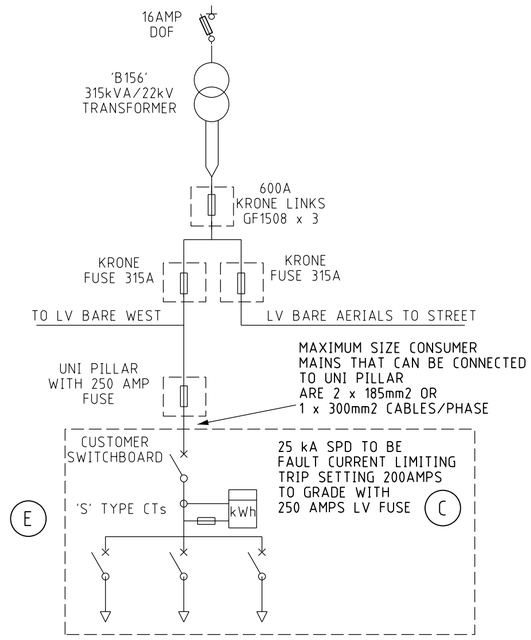
NTS



500mm to either side boundary

UNI-PILLAR L16/#8 CONNECTION DETAILS





SIMPLIFIED SINGLE LINE DIAGRAM

WATER CO. PIPELINES IN VICINITY OF WORKS

COMMUNICATION CABLES IN VICINITY OF WORKS

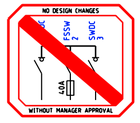
CUSTOMER CONTACT DETAILS

COMPANY NAME: SCOPE ELECTRICAL
CONTACT: SCOTT ROWE TEL: 0427 345 377

ISSUED FOR CONSTRUCTION

SAFETY - MAKE IT YOUR PRIMARY VALUE

GOLDEN SAFETY RULES ARE INDICATIVE ONLY - ALWAYS PERFORM A DETAILED HAZARD ANALYSIS.



DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER SCOPE OF WORK

(A) THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

(B) PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

(C) CUSTOMER ELECTRICAL CONTRACTOR TO ENSURE CUSTOMER'S SPD GRADES WITH WESTERN POWER'S UPSTREAM PROTECTION AS NOTED ON THE SIMPLIFIED LINE DIAGRAM.

(D) CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

(E) CUSTOMER TO SUPPLY AND INSTALL SMSB, PRIOR TO THE ENERGISATION OF THE POINT OF SUPPPLY. CUSTOMER MUST INSTALL SMSB WITHIN 30m FROM POINT OF SUPPLY. CUSTOMER TO SUPPLY AND INSTALL 'S' TYPE METERING CT's.

WESTERN POWER SCOPE OF WORK

DETAIL A

(1) INSTALL NEW UNI-PILLAR ON L16/#8 SOUTH WESTERN HWY AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
CU: LU11
DCSH: U09

(2) INSTALL NEW 240LV CABLE FROM POLE#S705723 AT L101/#72 BOYANUP PICTON RD TO NEW UNI-PILLAR VIA 110mm POLYPIPE.
CU: CN45 x 160m, CN54 x 6, CN73_LVMAINS x 3
DCSH: U16-1

(3) AT UNI-PILLAR (L16/#8 SOUTH WESTERN HWY)
- TERMINATE NEW 240LV STREET MAINS CABLE
- TERMINATE CUSTOMER'S CABLE TO THE TOP BAR VIA 250A FUSES.
CU: LU7 x 1
NON-CU: GF0560 x 3, GF2129 x 3
DCSH: U09

DETAIL B

(4) AT POLE#S705723
- REMOVE 600A DISCONNECTORS AND ASSOCIATED TAPPING FROM EXISTING IN-LINE STRAIN XARM.
- INSTALL MASTER KRONE FUSE BOX WITH 600A LINKS AS PER DCSH R12-3
- INSTALL 2 x NEW KRONE FUSE BOX WITH 315A FUSES.
- TAP PTTX LV BUSHING ONTO MASTER KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE EAST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TAP LV BARE WEST TO MASTER KRONE VIA SLAVE KRONE AS PER SLD - TO BE 'ON'.
- TERMINATE NEW 240LV ONTO POLE AND TAP TO LV BARE WEST
CU: CN39_150/4M x 5, CN11_150 x 4, CN70_150 x 8, CN9_7/475AA x 8, LU6_240O x 1, LU6_240U x 1
NON CU: GF1508 x 3, OZ0012 x 1, OZ0017 x 1, GF1101 x 6, GS0111 x 2, GS0109 x 2
DCSH: U17, R12-3

(5) AT POLE#S724667, REMOVE EXISITNG OVERHEAD SERVICE TO L16/#8

(6) ENVIRONMENTAL FILTER CHECKED.
THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

HERITAGE COUNCIL OF WA
MUNICIPAL INVENTORY
BOYANUP GENERAL STORE

ACID SULPHATE SOILS (ASS)
EXPOSURE TO AIR CAUSES ASS TO CREATE SULPHURIC ACID.
IN AREAS WHERE THIS IS A RISK YOU NEED TO:
- BACKFILL EXCAVATIONS WITHIN 18 HOURS OF OPENING.
- IF DEWATERING OR EXCAVATING 100M3 OR MORE IN TOTAL FOR THE PROJECT AN ACID SULPHATE SOIL MANAGEMENT PLAN MAY BE DEVELOPED

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	08/24

FILE REFERENCE DETAILS
SHE FILTER ASSESSMENT EDM#: 69329479
HAZARD MANAGEMENT REGISTER EDM#: 69330152

200A COMMERCIAL SUPPLY UPGRADE

NETWORK VOLTAGE: 22kV/415V

8A WESTERN HWY, BOYANUP WA 6237

DESIGN DRAWING

DESIGNER: CCD
DESIGN TEAM: CCS SBM QAT - RASP
EMAIL: ccs.sbm.qat@westernpower.com.au
TEL: 13 10 87

DATE: 08/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 33° 29' 00" S
Long. - 115° 43' 40" E

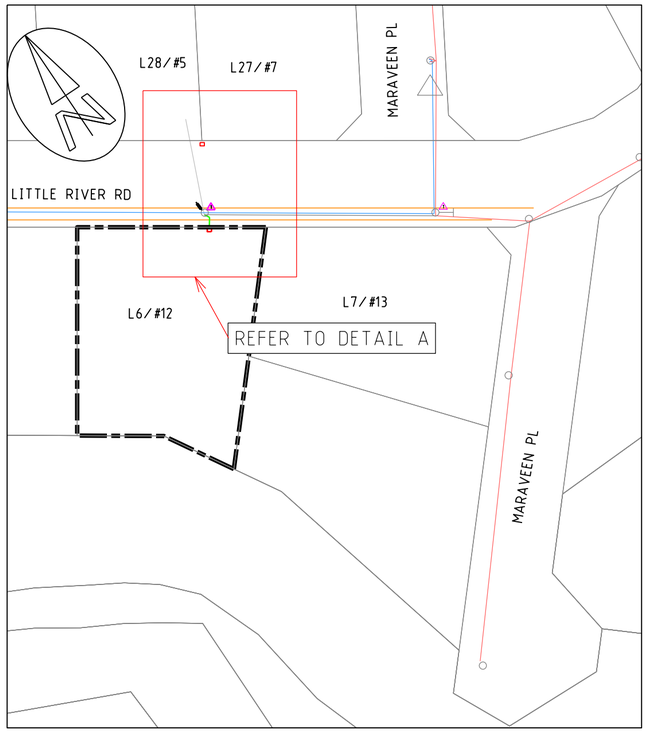
westernpower

DRAWING NUMBER

SP051462

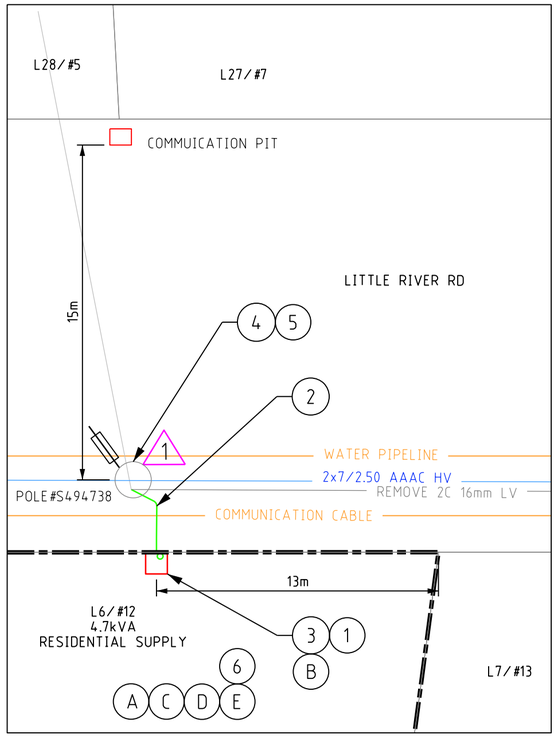
SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2



LOCATION PLAN

1:1000 20 10 0 20 40 60 80



DETAIL A

1:200 4 2 0 4 8 12 16

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

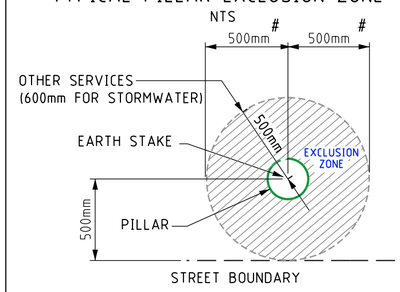
COMMUNICATION CABLES IN VICINITY OF WORKS

WATER PIPELINES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

TYPICAL PILLAR EXCLUSION ZONE

NTS

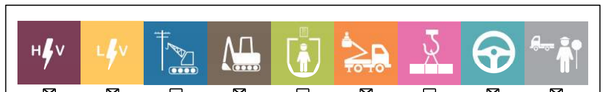


500mm to either side boundary

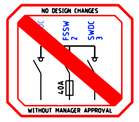
CUSTOMER CONTACT DETAILS

COMPANY NAME:

CONTACT: JACINTA CLITHEROE TEL: 0427774120

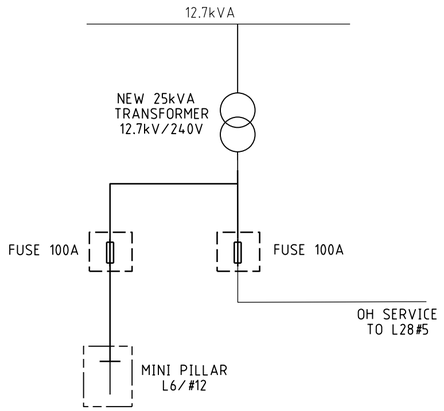


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SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED



SIMPLIFIED LINE DIAGRAM

LEGEND

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON SOUTH EAST BOUNDARY OF L6/#12 LITTLE RIVER RD AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10
   DCSH U8-4
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S494739.
   CU: CN42 x 20m, CN73_LV25 x 3, CN54 x 2
3. AT MINI-PILLAR (L6/#12 LITTLE RIVER RD)
   - TERMINATE NEW 25LV CABLE.
   DCSH: U8-4
4. AT POLE#S494738
   - INSTALL NEW 12.7kV/25kVA POLE TOP TRANSFORMER VIA FOR 240V.
   - TERMINATE NEW 25LV CABLE TO NEW TRANSFOMER VIA FLOWLINE FUSES.
   - DISCONNECT EXISTING OVERHEAD AERIAL SERVICE AND REMOVE EXISTING BAY OF 2C 16LV CONDUCTOR AND ASSOCIATED EQUIPMENT BETWEEN WP POLES #S494738 AND S494739
   - TERMINATE EXISTING AERIAL SERVICE TO NEW TRANSFORMER VIA FLOWLINE FUSE.
   CU: TX5_25/12.7 x 1, LU61_25_2400 x 1, LU61_25_240U x 1, LV8_1P x 1, CN9_7/250AC x 1, CN35_POLE x 1, CN35_100 x 2
   DCSH: H46, U19-3, R12-1, R11-3, R08-5, H49
5. ENSURE POLE TOP TRANSFORMER EARTH IS LOCATED A MINIMUM OF 15m AWAY FROM THE EXISTING TELSTRA COMMS PIT LOCATED NEAR THE COMMON LOT BOUNDARY OF #5 AND #7 LITTLE RIVER RD. REFER TO DETAIL A.
6. ENVIRONMENTAL FILTER CHECKED.
   THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:

   HERITAGE COUNCIL OF WA
   MUNICIPAL INVENTORY
   ABORIGINAL HERITAGE SITES
   IF ANY OBJECTS ARE FOUND DURING EARTHWORKS THAT ARE SUSPECTED OF BEING OF ABORIGINAL ORIGIN, WORKS MUST STOP IMMEDIATELY, ESTABLISH A 'NO-WORK' ZONE AND REPORT THE INCIDENT TO THE INCIDENT HOTLINE ON 1300 225 597 (CALL WP) AND REFER TO THE ENVIRONMENT ADN LAND ACCESS IN OPERTATIONS WORK INSTRUCTION (EDM 4150794)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

   1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.

   2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.

   3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.

   4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.

   5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WP, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENGERGISED.

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/24

FILE REFERENCE DETAILS

SHE FILTER ASSESSMENT EDM#: 69243351

HAZARD MANAGEMENT REGISTER EDM#: 69244341

4.7kVA ADMD RESIDENTIAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

12 LITTLE RIVER RD, OCEAN BEACH WA 6333

DESIGN DRAWING

DESIGNER: CCD

DESIGN TEAM: CCS SBM QAT - RASP

EMAIL: ccs.sbm.qat@westernpower.com.au

TEL: 13 10 87

DATE: 10/24 SCALE: AS NOTED @ A2

GEO REF: Lat. - 34° 59' 35" S
Long. - 117° 20' 05" E

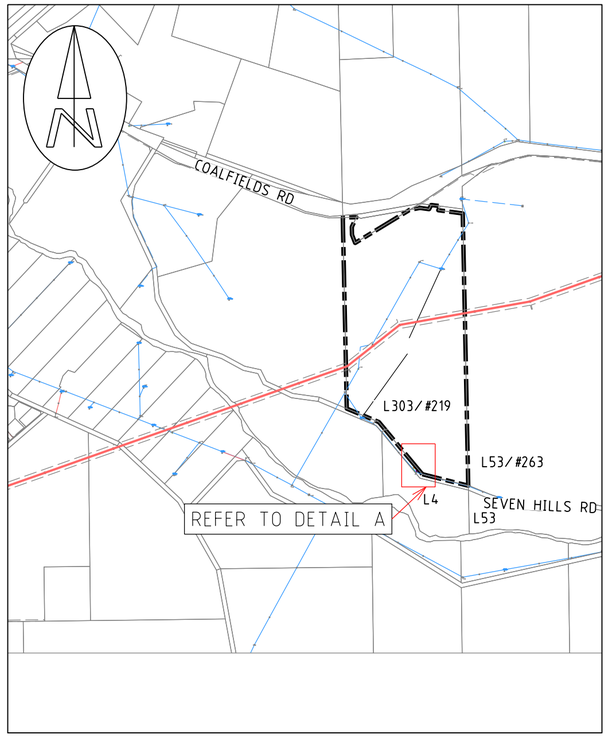
westernpower

DRAWING NUMBER

SP051506

SHEET 01 OF 01

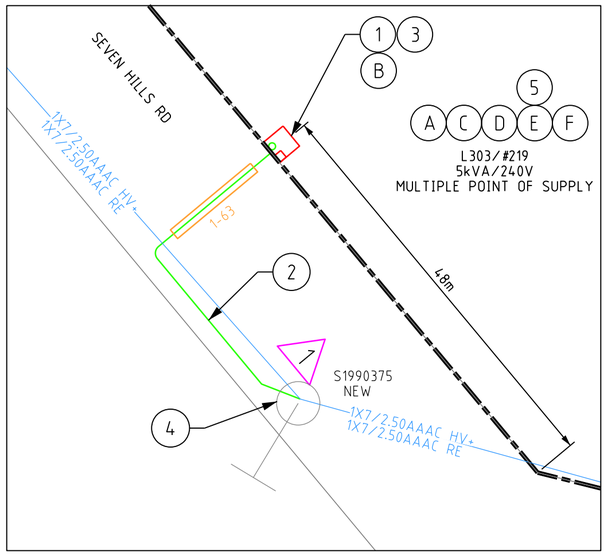
REV: A ORIGINAL SIZE - A2



LOCATION PLAN

400 200 0 400 800 1200 1600

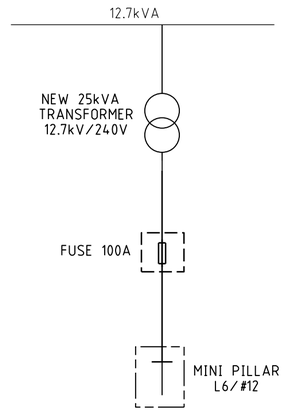
1:20,000



DETAIL A

10 5 0 10 20 30 40

1:500



SIMPLIFIED LINE DIAGRAM

WESTERN POWER SCOPE OF WORK

1. INSTALL NEW MINI-PILLAR ON L303 SEVEN HILLS RD WHERE PEGGED BY CUSTOMER AS PER ZONE EXCLUSION DETAIL (WASIR CL. 12.5.3)
   CU: LU10
   DCSH U8-4
2. INSTALL NEW 25LV CABLE FROM NEW MINI-PILLAR TO POLE#S494739.
   CU: CN42 x 60m, CN73_LV25 x 3, CN54 x 2
3. AT NEW MINI-PILLAR.
   - TERMINATE NEW 25LV CABLE.
   DCSH: U8-4
4. AT POLE#S494738
   - REPLACE EXISTING HV INTERMEDIATE WITH STRAIN.
   - INSTALL NEW 12.7kV/25kVA POLE TOP TRANSFORMER FOR 240V. AND TAP ON HV BARE.
   - INSTALL NEW RUNNING EARTH ANGLE.
   - TERMINATE NEW 25LV CABLE TO NEW TRANSFOMER VIA FLOWLINE FUSES.
   CU: TX5_25/12.7 x 1, LU61_25_240O x 1, LU61_25_240U x 1, HV30_H, RE2_7/250A, CN9_7/250AC x 2
   DCSH: H47-3, U19-3, H49
5. ENVIRONMENTAL FILTER CHECKED.
   NO ENVIRONMENTAL ISSUES IDENTIFIED GIVEN THE NATURE OF THE PROPOSED WORKS. THE FOLLOWING POTENTIAL ENVIRONMENTAL RISK LAYERS ARE PRESENT WITHIN THE AREA OF WORKS:
   MUNICIPAL INVENTORY
   ACID SULFATE SOILS
   ACID SULFATE SOIL RISK MAP.
   SWAN COASTAL PLAIN (DWER-055)
   LOCAL PLANNING SCHEME - ZONES AND RESERVES (DPLH-071)
   REGION SCHEME - ZONES AND RESERVES (DPLH-023)
   INDIGENOUS LAND USE AGREEMENTS (ILUAS)
   NATIVE TITLE BOUNDARIES
   RIWI ACT, SURFACE WATER AREAS AND IRRIGATION DISTRICTS (DWER-037)
   WATERWAYS CONSERVATION ACT MANAGEMENT AREAS (DWER-072)
   PRE-EUROPEAN VEGETATION (DPIRD-006)

   REFER TO environment@westernpower.com.au

LEGEND

- SCHEME BDY.
- RETAINING WALL
- RESTRICTED ZONE
- EASEMENT

LV-SL CABLES

- CN45 - 240mm2 (1x3c)
- CN44 - 185mm2 (1x3c)
- CN43 - 120mm2 (1x3c)
- CN42 - 25mm2 (1x3c)
- CN41 - 16mm2 SL (1c)
- 40 C PILOT
- EXISTING LV CABLES (SIZE AND TYPE INDICATED)

PILLARS / PITS / ETC

- MINI (WORKING END)
- UNIVERSAL
- MINI
- LINK
- CABLE MARKER U/G
- UMS PIT
- UTILITIES PIT
- BGSP PIT
- 100/200AMP CUTOUT
- CONDUIT / POLY PIPE
- SUDI Serviced Uncleared De-energised Lot

AERIAL CONDUCTORS AND POLES

- NEW POLE (ANY TYPE AS INDICATED)
- EXISTING POLE (ANY TYPE AS INDICATED)
- HV LV EXISTING AERIAL CONDUCTORS (SIZE AND TYPE INDICATED)

CUSTOMER SCOPE OF WORK

A. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS.

B. PILLAR EXCLUSION ZONE TO BE PREPARED AS PER THE WESTERN AUSTRALIAN SERVICE AND INSTALLATION REQUIREMENTS CLAUSE 12.5.3. THIS INCLUDES, BUT IS NOT LIMITED TO THE FOLLOWING:

1. PEGGING THE PILLAR LOCATION AND ADJACENT BOUNDARY LOCATION PRIOR TO CONSTRUCTION.
2. ONLY EASILY REMOVABLE SURFACE TREATMENT IS TO BE INSTALLED IN THIS AREA, SUCH AS BRICK PAVING.
3. CLEAR OF ALL SERVICES INCLUDING CONTAMINATED SOIL, ASBESTOS ETC.
4. LOCATED 300mm ABOVE 100 YEAR FLOOD LEVEL AND NOT IN A PRECARIOUS POSITION.
5. CUSTOMER TO ENSURE PILLAR EXCLUSION ZONE CLEAR OF VEGETATION TO THE SATISFACTION OF THE WP SCOPER/PNO.

C. CUSTOMER MUST COMPLY WITH POWER QUALITY LIMITS AS PER AS/TS IEC 61000.3.5 AND WESTERN POWER TECHNICAL REQUIREMENTS.

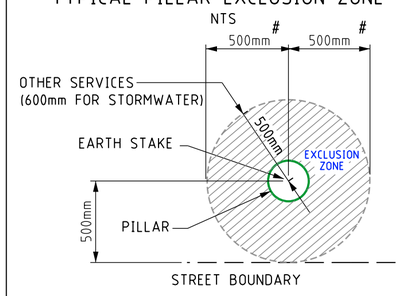
D. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WESTERN POWER, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR, TO TERMINATE ALL CONSUMER MAINS TO THE MINI-PILLAR. MAX SIZE CONSUMER MAINS THAT CAN BE TERMINATED IS 35mm2/PHASE.

E. CUSTOMER IS RESPONSIBLE TO ARRANGE FOR WP, OR AN APPROPRIATELY ACCREDITED ELECTRICAL CONTRACTOR TO INSTALL DIRECT METERING IN THE SWITCHBOARD PRIOR TO SUPPLY BEING ENGERGISED.

F. INSTALLATION SHALL COMPLY WITH THE WAER CLAUSE 3.6.7 FOR MULTIPLE POINTS OF SUPPLY FOR RURAL / SEMI-RURAL LOTS. ALL REQUIREMENTS OF THE WAER CONDITIONS, WHICH INCLUDES 200m SEPARATION FROM EACH POINT OF SUPPLY AND 50m SEPARATION FROM EACH SUPPLIES ELECTRICAL INSTALLATION SHALL BE UNDERTAKEN PRIOR TO REQUESTING ENERGISATION. PLEASE ENSURE THE ELECTRICAL CONTRACTOR IS AWARE OF THE REQUIREMENTS.

TYPICAL PILLAR EXCLUSION ZONE

NTS



500mm to either side boundary

NO NATIVE VEGETATION MAY BE DISTURBED WITHOUT A PERMIT OR OUTSIDE OF THE PERMIT AREA. IF IN DOUBT, PLEASE CONTACT THE RELEVANT MANAGER WHO WILL CONTACT WESTERN POWER ENVIRONMENTAL TEAM.

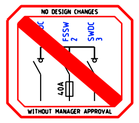
PLEASE NOTE: HEAVY PENALTIES MAY APPLY TO INDIVIDUALS AND BUSINESSES FOR DAMAGE TO NATIVE VEGETATION WITHOUT A PERMIT.

COMMUNICATION CABLES IN VICINITY OF WORKS

ISSUED FOR CONSTRUCTION

CRITICAL RISK MANAGEMENT

☒	☒	☒	☒	☒	☒	☐	☒	☐



SAFETY - MAKE IT YOUR PRIMARY VALUE

DESIGNED TO BE CONSTRUCTED DE-ENERGISED UNLESS RISK ASSESSED

CUSTOMER CONTACT DETAILS

COMPANY NAME: Swintec Pty Ltd

CONTACT: GRAHAM MACNISH TEL: 0439091471

Rev.	REVISIONS	DRN	DATE
A	ISSUED FOR CONSTRUCTION	I.P.	10/25

ENVIRONMENTAL ASSESSMENT
HAZARD MANAGEMENT REGISTER
FILE REFERENCE DETAILS

5kVA STANDARD RURAL SUPPLY

NETWORK VOLTAGE: 12.7kV/240V

L303/#219 SEVEN HILLS RD, ROELANDS WA 6226

DESIGN DRAWING

DESIGNER: CUSTOMER CONNECTION DESIGN
DESIGN TEAM: CCS RESI. SOL - RASP
EMAIL: ccs.design.resi.solution@westernpower.com.au
TEL: 13 10 87
DATE: 10/25 SCALE: AS NOTED @ A2
GEO REF: Lat. - 33° 18' 57" S Long. - 115° 50' 27" E

westernpower

DRAWING NUMBER

SP052694

SHEET 01 OF 01

REV: A ORIGINAL SIZE - A2